

Global Structure of Minor-Free Graphs and Beyond

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Abstract

This thesis explores the global structure of minor-free graphs through the lenses of graph products and Assouad–Nagata dimension. We also investigate graphs that are metrically similar to minor-free graphs.

Regarding graph products, we are especially interested in representing n -vertex K_h -minor-free graphs as $O_h(\sqrt{n})$ -blowups of bounded treewidth graphs. Such results qualitatively strengthen a famous result of Alon, Seymour, and Thomas (1990), which says that these graphs have treewidth $O_h(\sqrt{n})$. This thesis contributes to this study by showing that these graphs are $O_h(\sqrt{n})$ -blowups of treewidth 3 graphs. This improves upon existing results by Illingworth, Scott, and Wood (2024) and Distel, Dujmović, Eppstein, Hickingbotham, Joret, Micek, Morin, Seweryn, and Wood (2024), who showed that these graphs are $O_h(\sqrt{n})$ -blowups of treewidth $h - 2$ graphs and treewidth 4 graphs respectively. Allowing the blowup factor to include a polylog term, we further decrease the treewidth to 2. In fact, we show that these graphs are $O_h(\sqrt{n} \log(n)^2)$ -blowups of a (singular) fan, which has pathwidth 2 and gives a universal structure for n -vertex K_h -minor-free graphs.

We show that the Assouad–Nagata dimension of K_h -minor-free graphs is 2, dropping to 1 if and only if the graphs also have bounded treewidth. This improves on a result by Bonamy, Bousquet, Esperet, Groenland, Liu, Pirot, Scott (2024), who showed that the same result but with asymptotic dimension rather than Assouad–Nagata dimension. We also classify when a subdivision-closed class has bounded Assouad–Nagata dimension, being exactly when it is minor-free.

Regarding graphs that are metrically similar to minor-free graphs, we are particularly interested in graphs that are quasi-isometric to graphs of bounded treewidth. This thesis presents a new method for constructing a quasi-isometry between graphs of bounded cliquewidth and graphs of bounded treewidth. The existence of such a quasi-isometry is implicit in the work of Nguyen, Scott, and Seymour (2025) and Hickingbotham (2025), however ours is substantially improved. We reduce the coarseness factor to a constant (3), and for cliquewidth k , reduce the treewidth from $4k + 4$ to $k - 1$. In fact, we prove a stronger result, namely that these graphs admit partitions whose quotients have treewidth $k - 1$ and such that each part is contained within the neighbourhood of a vertex. We further show that the treewidth bound is within 2 of being tight, as there exists graphs of cliquewidth k that are not quasi-isometric to graphs of treewidth $k - 3$.

This inspires a characterisation of graphs that are quasi-isometric to graphs of treewidth k . We show that a graph admits such a quasi-isometry if and only if it admits a partition into parts of bounded weak diameter whose quotient has treewidth k . The primary contribution of this characterisation is a structural description of graphs that are known to admit such a quasi-isometry. This complements a known characterisation of these graphs by Nguyen, Scott, and Seymour (2025) and Hickingbotham (2025), whose primary contribution is to construct a quasi-isometry.

Declaration

This thesis is an original work of my research and contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, or any use of generative artificial intelligence technologies, except where due reference is made in the text of the thesis.

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Chapter 1

Introduction

A graph consists of a set of vertices, representing objects, and edges between pairs of vertices, representing connections between these objects. Graphs are used to model networks, such as road networks, transportation networks, social networks, computer networks, and many more. They can also be used to model complex problems such as disease transmission and molecular interactions.

1.1 Overview

Graph minors are a type of substructure within a larger graph. Thus, a minor-free graph excludes a particular substructure, and is therefore limited in its overall shape. See Section 1.5 for the definition. Minor-free graphs include many well-studied classes of graphs, and are of significant interest. Examples include bounded treewidth graphs, planar graphs, and graphs of bounded Euler genus. (See Section 1.7 for the definitions of treewidth and tree-decompositions, and Section 1.6 for the definitions of embedded graphs, Euler genus, and planar graphs.)

This thesis studies the structure of minor-free graphs. Specifically, we are interested in the global structure of such graphs. To this end, we consider parameters that describe these graphs at large scale, and are less interested in local details or obstructions. We study two specific perspectives, namely graph products and Assouad–Nagata dimension.

The product of two (or more) graphs is a graph that contains the graphs in the product as orthogonal substructures. See Section 2.1 for the definition. We seek to describe a minor-free graph as a subgraph of the product of ‘simple’ graphs. We study graph products from two specific angles, namely row-treewidth and graph blowups. The

former describes a graph as a subgraph of the product of a path and a graph of small treewidth. The latter describes a graph as a subgraph of the product of some base graph and a clique of relatively small size. We have specific interest in the case when the base graph has small treewidth. See Section 2.1.1 and Section 2.1.2 respectively for an exploration of known results on these topics.

This thesis contributes two results to the study of graph blowups. Specifically, we describe minor-free graphs as blowups of fans and as blowups of treewidth 3 graphs. The former is proven in Chapter 6, whereas the proof of the latter is spread across Chapter 6 and Chapter 7.

Assouad–Nagata dimension is an integer-valued measure of the large-scale shape of a graph that attempts to generalise classical notions of dimension. It can be thought of as a colouring problem, where the colour classes are divided into clusters such that different clusters of the same colour can be made arbitrarily far apart, while maintaining that the clusters are also small (in terms of distances). See Section 2.2 for the definition, and a study of known results regarding Assouad–Nagata dimension and a related parameter ‘asymptotic dimension’.

This thesis contributes to this study by showing that minor-free graphs have Assouad–Nagata dimension 2, and that bounded treewidth graphs have Assouad–Nagata dimension 1. The proof is the subject of Chapter 5. This result was also independently proven by Liu [101].

One of the core tools that enables our study of minor-free graphs is the ‘Graph Minor Structure Theorem’ of Robertson and Seymour [121] (Theorem 1.2 in this thesis). This describes minor-free graphs using several building blocks, which include graphs on surfaces and tree-decompositions. See Section 1.9 for a precise description of these building blocks.

Dujmović, Joret, Micek, Morin, Ueckerdt, and Wood [58] showed that some of these building blocks can be expressed in terms of row-treewidth, leading to a variation of the Graph Minor Structure Theorem. This thesis uses this variant to prove that minor-free graphs are blowups of fans. Bonamy, Bousquet, Esperet, Groenland, Liu, Pirot, and Scott [22] also used a very similar variant (that uses a parameter called ‘layered treewidth’) to prove a result about the asymptotic dimension of minor-free graphs.

Row-treewidth was also relevant to a result of Distel, Dujmović, Eppstein, Hickingbotham, Joret, Micek, Morin, Seweryn, and Wood [45], who described minor-free graphs as blowups of treewidth 4 graphs. (This result is the precursor to the treewidth 3 blowup

result presented in this thesis.) Specifically, they used the Graph Minor Structure Theorem, along with a structural description of graphs on surfaces proven by Distel, Hickingbotham, Huynh, and Wood [49] that was originally used to show that graphs on surfaces have row-treewidth at most 3.

This thesis also uses the Graph Minor Structure Theorem to prove our results about the Assouad–Nagata dimension and treewidth 3 blowup structure of minor-free graphs. For the former, we also use a result of Bonamy et al. [22] about the Assouad–Nagata dimension of graphs on surfaces, and for the latter, we use a (variant of a) result of Distel et al. [45] about the treewidth 2 blowup structure of graphs on surfaces.

We also study several non-minor-free graphs that resemble minor-free graphs. In particular, we consider two ways a graph can globally resemble a minor-free graph, namely quasi-isometry and fat minors.

Quasi-isometry measures how similar two graphs are with respect to their metrics. See Section 2.3.1 for the definition. Intuitively, we expect graphs that are similar to minor-free graphs to still behave like minor-free graphs at a global scale, when the local details and differences are less important. Thus, the study of the global structure of minor-free graphs also provides insight to the global structure of such graphs.

Quasi-isometry is strongly related to Assouad–Nagata (and asymptotic) dimension, since if two classes of graphs are related by quasi-isometry, then the Assouad–Nagata dimensions of the classes are tied (and the same for the asymptotic dimension). See Observation 2.21. This provides a method for bounding the Assouad–Nagata dimension of a class, using the known Assouad–Nagata dimension of a class related by quasi-isometry.

Fat minors are a way of identifying a minor substructure that is global. A graph that excludes a fat minor can have an undesirable substructure, but only locally. See Section 2.3.2 for more details. It is natural to expect that these local substructures should not dramatically change the global structure of a graph. Thus, one may expect these graphs to have similar structure to minor-free graphs. In Section 2.3.3, we explore the ways this intuition succeeds, but also areas of difficulty. In particular, we investigate conjectures related to whether these graphs are quasi-isometric to minor-free graphs, and whether these graphs have bounded Assouad–Nagata dimension.

Of particular interest in this thesis are graphs that are similar to graphs of bounded treewidth. We explore such notions of similarity in Section 2.4. These include graphs of bounded cliquewidth, graphs that are quasi-isometric to graphs of small treewidth, and graphs that admit tree-decompositions where each bag is the union of a few bounded

weak-diameter sets (see Section 1.4 for the definition of weak-diameter). These types of graphs are already known to be related by work of other authors [84, 108]. This thesis builds upon this relationship by constructing an improved quasi-isometry between bounded cliquewidth graphs and bounded treewidth graphs. This is the subject of Chapter 3. The existence of this quasi-isometry (which was first implicitly shown by Hickingbotham [84]) implies that bounded cliquewidth graphs have Assouad–Nagata dimension 1, by the properties of quasi-isometry and our result about the Assouad–Nagata dimension of bounded treewidth graphs.

We also characterise graphs that are quasi-isometric to graphs of bounded treewidth. Specifically, we show that a graph admits such a quasi-isometry if and only if it admits a partition into nonempty parts of bounded weak diameter whose quotient has bounded treewidth. (See Section 1.8 for definitions related to partitions.) This contrasts with a known characterisation by Nguyen, Scott, and Seymour [108] and Hickingbotham [84], whose primary purpose is to construct a quasi-isometry. As such, these characterisations are complementary. See Sections 2.4.1 and 2.4.3 for more detail on these characterisations. The proof of our characterisation is the subject of Chapter 4. The proof takes inspiration from a concept by Distel, Hickingbotham, Seweryn, and Wood [50], which was used to give an improved bound on the row-treewidth of (g, k) -planar graphs (see Section 1.6 for the definition of (g, k) -planar graphs).

Our characterisation is inspired by a common method for constructing a quasi-isometry. It is well-known that partitions into nonempty parts of bounded weak diameter give rise to a quasi-isometry to the quotient. This method was used by Albrechtsen, Distel, and Georgakopoulos [2] to show a quasi-isometry between graphs that exclude $K_{2,t}$ as a fat minor and graphs that exclude $K_{2,t}$ as a minor, and by Hickingbotham [84] to show the original characterisation of graphs that are quasi-isometric to graphs of bounded treewidth. It is also used in this thesis to find our improved quasi-isometry between graphs of bounded cliquewidth and graphs of bounded treewidth. Our characterisation says that in the case of quasi-isometries to bounded treewidth graphs, this method suffices, in the sense that if a such quasi-isometry exists, then there is also a (possibly worse) quasi-isometry using only this method.

The remainder of this chapter is devoted to (relatively) basic definitions and concepts, along with the statement of the Graph Minor Structure Theorem. We cover the key concepts and formally introduce the main results of this thesis in Chapter 2

1.2 Preliminary Definitions

Let $\mathbb{N} := \{0, 1, 2, \dots\}$. For $c \in \mathbb{R}$, define $\mathbb{N}^{>c} := \mathbb{N} \cap [c, \infty)$, $\mathbb{R}^{>c} := \mathbb{R} \cap [c, \infty)$, and $\mathbb{R}^{>c} := \mathbb{R} \cap (c, \infty)$.

A ‘graph’ refers to a finite, undirected, and unweighted graph without parallel edges or loops. All definitions are given as such, although most extend naturally to infinite graphs, pseudographs, and weighted graphs. Indeed, Chapters 5 and 7 require weighted graphs and pseudographs. We mention any relevant changes to definitions when needed.

We use standard graph-theoretic definitions. See Diestel [36] for undefined terms.

The empty graph is considered connected. Walks and paths are considered to be a sequence of vertices rather than a subgraph. The ‘edges’ of a walk/path refers to the edges between consecutive vertices in the walk/path. Consecutive vertices in a walk are allowed to be identical. The *interior* of a walk/path is the subpath/subwalk consisting of all vertices except the endpoints.

Say that a graph H is *contained* in a graph G if H is isomorphic to a subgraph of G .

A set of vertices S is a *clique* in a graph G if the vertices in S are pairwise adjacent in G . The empty set is considered to be a clique. The *clique number* of G is the maximum size of a clique in G . The *degeneracy* of G is the minimum $k \in \mathbb{N}$ such that there exists a total ordering $\preceq$ of the vertices of G such that each $v \in V(G)$ is adjacent to at most k vertices $u \in V(G)$ with $u \prec v$.

Given an edge uv in a graph G , to *subdivide* uv $n \in \mathbb{N}$ times is to replace uv with a path of length $n + 1$ (n internal vertices) whose endpoints are u and v and whose internal vertices are new. The *n -subdivision* of G is the graph obtained from G by subdividing each edge exactly n times, and a *$\geq n$ -subdivision* is any graph obtained from G by subdividing each edge at least n times. A graph is a *subdivision* of G if it is a ≥ 0 -subdivision of G .

The *closed neighbourhood* of $S \subseteq V(G)$ in a graph G , denoted $N_G(S)$, is the set of vertices $v \in V(G)$ such that either $v \in S$, or there exists $s \in S$ such that $vs \in E(G)$. (The closed neighbourhood is often denoted in the literature as $N_G[S]$, with $N_G(S)$ used to mean the open neighbourhood, the set of vertices not in S but adjacent to some $s \in S$. This thesis never refers to the open neighbourhood, so this notation is used to be consistent with the ℓ -neighbourhood; see Section 1.4.) If $S = \{v\}$, we instead write $N_G(v)$. S is *dominated* in G if $S \subseteq N_G(v)$ for some $v \in V(G)$. Say that disjoint $S, T \subseteq V(G)$ are *adjacent* in G if $N_G(S) \cap T \neq \emptyset$ (equivalently, $N_G(T) \cap S \neq \emptyset$).

A *separation* of a graph G is a pair of subgraphs (A, B) of G such that $G = A \cup B$. The set $V(A \cap B)$ is called the *separator* of the separation. For $X, Y, S \subseteq V(G)$, say that S *separates* X from Y in G if there exists a separation (A, B) of G such that $X \subseteq V(A)$, $Y \subseteq V(B)$, and $S \subseteq V(A \cap B)$. Equivalently, every path in G from X to Y intersects S .

A *star* is a tree with at most one vertex of degree at least 2. The *centre* of a star is the unique vertex of degree at least 2, provided such a vertex exists. Otherwise, we consider each vertex to be a centre. Note that the centre is uniquely defined unless the star has exactly two vertices. A *spider* is a tree with at most one vertex of degree at least 3 (equivalently, a subdivision of a star). A *fan* is a graph G for which there exists a vertex v (called a *centre*) such that $V(G) \subseteq N_G(v)$ and $G - v$ is a path.

Fix $d \in \mathbb{N}$, and let $n_1, \dots, n_d \in \mathbb{N}$. The $n_1 \times n_2 \times \dots \times n_d$ -*grid* is the graph with vertex set $\{1, \dots, n_1\} \times \{1, \dots, n_2\} \times \dots \times \{1, \dots, n_d\}$ and edges between two vertices if they differ by exactly 1 in exactly one co-ordinate, and are the same in every other co-ordinate. For $m \in \mathbb{N}$, the (d, m) -*grid* is the $n_1 \times n_2 \times \dots \times n_d$ -grid with $n_1 := n_2 := \dots := n_d := m$. A *d-dimensional grid* is a (d, m) -grid for some $m \in \mathbb{N}$. A ‘grid’, without specifying d , means a 2-dimensional grid.

A *class* of graphs $\mathcal{G}$ is a collection of graphs that is closed under isomorphism. $\mathcal{G}$ is *proper* if it is not the class of all graphs. $\mathcal{G}$ is *hereditary* if $\mathcal{G}$ is closed under taking induced subgraphs. $\mathcal{G}$ is *subdivision-closed* if every subdivision of a graph in $\mathcal{G}$ is contained in $\mathcal{G}$.

1.3 Colourings

Let G be a graph and $S \subseteq V(G)$. A *colouring* of S with *colours* C is a map $c : S \mapsto C$. Say each vertex $v \in S$ is *coloured* $c(v)$ under c . Say that a colour $i \in C$ is *used* by c if at least one vertex of S is coloured i by c (the pre-image of i under c is nonempty).

A (vertex) *colouring* of a graph G (with colours C) is a colouring c of $V(G)$ (with colours C). For $m \in \mathbb{N}$, an m -colouring of G is a colouring of G with the colours $\{1, \dots, m\}$. Note that for $n \in \mathbb{N}^{\geq m}$, we can consider an m -colouring to be an n -colouring.

For $S \subseteq T \subseteq V(G)$, a colouring c_T of T *extends* a colouring c_S of S if c_S is the restriction of c_T to S . Note that if $S_1, \dots, S_n \subseteq V(G)$, and c_i is a colouring of S_i for each $i \in \{1, \dots, n\}$ such that $c_i = c_j$ on $S_i \cap S_j$ for any $i, j \in \{1, \dots, n\}$, then there exists a colouring c_G of G that extends $c_1, \dots, c_n$ simultaneously. Furthermore, if $\bigcup_{i=1}^n S_i = V(G)$, then this colouring c_G is unique, and is denoted by $\bigcup_{i=1}^n c_i$.

Under a colouring c of a graph G , for any colour i , a set of vertices $S \subseteq V(G)$ is said to be *i -monochromatic* under c if every vertex of S received the colour i under c . S is *monochromatic* under c if it is i -monochromatic under c for some colour i . Say that S is *rainbow* under c if each colour used by c is used by $c|_S$. That is, if there is a vertex of G coloured some colour i , then there is also a vertex of S coloured i .

1.4 Graphs Metrics

The *length* of a path P in a graph G is the number of edges in P (number of vertices minus 1). P is *shortest* if every other path with the same endpoints has length at least the length of P . Since G is finite, if two vertices are in the same connected component then a shortest path between them exists. The *distance* between vertices $u, v \in V(G)$, denoted $\text{dist}_G(u, v)$, is the length of a shortest path between them, or ∞ if no path exists.

The *distance* between two nonempty sets $S, T \subseteq V(G)$ in a graph G , denoted $\text{dist}_G(S, T)$, is the minimum distance between some $s \in S$ and $t \in T$ (this minimum exists, since G is finite). If T is a singleton $\{v\}$, we instead write $\text{dist}_G(v, S)$. For $\ell \in \mathbb{R}^{\geq 0}$, $S, T \subseteq V(G)$ are *ℓ -disjoint* in G if $\text{dist}_G(S, T) > \ell$.

For $\ell \in \mathbb{R}^{\geq 0}$, the *ℓ -neighbourhood* of $S \subseteq V(G)$ in a graph G , denoted $N_G^\ell(S)$, is $\{v \in V(G) : \text{dist}_G(v, S) \leq \ell\}$. Note that $N_G^\ell(S)$ is empty if $\ell < 0$, and that $S \subseteq N_G^\ell(S)$ if $\ell \geq 0$. If $S = \{v\}$ for some $v \in V(G)$, we instead write $N_G^\ell(v)$. For $k \in \mathbb{N}$, we say that $Z \subseteq V(G)$ is *(k, ℓ) -centred* in G if there exists $S \subseteq V(G)$ with $|S| \leq k$ such that $Z \subseteq N_G^\ell(S)$. We call S a *(k, ℓ) -centre* of Z .

The *radius* of a graph G is the minimum ℓ for which there exists a $v \in V(G)$ such that v is at distance at most ℓ from each $x \in V(G)$. Equivalently, $V(G)$ is a $(1, \ell)$ -centred set in G (with $\{v\}$ being a $(1, \ell)$ -centre of $V(G)$). The *diameter* of G is the maximum distance between any pair of vertices in G . For $S \subseteq V(G)$, the *weak diameter* of S in G is the maximum distance between any pair of vertices in G (as measured in G , not $G[S]$). The weak diameter of S in G can be arbitrarily smaller than the diameter of $G[S]$. For example, a path can have arbitrarily large diameter, but its vertices have weak diameter 2 within a fan.

For a graph G , a graph H with $V(H) \subseteq V(G)$ is *isometric* in G if $\text{dist}_H(u, v) = \text{dist}_G(u, v)$ for all $u, v \in V(H)$. An *isometric subgraph* of G is a subgraph H of G that is isometric in G .

For $\ell \in \mathbb{R}^{\geq 0}$ and a graph G , a collection $\mathcal{C}$ of subsets of $V(G)$ is *ℓ -disjoint* in G if the

sets in $\mathcal{C}$ are pairwise ℓ -disjoint. The ℓ -th power of G , denoted G^ℓ , is the graph with vertex set $V(G)$ where distinct $u, v \in V(G)$ are adjacent if $\text{dist}_G(u, v) \leq \ell$. An ℓ -walk [respectively ℓ -path] P from $u \in V(G)$ to $v \in V(G)$ in G is a sequence of vertices of G that form a walk [respectively path] from u to v in G^ℓ . This is equivalent to requiring that consecutive vertices in the walk [respectively path] are at distance at most ℓ . Recall that we permit walks, and thus ℓ -walks, to have equal consecutive vertices. Note that if the interior of an ℓ -path P has weak diameter in G at most d , then P has weak diameter in G at most $d + 2\ell$.

For $Z \subseteq V(G)$, an ℓ -subwalk [respectively ℓ -subpath] of $P \setminus Z$ is a subwalk [respectively subpath] of P (as a walk/path in G^ℓ) disjoint to Z . In particular, an ℓ -subwalk [respectively ℓ -subpath] of P is subwalk [respectively subpath] of P (as a walk/path in G^ℓ).

1.5 Graph Minors

To *contract* an edge uv in a graph G is to delete u and v from G and add a new vertex adjacent to the vertices in $N_G(\{u, v\}) \setminus \{u, v\}$. A graph H is a *minor* of G if a graph isomorphic to H can be obtained from G by performing any sequence of contractions, vertex deletions, and edge deletions. This is equivalent (up to isomorphism) to first taking some subgraph of G , and then only performing contractions. If H is a minor of G , we write $H \leq G$. Otherwise, we say that G *excludes* H as a minor, or that G is *H -minor-free*. Note that if $H \subseteq G$, then $H \leq G$.

By concatenating the sequence of operations performed to find a minor, it follows that the relation $\leq$ is transitive. Thus, if a graph G excludes a graph H as a minor, then G excludes any graph H' with $H \leq H'$ as a minor. In particular, G excludes $K_{|V(H)|}$ as a minor (since $H \subseteq K_{|V(H)|}$). In this thesis, when we discuss minor-free graphs, we are usually not particularly interested in the structure of H , beyond knowing that it is a fixed graph. Thus, it suffices to only consider excluding some complete graph as a minor.

An important result about minor-free graphs is that they have a precise structural description due to Robertson and Seymour [121]. This description enables the study of these graphs in this thesis. See Section 1.9 for more details.

A class of graphs $\mathcal{G}$ is *minor-closed* if every minor of a graph in $\mathcal{G}$ is contained in $\mathcal{G}$. $\mathcal{G}$ is *H -minor-free* (or *excludes H as a minor*) if each $G \in \mathcal{G}$ is H -minor-free. If $\mathcal{G}$ is minor-closed and H is a graph with $H \notin \mathcal{G}$, then $\mathcal{G}$ excludes H as a minor. Thus, every

proper minor-closed class excludes some graph H as a minor, and thus some complete graph $(K_{|V(H)|})$ as a minor.

A *model* of a graph H in a graph G is a collection $((X_v : v \in V(H)), (P_{uv} : uv \in E(H)))$, where $(X_v : v \in V(H))$ is a collection of pairwise disjoint nonempty subsets of $V(G)$ that induce connected subgraphs of G , and $(P_{uv} : uv \in E(H))$ is a collection of pairwise internally disjoint paths such that, for each $uv \in E(H)$, P_{uv} has its endpoints in X_u and X_v and is disjoint to X_w for $w \in V(G) \setminus \{u, v\}$. The X_v , $v \in V(H)$, are called *branch sets*, and the P_{uv} , $uv \in E(H)$, are called *branch paths*.

For each $uv \in E(H)$, we can find a subpath P'_{uv} of P_{uv} that is internally disjoint to X_u and X_v but has endpoints in X_u and X_v . By adding the interior of P'_{uv} to X_u and updating P_{uv} to only consist of the endpoints of the final edge, we obtain a model where X_u is adjacent to X_v . Repeating this for each edge of H , we find a model where branch sets corresponding to adjacent vertices of H are adjacent in G . We refer to such a model as a *proper model*, and we omit the branch paths when referring to these models. So a proper model is a set $(X_v : v \in V(H))$ of nonempty subsets (*branch sets*) of $V(G)$ that induce connected subgraphs of G , are pairwise disjoint, and are adjacent whenever $uv \in E(H)$. As noted, a proper model exists if and only if a model exists. Proper models are often taken to be the definition of a model. However, we prefer to include the branch paths, because they will be relevant to an extension of minors called fat minors. See Section 2.3.2.

By contracting each branch set to a single vertex, we find that if G admits a (proper) model of H , then $H \leq G$. Similarly, by reversing the sequence of contractions, if $H \leq G$, then G admits a (proper) model of H . Thus, admitting a model of H is equivalent to having H as a minor. In this way, a model identifies the structure of a minor inside a graph.

1.6 Graphs on Surfaces

This subsection briefly introduces graphs embedded on surfaces. See Mohar and Thomassen [106] for more details, proofs, and other standard definitions.

The *Euler genus* of a surface Σ with h handles and s crosscaps is $2h + s$. Recall that the sphere $\mathbb{S}^2$ minus a point is homeomorphic to the plane $\mathbb{R}^2$. As such, we will usually treat the plane as being the surface with Euler genus 0 (we will explain briefly how to convert any definitions when required).

A *drawing* σ of a graph on a surface Σ (or the plane $\mathbb{R}^2$) is a pair (σ_V, σ_E) where:

1. σ_V is an injection from $V(G)$ to Σ ,
2. σ_E is a map from $E(G)$ to 2^Σ such that for each $uv \in E(G)$, $\sigma_E(uv)$ is a curve from $\sigma_V(u)$ to $\sigma_V(v)$ in Σ that does not intersect itself or $\sigma_V(V(G) \setminus \{u, v\})$.

For technical reasons, we further require that for any pair of distinct $e, e' \in E(G)$, $|\sigma_E(e) \cap \sigma_E(e')| < \infty$, and that for any triple of distinct $e, e', e'' \in E(G)$, the interiors of the curves $\sigma_E(e)$, $\sigma_E(e')$, and $\sigma_E(e'')$ do not share a common point of intersection.

We use $\sigma(G)$ to denote $\sigma_V(V(G)) \cup \bigcup_{e \in E(G)} \sigma_E(e)$. A *crossing* between two distinct edges $uv, xy \in E(G)$ is a point in the intersection of $\sigma_E(uv) \setminus \{u, v\}$ and $\sigma_E(xy) \setminus \{x, y\}$; that is, a point of intersection between the interiors of $\sigma_E(uv)$ and $\sigma_E(xy)$. uv and xy *cross* if there is a crossing between them. A point on the interior of $\sigma_E(uv)$ is a *crossing* of uv if it is a crossing between uv and some $xy \in E(G) \setminus \{uv\}$.

Since G is finite, $\Sigma \setminus \sigma(G)$ is nonempty. Thus, any drawing on the sphere can be converted to a drawing on the plane that preserves crossings.

An *embedding* of G into Σ is a drawing where no pair of distinct edges cross. We refer to the pair (G, σ) as a graph *embedded* in Σ . By abuse of notation, we often drop the embedding $\sigma = (\sigma_V, \sigma_E)$ when referring to a graph embedded in Σ , using ‘ G ’ to refer to the pair ‘ (G, σ) ’. When we do this, we treat the graph G as if it is a subset of Σ . That is, we identify each $v \in V(G)$ with $\sigma_V(v)$, and each $e \in E(G)$ with $\sigma_E(e)$. We refer to σ as the *embedding* of G , if we need to explicitly identify it. We also define an *embedded graph* to be a graph embedded in some surface Σ , and refer to Σ as the *surface* G is embedded in, if we need to explicitly identify it. A *plane graph* is a graph embedded in the plane (equivalently up to homeomorphism, the sphere). An *embedded subgraph* of an embedded graph G is a subgraph H of G embedded in the same surface as G whose embedding is the restriction of the embedding of G to H . That is, if (σ_V, σ_E) is the embedding of G , then the embedding of H is $(\sigma_V|_{V(H)}, \sigma_E|_{E(H)})$. If G is a plane graph, we instead refer to H as a *plane subgraph*.

A *face* F of a graph G embedded in a surface Σ is a connected component of $\Sigma \setminus G$. The *boundary vertices* and *boundary edges* of F are the vertices and edges of G respectively that intersect the closure of F . We remark the entirety of each boundary edge is contained in the closure of F .

A graph G is *embeddable* in a surface Σ if there exists an embedding of G into Σ . The *Euler genus* of a graph G is the minimum $g \in \mathbb{N}$ such that g is embeddable in a surface Σ of Euler genus at most g . The Euler genus of G is at most $2|E(G)|$, since we can take any injection from $V(G)$ to $\mathbb{S}^2$ (of which, at least one exists, since G is finite), and then add a handle for every edge that attaches close to the endpoints of the edge. A graph

is *planar* if it has Euler graph 0; that is, it can be embedded in the plane.

Let G be a graph embedded in a surface Σ . Using the restriction of the embedding of G , each subgraph H of G is embeddable in Σ . Further, if G' is obtained from G by contracting an edge xy into a vertex v , by embedding v on xy and curving all the edges incident to x and y to follow xy to v , we can embed G' in Σ . It follows that the class of graphs embeddable in a surface Σ is minor closed. Thus, the class of graphs of Euler genus at most $g \in \mathbb{N}$ (and the class of planar graphs) is minor-closed.

Every n -vertex graph of Euler genus g has at most $3(|V(G)| + g - 2)$ edges, and at most $2(|V(G)| + g - 2)$ edges if it is bipartite. (This is derived from Euler's formula and a simple double counting argument for the number of edges per face and faces per edge.) Combined with the fact that Euler genus does not increase under taking minors, we find that each graph of Euler genus at most g excludes $K_{3,2g+3}$ as a minor. Further, each planar graph excludes K_5 and $K_{3,3}$ as minors. It was famously shown by Kuratowski [96] and Wagner [127] that this property characterises planar graphs.

Theorem 1.1 (Kuratowski–Wagner). *A graph is planar if and only if it excludes K_5 and $K_{3,3}$ as minors.*

In Section 1.9, we discuss the Graph Minor Theorem of Robertson and Seymour [122], which characterises a proper minor-closed class in a manner analogous to Theorem 1.1.

An *apex* graph is a graph G such that, for some $v \in V(G)$, $G - v$ is planar. For $g, k \in \mathbb{N}$, a graph is *(g, k) -planar* if it can be drawn in a surface of Euler genus at most g such that each edge has at most k crossings. In particular, each edge crosses at most k other edges. The $g = 0$ case is called a *k -planar* graph (so it can be drawn in the plane with at most k crossings per edge). The $k = 0$ case is a graph of Euler genus g .

It is natural to ask if (g, k) -planar graphs have an excluded minor structure, similar to how graphs of Euler genus g excluded $K_{3,2g+3}$ as a minor. However, one can take any graph G and any drawing of G in the plane (of which at least one exists), and then subdivide each edge more times than it has crossings. The resulting subdivided graph G' can be drawn such that each edge has at most one crossing. So G' is 1-planar, but by contracting the subdivided edges, we find that G is a minor of G' . So 1-planar graphs have no excluded minors. This extends to (g, k) -planar graphs whenever $k \in \mathbb{N}^{\geq 1}$, since any 1-planar graph is also (g, k) -planar.

1.7 Tree-Decompositions

This section introduces graph-decompositions. These are well-studied, and a certain amount of familiarity with them is assumed. As such, we only briefly touch on the key ideas of the proofs for many basic facts and properties of these decompositions. We direct the reader to Bodlaender [17] and Diestel [36] for further details and proofs of these results and many other basic results.

For graphs G and H , a *H -decomposition* of G is a collection $\mathcal{J} = (J_x : x \in V(H))$ of subsets of $V(G)$ such that:

1. for each $v \in V(G)$, the subgraph of H induced by the vertices $x \in V(H)$ with $v \in J_x$ induce a connected and nonempty subgraph of H , and
2. for each edge $uv \in E(G)$, there exists $x \in V(H)$ such that $u, v \in J_x$.

The sets J_x , $x \in V(H)$, are called *bags*. The *width* of $\mathcal{J}$ is the maximum size of a bag minus 1. The *H -width* of G is the minimum width of an H -decomposition of G .

A *tree-decomposition* of a graph G is a T -decomposition of G for some tree T . A tree-decomposition is a *path-decomposition* or a *star-decomposition* respectively if T is a path or a star respectively. For star-decompositions, a *centre* of a star-decomposition is a centre of the star T . The *treewidth* and *pathwidth* of G , denoted $\text{tw}(G)$ and $\text{pw}(G)$ respectively, are the minimum width of a tree-decomposition and a path-decomposition of G respectively.

While graph minors identify a precise substructure of a graph (using a model), tree-decompositions identify a much looser tree-like structure, and the width is in some ways the error of this structure. So treewidth measures how far the graph is from a tree. A connected graph has treewidth 1 if and only if it is a tree. Since cycles have K_3 -minors, it follows that a graph has treewidth at most 1 if and only if it excludes K_3 as a minor. Further, a graph has treewidth at most 2 if and only if it is K_4 -minor-free [15].

One might naively expect this connection between being complete-minor-free and having bounded treewidth to continue; however, it fails at K_5 . Grids are planar and are thus K_5 -minor-free by the Kuratowski–Wagner Theorem (Theorem 1.1 [96, 127]), but for each $n \in \mathbb{N}^{\geq 2}$, the $n \times n$ grid has treewidth exactly n . So even being K_5 -minor-free is not enough to bound the treewidth. However, the famous ‘Grid Minor Theorem’ of Robertson and Seymour [118] shows that grids are exactly the obstruction to having large treewidth, in the sense that excluding a grid as a minor bounds the treewidth. (This is equivalent to excluding a planar graph as a minor, since every planar graph is a minor of a sufficiently large grid [119, 124].) See also [57] for a qualitative strengthening

of the Grid Minor Theorem in terms of graph blowups, which we introduce in Section 2.1.

A famous result by Alon, Seymour, and Thomas [7] (generalising a result by Lipton and Tarjan [100] regarding planar graphs) shows that each n -vertex K_h -minor-free graph has treewidth $O_h(\sqrt{n})$. This (non-trivially) implies that these graphs also have pathwidth $O_h(\sqrt{n})$. (This uses the fact that being minor-free is closed under subgraphs, that bounded treewidth gives ‘balanced’ separators, and that $\sum_{i=1}^{\infty} \sqrt{\alpha^i n} = O(\sqrt{n})$ for any $\alpha \in (0, 1)$.)

Treewidth does not increase under taking a minor (or subgraph). Specifically, one can take a proper model and ask which bags intersect which branch sets to get a tree-decomposition of the minor. Therefore, the graphs of bounded treewidth form a (proper) minor-closed class, and exclude a complete graph as a minor. In particular, K_{b+1} has treewidth exactly b for each $b \in \mathbb{N}$ (one can show that there exists a bag containing all vertices of the clique). So graphs of treewidth b have clique number at most $b + 1$, and exclude K_{b+2} as a minor.

Starting from an arbitrary tree-decomposition $(J_t : t \in V(T))$ of width at most $k \in \mathbb{N}$ of a graph G , repeatedly contract each edge tt' with $J_t = J_{t'}$ into a vertex t'' , and set $J_{t''} := J_t = J_{t'}$. We find that the resulting collection $(J'_t : t \in V(T'))$ is a tree-decomposition of G of width at most k , but has the extra property that, for each leaf $t \in V(T)$, J'_t contains a vertex v not contained in any other bag. Thus, v has degree at most k . Since treewidth is closed under subgraphs, it follows that every graph of treewidth k has degeneracy at most k .

Tree-decompositions $(J_t : t \in V(T))$ of a graph G of unbounded width can still be useful. For $tt' \in E(T)$, the *adhesion set* at tt' is $J_t \cap J_{t'}$. If T is rooted at $r \in V(T)$ and t' is the parent of t , we call this adhesion set the *parent adhesion set* at t and a *child adhesion set* at t' . We consider the *parent adhesion set* at r to be empty. The *adhesion* of $(J_t : t \in V(T))$ is the maximum size of an adhesion set. This is also the maximum intersection between any two bags (including bags indexed by non-adjacent vertices of T). For $t \in V(T)$, the *torso* of G at t (with respect to $(J_t : t \in V(T))$), denoted $G\langle J_t \rangle$, is the graph obtained from $G[J_t]$ by, for each $t' \in V(T)$ with $tt' \in E(T)$ and each pair of distinct vertices $u, v \in J_t \cap J_{t'}$, adding the edge uv if it does not already exist. So the adhesion sets at t are (possibly empty) cliques in $G\langle J_t \rangle$.

These parameters will be relevant when we explain the structural characterisation of minor-free graphs given by Robertson and Seymour [121] in Section 1.9. In Section 2.4, we discuss coarse analogues of treewidth that are bounded even in a beyond-minor-free setting.

1.8 Partitions

A *partition* $\mathcal{P}$ of a graph G is a collection of disjoint subsets of $V(G)$ whose union is $V(G)$. The elements of $\mathcal{P}$ are called *parts*. For technical reasons, we allow empty parts. We also allow duplicate parts, so $\mathcal{P}$ is not a set. However, we remark that the only duplicate parts that can exist are empty. We say that a partition is *proper* if each part is nonempty. In this case, $\mathcal{P}$ is a set.

The *width* of a partition $\mathcal{P}$ of a graph G is the maximum size of a part of $\mathcal{P}$. $\mathcal{P}$ is *dominated* in G if each part is dominated in G . $\mathcal{P}$ is *connected* if each $P \in \mathcal{P}$ induces a connected subgraph of G . The *quotient* of G by $\mathcal{P}$, denoted $G/\mathcal{P}$, is the graph with vertex set $\mathcal{P}$ (allowing for multiple vertices to have the duplicate label $\emptyset$), where distinct parts are adjacent if and only if they are adjacent in G . Note that if $\mathcal{P}$ is connected, then $G/\mathcal{P}$ is a minor of G (the parts give a proper model). If $G/\mathcal{P}$ is contained in a path, then $\mathcal{P}$ is a *layering* of G , and the parts are called *layers*. Say that the *treewidth* of $\mathcal{P}$ is the treewidth of $G/\mathcal{P}$. Note that bags may contain duplicates labelled $\emptyset$, so they are no longer sets. However, this makes little difference, as empty parts are isolated in the quotient, and thus do not increase the treewidth (beyond 0).

Given a colouring c and partition $\mathcal{P}$ of a graph G , $\mathcal{P}$ is *monochromatic* under c if each part in $\mathcal{P}$ is monochromatic under c . Note that if $\mathcal{P}$ is monochromatic, then there is an *induced colouring* on the quotient $G/\mathcal{P}$, where each $P \in \mathcal{P}$ is coloured i if P is i -monochromatic under c .

1.9 The Graph Minor Structure Theorem

The ‘Graph Minor Structure Theorem’ of Robertson and Seymour [121] gives a structural description of K_h -minor-free graphs. We now work towards stating this result.

Given graphs H, U with $V(U) \subseteq V(H)$, say that a U -decomposition $(J_x : x \in V(U))$ of H is *planted* if $x \in J_x$ for each $x \in V(U)$.

Given a graph G embedded in a surface Σ , a closed disc D in Σ is *G -clean* if the interior of D is disjoint to G and the boundary of D is disjoint to the interior of each edge of G . A *G -clean disc* for some embedded graph G means a closed disc in the same surface that G is embedded in that is G -clean. For an embedded graph G and a G -clean disc D , let $B(D, G) := D \cap V(G)$. We call $B(D, G)$ the *boundary vertices* of D in G . We define the *underlying cycle* of D in G , denoted $U(D, G)$, to be the graph with vertex set $B(D, G)$ and edges between distinct $u, v \in B(D, G)$ if they appear consecutively

following the boundary of D . If $|B(D, G)| \geq 3$, then $U(D, G)$ is a cycle (hence the name), but if $|B(D, G)| \leq 2$, then $U(D, G)$ is instead a path. We say that D is *snug* in G if $U(D, G) \subseteq G$.

Following the boundary of D clockwise gives a cyclic ordering of $B(D, G)$, which we call the cyclic ordering of $B(D, G)$ *induced* by D . Observe that this cyclic ordering is precisely the order the vertices appear in $U(D, G)$ (up to orientation of the cycle and starting point).

For an embedded graph G and a G -clean disc D , a graph H is a (D, G) -*vortex* if $V(G \cap H) = B(D, G)$. (The vertices $V(G \cap H) = B(D, G)$ are often called *society vertices* in the literature.) A (D, G) -*vortex-decomposition* of a (D, G) -vortex H is a planted $U(D, G)$ -decomposition of H . The (D, G) -*vortex-width* of a (D, G) -vortex H is the minimum width of a (D, G) -vortex-decomposition of H . Note that if H has (D, G) -vortex-width k , then $\text{pw}(H) \leq 2k + 1$ (if $U(D, G)$ is a cycle, delete any edge to obtain a path, take one of the endpoints of that edge, and put all the vertices of its bag into every other bag). Further, if D is snug in G , observe that a (D, G) -vortex-decomposition of H is also a planted $G[B(U, G)]$ -decomposition of H of the same width.

An *almost-embedding* Γ for a graph G is a tuple $(\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s))$ such that:

1. Σ is a surface, and $s \in \mathbb{N}$,
2. $G_0, \dots, G_s$ are subgraphs of G ,
3. $A \subseteq V(G)$, and $G - A = \bigcup_{i=0}^s G_i$,
4. G_0 is embedded in Σ ,
5. $G_1, \dots, G_s$ are pairwise disjoint,
6. $D_1, \dots, D_s$ are pairwise disjoint G_0 -clean discs in Σ , and
7. for each $i \in \{1, \dots, s\}$, G_i is a (D_i, G_0) -vortex.

We call Σ the *surface*, G_0 the *embedded subgraph*, $G_1, \dots, G_s$ the *vortices*, the vertices in A the *apices*, and $D_1, \dots, D_s$ the *discs*. We call $\bigcup_{i=1}^s G_i$ the *vortex-union subgraph* of Γ , and denote it by $H(\Gamma)$. We call $\bigcup_{i=1}^s B(D_i, G) = \bigcup_{i=1}^s V(G_0 \cap G_i) = V(G_0 \cap H(\Gamma))$ the *boundary* of Γ , and denote it by $B(\Gamma)$. Say that Γ is *nontrivial* if $V(G) \neq V(A)$. Say that Γ is *snug* if for each $i \in \{1, \dots, s\}$, D_i is snug in G .

For $g, p, k, a \in \mathbb{N}$, an almost-embedding Γ is a (g, p, k, a) -*almost-embedding* if the following holds:

1. Σ has genus at most g ,
2. $s \leq p$,

3. $|A| \leq a$, and
4. for each $i \in \{1, \dots, s\}$, G_i has (D_i, G_0) -vortex-width at most k .

In this case, all the vortices have pathwidth at most $2k+1$. Since the vortices are pairwise disjoint, the vortex-union subgraph $H(\Gamma)$ has pathwidth (and treewidth) at most $2k+1$. Further, if the embedding is snug, $H(\Gamma)$ admits a planted $G_0[B(\Gamma)]$ -decomposition of width at most k , obtained from the merging the vortex-decompositions. Here, we use that the discs have pairwise disjoint boundary vertices, since the discs are disjoint.

A graph G is *(g, p, k, a) -almost-embeddable* if it admits a (g, p, k, a) -almost-embedding. We further say that G is *snugly (g, p, k, a) -almost-embeddable* if G admits a snug (g, p, k, a) -almost-embedding.

We can now state the Graph Minor Structure Theorem [121].

Theorem 1.2. *For each $h \in \mathbb{N}$, there exists $g, p, k, a, j \in \mathbb{N}$ such that each K_h -minor-free graph G admits a tree-decomposition of adhesion at most j such that each torso is (g, p, k, a) -almost-embeddable.*

We remark that the original bounds on g, p, k, a , and j shown by Robertson and Seymour [121] grow rapidly in h . However, these have since been dramatically improved by Kawarabayashi, Thomas, and Wollan [95] and then Gorsky, Seweryn, and Wiederrecht [79], the latter of which got the growth rate down to polynomial. Theorem 1.2 is also a rough characterisation of complete-minor-free graphs, in the sense that for each $g, p, k, a, j \in \mathbb{N}$ and graph G that admits a tree-decomposition as in Theorem 1.2, there exists $h \in \mathbb{N}$ such that G excludes K_h as a minor. A proof of this appears in Joret and Wood [92], who also obtained tight bounds on h .

Robertson and Seymour [122] used Theorem 1.2 to show the ‘Graph Minor Theorem’, which says that the minor operation forms a well-quasi-ordering. This means that for any (infinite) sequence of graphs $G_0, G_1, G_2, \dots$, there exists $i, j \in \mathbb{N}$ with $i < j$ such that $G_i \leq G_j$. A simple consequence of this is that any set of graphs that are pairwise not minors of each other must be finite.

For a proper minor-closed class $\mathcal{G}$, a *minimal excluded minor* of $\mathcal{G}$ is a graph H with $H \notin \mathcal{G}$ such that for each proper minor of H is in $\mathcal{G}$. $\mathcal{G}$ is completely determined by its minimal excluded minors. That is, given the list $\mathcal{F}$ of minimal excluded minors, one can determine if any given graph G is in $\mathcal{G}$, specifically by checking if there does not exist $H \in \mathcal{F}$ with $H \leq G$. It follows from the Graph Minor Theorem that $\mathcal{F}$ is finite (up to isomorphism). Thus, for any proper minor-closed class $\mathcal{G}$, there is a characterisation of graphs in $\mathcal{G}$ in the form of a finite list of excluded minors. So there is a Kuratowski–Wagner-like (Theorem 1.1 [96, 127]) theorem for any proper minor-closed

class.

Other applications of Theorem 1.2 include queue number [58], non-repetitive chromatic number [55], clustered colouring [103, 111], low treewidth 2-colourings [34], linkages [120, 123], and testing if a graph contains a fixed graph as a minor [120], amongst many others.

Theorem 1.2 is crucial to all the contributions of this thesis that relate to minor-free graphs.

Chapter 2

Global Structure of Graphs

In this chapter, we introduce the main concepts of this thesis, including graph products (Section 2.1), Assouad–Nagata dimension (Section 2.2), and quasi-isometries (Section 2.3.1). We discuss several key results in the literature, and introduce the main results of this thesis.

2.1 Graph Products

The *strong product* of a list of graphs $G_1, \dots, G_n$, denoted $G_1 \boxtimes G_2 \boxtimes \dots \boxtimes G_n$, is the graph with vertex set $V(G_1) \times V(G_2) \times \dots \times V(G_n)$ and edges between distinct vertices $(u_1, \dots, u_n), (v_1, \dots, v_n) \in V(G_1) \times \dots \times V(G_n)$ if, for each $i \in \{1, \dots, n\}$, either $u_i = v_i$ or u_i is adjacent to v_i in G_i . Up to isomorphism, the strong product is associative and commutative, and that the strong products of two or more strong products is the strong product of all the graphs involved in the products.

Graph product structure theory is the study of when a graph G is contained in the strong product of two (or more) ‘simple’ graphs H_1, H_2 .

One of the simplest types of products is when one of the terms is a clique. For a graph H and $c \in \mathbb{N}$, a graph G is a *c-blowup* of H if G is contained in $H \boxtimes K_c$. We refer to c as the *blowup factor* and H as the *base graph*. Note that G is a c -blowup of H if and only if G admits a partition of width at most c such that $G/\mathcal{P}$ is contained in H .

Graphs blowups can be used to bound the treewidth of a graph. Specifically, if a graph G is a c -blowup of a graph H , then $\text{tw}(G) \leq c(\text{tw}(H) + 1) - 1$. This can be seen by taking the tree-decomposition of H , and in each bag replace each vertex by its associated clique. However, the structure of G is far more limited than the structure of

an arbitrary graph of treewidth $c(\text{tw}(H) + 1) - 1$. This is reflected in the fact that many applications of graph blowups have fast-growing (often exponential) dependence on $\text{tw}(H)$ but slow-growing (often linear) dependence on c . These include queue number [58], nonrepetitive chromatic number [59], vertex rankings [25], and centred colourings [52].

2.1.1 Row-Treewidth

Dujmović, Joret, Micek, Morin, Ueckerdt, and Wood [58] showed that planar graphs admit the following product structure.

Theorem 2.1. *Every planar graph is contained in $H \boxtimes P$ for some path P and graph H with $\text{tw}(H) \leq 8$.*

Dujmović et al. [58] used Theorem 2.1 to solve a long-standing open problem regarding the ‘queue number’ of planar graphs. It has since seen application to various other problems, including non-repetitive chromatic number [55], clustered colourings [56], centred colourings [52], adjacency labelling [23, 54, 70], vertex rankings [25], twin-width [12, 24], and universal graphs [89]. As such, it motivated the introduction of the following graph parameter.

The *row-treewidth* of a graph G is the minimum $k \in \mathbb{N}$ for which there exists a graph H with $\text{tw}(H) \leq k$ and a path P such that G is contained in $H \boxtimes P$. So Theorem 2.1 says that planar graphs have row-treewidth at most 8. This bound was improved to 6 by Ueckerdt, Wood, and Yi [126]. Dujmović et al. [58] also showed that planar graphs admit the following product structure, which is a slight variation of Theorem 2.1.

Theorem 2.2. *Every planar graph is contained in $H \boxtimes P \boxtimes K_3$ for some path P and graph H with $\text{tw}(H) \leq 3$.*

Theorem 2.2 implies that planar graphs have row-treewidth at most 11. In this sense, Theorem 2.2 is weaker than Theorem 2.1. However, as mentioned earlier, the structural complexity of a blowup (and thus the dependence in application) depends far more on the base graph than the blowup factor. So in some regards, Theorem 2.2 is stronger than Theorem 2.1. This motivates the following definition.

For $c \in \mathbb{N}$, the *c-row-treewidth* of a graph G is the minimum $k \in \mathbb{N}$ for which there exists a graph H with $\text{tw}(H) \leq k$ and a path P such that G is contained in $H \boxtimes P \boxtimes K_c$. Note that each graph of c -row-treewidth at most k has row-treewidth at most $c(k + 1) - 1$.

So Theorem 2.2 says that planar graphs have 3-row-treewidth at most 3. Vida Dujmović observed that a variation of the proof of Ueckerdt et al. [126] shows that planar graphs have 2-row-treewidth at most 4 [26, 126].

Dujmović et al. [58] also showed that for each $c \in \mathbb{N}$, there exists a planar graph of c -row-treewidth at least 3. So the bound on the treewidth in Theorem 2.2 is the best possible, regardless of the blowup factor. It is natural to ask if replacing P by a slightly more complex graph would allow us to bypass this limit of 3 on the treewidth. Distel, Hendrey, Karol, Wood, and Yip [48] showed that this can be done by replacing P with a graph of treewidth 2.

Theorem 2.3. *For each planar graph G , there exist graphs H_1 and H_2 of treewidth at most 2 such that G is contained in $H_1 \boxtimes H_2 \boxtimes K_2$.*

Distel et al. [48] also showed that replacing P with a tree fails.

Theorem 2.4. *For each $c \in \mathbb{N}$, there exists a planar graph G such that for any tree T and graph H with $\text{tw}(H) \leq 2$, G is not contained in $H \boxtimes T \boxtimes K_c$.*

It is also natural to wonder if H can be a graph of bounded pathwidth instead of bounded treewidth. However, Liu, Norin, and Wood [102] showed that this also fails (for K_4 -minor-free graphs, which are planar), even if P is also replaced with a graph of bounded pathwidth.

Theorem 2.5. *For each $c, k \in \mathbb{N}$, there exists a planar graph G such that for any graphs H_1, H_2 with $\text{pw}(H_1), \text{pw}(H_2) \leq k$, G is not contained in $H_1 \boxtimes H_2 \boxtimes K_c$.*

Row-treewidth has also been studied for graphs embedded on higher surfaces. Dujmović et al. [58] showed that for each $g \in \mathbb{N}$, each graph of Euler genus g has row-treewidth at most $2g + 8$, $\max(2g, 1)$ -row-treewidth at most 9, and $\max(2g, 3)$ -row-treewidth at most 4. Ueckerdt et al. [126] improved the bound on the row-treewidth to $2g + 6$, and Distel, Hickingbotham, Huynh, and Wood [49] improved the bound on the $\max(2g, 3)$ -row-treewidth to 3.

Dujmović, Morin, and Wood [62] built upon the above work to show that for each $k, g \in \mathbb{N}$, each (g, k) -planar graph has c -row-treewidth at most $O(k^5)$, where $c \in O(gk^2)$. Hickingbotham and Wood [87] studied the row-treewidth for a variety of non-minor-closed classes, and in the process, they showed that each (g, k) -planar graph has c' -row-treewidth at most $O(k^5)$, where $c' \in O(gk^2)$ is an asymptotic factor of 3 smaller than the c obtained by Dujmović et al. [62]. Distel, Hickingbotham, Seweryn, and Wood [50] showed that each (g, k) -planar graph has c -row-treewidth at most 10^9 for

some $c = f(g, k) \in \mathbb{N}$. Further, Distel et al. [49] showed that each $(g, 1)$ -planar graph has $\max(4g, 7)$ -row-treewidth at most 7 (the $g = 0$ case was first shown by Dujmović et al. [62]), and Distel et al. [50] showed that each k -planar graph has c -row-treewidth at most $2 \cdot 10^7$ for some $c = f(k) \in \mathbb{N}$.

As for other minor-closed-classes, Dujmović et al. [58] showed the following.

Theorem 2.6. *For each minor-closed class of graphs $\mathcal{G}$, there exists $c \in \mathbb{N}$ such that each $G \in \mathcal{G}$ has row-treewidth at most c if and only if some apex graph is not in $\mathcal{G}$.*

(Recall that an apex graph contains a vertex whose deletion gives a planar graph.)

So excluding some apex graph as a minor is necessary to bound the row-treewidth. In particular, one cannot describe every minor-closed class of graphs in terms of row-treewidth. However, Dujmović et al. [58] did show that row-treewidth could be integrated with the Graph Minor Structure Theorem (Theorem 1.2 [121]), giving the following variant.

Theorem 2.7. *For each $h \in \mathbb{N}$, there exists $a, b, j \in \mathbb{N}$ such that each K_h -minor-free graph G admits a tree-decomposition $(J_t : t \in V(T))$ of adhesion at most j such that each $t \in V(T)$, there exists $A_t \subseteq J_t$ with $|A_t| \leq a$ such that $G \langle J_t \rangle - A_t$ has row-treewidth at most b .*

Unlike the Graph Minor Structure Theorem, Theorem 2.7 is not a rough characterisation of K_h -minor-free graphs, since 1-planar graphs have bounded row-treewidth [50, 62, 87] and thus admit a tree-decomposition as in Theorem 2.7 with $a = j = 0$, but have arbitrarily large complete minors (as observed in Section 1.6).

2.1.2 Blowups of Bounded Treewidth Graphs

One of the simplest types of blowups are blowups of trees, also known as tree-partitions (of bounded width). Introduced independently by Seese [125] and Halin [82], tree-partitions have been widely studied [18–20, 38, 39, 51, 67, 129, 130]. Applications include graph drawings [29, 35, 60, 63, 131], nonrepetitive colourings [10], clustered colourings [6, 103], monadic second-order logic [97], graphs of linear growth [28], network emulations [15, 16, 21, 72], statistical learning theory [132], size Ramsey numbers [53, 93], and the edge-Erdős-Pósa property [30, 78, 116].

The wide variety of usages of tree-partitions motivate the following definition. The *tree-partition-width* of a graph G , denoted $\text{tpw}(G)$, is the minimum $c \in \mathbb{N}$ such that G is a c -blowup of a tree.

Fans have treewidth 2 but unbounded tree-partition-width [19, 39]. So the tree-partition-width is not bounded by any function of the treewidth. However, fans have unbounded maximum degree. If the degree is bounded, then the tree-partition-width is bounded by a function of treewidth, as shown by Ding and Oporowski [38]. The referee of this [38] optimised this bound, giving the following:

Theorem 2.8. *For each graph G , $\text{tpw}(G) \in O(\Delta(G) \text{tw}(G))$.*

Wood [130] proved that $O(\Delta(G) \text{tw}(G))$ is the best possible, while improving the multiplicative constant. Distel and Wood [51] further showed that G is a $O(\Delta(G) \text{tw}(G))$ -blowup of a tree T with $\Delta(T) \in O(\Delta(G))$.

Recalling that $\text{tw}(G) \leq c(\text{tw}(H) + 1) - 1$ for any graphs G, H such that G is a c -blowup of H , we obtain $\text{tw}(G) \leq 2 \text{tpw}(G) - 1$. This was first noted by Seese [125]. Thus, if the degree is bounded, then by Theorem 2.8, the treewidth and tree-partition-width are tied. In this sense, we can approximate the treewidth with the tree-partition-width. This is valuable, because as mentioned before, many applications care more about the treewidth of H than the blowup factor c . So if the tree-partition-width is a good approximation of the treewidth, then expressing G as a $\text{tpw}(G)$ -blowup of a tree gives substantially better bounds in application than expressing G as a graph of treewidth at most $\text{tw}(G)$.

To tie the treewidth and tree-partition-width, we required that the degree was bounded. This requirement is annoying, but necessary, as seen with the fans [19, 39]. To remove this dependence on the degree, we consider a wider variety of blowups. Specifically, we allow the base graph to have some fixed treewidth greater than 1. This motivates the following definition.

For $k \in \mathbb{N}$, the *k -tree-partition-width* of a graph G , denoted $\text{tpw}_k(G)$, is the minimum $c \in \mathbb{N}$ such that there exists a graph H with $\text{tw}(H) \leq k$ such that G is a c -blowup of H . We remark that a *k -tree* is an edge-maximal graph of treewidth k . One can always assume that H is edge-maximal, hence the name. Further, the 1-trees are precisely the trees, so the 1-tree-partition-width is the tree-partition-width. Note also that $\text{tw}(G) \leq (k + 1) \text{tpw}_k(G) - 1$.

For the purposes of application, we want to minimise k while keeping the k -tree-partition-width reasonable. In particular, we want that $\text{tpw}_k(G)$ is small enough so that it is outweighed by the fast-growing dependence on k in application.

Campbell, Clinch, Distel, Gollin, Hendrey, Hickingbotham, Huynh, Illingworth, Tamitegama, Tan, and Wood [27] showed that for each $h \in \mathbb{N}$ and each K_h -minor-free graph G , $\text{tpw}_{h-2}(G) \in O_h(\text{tw}(G)^2)$. Illingworth, Scott, and Wood [90] improved this

to $\text{tpw}_{h-2}(G) \leq \text{tw}(G) + 1$. A variant of their proof showed that for each n -vertex K_h -minor-free graph G , $\text{tpw}_{h-2}(G) \in O_h(\sqrt{n})$. From this, one obtains $\text{tw}(G) \in O_h(\sqrt{n})$, mirroring the result of Alon et al. [7]. In fact, the constant obtained by Illingworth et al. [90] was tight enough that they could match the original bounds of Alon et al. [7].

However, Campbell et al. [27] also showed that there is no function f such that each K_h -minor-free graph G has $(h - 3)$ -tree-partition-width at most $f(\text{tw}(G))$. Thus, the k -tree-partition-width is bounded by the treewidth only if $k \geq h - 2$. However, the counterexample has low treewidth, rather than high treewidth. This suggests that if one is only interested in graphs of high treewidth (or a bound for the class), then this issue can be sidestepped. Inspired by the results of Alon et al. [7] and Illingworth et al. [90], we now aim for k -tree-partition-width at most $O_h(\sqrt{n})$ for each n -vertex K_h -minor-free graph G . Illingworth et al. [90] asked if this can be done with k a constant independent of h . Distel, Dujmović, Eppstein, Hickingbotham, Joret, Micek, Morin, Seweryn, and Wood [45] answered this in the affirmative, showing that $\text{tpw}_4(G) \in O_h(\sqrt{n})$. As the proof uses Graph Minor Structure Theorem (Theorem 1.2 [121]), the multiplicative constant is worse than the ones obtained by Alon et al. [7] and Illingworth et al. [90]. The proof of this result makes use of a structural result about graphs on surfaces showed by Distel et al. [49] that was originally used to show that these graphs have row-treewidth at most 3.

This thesis goes one step further, and shows the following.

Theorem 2.9. *For each $h, n \in \mathbb{N}$, each n -vertex K_h -minor-free graph satisfies $\text{tpw}_3(G) \in O_h(\sqrt{n})$.*

Theorem 2.9 uses the Graph Minor Structure Theorem, so the exact bounds again do not match those of Alon et al. [7]. We prove Theorem 2.9 in Chapters 6 and 7. This result first appeared in [42].

Distel et al. [45] observed that a result of Linial, Matoušek, Sheffet, and Tardos [99] implies that there exists n -vertex planar graphs of increasing size with $\text{tpw}(G) \in \Omega(n^{2/3})$. Thus, the ‘3’ in Theorem 2.9 cannot be replaced with a ‘1’ for $t \geq 5$, meaning it is only 1 off the theoretical best-possible of 2. However, Distel et al. [45] also showed the following.

Theorem 2.10. *For each $g, n \in \mathbb{N}$, each n -vertex graph of Euler genus at most g satisfies $\text{tpw}_2(G) \in O_g(\sqrt{n})$.*

(Theorem 2.10 improves a result of Illingworth et al. [90], who showed that $\text{tpw}_3(G) \in O_g(\sqrt{n})$.)

It is still possible that $\text{tpw}_2(G) \in O_h(\sqrt{n})$ for n -vertex K_h -minor-free graphs G . As such, this thesis makes extra effort to generalise our results, with the hope that parts of the proof can be recycled if any further breakthroughs are made in this topic. See Section 6.2, along with Lemma 6.13 and Theorem 7.4.

This thesis also bounds $\text{tpw}_2(G)$. Specifically, we show that $\text{tpw}_2(G) \in O_h(\sqrt{n} \log(n)^2)$. In fact, we show the following stronger result.

Theorem 2.11. *For each $h, n \in \mathbb{N}$, each n -vertex K_h -minor-free graph is a $O_h(\sqrt{n} \log(n)^2)$ -blowup of a fan.*

Theorem 2.11 implies that $\text{tpw}_2(G) \in O_h(\sqrt{n} \log(n)^2)$, since fans have treewidth 2. However, it is much stronger than this, for several reasons. Firstly, fans also have pathwidth 2. So Theorem 2.11 implies that G is a $O_h(\sqrt{n} \log(n)^2)$ -blowup of a graph of bounded pathwidth (at most 2). Up to a polylog factor, this answers a question of Distel et al. [45], who asked if G is a $O_h(\sqrt{n})$ of a graph of bounded pathwidth. Secondly, there is only one fan with a fixed number of vertices (up to isomorphism). It is easy to see that the fan needs at most n vertices, so the fan in Theorem 2.11 can be the same fan for each graph G (for a fixed n). This provides a universal structure for n -vertex K_h -minor-free graphs. That is, for each $h, n \in \mathbb{N}$, there exists a fan F and $c \in O_h(\sqrt{n} \log(n)^2)$ such that each n -vertex K_h -minor-free graph is contained in $F \boxtimes K_c$.

The proof uses the row-treewidth variant of the Graph Minor Structure Theorem (Theorem 2.7 [58]), and builds upon the work of Dujmović et al. [59], who showed the following.

Theorem 2.12. *For each $b, n \in \mathbb{N}$, each n -vertex graph of row-treewidth at most b is a $O_b(\sqrt{n} \log(n)^2)$ -blowup of a fan.*

Theorem 2.11 first appeared in [40]. This paper was merged with [59] to make a single paper [46]. We prove Theorem 2.11 in Chapter 6. The extra $\log(n)^2$ dependence in Theorem 2.11 comes completely from Theorem 2.12. As such, we write our proof as generally as possible so that any improvements made to Theorem 2.12 (or more precisely, a slight strengthening of Theorem 2.12; see Theorem 6.8) can be used to improve Theorem 2.11.

We remark that the existence of a layering of bounded width is tied to a parameter called ‘bandwidth’. Specifically, a graph admits a layering of width c (equivalently, is a c -blowup of a path) if it has bandwidth at most c and only if it has bandwidth at most $2c$. Since deleting the centre of a fan gives a path, Theorems 2.11 and 2.12

can be expressed as ‘each n -vertex minor-free/bounded row-treewidth graph G has a set X of at most $O(\sqrt{n} \log(n)^2)$ vertices such that $G - X$ has bandwidth at most $O(\sqrt{n} \log(n)^2)$ ’.

2.2 Asymptotic and Assouad–Nagata Dimension

Asymptotic dimension, denoted by asdim , is a measure of the large-scale shape of a metric space. First introduced by Gromov [80] for the geometric study of groups, it has since been studied for metrics induced by graphs. While the general definition is somewhat technical, on graphs it can be reduced to a colouring problem. (See Observation 2.13, along with [22, Proposition 1.17].)

Given a graph G and $n \in \mathbb{N}$, a function $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ is an *n -dimensional control function* for G if, for every $r \in \mathbb{R}^{\geq 0}$, there exist $n + 1$ collections $\mathcal{C}_1, \dots, \mathcal{C}_{n+1}$ of subsets of $V(G)$ such that:

- (CF1) $\bigcup_{i=1}^{n+1} \bigcup_{S \in \mathcal{C}_i} S = V(G)$,
- (CF2) for each $i \in \{1, \dots, n + 1\}$, $\mathcal{C}_i$ is r -disjoint in G , and
- (CF3) for each $i \in \{1, \dots, n + 1\}$ and each $S \in \mathcal{C}_i$, S has weak diameter in G at most $f(r)$.

Note that if f is an n -dimensional control function for G , it is also an m -dimensional control function for G for each $m \in \mathbb{N}$ with $m \geq n$.

This is the standard definition used for general metric spaces, but it is rather cumbersome. Since we are only interested in graphs, we can instead reformulate the definition in terms of graphs colourings.

Let c be a colouring of a graph G , and let $r \in \mathbb{R}^{\geq 0}$. Say that a *monochromatic r -component* of G under c is a maximal nonempty connected subgraph M of G^r such that $V(M)$ is monochromatic under c . Specifically, for a colour i , M is an *i -monochromatic r -component* of G under c if $V(M)$ is i -monochromatic under c . For $m \in \mathbb{N}$ and $d \in \mathbb{R}^{\geq 0}$, say that c is a *(m, r, d) -colouring* if c is an m -colouring and, for each monochromatic r -component M of G under c , $V(M)$ has weak diameter in G at most d .

Observation 2.13. *Let $n \in \mathbb{N}$, and let G be a graph. Then $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ is an n -dimensional control function for G if and only if, for every $r \in \mathbb{R}^{\geq 0}$, G admits an $(n + 1, r, f(r))$ -colouring.*

The proof idea is simple. To go from a control function to the colouring, each $v \in V(G)$

is given the colour i for some $i \in \{1, \dots, n+1\}$ such that $v \in \bigcup_{S \in \mathcal{C}_i} S$ (which exists by (CF1)). For the reverse direction, for each $i \in \{1, \dots, n+1\}$, $\mathcal{C}_i$ is $\{V(M) : M \text{ an } i\text{-monochromatic } r\text{-component under } c\}$. See the appendix of [44] for a proof, and also the proof of [22, Proposition 1.17] (which is a conceptually similar result). We use Observation 2.13 as the working definition of control functions from now on.

An $(m, 1, D)$ -colouring is also known as a ‘weak diameter colouring’ (specifically, an m -colouring with weak diameter D) in the literature. These colourings have been studied by various authors [9, 23, 64, 65]. A control function can also be characterised in terms of the existence of weak-diameter colourings for each power of a graph. (Specifically, $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ is an n -dimensional control function for G if for each $r \in \mathbb{R}^{>0}$, G^r admits an $n+1$ -colouring with weak diameter $f(r)/r$, and only if for each $r \in \mathbb{R}^{>0}$, G^r admits $n+1$ -colouring with weak diameter $\lceil f(r)/r \rceil$.)

Observe that for any graph H that is isometric in G and any (m, r, d) -colouring c of G , $c|_{V(H)}$ is an (m, r, d) -colouring of H . We also remark that two vertices are contained in the same monochromatic r -component if and only if there is a monochromatic r -path between them. This allows us to re-characterise (m, r, d) -colourings as follows.

Observation 2.14. *Let $m \in \mathbb{N}$ and $r, d \in \mathbb{R}^{\geq 0}$. Then a colouring c of a graph G is an (m, r, d) -colouring if and only if every monochromatic r -path has weak diameter in G at most d .*

We use Observation 2.14 implicitly from now on.

In this thesis, sometimes we find that a function only acts as a control function (in the sense we can find collections satisfying (CF1)–(CF3)) when r is sufficiently large. As such, for $\ell \in \mathbb{R}^{\geq 0}$, we say that f is an *ℓ -almost n -dimensional control function* for a graph G if G admits an $(n+1, r, f(r))$ -colouring for every $r \in \mathbb{R}^{\geq \ell}$. Note that a 0-almost n -dimensional control function is an n -dimensional control function.

Given a class of graphs $\mathcal{G}$ and $n \in \mathbb{N}$, a function $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ is an *n -dimensional control function* for $\mathcal{G}$ if f is an n -dimensional control function for each $G \in \mathcal{G}$. The *asymptotic dimension* of $\mathcal{G}$, denoted $\text{asdim}(\mathcal{G})$, is the smallest $n \in \mathbb{N}$ such that $\mathcal{G}$ admits an n -dimensional control function, or ∞ otherwise.

We define asymptotic dimension for classes of graphs, rather than individual graphs, since each (finite) graph admits a 0-dimensional control function, namely the constant map to the maximum diameter of a connected component.

One downside of asymptotic dimension is that the control function f can (and sometimes must [112]) grow rapidly in r . It is natural to wonder how restricting the growth of f

can restrict the dimension. Specifically, a common variant of asymptotic dimension, Assouad–Nagata dimension restricts the growth of f to be proportional to r .

A function $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ is a *dilation* if there exists $c \in \mathbb{R}^{\geq 0}$ such that $f(r) \leq cr$ for every $r \in \mathbb{R}^{\geq 0}$. The *Assouad–Nagata dimension* of a class of graphs $\mathcal{G}$ is the smallest $n \in \mathbb{N}$ such that $\mathcal{G}$ admits an n -dimensional control function f that is also a dilation, or ∞ otherwise. Observe that the asymptotic dimension of a class of graphs is at most the Assouad–Nagata dimension. However, no bound on the reverse is true [22] (see Section 2.2.1).

We remark that Assouad–Nagata dimension was originally introduced by Nagata [107] prior to the work of Gromov [80] (which introduced asymptotic dimension), and was defined analogously to Lebesgue dimension. It is due to Assouad [8] that we have the equivalent definition analogous to asymptotic dimension presented above.

Asymptotic (and Assouad–Nagata) dimension have applications in geometric group theory [66, 80, 88], for embeddings into Banach spaces [75, 98], and to clustered colouring [22]. Furthermore, Assouad–Nagata dimension has applications to sparse partitions [22, 80], the travelling salesman problem [69, 91], Steiner trees [71, 91], and set covers [91].

2.2.1 Key Results

Gromov [80] showed that for each $d \in \mathbb{N}^{\geq 1}$, the class of all d -dimensional grids has asymptotic and Assouad–Nagata dimension d . Bounding the degree is not sufficient to bound the asymptotic (or Assouad–Nagata) dimension [11, 80].

Bonamy, Bousquet, Esperet, Groenland, Liu, Pirot, and Scott [22] established the following breakthrough result, improving or generalising previous work in [13, 73, 113].

Theorem 2.15 ([22]). *For every proper minor-closed class of graphs $\mathcal{G}$,*

- (a) $\text{asdim}(\mathcal{G}) \leq 2$, and
- (b) $\text{asdim}(\mathcal{G}) \leq 1$ if and only if there exists $k \in \mathbb{N}$ such that $\text{tw}(G) \leq k$ for each $G \in \mathcal{G}$.

We remark that Theorem 2.15(a) is proven using a version of the Graph Minor Structure Theorem (Theorem 1.2 [121]) very similar to Theorem 2.7 [58] (the row-treewidth variant). (This version instead uses layered treewidth, which is slightly more general than row-treewidth, in the sense that bounded row-treewidth implies bounded layered treewidth.) It is also implicit from Bonamy et al. [22] that for each $b \in \mathbb{N}$, the class

of graphs of row-treewidth at most b has asymptotic dimension at most 2. Since (g, k) -planar graphs have bounded row-treewidth for each $g, k \in \mathbb{N}$ [50, 62, 87], this implies that the class of (g, k) -planar graphs have asymptotic dimension 2.

Asymptotic dimension has also been studied for various other non-minor-closed classes in [22, 64, 115].

Regarding Assouad–Nagata dimension, Bonamy et al. [22] showed the following.

Theorem 2.16. *For each $g \in \mathbb{N}$, the class of graphs of Euler genus at most g has Assouad–Nagata dimension 2.*

However, unlike with asymptotic dimension, Theorem 2.16 does not extend to (g, k) -planar graphs for $k \in \mathbb{N}^{\geq 1}$, since Bonamy et al. [22] also showed:

Theorem 2.17. *The class of $(0, 1)$ -planar graphs has infinite Assouad–Nagata dimension.*

Thus, Theorem 2.17 creates a key distinction between asymptotic and Assouad–Nagata dimension.

The following question arises naturally from the work of Bonamy et al. [22]: Can Theorem 2.15 be strengthened by replacing asymptotic dimension with Assouad–Nagata dimension. While it was already known that the class of K_h -minor-free graphs has Assouad–Nagata dimension at most $4^h - 1$ [113], in this thesis, we show that the bound in Theorem 2.15 can be matched exactly in the stronger setting of Assouad–Nagata dimension.

Theorem 2.18. *For every proper minor-closed class of graphs $\mathcal{G}$,*

- (a) $\text{ANdim}(\mathcal{G}) \leq 2$, and
- (b) $\text{ANdim}(\mathcal{G}) \leq 1$ if and only if there exists $k \in \mathbb{N}$ such that $\text{tw}(G) \leq k$ for each $G \in \mathcal{G}$.

Our proof of Theorem 2.18 first appeared in [44], and uses Theorem 2.16 (or more precisely, a version of Theorem 2.16 that allows edge-weights, see Theorem 5.16). Theorem 2.18 was also shown independently by Liu [101]. The bounded degree case of Theorem 2.18(b) is implicit in Benjamini, Schramm, and Timár [14].

We also give an exact characterisation of when subdivision-closed classes of graphs have bounded Assouad–Nagata dimension.

Theorem 2.19. *Let $\mathcal{G}$ be a subdivision-closed graph class. Then $\mathcal{G}$ has bounded Assouad–Nagata dimension if and only if there is a graph H such that $\mathcal{G}$ is H -minor-free.*

Theorem 2.19 also first appeared in [44]. Theorem 2.19 implies Theorem 2.17, since the class of (g, k) -planar graphs are subdivision-closed and admits every graph H as a minor when $k \geq 1$.

We prove Theorems 2.18 and 2.19 in Chapter 5.

2.3 Beyond Minor-Free Graphs

2.3.1 Quasi-Isometry

For $c \in \mathbb{R}^{\geq 1}$, a map $\phi : V(G) \mapsto V(H)$ is a *c -quasi-isometry* from a graph G to a graph H if:

1. for each pair of distinct $u, v \in V(G)$,
 - (a) $\text{dist}_G(u, v) \leq c \text{dist}_H(\phi(u), \phi(v)) + c$, and
 - (b) $\text{dist}_H(\phi(u), \phi(v)) \leq c \text{dist}_G(u, v) + c$, and
2. for each $h \in V(H)$, there exists $v \in V(G)$ such that $\text{dist}_H(h, \phi(v)) \leq c$.

We refer to c as the *coarseness factor*. The first condition can be rewritten as

$$\text{dist}_G(u, v)/c - 1 \leq \text{dist}_H(\phi(u), \phi(v)) \leq c \text{dist}_G(u, v) + c.$$

The second condition is to ensure that the map is ‘almost’ surjective. As such, we refer to it as the *quasi-surjectivity* condition. It gives rise to a c' -quasi-isometry from H to G , where c' is not much larger than c ($c' := 3c^2$ suffices).

A graph G is *c -quasi-isometric* to a graph H if there exists a c -quasi-isometry from G to H . By quasi-surjectivity, H is c' -quasi-isometric to G .

Quasi-isometries measure how similar two graphs are with respect to their metrics. In this sense, we expect graphs that are tied via quasi-isometry to share the same coarse properties. One way this intuition is validated is with asymptotic and Assouad–Nagata dimension. Specifically, for classes $\mathcal{G}, \mathcal{H}$ of graphs, if each $G \in \mathcal{G}$ is c -quasi-isometric to some $H \in \mathcal{H}$, then control functions for $\mathcal{H}$ give rise to control functions for $\mathcal{G}$ as follows.

Observation 2.20. *Let $c \in \mathbb{R}^{\geq 1}$, and let $\mathcal{G}, \mathcal{H}$ be classes of graphs such that each $G \in \mathcal{G}$ is c -quasi-isometric to some $H \in \mathcal{H}$. Then for each $n \in \mathbb{N}$, and n -dimensional control function $f' : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ for $\mathcal{H}$, the function $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ defined by $f(r) := cf'(2cr) + cr$ for all $r \in \mathbb{R}^{\geq 0}$ is an n -dimensional control function for $\mathcal{G}$. Further, if f' is a dilation, then so is f .*

We remark that the $(m, r, f(r))$ -colourings for the graphs in $\mathcal{G}$ are just obtained from taking $(m, 2cr, f'(2cr))$ -colourings for graphs in $\mathcal{H}$, and using the colours of the image under a c -quasi-isometry. See [43] for a full proof, and also the proof of Observation 5.25 (which is conceptually very similar).

This leads to the following as an easy corollary.

Observation 2.21. *Let $c \in \mathbb{R}^{\geq 1}$, and let $\mathcal{G}, \mathcal{H}$ be classes of graphs such that each $G \in \mathcal{G}$ is c -quasi-isometric to some $H \in \mathcal{H}$. Then $\text{asdim}(\mathcal{G}) \leq \text{asdim}(\mathcal{H})$ and $\text{ANdim}(\mathcal{G}) \leq \text{ANdim}(\mathcal{H})$.*

One simple way to construct a quasi-isometry is to find a proper partition where each part has bounded weak diameter. Such a partition gives rise to a quasi-isometry to the quotient.

Observation 2.22. *Let G be a graph, let $d \in \mathbb{R}^{\geq 0}$, and let $\mathcal{P}$ be a proper partition of G such that each part has weak diameter in G at most d . Then G is $(d+1)$ -quasi-isometric to $G/\mathcal{P}$.*

Each vertex is just mapped to the part containing it. The ‘proper’ condition is needed for quasi-surjectivity. See [43, Lemma 3.7] or [2, Lemma 2.2] for a full proof.

2.3.2 Fat Minors

A minor of a graph G may be contained in a small region of G , and thus be unrepresentative of the global structure of G . Indeed, a graph G could be structurally simple save for the presence of one large clique, which disqualifies the graph from being minor-free. For example, for some even integer n , G could be the graph obtained from the path of length $n/2 + 1$ by replacing the last vertex with a clique of size $n/2$. Just saying that G does not exclude any graph as a minor, while correct, is a disservice to its structural simplicity. As such, we are motivated to come up with a global version of a minor, in which the structure of the minor is spread out across the graph. This leads to the following definition.

For $q \in \mathbb{R}^{\geq 0}$, a model $((X_h : h \in V(H)), (P_e : e \in E(H)))$ of a graph H in a graph G is *q -fat* if:

1. for each pair of distinct $u, v \in V(H)$, $\text{dist}_G(X_u, X_v) \geq q$,
2. for each pair of distinct $e, f \in E(H)$, $\text{dist}_G(P_e, P_f) \geq q$, and
3. for each $v \in V(H)$ and $e \in E(H)$ that is not incident to v in H , $\text{dist}_G(X_v, P_e) \geq q$.

H is a *q -fat minor* of G if there exists a q -fat-model of H in G . q is referred to as the *fatness* of the fat minor. Otherwise, we say that G *excludes* H as a q -fat minor. Note that every model is 0-fat, so the 0-fat minors of G are precisely the minors of G . If H is a q -fat minor of G , then any minor H' of H is also a q -fat minor of G (we obtain the fat model of H' from the fat model of H by merging and removing branch paths and sets accordingly). So if we are only interested in the fact that G excludes some graph as a fat minor, it suffices to consider graphs that exclude complete graphs as fat minors.

Fat minors provide a coarse analogue to the standard concept of minors. The $n \times n \times 2$ -grid is a helpful example of how fat minors can differ from minors. One can construct a model of K_n where, for each $i \in \{1, \dots, n\}$, the i -th branch set consists of the i -th row in the bottom level and the i -th column in the top level. However, one cannot do the same with a 2-fat model. For any 2-fat-model of a graph H , each vertical path can only consist of one branch path/set, or a branch path and set that are incident in H . This allows us to project the 2-fat-model of H in the $n \times n \times 2$ -grid down to a model of H in the $n \times n$ -grid. Thus, the 2-fat minors of the $n \times n \times 2$ -grid are minors of the $n \times n$ -grid, and are thus planar. In particular, we find that K_5 and $K_{3,3}$ are not 2-fat minors of the $n \times n \times 2$ -grid. Thus, in some sense, the $n \times n \times 2$ grid is coarsely planar. This formalises the intuition that the $n \times n \times 2$ -grid can be squished back onto the plane.

One advantage of fat minors over minors is that (up to a decrease in fatness) they are preserved under quasi-isometry. Specifically, for each $q \in \mathbb{R}^{\geq 0}$ and $c \in \mathbb{R}^{\geq 1}$, there exists $c' \in \mathbb{R}^{\geq 1}$ ($c' := c(q + 2c + 1)$ suffices) such that if H is a c' -fat minor of a graph G , then H is a q -fat minor of any graph G' that is c -quasi-isometric to G . This can be seen by taking the image of a c' -fat-model of H in G , and then reconnecting the images of the branch sets and branch paths. See Observation 3.6 for a proof of a similar result.

2.3.3 Coarse Conjectures

A central area of interest in the emerging area of coarse graph theory is to establish coarse analogues of classical results. This study was initiated by Georgakopoulos and Papasoglu [77]; some of the conjectures they asked or generalised versions thereof asked by other authors include the following [4, 77, 110].

Conjecture 2.23 (Weak Coarse Menger's [110]). *For each $k, d \in \mathbb{N}$, there exists $n, \ell \in \mathbb{N}$ such that for each graph G and $S, T \subseteq V(G)$, there exists either:*

1. *a set $X \subseteq V(G)$ with $|X| \leq n$ such that $N_G^\ell(X)$ intersects each path from S to T ,*
or
2. *k paths from S to T that are pairwise at distance at least d .*

There is special interest in whether Conjecture 2.23 holds with $n = k - 1$, which would be a true analogue of the classical Menger’s Theorem [105]. This stronger version was conjectured by Georgakopoulos and Papasoglu [77] and Albrechtsen et al. [4], and we refer to it as the ‘Coarse Menger’s Conjecture’.

Conjecture 2.24 (Coarse Grid Minor [77]). *For each $n, q \in \mathbb{N}$, there exists $k, c \in \mathbb{N}$ such that each graph that excludes the $n \times n$ -grid as a q -fat minor is c -quasi-isometric to a graph of treewidth at most k .*

For the reverse direction, it can be seen that graphs that are quasi-isometric to graphs of small treewidth exclude a large grid as a fat minor. This is because fat minors are preserved under quasi-isometry, and since graphs of small treewidth exclude a large grid minor (including a fat grid minor). Thus, Conjecture 2.24 is a coarse analogue to the classical Grid Minor Theorem [118].

Conjecture 2.25 (Coarse Kuratowski–Wagner [77]). *For each $q \in \mathbb{N}$, there exists $c \in \mathbb{N}$ such that each graph that excludes K_5 and $K_{3,3}$ as q -fat minors is c -quasi-isometric to a planar graph.*

Conjecture 2.25 is a coarse analogue of the Kuratowski–Wagner Theorem (Theorem 1.1 [96, 127]).

Conjecture 2.26 (Weak Fat Minor Conjecture [33]). *For each $q \in \mathbb{N}$ and graph H , there exists $c \in \mathbb{N}$ and graph H' such that each graph that excludes H as a q -fat minor is c -quasi-isometric to a graph that excludes H' as a minor.*

There is special interest in whether Conjecture 2.26 holds with $H' = H$. This stronger version was conjectured by Georgakopoulos and Papasoglu [77], and we refer to it as the ‘Fat Minor Conjecture’. If true, this would imply Conjecture 2.24. Further, if it holds for any list of excluded fat minors simultaneously, it would imply Conjecture 2.25.

If the above conjectures were true, it would demonstrate a deep and powerful link between classical and coarse graph theory. Unfortunately, except for Conjecture 2.25 (which is open), they have all been shown to be false. The Coarse Menger’s Conjecture was the first to fall, with Nguyen, Scott, and Seymour [110] showing that it is false when $k, d \geq 3$. Figure 2.1, shows the counterexample they obtained, which in this thesis we refer to as the *NSS graphs*. In a follow-up paper, Nguyen, Scott, and Seymour [109] showed that even the weaker Conjecture 2.23 fails when $k, d \geq 3$, with the counterexample being a multidimensional version of the NSS graphs. We refer to these graphs as the *multi-dimensional NSS graphs*.

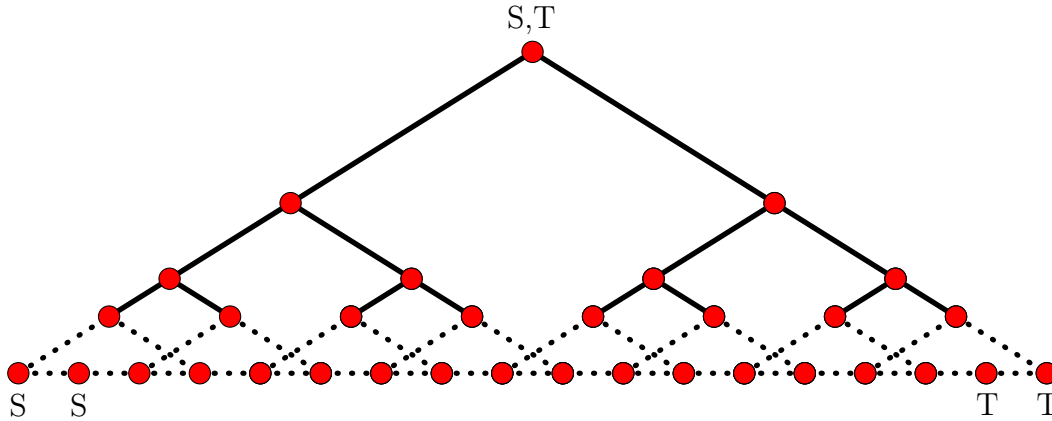


Figure 2.1: The NSS graph of height 5. Larger examples are obtained from increasing the height of the binary tree. Dotted edges represent long paths. The sets S, T that provide a counterexample to the Coarse Menger’s Conjecture are also depicted.

Davies et al. [33] used the NSS graphs as a building block to construct a counterexample to the Fat Minor Conjecture. Specifically, their graph H was obtained from two disjoint copies of K_{15} by adding a matching of three edges, and their graph G was obtained from a large subdivision of H by replacing the three paths corresponding to the matching with a large NSS graph, where the endpoints of the paths match the sets S, T for which the Coarse Menger’s Conjecture failed. They then used the fact that the Coarse Menger’s Conjecture failed in this construction to show that this construction violated the Fat Minor Conjecture.

Albrechtsen and Davies [1] used the multidimensional NSS graphs to disprove both Conjecture 2.24 and Conjecture 2.26. Specifically, they showed that these graphs exclude a fixed grid as a fat minor, but any graph quasi-isometric to them (for a sufficiently large starting graph) contains arbitrarily large complete minors.

However, before these counterexamples were known, there was some positive progress. We argue that because these conjectures are false in general, these positive examples are even more important, since knowing that these powerful conjectures hold in certain scenarios can be extremely helpful.

We refer to the graphs H that satisfy the (strong version of the) Fat Minor Conjecture as *suppressible*, and the graphs that violate this conjecture as *insuppressible*. So Davies et al. [33] showed the existence of an insuppressible graph on 30 vertices, and Albrechtsen and Davies [1] showed that large grids (and also large complete graphs) are insuppressible. By contrast, cycles [77, 104], stars [77], K_4 [5], and $K_{2,3}$ [5, 74] are known to be suppressible. Albrechtsen, Distel, and Georgakopoulos [2] expanded on this to show that $K_{2,t}$ is suppressible for any $t \geq 4$. However, in a follow-up paper,

Albrechtsen, Distel, and Georgakopoulos [3] showed that $K_{s,t}$ and K_n are insuppressible for any $t \geq s \geq 4$ and $n \geq 6$. This leaves K_5 and $K_{3,t}$ ($t \geq 3$), as critical unsolved cases, further motivated by Conjecture 2.25 also being unsolved.

Albrechtsen et al. [3] also showed that $K_{2,2,2}$ is insuppressible, which is of special interest due to being planar. Thus, being planar is not a sufficient condition to guarantee suppressibility, despite all known examples being planar. Interestingly, the examples that show that the $K_{2,2,2}$ is not suppressible are exactly the NSS graphs. In fact, these graphs also witness K_6 and $K_{4,t}$, $t \geq 4$, as being insuppressible. This further highlights that the NSS graphs are critical examples.

Albrechtsen et al. [4] and Georgakopoulos and Papasoglu [77] showed that the (strong version of) Coarse Menger's Conjecture holds when $k = 2$. That is, either there are two paths that are far apart, or one ball that intersects every path. See also [110] for a shorter proof. Interestingly, this fact was instrumental in the proof of Albrechtsen et al. [3] showing that $K_{2,2,2}$ (and other graphs) are insuppressible. In fact, this was used to show a deep fact about the NSS graphs, which essentially says that they are, to a certain extent, unchanged by quasi-isometry. So the 2-paths version of the Coarse Menger's Conjecture is used to prove a deep fact about a counterexample for the Coarse Menger's Conjecture for three or more paths.

An interesting variant of Menger's theorem replaces the requirement that paths join two sets (S and T) with the requirement that paths join distinct vertices in a single set A . Such a path is called an *A-path*. The goal is to find either k pairwise disjoint A -paths, or a set of at most $f(k)$ vertices that intersects each A -path. Gallai [76] showed that this can be done with $f(k) = 2(k - 1)$, which is the best possible. Geelen (at the Barbados Graph Theory Workshop held at the Bellairs Research Institute in March 2024) conjectured that this could be extended to a coarse setting.

Conjecture 2.27 (Weak Coarse Gallai's). *For each $k, d \in \mathbb{N}$, there exists $n, \ell \in \mathbb{N}$ such that for each graph G and $A \subseteq V(G)$, there exists either:*

1. *a set $X \subseteq V(G)$ with $|X| \leq n$ such that $N_G^\ell(X)$ intersects each A -path, or*
2. *k A -paths that are pairwise at distance at least d .*

Unlike the Weak Coarse Menger's Conjecture (Conjecture 2.23), Conjecture 2.27 is true. Distel, Giocanti, Hodor, Legrand-Duchesne, and Micek [47] showed that Conjecture 2.27 holds with $n = 4(k - 1)$ and $\ell = g(k, d)$, where g is a function that is linear in d but exponential in k . Thus, the falseness of Conjecture 2.23 depends on the fact that paths join distinct sets S and T . See also Hickingbotham and Joret [86] for an earlier proof of the $d = 2$ case of Conjecture 2.27 with $n = 78(k - 1)$ and $\ell = 1$, or with $n = 4(k - 1)$

and $\ell = 4$. (A proof of the $d = 2$ case of Conjecture 2.27 by Albrechtsen, Knappe, and Wollan was also announced in an online lecture given at the Banff workshop ‘New Perspectives in Colouring and Structure (24w5272)’ [128].)

Another interesting conjecture regarding fat minors is the following [77]:

Conjecture 2.28. *For each $h \in \mathbb{N}$, there exists $n \in \mathbb{N}$ such that for each $q \in \mathbb{N}$, the class of q -fat K_h -minor-free graphs has Assouad–Nagata dimension at most n .*

This question was first asked in terms of asymptotic dimension by Bonamy et al. [22], and then appeared as a conjecture by Georgakopoulos and Papasoglu [77] with $n = h - 2$. However, we suggest (and this was also suggested by Georgakopoulos and Papasoglu [77]) that $n = 2$ is the correct target, to match Theorem 2.18 (the Assouad–Nagata dimension of proper minor-closed classes). If Conjecture 2.26 were true, then Conjecture 2.28 would follow by Theorem 2.18 and Observation 2.21 (the latter of which says that quasi-isometry preserves Assouad–Nagata dimension). However, the falseness of Conjecture 2.26 even in the weakest sense makes Conjecture 2.28 even more interesting. If $n = 2$ in Conjecture 2.28 is possible, it would suggest that graphs that exclude a fat minor still have a similar global structure to minor-free graphs, even if they cannot be directly reduced to minor-free graphs by quasi-isometry. However, if Conjecture 2.28 requires $n \geq 3$ or is even false, it would demonstrate that excluding a fat minor gives a fundamentally different global structure to excluding a minor.

The only progress towards Conjecture 2.28 has been by Hickingbotham [85], who showed that if the maximum degree is bounded and the class is hereditary, then the asymptotic dimension is at most 2. However, the method used relies heavily on both assumptions. It also fails for Assouad–Nagata dimension, as Hickingbotham [85] showed that for such a class, the ‘Baker-treewidth’ is bounded, and used that to bound the asymptotic dimension. Bounded row-treewidth implies bounded Baker-treewidth; since bounding the row-treewidth does not bound the Assouad–Nagata dimension (Theorem 2.17 [22]), neither does bounding the Baker-treewidth. Thus, Conjecture 2.28 remains wide open in both the general setting and in terms of Assouad–Nagata dimension.

2.4 Coarse Analogues of Treewidth

2.4.1 Coarse Treewidth

It is natural to ask how treewidth can be extended to a coarse setting. One approach is to, instead of counting the number of vertices in a bag, count the fewest number of sets

of some fixed weak diameter needed to cover a bag. Nguyen, Scott, and Seymour [108] showed the following.

Theorem 2.29. *Let $k \in \mathbb{N}$ and $\ell \in \mathbb{R}^{\geq 0}$. If a graph G admits a tree-decomposition such that each bag is the union of at most k sets of weak diameter at most ℓ , then G is $2(k+2)\ell$ -quasi-isometric to a graph of treewidth at most k .*

The exact bounds for the coarseness in Theorem 2.29 are not directly stated by Nguyen et al. [108], but can be deduced from their proof (specifically, using [108, (2.3)] and the observation in the paragraph preceding it with $r := \ell$). Hickingbotham [84] independently showed a result similar to Theorem 2.29, but with a weaker bound on the treewidth (namely $2k - 1$), and a different value for the coarseness.

In Theorem 2.29, the bags are a union of k sets, but the quasi-isometry is to a graph of treewidth k . So the bags contain $k+1$ vertices. This increase from k to $k+1$ is annoying, but necessary for all $k \in \mathbb{N}^{\geq 1}$, as noted by Nguyen et al. [108]. However, Nguyen et al. [108] did show that by tweaking the definition of treewidth slightly (obtaining a new parameter called ‘pseudo-treewidth’), one can obtain an exact equality of the constants (pseudo-treewidth at most $k - 1$).

Up to a change in constants, the reverse direction also holds. Specifically, it is easily shown (and was shown by both Nguyen et al. [108] and Hickingbotham [84], albeit with slightly different bounds on the weak diameter) that if a graph is c -quasi-isometric to a graph of treewidth at most k , then it admits a tree-decomposition where each bag is the union of at most $k+1$ sets of weak diameter at most $2c^2 + c$. (To get the exact bound on the diameter, follow the proof of [84, Lemma 6], but use the precise bound $\text{dist}_H(\phi(u), \phi(v)) \leq c \text{dist}_G(u, v) + c$ rather than the cruder bound $\text{dist}_H(\phi(u), \phi(v)) \leq c \text{dist}_G(u, v) + c^2$.)

As such, Theorem 2.29 provides a characterisation of graphs that are quasi-isometric to graphs of bounded treewidth. The ease at which the reverse direction is obtained means that the primary contribution of this result is the ability to construct a quasi-isometry to a graph of bounded treewidth (by finding such a tree-decomposition).

Nguyen et al. [108] showed a result analogous to Theorem 2.29 for path-decompositions.

Theorem 2.30. *Let $k \in \mathbb{N}$ and $\ell \in \mathbb{R}^{\geq 0}$. If a graph G admits a path-decomposition where each bag is the union of at most k sets of weak diameter at most ℓ , then G is $2(k+2)\ell$ -quasi-isometric to a graph of pathwidth at most k .*

The reverse direction of Theorem 2.30 also holds with the same constants as the treewidth case (and using the same argument). So this gives a characterisation of

graphs that are quasi-isometric to graphs of bounded pathwidth, with the primary contribution being the ability to construct a quasi-isometry to a graph of bounded pathwidth.

2.4.2 Cliquewidth

Another analogue for treewidth on dense graphs is cliquewidth, which we now work towards defining.

Recall that a colour i is used by a colouring if at least one vertex is coloured i .

Let G be a graph, let $k \in \mathbb{N}^{\geq 1}$, and let c be a k -colouring of G . We say that (G, c) is a *cliquewidth- k pair* if either $|V(G)| \leq 1$, or if at least one of the following holds:

- (OP1) there exist cliquewidth- k pairs $(G_1, c_1), (G_2, c_2)$ such that G_1 and G_2 are nonempty, G is the disjoint union of G_1 and G_2 , and $c = c_1 \cup c_2$,
- (OP2) there exists a colouring c' of G and distinct $i, j \in \{1, \dots, k\}$ such that (G, c') is a cliquewidth- k pair, i, j are used by c' , and c is obtained from c' by changing the colour of every vertex coloured i under c' to j ,
- (OP3) there exists a proper subgraph G' of G and distinct $i, j \in \{1, \dots, k\}$ such that (G', c) is a cliquewidth- k pair, and G is obtained from G' by adding an edge between each pair of (non-adjacent) vertices $u, v \in V(G')$ with u coloured i and v coloured j .

We consider this definition to be a list of operations that build cliquewidth- k pairs.

The *cliquewidth* of a graph G is the minimum $k \in \mathbb{N}^{\geq 1}$ such that G admits a k -colouring c for which (G, c) is a cliquewidth- k pair. Note that the cliquewidth does not increase under taking induced subgraphs (this can be seen by taking the intersection/restriction at each stage of the construction).

There are some slight technical differences between our definition of cliquewidth and that of other authors. Specifically, we add the ‘nonempty’ condition in (OP1), ‘used’ condition in (OP2), and ‘proper’ condition in (OP3). These conditions prevent an infinite loop of operations, allowing for induction to be performed on the number of operations used to form a cliquewidth- k pair. Removing these conditions do not create any new cliquewidth- k pairs (it is easy to check that cliquewidth- k pairs are preserved under relabelling vertices and permuting the colours, so removing the ‘used’ condition in (OP2) does not create any new pairs), so they can be included without loss of generality to simplify technical aspects of some proofs that appear later in this thesis.

Unlike treewidth, cliquewidth can be bounded on dense graphs. In fact, cliquewidth is only meaningfully distinct from treewidth for dense graphs, since Gurski and Wanke [81] showed that if a graph excludes $K_{t,t}$ as a subgraph and has cliquewidth at most k , then it has treewidth at most $3k(t-1)-1$. For the reverse direction, Corneil and Rotics [31] showed that any graph of treewidth at most k has cliquewidth at most $3 \cdot 2^{k-1}$ (see [32] for an earlier bound).

This analogy can be furthered using the language of quasi-isometry. It can be concluded from Theorem 2.29 [84, 108] and the work of Hickingbotham [84] that graphs of bounded cliquewidth are quasi-isometric to graphs of bounded treewidth. In particular, every graph of cliquewidth k is $(4k+4)$ -quasi-isometric to a graph of treewidth at most $6k$. This is proved by using a different parameter called ‘sim-width’, which is at most the clique-width (see [114, Proposition 6.3] and [94, Lemma 2.2]). Combined with [84, Lemma 15] and [108, (1.2)], it can be deduced that each graph of cliquewidth k is $f(k)$ -quasi-isometric to a graph of treewidth at most $6k$. A slightly deeper dive into the proof of [108, (1.2)] (specifically, using [108, (2.3)] and the observation in the paragraph preceding it with $r = 2$) reveals that $f(k) = 4k + 4$ suffices.

This thesis improves on this result in the following ways: (1) we provide a direct and algorithmic construction using just the definitions of cliquewidth and treewidth, (2) the quasi-isometry parameter is smaller and independent of k , and (3) the bound on the treewidth is smaller. Specifically, we obtain the following.

Theorem 2.31. *For any $k \in \mathbb{N}^{\geq 1}$, each graph of cliquewidth at most k is 3-quasi-isometric to a graph of treewidth at most $k - 1$.*

We also show that the bound on the treewidth in Theorem 2.31 is within 2 of being tight.

Theorem 2.32. *For every $c \in \mathbb{R}^{\geq 1}$ and $k \in \mathbb{N}^{\geq 1}$, there exists a graph of cliquewidth at most k that is not c -quasi-isometric to any graph of treewidth less than $k - 3$.*

We actually prove the following strengthening of Theorem 2.31.

Theorem 2.33. *For any $k \in \mathbb{N}^{\geq 1}$, each graph G of cliquewidth at most k admits a dominated proper partition $\mathcal{P}$ of treewidth at most $k - 1$.*

Recall that a dominated partition is a partition in which each part is dominated. So each part is contained within the closed neighbourhood of a vertex, and thus has weak diameter at most 2. Since proper partitions into parts of bounded weak diameter give rise to quasi-isometries by Observation 2.22 (specifically with a coarseness factor 1

greater than the bound on the weak diameter), Theorem 2.31 follows from Theorem 2.33 and Observation 2.22.

Theorem 2.31, Theorem 2.18(b) (bounded treewidth graphs have Assouad–Nagata dimension 2), and Observation 2.21 (the existence of quasi-isometries bound the Assouad–Nagata dimension) together imply the following.

Theorem 2.34. *For any $k \in \mathbb{N}^{\geq 3}$, the class of graphs of cliquewidth at most k has asymptotic and Assouad–Nagata dimension 1.*

The lower bound follows from the fact that any class containing graphs of arbitrarily large diameter has asymptotic (and Assouad–Nagata) dimension at least 1, and the fact that any path has cliquewidth at most 3 (this is well-known and easy to show, but see Corollary 3.3 for a proof). We also remark that each connected component of a graph of cliquewidth at most 2 has diameter at most 3 (as it can be checked that P_4 has cliquewidth exactly 3). So the class of all graphs of cliquewidth at most 2 has asymptotic and Assouad–Nagata dimension 0.

Since the control functions for graphs of bounded treewidth grow with the treewidth, the corresponding control functions for graphs of bounded cliquewidth grows with the bound on the treewidth of the graphs they are quasi-isometric to (see Observation 2.20). Thus, our contribution improves the control function for bounded cliquewidth graphs. Since the known control functions for bounded treewidth graphs grows rapidly in the treewidth, this difference is significant, although the final control function is still very large.

We prove Theorems 2.32 and 2.33 in Chapter 3. These results first appeared in [43].

2.4.3 Characterisations of Quasi-Isometries

Recall that to construct a quasi-isometry between graphs of bounded cliquewidth and graphs of bounded treewidth (Theorem 2.31), we found a proper partition into parts of bounded weak diameter (Observation 2.22). The same method was used in Hickingbotham [84] and Albrechtsen et al. [2]. An interesting question is when does this type of quasi-isometry suffice. That is, given a quasi-isometry from a graph G to a graph H in a class $\mathcal{H}$, is there a proper partition of G into parts of bounded weak diameter such that the quotient is in $\mathcal{H}$. We do not insist that the quotient is isomorphic to H , as we are only interested in maintaining the structural properties of H . This thesis shows that this can be done when $\mathcal{H}$ is the class of graphs of treewidth (or pathwidth) at most $k \in \mathbb{N}$.

Theorem 2.35. *Let $c \in \mathbb{R}^{\geq 1}$ and $k \in \mathbb{N}$, and let G be a graph that is c -quasi-isometric to a graph H of treewidth [respectively pathwidth] at most k . Then G admits a proper partition $\mathcal{P}$ such that each part has weak diameter in G at most $(4(k+1)c^2 + c)$ whose quotient has treewidth [respectively pathwidth] at most k .*

The partition $\mathcal{P}$ is not necessarily connected, so $G/\mathcal{P}$ may not be a minor of G . The tree indexing the tree-decomposition of $G/\mathcal{P}$ is the same tree as the one indexing the tree-decomposition of H .

The proof of Theorem 2.35 is the subject of Chapter 4. Theorem 2.35 is a simple consequence of a new structural result about graphs of bounded treewidth/pathwidth (Theorem 4.4), which may be of independent interest. Section 4.1 explains the idea and motivation behind this new result, which takes inspiration from the ‘blocking partitions’ introduced by Distel et al. [50]. This proof first appeared in [41].

Combined with Observation 2.22 (to construct a quasi-isometry from a proper partition into bounded weak diameter parts), Theorem 2.35 provides a characterisation of graphs that are quasi-isometric to graphs of bounded treewidth/pathwidth. This invites comparison with Theorem 2.29 and Theorem 2.30, which characterised these graphs in terms of graphs that admit tree/path-decompositions in which each bag is the union of k sets of bounded weak diameter. The primary contribution of this characterisation is the ability to construct a quasi-isometry to a graph of bounded treewidth/pathwidth. By contrast, the primary contribution of Theorem 2.35 is to provide a structural description of graphs that are known to admit such a quasi-isometry. As such, Theorem 2.35 is complementary to Theorem 2.29. Further, neither Theorem 2.35 nor Theorem 2.29/Theorem 2.30 easily imply the other.

Chapter 3

Quasi-Isometry for Graphs of Bounded Cliquewidth

This chapter proves results regarding quasi-isometries between graphs of bounded cliquewidth and graphs of bounded treewidth, namely Theorems 2.31 and 2.32. Theorem 2.31 says that each graph of cliquewidth k is 3-quasi-isometric to a graph of treewidth at most $k - 1$. Theorem 2.32 says that there exists graphs of cliquewidth k that are not quasi-isometric to graphs of treewidth less than $k - 3$.

We prove Theorem 2.31 via Theorem 2.33, which asserts that graphs of cliquewidth k admit a dominated proper partition whose quotient has treewidth $k - 1$. The parts in a dominated partition have weak diameter at most 2, and thus this partition gives rise to a 3-quasi-isometry to the quotient (by Observation 2.22), as desired.

We prove Theorem 2.33 (and thus Theorem 2.31) in Section 3.1, and Theorem 2.32 in Section 3.2.

Recall that a graph has cliquewidth at most k if it admits a k -colouring c for which (G, c) is a cliquewidth- k pair. For ease of reference, we repeat the operations that define a cliquewidth- k pair here. (Recall that the base case is when $|V(G)| = 1$.)

- (OP1) there exist cliquewidth- k pairs $(G_1, c_1), (G_2, c_2)$ such that G_1 and G_2 are nonempty, G is the disjoint union of G_1 and G_2 , and $c = c_1 \cup c_2$,
- (OP2) there exists a colouring c' of G and distinct $i, j \in \{1, \dots, k\}$ such that (G, c') is a cliquewidth- k pair, i, j are used by c' , and c is obtained from c' by changing the colour of every vertex coloured i under c' to j ,
- (OP3) there exists a proper subgraph G' of G and distinct $i, j \in \{1, \dots, k\}$ such that (G', c) is a cliquewidth- k pair, and G is obtained from G' by adding an edge

between each pair of (non-adjacent) vertices $u, v \in V(G')$ with u coloured i and v coloured j .

Recall that the operations cannot loop, and thus we can perform induction on the number of operations needed.

3.1 An Improved Quasi-Isometry

Recall that a set S is rainbow under a colouring if for each used colour i , at least one vertex of S is coloured i . Recall that a partition is dominated if each part is contained within the closed neighbourhood of a vertex (this includes the vertex itself), and monochromatic if for each part P , there exists a colour i such that each vertex in P is coloured i . This induces a colouring of $G/\mathcal{P}$, where P is coloured i .

The proof of Theorem 2.33 inductively constructs a partition as we perform the operations, while maintaining that each part is monochromatic. The main obstacle in preserving the treewidth is (OP3). Whenever (OP3) is performed, we merge all the parts of colour i into a single part (and the same for all the parts of colour j), which is dominated because both i, j must have been used by c (as G' is a proper subgraph). We inductively ensure the existence of a rainbow bag, and maintain that all the parts of a given colour are contained in bags indexed by vertices that induce a subtree. Thus, even after merging all the parts coloured i/j and possibly adding a new edge to the quotient, we still have a tree-decomposition (as the new edge is handled by the rainbow bag).

This gives the following lemma, from which Theorem 2.33 is an immediate corollary.

Lemma 3.1. *For any $k \in \mathbb{N}^{\geq 1}$ and any cliquewidth- k pair (G, c) , G admits a dominated proper partition $\mathcal{P}$ that is monochromatic under c and whose quotient $G/\mathcal{P}$ admits a tree-decomposition $(B_t : t \in V(T))$ of width at most $k - 1$ such that:*

- (1) *there exists $q \in V(T)$ such that B_q is rainbow under the induced colouring $c_{\mathcal{P}}$ of $G/\mathcal{P}$, and*
- (2) *for each colour $i \in \{1, \dots, k\}$ that is used by c , the set of vertices $t \in V(T)$ such that B_t contains a part coloured i under c induces a nonempty subtree of T .*

Proof. We perform induction on the number of operations needed to build (G, c) .

In the base case, $|V(G)| \leq 1$. Let $\mathcal{P}$ be the partition of G into singletons, which is trivially proper, dominated in G , and monochromatic under c . Take T be a tree on a

single vertex t , and $B_t := \mathcal{P}$. $(B_t : t \in V(T))$ is trivially a tree-decomposition of $G/\mathcal{P}$ of width $|V(G)| - 1 \leq 0 \leq k - 1$. Further, B_t is rainbow under $c_{\mathcal{P}}$, and (2) is trivially satisfied.

So we may proceed to the inductive step, where at least one operation is used, and in particular, there is a last operation used. Let (G, c) be a cliquewidth- k pair, and assume the lemma is true for cliquewidth- k pairs that can be built with fewer operations. We break into cases depending on what the last operation used to obtain (G, c) is.

Case (OP1):

So there exist cliquewidth- k pairs (G_1, c_1) , (G_2, c_2) that can be constructed with fewer operations such that G_1 and G_2 are nonempty, G is the disjoint union of G_1 and G_2 , and $c = c_1 \cup c_2$. As (G_1, c_1) and (G_2, c_2) can both be built with fewer operations, induction applies. For $i \in \{1, 2\}$, let $\mathcal{P}_i$ and $(B_t^i : t \in V(T_i))$ be the proper partition of G_i and tree-decomposition of $G_i/\mathcal{P}_i$ respectively obtained via induction. So $\mathcal{P}_i$ is dominated in G_i , is c_i -monochromatic, and $(B_t^i : t \in V(T_i))$ has width at most $k - 1$ and obeys (1) and (2).

Let $\mathcal{P} := \mathcal{P}_1 \cup \mathcal{P}_2$. So $\mathcal{P}$ is a partition of G , and further it is proper, dominated in G , and monochromatic under c since, for each $i \in \{1, 2\}$, $\mathcal{P}_i$ is proper, dominated in G_i , and is c_i -monochromatic. So it remains to find a tree-decomposition of $G/\mathcal{P}$ that satisfies (1) and (2).

For each $i \in \{1, 2\}$, let $q_i \in V(T_i)$ be the vertex that satisfies (1) (with respect to (G_i, c_i)). Let T be the graph obtained from the disjoint union of T_1 and T_2 by adding a new vertex adjacent to q_1 and q_2 . So T is a tree.

Observe that for each colour $i \in \{1, \dots, k\}$ that is used by c , i is used by least one of c_1 or c_2 . Thus, and by definition of q_1, q_2 , $B_{q_1}^1 \cup B_{q_2}^2$ is rainbow under $c_{\mathcal{P}}$. Let B_q be a minimal subset of $B_{q_1}^1 \cup B_{q_2}^2$ that is rainbow. Note that no two parts in B_q share a colour, and thus $|B_q| \leq k$. Now, for each $t \in V(T) \setminus \{q\}$, let $B_t := B_t^i$ for the $i \in \{1, 2\}$ such that $t \in V(T_i)$.

We now show that $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$. Note that, subject to this being true, (1) follows immediately by definition of B_q , and that the width is at most $k - 1$ since $|B_q| \leq k$ and since, for each $i \in \{1, 2\}$, $(B_t^i : t \in V(T_i))$ has width at most $k - 1$.

For each $P \in \mathcal{P}$, since $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$, there exists $i \in \{1, 2\}$ such that $P \in \mathcal{P}_i$. Observe that the set S_P of vertices $t \in V(T)$ such that $P \in B_t$ is precisely the set of vertices of $t \in T_i$ such that $P \in B_t^i$, plus possibly q . Additionally, observe that $q \in S_P$ only if

$q_i \in S_P$. Thus and by induction, $T[S_P]$ is connected and nonempty, as desired.

For each $PP' \in E(G/\mathcal{P})$, there exists $xy \in E(G)$ such that $x \in P$ and $y \in P'$. Since G is the disjoint union of G_1 and G_2 , there exists $i \in \{1, 2\}$ such that $xy \in E(G_i)$. Observe that $P, P' \in \mathcal{P}_i$, and thus $PP' \in E(G_i/\mathcal{P}_i)$. By induction, there exists $t \in V(T_i) \subseteq V(T)$ such that $x, y \in B_t^i = B_t$.

Hence, $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$ of width at most $k - 1$. It remains only to show (2).

For each $i \in \{1, \dots, k\}$ used by c , and each $j \in \{1, 2\}$, let S_i^j be the set of vertices $t \in V(T_j)$ such that B_t^j contains a part coloured i . Observe that either i is not used by c_j , in which case S_i^j is empty, or S_i^j induces a connected subgraph of T_j (by (2)) and $q_j \in S_i^j$. Let S_i be the set of vertices $t \in V(T)$ such that B_t contains a part coloured i . Since i is used by c , observe that $S_i = S_i^1 \cup S_i^2 \cup \{q\}$. By previous observations and since q is adjacent to q_1 and q_2 , it follows that $T[S_i]$ is connected. So (2), and the lemma, hold.

Case (OP2):

So there exists a cliquewidth- k pair (G, c') that can be constructed with fewer operations and distinct $i, j \in \{1, \dots, k\}$, such that i, j are used by c' and c is obtained from c' by changing the colour of every vertex of colour i to j . As (G, c') can be built with fewer operations, induction applies. Let $\mathcal{P}$ and $(B_t : t \in V(T))$ be the proper partition of G and tree-decomposition of $G/\mathcal{P}$ respectively obtained via induction. So $\mathcal{P}$ is dominated in G , is c' -monochromatic, and $(B_t : t \in V(T))$ has width at most $k - 1$ and obeys (1) and (2) with respect to c' .

Observe that $\mathcal{P}$ is also monochromatic under c . Thus, we need only show that $(B_t : t \in V(T))$ obeys (1) and (2) with respect to c .

For (1), using induction, let $q \in V(T)$ be from (1) with respect to (G, c') . Let C be the set of colours that are used by c , and let C' be the set of colours that are used by c' . Since j is used by c' , observe that $C = C' \setminus \{i\}$. By choice of q , for each $\ell \in C$, there exists a part P_ℓ in B_q coloured ℓ under c' . Since $\ell \neq i$, P_ℓ is also coloured ℓ under c . It follows that B_q is rainbow under c_P . So (1) is satisfied.

For (2), by induction and since i is not used by c , observe that it suffices to consider only the colour j . For each $\ell \in \{i, j\}$, let S_ℓ be the set of vertices $t \in V(T)$ such that B_t contains a part coloured ℓ under c' . By (2), $T[S_\ell]$ is a nonempty subtree of T . Let S be the set of vertices $t \in V(T)$ such that B_t contains a part coloured j under c . Observe that $S = S_i \cup S_j$. Further, observe that $q \in S_i \cap S_j$ by (1) as i, j are used by

c' . Thus, since $T[S_i]$ and $T[S_j]$ induce nonempty subtrees of T with a common vertex, $T[S]$ induces a nonempty subtree of T , as desired. So (2), and the lemma, hold.

Case (OP3):

So there exists a cliquewidth- k pair (G', c) that can be constructed with fewer operations and distinct $i, j \in \{1, \dots, k\}$, such that G' is a proper subgraph of G and G is obtained from G' by adding an edge between each pair of (non-adjacent) vertices u, v with u coloured i and v coloured j . Note that because G' is a proper subgraph of G , both i and j are used by c .

As (G', c) can be built with fewer operations, induction applies. Let $\mathcal{P}'$ and $(B'_t : t \in V(T))$ be the proper partition of G' and tree-decomposition of $G'/\mathcal{P}'$ respectively obtained via induction. So $\mathcal{P}'$ is dominated in G' (and G), is monochromatic under c , and $(B'_t : t \in V(T))$ has width at most $k - 1$ and obeys (1) and (2).

For $\ell \in \{i, j\}$, let $\mathcal{P}_\ell \subseteq \mathcal{P}'$ be the set of parts coloured ℓ under c , and let $P_\ell := \bigcup_{P \in \mathcal{P}_\ell} P$. Since ℓ is used by c , P_ℓ is nonempty.

Set $\mathcal{P} := (\mathcal{P}' \setminus (\mathcal{P}_i \cup \mathcal{P}_j)) \cup \{P_i, P_j\}$. Observe that $\mathcal{P}$ is a proper partition of G (as P_i and P_j are nonempty).

We first show that $\mathcal{P}$ is dominated in G and monochromatic under c . By induction, it suffices to show that P_i, P_j are dominated in G and monochromatic under c . The latter property is immediate; by definition, each vertex of P_i is coloured i under c , and each vertex of P_j is coloured j under c . So we focus on the former property. Let $\ell \in \{i, j\}$, and let ℓ' be the unique element of $\{i, j\} \setminus \{\ell\}$. Since ℓ' is used by c , there is at least one vertex $v_{\ell'} \in V(G)$ coloured ℓ' under c . Every vertex of P_ℓ is coloured ℓ under c , and is therefore adjacent to $v_{\ell'}$ in G by construction. Thus, P_ℓ is dominated in G , as desired.

We move to finding the desired tree-decomposition of $G/\mathcal{P}$. For each $t \in V(T)$, let B_t be obtained from B'_t by replacing each instance of a part $P \in \mathcal{P}_i$ with P_i , and each instance of a part $P \in \mathcal{P}_j$ with P_j , deleting duplicates as necessary.

Before we show that $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$, we first show that (1) holds. Using induction, let $q \in V(T)$ be from (1) with respect to (G', c) . By choice of q and since i, j are used by c , there exists a part of $\mathcal{P}'$ coloured i and a part of $\mathcal{P}'$ coloured j in B'_q . Hence, $P_i, P_j \in B_q$. From this and by induction, it follows that B_q is rainbow under $c_{\mathcal{P}}$. So (1) holds.

We now show that $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$. Note that, by construction and subject to the previous statement being true, $(B_t : t \in V(T))$ has width at most the width of $(B'_t : t \in V(T))$, which is at most $k - 1$ by induction.

For each $P \in \mathcal{P}$, let S_P be the set of vertices $t \in V(T)$ such that $P \in B_t$. If $P \notin \{P_i, P_j\}$, observe that S_P is precisely the set of vertices $t \in V(T)$ such that $P \in B'_t$. In this case, $T[S_P]$ is nonempty and connected by induction. So we may assume that $P = P_\ell$ for some $\ell \in \{i, j\}$. In this case, observe that S_P is precisely the set of vertices $t \in V(T)$ such that B'_t contains a part coloured ℓ . By induction with (2), $T[S_P]$ induces a nonempty subtree of T .

Fix $PP' \in E(G/\mathcal{P})$, there exists $xy \in E(G)$ such that $x \in P$ and $y \in P'$. Observe that if neither P nor P' is in $\{P_i, P_j\}$, then neither x nor y is coloured either i or j . It follows that $xy \in E(G')$. Further, observe $P, P' \in \mathcal{P}'$, and thus $PP' \in E(G'/\mathcal{P}')$. In this case, by induction, there exists $t \in V(T)$ such that $P, P' \in B'_t$. Noting that neither P nor P' is $\mathcal{P}_i \cup \mathcal{P}_j$, we obtain $P, P' \in B_t$, as desired. So we may assume that at least one of P, P' is in $\{P_i, P_j\}$. Without loss of generality, we may assume that $P = P_i$ (noting that the roles of i, j are symmetric). Observe that there exists $Q \in \mathcal{P}_i$ such that $x \in Q$.

If $P' \neq P_j$, then observe that y is coloured neither i nor j , and thus $xy \in E(G')$. Note also that $P' \in \mathcal{P} \setminus \{P_i, P_j\} = \mathcal{P}' \setminus (\mathcal{P}_i \cup \mathcal{P}_j)$. So $QP' \in E(G'/\mathcal{P}')$, and thus by induction, there exists $t \in V(T)$ such that $Q, P' \in B'_t$. Since $Q \in \mathcal{P}_i$ and $P' \notin \mathcal{P}_i \cup \mathcal{P}_j$, we find that $P = P_i, P' \in B_t$, as desired. Otherwise, $P = P_i$ and $P' = P_j$. Recall that $P_i = P, P_j = P' \in B_q$, where $q \in V(T)$ is the vertex satisfying (1) with respect to $\mathcal{P}'$. It follows that $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$ of width at most $k - 1$.

Since we already showed that (1) holds, it remains only to show that (2) holds. By induction, observe that it suffices to only consider the colours i and j . However, P_i is the only part of $\mathcal{P}$ coloured i , and P_j is the only part of $\mathcal{P}$ coloured j . Since $P_i, P_j \in V(G/\mathcal{P})$ and since $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$, (2) follows, and the lemma is true.

We have shown that the lemma holds in all three cases, which completes the proof. $\square$

3.2 Lower Bounds on Treewidth

This section proves Theorem 2.32 which says that there exists graphs of cliquewidth k that are not quasi-isometric to graphs of treewidth less than $k - 3$. Specifically, we show that a sufficiently subdivided complete graph has the desired properties. The proof is a little technical, and is aided by including some preparatory lemmas, namely Lemmas 3.2 and 3.4. The goal of these lemmas is to find cliquewidth-pairs where the graphs are paths and spiders respectively and the colourings have specific properties. Specifically, in Lemma 3.2 the colouring of the path is such that the endpoints and

interior receive fixed colours. The colours of the endpoints must be distinct, but need not be distinct from the colour of the interior. In Lemma 3.4, the leaves of the spider receive pairwise distinct colours, and the interior receives a colour distinct from the colours of the leaves. The cliquewidth construction for subdivided complete graphs then comes from attaching new paths between the leaves of the spider, using the specific colourings to ensure that no extra edges are added.

Lemma 3.2. *Let $n \in \mathbb{N}^{\geq 3}$, and let $i, j, k \in \{1, \dots, n\}$ be such that $j \notin \{i, k\}$ and $\{1, \dots, n\} \setminus \{i, j, k\} \neq \emptyset$. Let P be a path on at least two vertices, and let c be a colouring of P obtained from giving one endpoint x the colour i , the other endpoint y the colour j , and all other vertices the colour k . Then (P, c) is a cliquewidth- n pair.*

Proof. By induction on $|V(P)|$. The base case is when $|V(P)| = 2$. Since $i \neq j$ and since $i, j \in \{1, \dots, n\}$, this case is trivial; start with the respective singleton graphs with the correct colours, and then use (OP1) and (OP3). So we proceed to the inductive step with $|V(P)| \geq 3$.

Since $\{1, \dots, n\} \setminus \{i, j, k\} \neq \emptyset$, there exists some $\ell \in \{1, \dots, n\}$ with $\ell \notin \{i, j, k\}$. Note that since $j \notin \{i, k\}$, $j \in \{1, \dots, n\} \setminus \{i, k, \ell\}$.

Let $P' := P - y$. Since $|V(P)| \geq 3$, $|V(P')| \geq 2$. Thus, P' has an endpoint y' distinct from x . Let c' be the colouring of P' obtained from giving x the colour i , y' the colour ℓ , and all other vertices the colour k . By induction (P', c') is a cliquewidth- n pair.

Observe that (P, c) can be obtained from (P', c') by adding a new vertex y coloured j , adding the edge yy' , and then recolouring y' to k . We can achieve the first step with (OP1), and then the second step with (OP3) with the colours j and ℓ as $j \neq \ell$, and since y is the only vertex coloured j and y' is the only vertex coloured ℓ . Finally, we can achieve the third step with (OP2), changing all the colour of all vertices coloured ℓ to k , as y' is the only vertex coloured ℓ . Thus, (P, c) is a cliquewidth- n pair, as desired. $\square$

We remark that Lemma 3.2 gives the following (well-known) result as a corollary.

Corollary 3.3. *Every path has cliquewidth at most 3.*

Proof. This is trivial for paths of length 1. For length at least 2, apply Lemma 3.2 with $n = 3$, $i = k = 1$, and $j = 2$. $\square$

Lemma 3.4. *Let $t \in \mathbb{N}^{\geq 3}$, let S be a spider with exactly t leaves, and let c be the colouring of S defined by giving each leaf a distinct colour in $\{1, \dots, t\}$, and all other vertices the colour $t + 1$. Then (S, c) is a cliquewidth- $(t + 3)$ pair.*

Proof. Let r be the unique vertex of degree at least 3 (which must exist as S has at least three leaves). Assume that the leaves are labelled $1, \dots, t$, matching the colour they received. Let L be the set of leaves adjacent to r .

For each $i \in \{1, \dots, t\}$, let $P'_i \subseteq S$ be the path from r to i , and let $P_i := P'_i - r$. If $|V(P_i)| = 1$ (which occurs if and only if $i \in L$), let $y_i := i$, and let c_i be the colouring of P_i obtained from giving i the colour i . In this case, (P_i, c_i) is trivially a cliquewidth- $(t+3)$ pair. Otherwise, let y_i be the endpoint of P_i different from i , and let c_i be the colouring of P_i obtained from giving i the colour i , y_i the colour $t+2$, and all other vertices the colour $t+1$. Observe that $\{1, \dots, t+2\} \setminus \{i, t+1, t+2\}$ is nonempty as $t \geq 3$, and that $t+2 \notin \{i, t+1\}$. Thus, by Lemma 3.2 with $n := t+2$, $x := i$, $y := y_i$, $j := t+2$, and $k := t+1$, (P_i, c_i) is a cliquewidth- $(t+2)$ pair (and thus also a cliquewidth- $(t+3)$ pair).

Let P_0 be the graph consisting of only the vertex r , and let c_0 be the colouring of P_0 where r is coloured $t+3$. (P_0, c_0) is trivially a cliquewidth- $(t+3)$ pair. Let S' be the disjoint union of $P_0, P_1, \dots, P_t$, and let $c' := c_0 \cup c_1 \cup \dots \cup c_t$. By repeated use of (OP1), (S', c') is a cliquewidth- $(t+3)$ pair. Observe that S can be obtained from S' by adding, for each $i \in \{1, \dots, t\}$, an edge ry_i , and that c can be obtained from c' by recolouring r to $t+1$, and, for each $i \in \{1, \dots, t\} \setminus L$, recolouring y_i to $t+1$. We can achieve the former by use repeated use of (OP3). First, with the colours $t+3$ and $t+2$, as r is the only vertex coloured $t+3$, and the set of vertices coloured $t+2$ is precisely $\{y_i : i \in \{1, \dots, t\} \setminus L\}$. Then, for each $i \in L$, with the colours $t+3$ and i , as $i = y_i$ is the only vertex coloured i . Finally, we can achieve the latter by two uses of (OP2), changing the colour all the vertices coloured $t+3$, and then $t+2$, to $t+1$. So (S, c) is a cliquewidth- $(t+3)$ pair, as desired. $\square$

We can establish the first property in Theorem 2.32 (the bound on the cliquewidth).

Lemma 3.5. *For each $n \in \mathbb{N}^{\geq 1}$, any subdivision G of K_n has cliquewidth at most $n+2$.*

Proof. Any subdivision of K_1 (a graph on a single vertex) has cliquewidth 1 (as $|V(G)| = 1$). Any subdivision of K_2 (a path) has cliquewidth at most 3 by Corollary 3.3. Taking $n := 4$ and i, j, k to be distinct in Lemma 3.2, and then adding the edge xy using (OP3) (with the colours i and j), we find that any subdivision of K_3 (a cycle) has cliquewidth at most 4. Thus, the lemma holds if $n \leq 3$. So we may assume that $n \geq 4$.

Let X be the vertices of G of degree at least 3. We consider the vertices in X to be labelled $1, \dots, t$.

Let S be the spider obtained from taking the union of all the paths from n to $\{1, \dots, n-$

1} with no internal vertices in $\{1, \dots, n\}$. So S is a spider with $n - 1 \geq 3$ leaves. Let c be the colouring of S obtained from giving each $i \in \{1, \dots, n\}$ the colour i and all other vertices the colour n . By Lemma 3.4 with $t := n - 1$, S is a cliquewidth- $(n + 2)$ pair.

For each pair $i, j \in \{1, \dots, n - 1\}$ with $i < j$, let $P_{i,j}$ be the path from i to j with no internal vertices in $\{1, \dots, n\}$, and let $P'_{i,j}$ be the (possibly empty) interior of $P_{i,j}$. If $|V(P'_{i,j})| = 1$, let v be the unique vertex of $P'_{i,j}$, set $x_{i,j} := y_{i,j} := v$, and let $c'_{i,j}$ be the colouring of $P'_{i,j}$ obtained by giving v the colour $n + 1$. It is immediate that $(P'_{i,j}, c'_{i,j})$ is a cliquewidth- $(n + 2)$ pair. If instead $|V(P'_{i,j})| \geq 2$, then let $c'_{i,j}$ be the colouring of $P'_{i,j}$ obtained from giving the vertex $x_{i,j}$ adjacent to i the colour $n + 1$, the vertex $y_{i,j}$ adjacent to j the colour $n + 2$, and every other vertex the colour n . Since $\{1, \dots, n + 2\} \setminus \{n, n + 1, n + 2\} \neq \emptyset$ as $n \geq 4$, by Lemma 3.2 with $n := n + 2$, $x := x_{i,j}$, $y := y_{i,j}$, $i := n + 1$, $j := n + 2$, and $k := n$, we obtain that $(P'_{i,j}, c'_{i,j})$ is a cliquewidth- $(n + 2)$ pair.

For each pair $i, j \in \{0, \dots, n - 1\}$ with $i \leq j$, let $G_{i,j} := S \cup \bigcup_{k=1}^i \bigcup_{\ell=k+1}^j P_{k,\ell}$, and let $c_{i,j}$ be the colouring of $G_{i,j}$ obtained by giving each $i \in \{1, \dots, n\}$ the colour i , and all other vertices the colour n . Observe that $G_{n-1,n-1} = G$ and that, for each $j \in \{0, \dots, n - 1\}$, $G_{0,j} = S$ and $c_{0,j} = c$. Note also that for each $i \in \{1, \dots, n - 1\}$, $G_{i-1,i} = G_{i,i}$ and $c_{i-1,i} = c_{i,i}$. Finally, observe that, for each pair $i, j \in \{1, \dots, n - 1\}$ with $i < j$, if $|V(P'_{i,j})| \geq 1$, then $G_{i,j}$ can be obtained from the disjoint union of $G_{i,j-1}$ and $P'_{i,j}$ by adding the edges $ix_{i,j}$ and $jy_{i,j}$, and $c_{i,j}$ can be obtained from $c_{i,j-1} \cup c'_{i,j}$ by recolouring both $x_{i,j}$ and $y_{i,j}$ to n . If instead $|V(P'_{i,j})| = 0$, then $G_{i,j}$ can be obtained from $G_{i,j-1}$ by adding the edge ij , and $c_{i,j-1} = c_{i,j}$.

Claim 1. *For each pair $i, j \in \{0, \dots, n - 1\}$ with $i \leq j$, $(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n + 2)$ pair.*

Proof. By induction on $i + j$. Observe that $i + j \geq 0$, and if $i + j = 0$, then $i = j = 0$. In this case, $G_{i,j} = S$ and $c_{i,j} = c$. Thus, $(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n + 2)$ pair, as desired. Additionally, observe that the same argument applies anytime $i = 0$. So we proceed to the inductive step, while assuming that $i \geq 1$.

If $i = j$, then set $i' := i - 1$. Observe that $i' + j < i + j$, so induction applies to $(G_{i',j}, c_{i',j})$. However, since $i = j = i' + 1$ and as observed above, $(G_{i',j}, c_{i',j}) = (G_{i,j}, c_{i,j})$. So $(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n + 2)$ pair, as desired.

Otherwise, assume $i < j$. We need to consider three cases based on $|V(P'_{i,j})|$.

If $|V(P'_{i,j})| = 0$, then, as observed above, $c_{i,j-1} = c_{i,j}$, and $G_{i,j}$ can be obtained from $G_{i,j-1}$ by adding the edge ij . Since i is the only vertex coloured i and since j is the only vertex coloured j , this can be achieved using (OP2) (as $i \neq j$). It follows that

$(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n+2)$ pair, as desired.

If $|V(P'_{i,j})| = 1$, then $G_{i,j}$ can be obtained from the disjoint union of $G_{i,j-1}$ and $P'_{i,j}$ by adding the edges $ix_{i,j}$ and $jy_{i,j} = jx_{i,j}$, and $c_{i,j}$ can be obtained from $c_{i,j-1} \cup c'_{i,j}$ by recolouring $x_{i,j} = y_{i,j}$ to n . Since i is the only vertex coloured i , since j is the only vertex coloured j , and since $x_{i,j}$ is the only vertex coloured $n+1$, the former can be achieved by (OP1) and then two uses of (OP3); first with the colours i and $n+1$, and then with the colours j and $n+1$. We can then achieve the latter with (OP2), changing the colour of each vertex coloured $n+1$ to n . It follows that $(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n+2)$ pair, as desired.

If $|V(P'_{i,j})| \geq 2$, then $G_{i,j}$ can be obtained from the disjoint union of $G_{i,j-1}$ and $P'_{i,j}$ by adding the edges $ix_{i,j}$ and $jy_{i,j}$, and $c_{i,j}$ can be obtained from $c_{i,j-1} \cup c'_{i,j}$ by recolouring both $x_{i,j}$ and $y_{i,j}$ to n . Since i is the only vertex coloured i , since j is the only vertex coloured j , since $x_{i,j}$ is the only vertex coloured $n+1$, and since $y_{i,j}$ is the only vertex coloured $n+2$, the former can be achieved by (OP1) and then two uses of (OP3); first with the colours i and $n+1$, and then with the colours j and $n+2$. We can then achieve the latter with two uses of (OP2), first changing the colour of every vertex coloured $n+1$ to n , and then changing the colour of every vertex coloured $n+2$ to n . It follows that $(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n+2)$ pair, as desired.

Thus, in all cases, $(G_{i,j}, c_{i,j})$ is a cliquewidth- $(n+2)$ pair. This completes the inductive step, and the proof of the claim. $\square$

The lemma follows from Claim 1 as $G_{n-1,n-1} = G$. $\square$

It remains to show that the second property in Theorem 2.32 (the lower bound on the treewidth of any graph that a subdivision of the complete graph is quasi-isometric to). We first need the following observation, which we suspect is well-known, but include for completeness.

Observation 3.6. *Let $c \in \mathbb{R}^{\geq 1}$, and let H, H', G be graphs such that H' is a $\geq (4c(c+1) - 1)$ -subdivision of H and c -quasi-isometric to G . Then G contains H as a minor.*

Proof. Note that adjacent vertices in H are at distance at least $4c(c+1)$ in H' . Let $\phi : V(H') \mapsto V(G)$ be a c -quasi-isometry from H' to G .

For each $v \in V(H)$, let $X_v := N_{H'}^{c(c+1)}(v)$. Note that $H'[X_v]$ is connected. Further, for each pair of distinct $v, v' \in V(H)$, observe that $\text{dist}_{H'}(X_v, X_{v'}) \geq 4c(c+1) - 2c(c+1) = 2c(c+1)$.

For each $vv' \in E(H)$, let $P_{vv'}$ be the vertices of the unique path in H' between X_v and $X_{v'}$ with no internal vertices in $V(H)$. Note that $P_{vv'}$ intersects X_v and $X_{v'}$, and that $P_{vv'}$ is at distance exactly $c(c+1)$ from both v and v' . Thus, observe that for distinct $vv', ww' \in E(H)$, $\text{dist}_{H'}(P_{vv'}, P_{ww'}) \geq 2c(c+1)$. Note also that for each $vv' \in E(H)$ and each $w \in V(H)$ that is not an endpoint of ww' , $\text{dist}_{H'}(P_{vv'}, X_w) \geq 2c(c+1)$.

For each $v \in V(H)$, let $X'_v := N_G^c(\phi(X_v))$. Since $H'[X_v]$ is connected and since adjacent vertices are mapped to vertices at distance at most $c(1) + c = 2c$, $G[X'_v]$ is connected. Further, for each pair of distinct $v, v' \in V(H)$, observe that $2c(c+1) \leq \text{dist}_{H'}(X_v, X_{v'}) \leq c \text{dist}_G(f(X_v), f(X_{v'})) + c$, so $\text{dist}_G(f(X_v), f(X_{v'})) \geq 2(c+1) - 1 = 2c+1$. Thus, $\text{dist}_G(X'_v, X'_{v'}) \geq 2c+1 - 2c = 1$. In particular, X'_v and $X'_{v'}$ are disjoint.

For each $vv' \in E(H)$, let $P'_{vv'} := N_G^c(f(P_{vv'}))$. Using the same argument as above, $G[P'_{vv'}]$ is connected, and for each pair of distinct $vv', ww' \in E(H)$, $P'_{vv'}$ and $P'_{ww'}$ are disjoint. Similarly, for each $vv' \in E(H)$ and each $w \in V(H)$ that is not an endpoint of vv' , $P'_{vv'}$ and X'_w are disjoint. Finally, note that, for each $vv' \in E(H)$, $P'_{vv'}$ intersects both X'_v and $X'_{v'}$. Observe that within $P'_{vv'}$, we can find a path $P''_{vv'}$ that is internally disjoint from X'_v and $X'_{v'}$, but whose endpoints are in X'_v and $X'_{v'}$. It follows that $((X'_v : v \in V(H)), (P''_{vv'} : vv' \in E(H)))$ is a model of H in G . So $H \leq G$, as desired. $\square$

The following is an easy corollary of Observation 3.6.

Corollary 3.7. *For every $c \in \mathbb{R}^{\geq 1}$ and $n \in \mathbb{N}$, each $\geq (4c(c+1) - 1)$ -subdivision of K_n is not c -quasi-isometric to any graph of treewidth less than $n - 1$.*

Proof. Let H' be a $\geq (4c(c+1) - 1)$ -subdivision of K_n , and let G be a graph such that H' is c -quasi-isometric to G . By Observation 3.6, G contains K_n as a minor. Since K_n has treewidth $n - 1$, and treewidth does not increase under taking a minor, we have $\text{tw}(G) \geq n - 1$, as desired. $\square$

We can now complete the proof of Theorem 2.32.

Proof. If $k \leq 3$, then any graph G with one vertex has cliquewidth $1 \leq k$, but each graph that is c -quasi-isometric to G is nonempty and thus has treewidth at least $0 \geq k - 3$. Now, assume that $k \geq 4$.

Let G be the $(4c(c+1) - 1)$ -subdivision of K_{k-2} . By Lemma 3.5 (with $n := k - 2 \geq 1$), G has cliquewidth at most k . By Corollary 3.7, G is not c -quasi-isometric to any graph of treewidth less than $k - 3$. $\square$

Chapter 4

Quasi-Isometries to Bounded Treewidth Graphs

This chapter proves Theorem 2.35. This result characterises graphs that are c -quasi-isometric to graphs of treewidth [respectively pathwidth] at most k as the graphs that admit a proper-partition into parts of bounded weak diameter whose quotient has treewidth [respectively pathwidth] at most k .

4.1 Compressing Partitions

Given graphs G, H , say that an *H -indexed partition* of G is a pair $(\mathcal{P}, \psi)$, where $\mathcal{P}$ is a partition of G and ψ is a bijection from $V(H)$ to $\mathcal{P}$. We usually leave this bijection implicit, and write the partition $\mathcal{P}$ as $(P_h : h \in V(H))$ (where $P_h := \psi(h)$ for each $h \in V(H)$). We call P_x , $x \in V(H)$, the *parts* of $(P_h : h \in V(H))$, and we say that $(P_h : h \in V(H))$ is *proper* if $\mathcal{P}$ is proper (equivalently, each part is nonempty). Say that $(P_h : h \in V(H))$ is an *H -partition* if distinct $h, h' \in V(H)$ are adjacent in H whenever there exists $uv \in E(G)$ with $u \in P_h$ and $v \in P_{h'}$. In this case, if $(P_h : h \in V(H))$ is proper, then ψ is an isomorphism from a spanning subgraph of H to $G/\mathcal{P}$.

We think of a partition $\mathcal{P}$ of a graph G as being $G/\mathcal{P}$ -indexed, using the identity as the bijection. So $\mathcal{P}$ is a $G/\mathcal{P}$ -partition.

The launching point for Theorem 2.35 is the following proposition, which we do not use directly, but include for intuition.

Proposition 4.1. *Let $c \in \mathbb{R}^{\geq 1}$, and let G, H be graphs such that G is c -quasi-isometric to H . Then G admits an H^{2c} -partition such that each part has weak diameter in G at*

most c .

Proof. Let ϕ be a c -quasi-isometry from G to H . For each $h \in V(H)$, let $P_h := \{\phi^{-1}(h)\}$. Observe that $(P_h : h \in V(H)) = (P_h : h \in V(H^{2c}))$ is an H^{2c} -indexed partition of G .

Let distinct $h, h' \in V(H) = V(H^{2c})$ be such that there exists $uv \in E(G)$ with $u \in P_h$ and $v \in P_{h'}$. So $h = \phi(u)$ and $h' = \phi(v)$. Observe that $\text{dist}_H(h, h') \leq c \text{dist}_G(u, v) + c = c(1) + c = 2c$. Thus, h is adjacent to h' in H^{2c} . Hence, $(P_h : h \in V(H^{2c}))$ is an H^{2c} -partition of G .

Fix $h \in V(H)$, and consider any pair $u, v \in P_h$. So $\phi(u) = \phi(v) = h$, and thus $\text{dist}_H(\phi(u), \phi(v)) = 0$. Hence, we have $\text{dist}_G(u, v) \leq c \text{dist}_H(\phi(u), \phi(v)) + c = c(0) + c = c$. So P_h has weak diameter in G at most c . This completes the proof. $\square$

If graph powers preserved treewidth and pathwidth, then we could apply Proposition 4.1 (and delete vertices corresponding to empty parts and edges not in the quotient) to immediately obtain Theorem 2.35. However, if the degree is unbounded, then this is false. Every star has treewidth and pathwidth at most 1, but the second power is a complete graph, whose treewidth grows proportionally to the number of vertices in the clique (and thus the original star). So we need to eliminate the power.

For this, we draw inspiration from the concept of ‘blocking’ partitions, introduced by Distel et al. [50]. These are connected partitions in which every long (at least some fixed constant) path intersects some part at least twice. Such a partition trivially exists (by having one part for each connected component), so they aimed to construct such a partition with bounded width. Their main focus was (bounded degree) planar graphs, but more relevant to this present work, they also briefly considered graphs of bounded treewidth (and bounded degree).

A key feature of a blocking partition $\mathcal{P}_1$ is that each long path is compressed into a strictly shorter path in the quotient. More precisely, for a long path $v_0, \dots, v_n$, if $P_0, \dots, P_n$ is the sequence of parts containing $v_0, \dots, v_n$ respectively, then by assumption, $P_i = P_j$ for some pair of distinct $i, j \in \{0, \dots, n\}$. Thus, within $P_0, \dots, P_n$ there is a path from P_0 to P_n in $G/\mathcal{P}_1$ of length at most $n - 1$. Therefore, the structure of G^n can be described using the structure of $(G/\mathcal{P}_1)^{n-1}$.

Because the partition $\mathcal{P}_1$ is connected, the quotient is also planar/has bounded treewidth, so the argument can be repeated. This finds a connected partition $\mathcal{P}'_2$ of the quotient of bounded width, which can be thought of as a connected partition $\mathcal{P}_2$ of G of bounded width. Because both $\mathcal{P}_1$ and $\mathcal{P}'_2$ are blocking, every long path in G is reduced in length by 2 in $G/\mathcal{P}_2$. So the structure of G^n can be describing using the structure of $(G/\mathcal{P})^{n-2}$.

This argument is repeated inductively, giving a sequence $\mathcal{P}_1, \mathcal{P}_2, \dots$ of connected partitions of G of bounded width. Each partition $\mathcal{P}_i$ is such that every long path in G is reduced in length by i in $G/\mathcal{P}_i$. So the structure of G^n can be described using the structure of $(G/\mathcal{P}_i)^{n-i}$, provided that n is sufficiently large. In particular, since $G/\mathcal{P}_i$ is connected, the structure of G^n can be described using the structure of H_i^{n-i} for a planar/bounded treewidth graph H_i (namely $G/\mathcal{P}_i$).

The limiting factor is that a path of length $n - i + 1$ (the length in H_{i-1}) needs to be long, so that the blocking partition of H_{i-1} guarantees a reduction in the length of the path. Therefore, if a path needs to be length at least ℓ to be long, the structure of G^n can be described using the structure of $H^{\ell-1}$ for a planar/bounded treewidth graph H (namely $H_{n-\ell+1}$). We can summarise this by saying ‘large powers of planar/bounded treewidth graphs can be described using small powers of planar/bounded treewidth graphs’.

This is ideal for our purposes, since we want to reduce H^{2^c} to a graph H' of bounded treewidth. However, there are a few obstacles. Firstly, these partitions only exist when the degree is bounded. We fix this by relaxing the requirement that the parts have bounded size, instead asking that the parts have bounded weak diameter in G . This is natural, since when the degree is bounded, the parts have bounded size.

However, more critically, we cannot reduce beyond the maximum length for which a path is considered short. For bounded treewidth graphs, the partition obtained by Distel et al. [50] treats every path of length at least 3 as long, and hence the power can be reduced to 2. While this is a huge improvement, it still destroys the bound on the treewidth, as seen with stars. We need to remove the power completely.

To achieve this, we modify the concept of blocking partitions. For us, the key property of blocking partitions is that distances in the graph are shrunk in the quotient. We take this as a defining property, and then tweak it by instead asking that the distances decrease in some graph H for which $\mathcal{P}$ is an H -indexed partition. In fact, instead of just decreasing the power by 1 each step, we can directly demand that two vertices that are close (adjacent in the power) belong to adjacent parts. This leads to the following definition.

For a graph H and $\ell \in \mathbb{R}^{\geq 0}$, say that an H -indexed partition $(P_h : h \in V(H))$ of a graph G is *ℓ -compressing* (in G) if for each pair of distinct $h, h' \in V(H)$ with $\text{dist}_G(P_h, P_{h'}) \leq \ell$, h is adjacent to h' in H . Equivalently, $(P_h : h \in V(H))$ is an H -partition of G^ℓ .

Note that we have dropped the requirement that the partition is connected. Because we allow H to have more edges than the quotient, H may not be a minor of G even if the

quotient is. Thus, it is less important that the quotient is a minor of G . However, more importantly, we cannot ask both that $H = G/\mathcal{P}$ (with the identity map from H to $\mathcal{P}$) and that $\mathcal{P}$ is connected, as such a partition cannot be 2-compressing for large binary trees.

Proposition 4.2. *For each $d \in \mathbb{N}$, the binary tree T of height $d + 1$ does not admit a connected partition $\mathcal{P}$ that is 2-compressing in G (as a $(G/\mathcal{P})$ -indexed partition with the identity map) and such that each $P \in \mathcal{P}$ has weak diameter at most d .*

Proof. Presume otherwise. Let r be the root of T . So each leaf is at distance exactly $d + 1$ from r . Let P be the part containing r . Since $\mathcal{P}$ is connected, $G[P]$ is a subtree of T . Either $P = \{r\}$, in which case set $t := r$, or there exists a leaf t of $G[P]$ other than r . Since P has weak diameter at most d in G and since $r \in P$, t is not a leaf of T . Thus, t has two children c_1, c_2 in T , neither of which are in P (by choice of t , and since $G[P]$ is connected and contains r). Since $\mathcal{P}$ is connected and since $t \in P$, c_1, c_2 are in distinct parts of $\mathcal{P}$ that are non-adjacent in $G/\mathcal{P}$. However, $\text{dist}_G(c_1, c_2) = 2$, which contradicts the fact that $\mathcal{P}$ is 2-compressing in G . $\square$

We remark (without proof) that Proposition 4.2 can be extended to many other classes of trees and graphs of bounded treewidth.

Proposition 4.2 eliminates the main advantage of connected partitions, since we cannot take $H := G/\mathcal{P}$ to conclude $H \leq G$. So we find it more natural to drop the connected requirement and make no attempt to relate G and H . Instead, we construct a tree-decomposition of H directly. This allows for more flexibility in choice of H and $\mathcal{P}$, which is important for our proof.

Because we no longer relate the graphs G and H , it is natural to tie the existence of compressing partitions to a class of graphs. This leads to the following definition.

Let $\mathcal{H}$ be a class of graphs. Say that a function $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ is a *compressing function* for $\mathcal{H}$ if for each $\ell \in \mathbb{R}^{\geq 0}$ and $H \in \mathcal{H}$, there exists $A \in \mathcal{H}$ and an A -indexed partition of H that is ℓ -compressing and such that each part has weak diameter in H at most $f(\ell)$.

If a hereditary class $\mathcal{H}$ admits a compressing function, then being quasi-isometric to some $H \in \mathcal{H}$ is equivalent to admitting, for some $H' \in \mathcal{H}$, a proper H' -partition into bounded weak diameter parts.

Lemma 4.3. *Let $\mathcal{H}$ be a hereditary class of graphs, and let $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ be a compressing function for $\mathcal{H}$. Then for every $c \in \mathbb{R}^{\geq 1}$ and graph G that is c -quasi-isometric to some $H \in \mathcal{H}$, there exists $A \in \mathcal{H}$ and a proper A -partition of G such that each part has weak diameter at most $cf(2c) + c$ in G .*

Proof. Let ϕ be a c -quasi-isometry from G to H . Since f is a compressing function for $\mathcal{H}$, there exists a graph $A' \in \mathcal{H}$ and an A' -indexed partition $(P_a : a \in V(A'))$ of H that is $2c$ -compressing and such that each part has weak diameter at most $f(2c)$ in H .

For each $a \in V(A')$, let $P'_a := \phi^{-1}(P_a)$. Let A be the subgraph of A' induced by the vertices $a \in V(A')$ such that $P'_a \neq \emptyset$. Observe that $(P'_a : a \in V(A))$ is a proper A -indexed partition of G . Since $\mathcal{H}$ is hereditary and $A' \in \mathcal{H}$, we have $A \in \mathcal{H}$.

Let distinct $a, a' \in V(A)$ be such that there exists $uv \in E(G)$ with $u \in P'_a$ and $v \in P'_{a'}$. So $\phi(u) \in P_a$ and $\phi(v) \in P_{a'}$. Observe that $\text{dist}_H(\phi(u), \phi(v)) \leq c \text{dist}_G(u, v) + c = c(1) + c = 2c$. Thus, $\text{dist}_H(P_a, P_{a'}) \leq 2c$. Since $(P_a : a \in V(A'))$ is a $2c$ -compressing partition of H , we have that a and a' are adjacent in A' and thus A . It follows that $(P'_a : a \in V(A))$ is an A -partition of G .

Fix $a \in V(A)$, and consider any pair $u, v \in P'_a$. So $\phi(u), \phi(v) \in P_a$. As P_a has weak diameter at most $f(2c)$ in H , we have $\text{dist}_G(u, v) \leq c \text{dist}_H(\phi(u), \phi(v)) + c \leq cf(2c) + c$. So P'_a has weak diameter at most $cf(2c) + c$ in G . This completes the proof. $\square$

Observation 2.22 provides the reverse direction.

The main technical result of this chapter is as follows.

Theorem 4.4. *For each $k \in \mathbb{N}$, the function $\ell \mapsto 2(k+1)\ell$ is a compressing function for the class of graphs of treewidth at most k and the class of graphs of pathwidth at most k .*

Theorem 4.4 is also true if we replace ‘treewidth/pathwidth’ with ‘admits a T -decomposition of width at most k for some $T \in \mathcal{T}$ ’, where $\mathcal{T}$ can be any class of trees.

Theorem 2.35 follows by applying Theorem 4.4 and Lemma 4.3 (noting that treewidth does not increase under taking subgraphs). The remainder of this chapter is devoted to proving Theorem 4.4.

4.2 Proof Explanation

We now explain the idea behind the proof of Theorem 4.4.

Fix $\ell \in \mathbb{R}^{\geq 0}$, a tree-decomposition $(J_t : t \in V(T))$ of width k of a graph G , and a root $r \in V(T)$ for T . We want to find a graph H , a T -decomposition of width k of H , and

an ℓ -compressing H -indexed partition of G whose parts have weak diameter at most $2(k+1)\ell$ in G . If this can always be done, then Theorem 4.4 follows.

The key idea is relatively simple. Each part is contained in the $(k+1)\ell$ -neighbourhood of some vertex, which we will call the *centre* of the part. Critically, the centre of the part may not be in the part. The parts are indexed by these centres, so $V(H)$ is precisely the set of centres. Inductively working along T , the proof carefully picks new vertices to act as centres.

For each bag J_t of the tree-decomposition, there is a corresponding set of centres B_t of size at most $k+1$. The T -decomposition of H will be $(B_t : t \in V(T))$, and distinct centres are adjacent in H if and only if they are in a common ‘bag’ B_t .

The initial set of centres B_r is J_r . Then for $t \in V(T) \setminus \{r\}$, we inductively select the centres B_t based on the centres $B_{t'}$ for the parent t' .

To each centre x , we associate not just a part, but also two special zones, which we call the ‘security’ and ‘coverage’. The coverage is the $i\ell$ -neighbourhood of x , for a specially picked $i \in \{1, \dots, k+1\}$. These are the vertices that could go in the part P_x . However, the coverages are not disjoint, so some vertices in the coverage will be assigned to other parts.

For each $v \in V(G)$, let $r(v)$ be the (unique) vertex of $V(T)$ closest to r such that $v \in J_{r(v)}$. We can create a priority system (total ordering) for $V(G)$, such that $v \in V(G)$ has higher priority than $u \in V(G)$ if $r(v)$ is a strict ancestor of $r(u)$. We assign each $v \in V(G)$ to the part P_x of the highest priority centre x whose coverage contains v and such that $r(x)$ is an ancestor of $r(v)$ (such a centre will exist). From an inductive point of view, this centre is also the oldest centre, in the sense that the algorithm created it before (or at the same time as) any other alternatives. So we prioritise putting vertices in the oldest parts. Critically, this can (and frequently will) result in x not being assigned to the part P_x .

The security is the $(i-1)\ell$ -neighbourhood of x (where i is the same as for the coverage). So any vertex v in the ℓ -neighbourhood of the security is in the coverage, and could be assigned to P_x . In particular, if x is sufficiently old, v will be assigned to P_x .

Our goal is to ensure that whenever we discard a centre $x \in B_t \setminus B_{t'}$, every vertex v in the coverage is either (a) contained in the coverage of an older (higher priority) centre, or (b) is in the security of a vertex in $B_t \cap B_{t'}$. In the former case, v is not in the part P_x (and is thus not relevant), and in the latter case, we find that every vertex in the ℓ -neighbourhood of x is in the coverage of a vertex in $B_t \cap B_{t'}$, and thus gets assigned to one of these parts by our priority system.

However, there is an obvious problem: what if v is at the limit of the coverages of two (or more) centres, one of which we must discard? In this case, we must grow the coverage of the other centre so that v becomes secure (is contained in the security). If repeated, this can lead to the coverage (and thus the part) having unbounded weak diameter. This issue can be addressed by limiting the diameters of the security/coverage based on the age of the centre.

Since each bag J_t is a separator, we need only consider the $k + 1$ youngest centres (B_t). Starting with the ℓ -neighbourhood for the coverage (and 0-neighbourhood for the security) for the newest (lowest priority) centre, for each increase in age (priority), we add an extra ℓ -neighbourhood to the security and coverage. As we consider at most $k + 1$ centres at a time, at most k increases are performed. Thus, the coverage is at most the $(k + 1)\ell$ -neighbourhood.

We stress that we are now considering securities/coverages to be relative to a vertex t of the tree. A centre will start with only a small coverage and security, but as it gets older (we move along the tree to a descendant), it is allowed to grow. This growth need not be consistent either. On one child, the security and coverage may grow, on another they may stay the same, and on yet another the centre may be discarded. However, critically, the security and coverage does not shrink (provided the centre is kept). Once it is grown, that growth is permanent, for as long as that centre is kept.

We also assign each $v \in J_t$ to a centre $x \in B_{t'}$, which we say v ‘belongs to’. This is different to which part $P_{x'}$ v is assigned to, as we might have $x' \notin B_{t'}$ (either because x' was discarded in a previous step, or because v is too far from any of the existing centres and needs a new centre). To chose x , instead of just asking ‘what is the closest centre’ or ‘what is the oldest centre’, we instead ask ‘what is the closest centre, relative to its age’. Specifically, we minimise the quantity $\text{dist}_G(x, v) + i\ell$, where i is the ‘penalty’ of the centre, starting at 0 for the oldest (highest priority) centre and increasing by 1 with each increase in age. In the event of a tie, we break the tie using the priority.

Note that the penalties are the reverse order of the relative priorities. So for each decrease by 1 in the penalty, we add an extra ℓ -neighbourhood to the security and coverage.

The intuition behind the quantity $\text{dist}_G(x, v) + i\ell$ only applies when v is in the coverage of at least one centre in $B_{t'}$. (The other case is less important, as then v is far enough away from any existing centres.) In this case, we want to pick the centre x whose coverage extends as far past v as possible. That is, we maximise the distance to a vertex not in the coverage. This creates a buffer of vertices in the coverage that is as wide as possible. The centre that maximises the width of this buffer is precisely the

centre that minimises $\text{dist}_G(x, v) + i\ell$.

When we discard a centre, we decrease the penalties of all centres older than it by 1, and thus add an extra ℓ -neighbourhood to their securities and coverages. This means that in our ‘tied’ example before, the vertex v that was at the edge of two coverages is now in the security of the newer centre x' after the older centre x is discarded. (The newer centre being discarded is not a concern, as we would never put v in $P_{x'}$ over P_x .) See Figure 4.1.

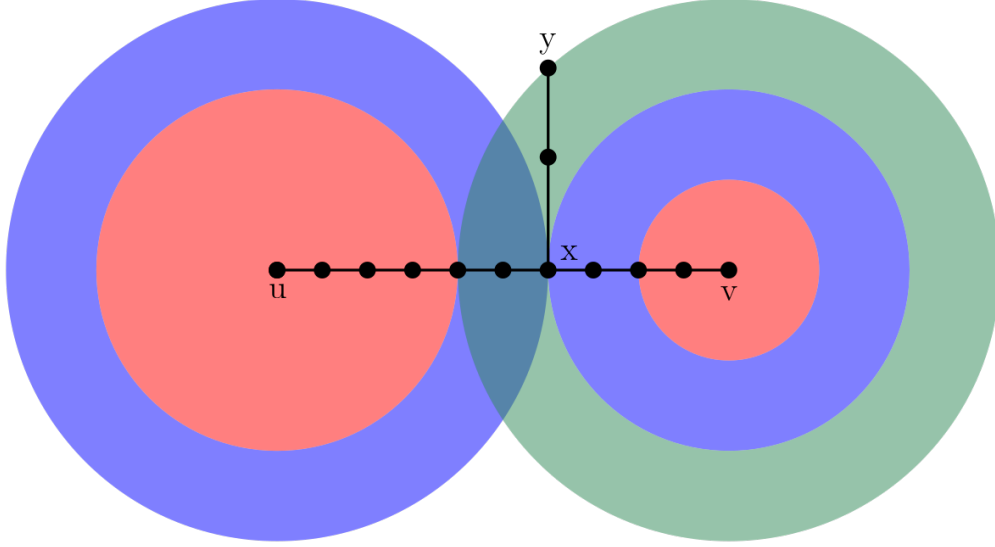


Figure 4.1: An example of how the algorithm grows the security/coverage. Here, $\ell = 2$, and u and v are centres, with u being older. The red regions are the securities, and the blue regions are the coverages. (For visual clarity, we have chosen to draw these regions bigger than what is correct for two centres; a third centre younger than u and v is needed to achieve the depicted regions.) When the algorithm reaches a separator (which will be an adhesion set in the tree-decomposition) such as x , it will decide which centres it needs to keep, and grow them accordingly. Here, it chose to discard u and keep v . Since v was younger, its securities and coverages grow; now the blue is the security, and the green is the coverage. Notably, x is now in the security, so every vertex at distance ℓ (2) from x , such as y , is now in the coverage. x will also become a new centre, as it was insecure (at the start of the step).

Occasionally, we must pick new centres. We do this if the bag contains insecure (not in a security) vertices (decided before the securities are grown). These vertices become new centres. From this, we can inductively ensure that each vertex in $J_{t'}$ is secure, so the new centres (the insecure vertices) come from $J_t \setminus J_{t'}$. Thus, we avoid discarding and then re-adding centres, ensuring that $(B_t : t \in V(T))$ is a tree-decomposition of H .

The vertices $J_t \cap J_{t'}$ are secure by induction, and belong to a centre using the penalty-

adjusted system above. Each of these centres will be kept (be in $B_t \cap B_{t'}$). This means that we need to keep at most $|J_t \cap J_{t'}|$ centres. Since the vertices in $J_t \cap J_{t'}$ are secure, none of them become new centres. So, at worst, we need to add at most $|J_t \setminus J_{t'}|$ centres. Thus, we need at most $|J_t|$ centres for the next step (B_t). For technical reasons, we prefer to avoid decreasing the number of centres. So we add extra centres from $B_{t'}$ to B_t until it has at least $|B_{t'}|$ but at most $\max(|J_t|, |B_{t'}|)$ centres. By induction, B_t has at most $k + 1$ centres.

There is one remaining concern: a vertex u that is close to (within distance ℓ of) a vertex v in the coverage of an old centre x that we discard ($B_{t'} \setminus B_t$). In such a case, could we have $u \in P_{x'}$ for a new centre x' (in $B_t \setminus B_{t'}$), while also having $v \in P_x$ (which is possible, as v is in the coverage of x). Since x and x' will not be adjacent in H (as they are not in a common bag B_{t^*}), this would be a problem. However, we show that this does not occur. For this, we exploit the penalty system and the fact $J_t \cap J_{t'}$ is a separator between J_t and $B_{t'}$, as each centre in $B_{t'}$ first appears in a bag J_{t^*} before t' (t^* is an ancestor of t').

Since $J_t \cap J_{t'}$ is a separator, there is a vertex $z \in J_t \cap J_{t'}$ on the shortest path from x to v . Note that z is also in the coverage of x . z belongs to a centre y we keep ($y \in B_t \cap B_{t'}$). By choice of how we pick y , the buffer zone in the coverage of y around z is wider than the buffer zone from x . This implies that v is also in the coverage of y .

If this centre y has higher priority than x , then we would never put v in P_x . If y has lower priority (and thus higher penalty), then in the next step, as we discarded x , the penalty of y will be decreased by at least 1. Hence, v will be in the security of y . Thus, the ℓ -neighbourhood of v , including u , is contained in the coverage of y .

Our priority system will always prioritise the kept centres over the new centres (by an easy inductive argument). Thus, we would never put u in a part belonging to a new centre (or any centre created thereafter). This means that the concerning scenario above cannot occur. From this, we derive that if $u, v \in V(G)$ are at distance at most ℓ , then the centres $x, x' \in V(H)$ with $u \in P_x$ and $v \in P_{x'}$ satisfy $u, v \in B_t$ for some $t \in V(T)$ (in particular, $t \in \{r(x), r(x')\}$). Then by definition of H , either $u = v$ or $uv \in E(H)$. So the partition is compressing.

The weak diameter of the parts is derived from the fact that the coverage is at most the $(k+1)$ -neighbourhood. Recall that we only add a vertex x as a centre in $B_t \setminus B_{t'}$ if it was insecure, and that once a vertex is secure, it stays secure. Therefore, we never discard a centre and re-add it later. It follows that $(B_t : t \in V(T))$ is a tree-decomposition of H .

4.3 Proof

Say that a *graph-priority pair* is a pair $Q = (G, \preceq)$ where G is a graph and $\preceq$ is a total ordering of $V(G)$. For distinct $u, v \in V(G)$, if $u \prec v$, then we say that u has *lower priority* than v in Q , and that v has *higher priority* than u in Q . We consider u to have neither higher nor lower priority than itself. For a binary condition β on $V(G)$ such that at least one vertex satisfies β , there exists exactly one vertex $v \in V(G)$ that satisfies β and has higher priority in Q than each $u \in V(G) \setminus \{v\}$ that also satisfies β . We call v the vertex of *maximum priority* in Q that satisfies β .

Let $Q = (G, \preceq)$ be a graph-priority pair, and let $B \subseteq V(G)$. The *penalty map* for B in Q , denoted σ_Q^B , is the map from B to $\{0, \dots, |B| - 1\}$ that sends each $x \in B$ to $|\{x' \in B : x \prec x'\}|$. So σ_Q^B counts the number of vertices in B of higher priority than x . Note that σ_Q^B is a bijection, and that $\sigma_Q^B(x') < \sigma_Q^B(x)$ whenever $x, x' \in B$ is such that x' has higher priority in Q than x . For each $x \in B$, we call σ_Q^B the *penalty* of x in Q (with respect to B).

For $\ell \in \mathbb{R}^{\geq 0}$, and $B' \subseteq B$, the *(B, ℓ) -security* of B' in Q , denoted $S_Q^{B, \ell}(B')$, is

$$\bigcup_{x \in B'} N_G^{(|B| - 1 - \sigma_Q^B(x))\ell}(x).$$

The *(B, ℓ) -coverage* of B' in Q , denoted $C_Q^{B, \ell}(B')$, is

$$\bigcup_{x \in B'} N_G^{(|B| - \sigma_Q^B(x))\ell}(x).$$

Note that $B' \subseteq S_Q^{B, \ell}(B')$ (as $\sigma_Q^B(x) \leq |B| - 1$), and that $N_G^\ell(S_Q^{B, \ell}(B')) \subseteq C_Q^{B, \ell}(B')$. If B' is a singleton $\{x\}$, we instead write $S_Q^{B, \ell}(x)$ and $C_Q^{B, \ell}(x)$ for the (B, ℓ) -security and (B, ℓ) -coverage respectively.

For $J \subseteq V(G)$ and $x \in B$, the *(B, ℓ) -leak* of x through J in Q , denoted $L_Q^{B, \ell}(x, J)$, is the set of vertices $v \in V(G)$ such that:

1. there exists $z \in J$ such that $\text{dist}_G(v, z) \leq (|B| - \sigma_Q^B(x))\ell - \text{dist}_G(x, z)$, and
2. there does not exist $x' \in B$ with higher priority than x in Q such that $v \in C_Q^{B, \ell}(x')$.

Before we start the main proof, we first need the following lemma, which says how to pick new centres.

Lemma 4.5. *Let $Q = (G, \preceq)$ be a graph-priority pair, let $\ell \in \mathbb{R}^{\geq 0}$, and let $B, J \subseteq V(G)$ be such that:*

1. $J \cap S_Q^{B,\ell}(B)$ separates B from J in G , and
2. for each $u \in B$ and $v \in J \setminus S_Q^{B,\ell}(B)$, u has higher priority than v in Q .

Then there exists $B' \subseteq V(G)$ such that:

1. $|B| \leq |B'| \leq \max(|B|, |J|)$,
2. $J \subseteq S_Q^{B',\ell}(B')$,
3. $B' \setminus B = J \setminus S_Q^{B,\ell}(B)$
4. for each $x \in B \setminus B'$, $L_Q^{B,\ell}(x, J) \subseteq S_Q^{B',\ell}(B \cap B')$.

Proof. For brevity, we define $\sigma := \sigma_Q^B$, and once B' is defined, we set $\sigma' := \sigma_Q^{B'}$. For $B^* \subseteq B$, we also define $(S(B^*), C(B^*)) := (S_Q^{B,\ell}(B^*), C_Q^{B,\ell}(B^*))$. If $B^* = \{x\}$ for some $x \in B$, we abbreviate $S(\{x\})$ as $S(x)$ and $C(\{x\})$ as $C(x)$. Once B' is defined, for $B^* \subseteq B'$, we similarly define $(S'(B^*), C'(B^*)) := (S_Q^{B',\ell}(B^*), C_Q^{B',\ell}(B^*))$. Again, if $B^* = \{x\}$ for some $x \in B'$, we abbreviate $S'(\{x\})$ as $S'(x)$ and $C'(\{x\})$ as $C'(x)$. Finally, for $x \in B$, we define $L(x) := L_Q^{B,\ell}(x, J)$.

Under these abbreviations, we have:

1. $S(B) \cap J$ separates B from J in G , and
2. for each $u \in B$ and $v \in J \setminus S(B)$, u has higher priority than v in Q ,

and we must show that:

1. $|B| \leq |B'| \leq \max(|B|, |J|)$,
2. $J \subseteq S'(B')$,
3. $B' \setminus B = J \setminus S(B)$, and
4. for each $x \in B \setminus B'$, $L(x) \subseteq S'(B \cap B')$.

If $B = \emptyset$, the lemma is satisfied with $B' := J$, as $J \subseteq S_Q^{J,\ell}(J) = S'(B')$ and $S(B) = \emptyset$ in this case. So we may assume $B \neq \emptyset$.

For each $v \in J$, let $a(v)$ be the $x \in B$ that minimises $\text{dist}_G(x, v) + \sigma(x)\ell$, and, subject to this, is of maximum priority in Q . Note that $a(v)$ exists as $B \neq \emptyset$. We first show that if a vertex $u \in J$ lies on the shortest path from v to $a(v)$, then $a(u) = a(v)$.

Claim 1. *If $u, v \in J$ are such that $\text{dist}_G(v, a(v)) = \text{dist}_G(u, v) + \text{dist}_G(u, a(v))$, then $a(u) = a(v)$.*

Proof. By assumption on u and v ,

$$\text{dist}_G(v, a(v)) + \sigma(a(v))\ell = \text{dist}_G(u, v) + \text{dist}_G(u, a(v)) + \sigma(a(v))\ell. \quad (4.1)$$

By triangle inequality,

$$\text{dist}_G(v, a(u)) + \sigma(a(u))\ell \leq \text{dist}_G(u, v) + \text{dist}_G(u, a(u)) + \sigma(a(u))\ell. \quad (4.2)$$

By choice of $a(u)$ and $a(v)$,

$$\text{dist}_G(u, a(u)) + \sigma(a(u))\ell \leq \text{dist}_G(u, a(v)) + \sigma(a(v))\ell, \text{ and} \quad (4.3a)$$

$$\text{dist}_G(v, a(v)) + \sigma(a(v))\ell \leq \text{dist}_G(v, a(u)) + \sigma(a(u))\ell. \quad (4.3b)$$

Applying (4.2), (4.3a), and (4.1) (in that order) and then comparing with (4.3b),

$$\text{dist}_G(v, a(u)) + \sigma(a(u))\ell = \text{dist}_G(v, a(v)) + \sigma(a(v))\ell. \quad (4.4)$$

By choice of $a(v)$, (4.4) implies that $a(u)$ does not have higher priority than $a(v)$ in Q .

Applying (4.1), (4.4), and (4.2) (in that order), cancelling out $\text{dist}_G(u, v)$, and then comparing with (4.3a),

$$\text{dist}_G(u, a(u)) + \sigma(a(u))\ell = \text{dist}_G(u, a(v)) + \sigma(a(v))\ell. \quad (4.5)$$

By choice of $a(u)$, (4.5) implies that $a(v)$ does not have higher priority than $a(u)$ in Q . Since for all pairs of distinct vertices, one will have higher priority in Q than the other, this implies $a(u) = a(v)$, as desired. $\blacksquare$

We next show that if a vertex v is in the security of B , then it is in the security of $a(v)$.

Claim 2. *For each $v \in S(B) \cap J$, $v \in S(a(v))$.*

Proof. Since $v \in S(B)$, there exists $x \in B$ such that $v \in S(x)$, and thus $\text{dist}_G(v, x) \leq (|B| - 1 - \sigma(x))\ell$. Hence, $\text{dist}_G(v, x) + \sigma(x)\ell \leq (|B| - 1)\ell$. Since $v \in J$, $a(v)$ is well-defined, and by definition we have $\text{dist}_G(v, a(v)) + \sigma(a(v))\ell \leq \text{dist}_G(v, x) + \sigma(x)\ell \leq (|B| - 1)\ell$. Thus, $\text{dist}_G(v, a(v)) \leq (|B| - 1 - \sigma(a(v)))\ell$, so $v \in S(a(v))$, as desired. $\blacksquare$

For each $x \in B$, let $d(x)$ be the set consisting of the $v \in J$ such that $a(v) = x$ and $v \in S(x)$. Let $D := \bigcup_{x \in B} d(x)$. Note that $D \subseteq J$.

Claim 3. $D = S(B) \cap J$.

Proof. For each $v \in D$, we have $v \in S(a(v)) \subseteq S(B)$. Since $D \subseteq J$, it follows that $v \in S(B) \cap J$. For each $v \in S(B) \cap J$, by Claim 2, $v \in S(a(v))$. So $v \in d(a(v))$, and thus $v \in D$. The claim follows. $\blacksquare$

If $D = J$, then by Claim 3, $J \subseteq S(B)$. Hence, in this case, the lemma is true with $B' := B$ (as $S'(B') = S(B)$). Thus and since $D \subseteq J$, we may assume that $D \subsetneq J$.

Let $A := \{x \in B : d(x) \neq \emptyset\}$. Noting that the $d(x)$, $x \in B$, are pairwise disjoint and disjoint to $J \setminus D$, observe that $|A| + |J \setminus D| \leq |J|$. Let $m := \min(|J \setminus D|, |B \setminus A|)$, and let X be a subset of $B \setminus A$ of size exactly m (which exists by choice of m). Let $B' := (B \setminus X) \cup (J \setminus D)$.

By Claim 3 and since $B \subseteq S(B)$, observe that $J \cap B \subseteq D$. Thus, $J \setminus D$ is disjoint to B . Hence and since $X \subseteq B$ and has size m , we have that $|B'| = |B| - m + |J \setminus D|$. If $m = |J \setminus D|$, then observe that $|B'| = |B|$. Otherwise, observe that $m = |B \setminus A| \leq |J \setminus D|$, and $|B'| = |B| - |B \setminus A| + |J \setminus D|$. As $|B \setminus A| \leq |J \setminus D|$, we obtain $|B| \leq |B'|$. However, we also have that $|B| - |B \setminus A| + |J \setminus D| = |A| + |J \setminus D| \leq |J|$ (since $A \subseteq B$). So $|B'| \leq |J|$. Hence, in either scenario, we find that $|B| \leq |B'| \leq \max(|B|, |J|)$.

Recalling that $J \setminus D$ is disjoint to B , we have $B \cap B' = B \setminus X$ and $B' \setminus B = J \setminus D = J \setminus S(B)$ by Claim 3. Note also that $B \setminus B' = X$.

We next show that σ' is smaller than σ on $B \cap B'$, and give a sufficient condition for it to be strictly smaller.

Claim 4. *For each $x \in B \cap B'$, $\sigma'(x) \leq \sigma(x)$. Further, if there exists $y \in B \setminus B'$ with higher priority in Q than x , then $\sigma'(x) \leq \sigma(x) - 1$.*

Proof. Recall that $\sigma(x)$ and $\sigma'(x)$ count the number of vertices in B and B' respectively with higher priority than x in Q . Since $x \in B \cap B' \subseteq B$ has higher priority in Q than each $v \in B' \setminus B = J \setminus S(B)$, it follows that the only vertices in B' with higher priority than x are contained in B . This gives $\sigma'(x) \leq \sigma(x)$. Further, if such a vertex y exists, then there is at least one fewer vertex in B' with higher priority in Q than x compared to B . This gives $\sigma'(x) \leq \sigma(x) - 1$. ■

We now show that D is contained in the new security of $B \cap B'$.

Claim 5. $D \subseteq S'(B \cap B')$.

Proof. Fix $v \in D$. So $v \in d(x)$ for some $x \in B$. By definition of $d(x)$, we find that $x = a(v)$ and $v \in S(x) = S(a(v))$. Since $v \in d(a(v))$, we have that $a(v) \in A$, and thus $a(v) \in B \setminus X \subseteq B'$ (as $X \subseteq B \setminus A$ and since $A \subseteq B$). So $a(v) \in B \cap B'$. Since $v \in S(a(v))$, we have $\text{dist}_G(v, a(v)) \leq (|B| - 1 - \sigma(a(v)))\ell$. By Claim 4, $\sigma'(a(v)) \leq \sigma(a(v))$. Recall also that $|B| \leq |B'|$. Thus, $\text{dist}_G(v, a(v)) \leq (|B'| - 1 - \sigma'(a(v)))\ell$. This gives $v \in S'(a(v)) \subseteq S'(B \cap B')$, as desired. ■

Noting that $J \setminus D \subseteq S'(J \setminus D) = S'(B' \setminus B)$ and observing that $S'(B') = S'(B' \setminus B) \cup S'(B' \cap B)$, it follows from Claim 5 that $J \subseteq S'(B')$.

Fix $x \in B \setminus B' = X$. We must show that $L(x) \subseteq S'(B \cap B')$. Let $v \in L(x)$. So:

1. there exists $z \in J$ such that $\text{dist}_G(v, z) \leq (|B| - \sigma(x))\ell - \text{dist}_G(x, z)$, and
2. there does not exist $x' \in B$ with higher priority than x in Q such that $v \in C(x')$.

Note that $a(z)$ is well-defined, and that $a(z) \in B$. Since $J \cap S(B)$ separates B from J , there exists $w \in J \cap S(B)$ such that $\text{dist}_G(z, a(z)) = \text{dist}_G(z, w) + \text{dist}_G(w, a(z))$. By Claim 1 (with $(u, v) := (w, z)$), $a(z) = a(w)$. By Claim 2, $w \in S(a(w))$. Hence, $w \in d(a(w))$, and thus $a(w) = a(z) \in A \subseteq B \setminus X = B \cap B'$. Since $x \in B \setminus B'$, this gives $a(z) \neq x$.

By definition of $a(z)$, we have that $\text{dist}_G(z, a(z)) + \sigma(a(z))\ell \leq \text{dist}_G(z, x) + \sigma(x)\ell$. Therefore, $(|B| - \sigma(x))\ell - \text{dist}_G(x, z) \leq (|B| - \sigma(a(z)))\ell - \text{dist}_G(a(z), z)$. Thus and since $v \in L(x)$, we obtain $\text{dist}_G(v, z) \leq (|B| - \sigma(a(z)))\ell - \text{dist}_G(a(z), z)$. Hence and by triangle inequality, we obtain $\text{dist}_G(v, a(z)) \leq \text{dist}_G(v, z) + \text{dist}_G(a(z), z) \leq (|B| - \sigma(a(z)))\ell$. Thus, $v \in C(a(z))$. Since $v \in L(x)$ (and since $x \neq a(z)$), this implies that x has higher priority than $a(z)$ in Q .

By Claim 4 (with $(x, y) := (a(z), x)$), $\sigma'(a(z)) \leq \sigma(a(z)) - 1$. Since $\text{dist}_G(v, a(z)) \leq (|B| - \sigma(a(z)))\ell \leq (|B'| - \sigma(a(z)))\ell$ (as $|B| \leq |B'|$), this gives $\text{dist}_G(v, a(z)) \leq (|B'| - 1 - \sigma'(a(z)))\ell$. So $v \in S'(a(z)) \subseteq S'(B \cap B')$. It follows that $L(x) \subseteq S'(B \cap B')$. This completes the proof of the lemma. $\square$

We can now get to the main proof.

Theorem 4.6. *Let $k \in \mathbb{N}$ and $\ell \in \mathbb{R}^{\geq 0}$, and let $(J_t : t \in V(T))$ be a tree-decomposition of a graph G of width at most k . Then there exists a graph H , a T -decomposition $(B_t : t \in V(T))$ of H of width at most k , and an ℓ -compressing H -indexed partition $(P_x : x \in V(H))$ of G whose parts each have weak diameter in G at most $2(k+1)\ell$.*

Proof. Fix a root r of T . For each $v \in V(G)$, let T_v be the subtree of T induced by the vertices $t \in V(T)$ such that $v \in J_t$, and let $r(v)$ be the (unique) vertex of T_v closest to r . We call $r(v)$ the root of v . Note that $r(v)$ is an ancestor of each $t \in V(T_v)$ (each $t \in V(T)$ with $v \in J_t$).

There exists a total ordering $\preceq$ of $V(G)$ such that for distinct $u, v \in V(G)$, if $r(u)$ is a strict ancestor of $r(v)$, then $v \prec u$. So $Q := (G, \preceq)$ is a graph-priority pair. By definition of $\preceq$, when distinct $u, v \in V(G)$ are such that $r(u)$ is a strict ancestor of $r(v)$, u has higher priority in Q than v .

For each $t \in V(T)$, we will define a set $B_t \subseteq V(G)$. For brevity, we define $\sigma_t := \sigma_Q^{B_t}$, which we call the penalty map at t . For $B^* \subseteq B_t$, we also define $(S_t(B^*), C_t(B^*)) := (S_Q^{B_t, \ell}(B^*), C_Q^{B_t, \ell}(B^*))$, which we call the security and coverage respectively of B^* at t . If $B^* = \{x\}$ for some $x \in B$, we abbreviate $S_t(\{x\})$ as $S_t(x)$ and $C_t(\{x\})$ as $C_t(x)$. Finally, for $x \in B_t$ and child t^* of t , we define $L_t(x, t^*) := L_Q^{B_t, \ell}(x, J_{t^*})$, which we call the leak from x through t^* at t .

For each $t \in V(T)$, B_t will satisfy the following properties:

1. $J_t \subseteq S_t(B_t)$, and
2. for each $x \in B_t$, $r(x)$ is an ancestor of t .

If $t \neq r$, let t' be the parent of t . We then further require that:

1. $|B_{t'}| \leq |B_t| \leq \max(|B_{t'}|, |J_t|)$,
2. $B_t \setminus B_{t'} = J_t \setminus S_{t'}(B_{t'})$, and
3. for each $x \in B_{t'} \setminus B_t$, $L_{t'}(x, t) \subseteq S_t(B_t \cap B_{t'})$.

We define B_t inductively on the distance from t to r . In the base case $t = r$, we set $B_r := J_r$, which trivially satisfies the above requirements (as $J_r \subseteq S_r(J_r)$). We now proceed to the inductive step. Note that $t \neq r$, so the parent t' of t is defined. Further, $B_{t'}$ is already defined, and has the above properties.

Since $J_{t'} \subseteq S_{t'}(B_{t'})$, observe that $J_t \cap J_{t'} \subseteq J_t \cap S_{t'}(B_{t'})$ and $J_t \setminus S_{t'}(B_{t'}) \subseteq J_t \setminus J_{t'}$. Since $r(x)$ is an ancestor of t' for each $x \in B_{t'}$, observe that $J_t \cap J_{t'}$, and thus $J_t \cap S_{t'}(B_{t'})$, separates $B_{t'}$ from J_t in G . For each $u \in B_{t'}$ and $v \in J_t \setminus S_{t'}(B_{t'}) \subseteq J_t \setminus J_{t'}$, observe that $r(v) = t$ and that $r(u)$ is an ancestor of t' . Hence, $r(u)$ is a strict ancestor of $r(v)$, so u has higher priority in Q than v . Thus, we can apply Lemma 4.5 with $(B, J) := (B_{t'}, J_t)$. Let $B_t := B'$ be the result. We have that:

1. $|B_{t'}| \leq |B_t| \leq \max(|B_{t'}|, |J_t|)$,
2. $J_t \subseteq S_t(B_t)$,
3. $B_t \setminus B_{t'} = J_t \setminus S_{t'}(B_{t'})$, and
4. for each $x \in B_{t'} \setminus B_t$, $L_{t'}(x, t) \subseteq S_t(B_t \cap B_{t'})$.

To satisfy the inductive conditions, it remains only to show that for each $x \in B_t$, $r(x)$ is an ancestor of t . If $x \in B_{t'}$, then $r(x)$ is an ancestor of t' and thus t . Otherwise, since $B_t \setminus B_{t'} = J_t \setminus S_{t'}(B_{t'}) \subseteq J_t \setminus J_{t'}$, we have $r(x) = t$. So $r(x)$ is trivially an ancestor of t . This completes the inductive construction of B_t .

Let H be the graph with vertex set $\bigcup_{t \in V(T)} B_t$ and edges between distinct vertices $u, v \in V(H)$ if and only if there exists $t \in V(T)$ such that $u, v \in B_t$. Note that

$$V(H) \subseteq V(G).$$

For each $v \in V(G)$, let $a(v)$ be the vertex $x \in V(H)$ of maximum priority in Q such that:

1. $r(x)$ is an ancestor of $r(v)$, and
2. there exists an ancestor $t^*(v) := t^*$ of $r(v)$ such that $x \in B_{t^*}$ and $v \in C_{t^*}(x)$.

We remark that such a vertex $a(v)$ exists, since $v \in J_{r(v)} \subseteq S_{r(v)}(B_{r(v)}) \subseteq C_{r(v)}(B_{r(v)})$, so there exists $y \in B_{r(v)}$ with $v \in C_{r(v)}(y)$. Hence, we could take $(x, t^*) := (y, r(v))$, since $r(y)$ is an ancestor of $r(v)$ as $y \in B_{r(v)}$.

For each $x \in V(H)$, let $P_x := \{v \in V(G) : a(v) = x\}$. Observe that $(P_x : x \in V(H))$ is an H -indexed partition of G . We show that $(H, (B_t : t \in V(T)), (P_x : x \in V(H)))$ satisfy the theorem. We first need to control the size of the ‘bags’ B_t , $t \in V(T)$.

Claim 1. *For each $t \in V(T)$ and each ancestor t^* of t , $|B_{t^*}| \leq |B_t| \leq k + 1$.*

Proof. By induction on $\text{dist}_T(r, t) + \text{dist}_T(r, t^*)$. In the base case, $r = t^* = t$. Thus, $B_{t^*} = B_t = B_r = J_r$, so $|B_{t^*}| = |B_t| = |J_r| \leq k + 1$ as $(J_t : t \in V(T))$ has width at most k . So we may proceed to the inductive step. In particular, observe that $t \neq r$, so the parent t' of t is well-defined.

If $t = t^*$, then observe that $\text{dist}_T(r, t) + \text{dist}_T(r, r) < \text{dist}_T(r, t) + \text{dist}_T(r, t^*)$ (since $t^* = t \neq r$). So we may apply induction with $(t, t^*) := (t, r)$. We find that $|B_r| \leq |B_t| \leq k + 1$. Thus, $|B_{t^*}| = |B_t| \leq k + 1$, as desired.

So we may assume that t^* is a strict ancestor of t . Thus, t^* is an ancestor of t' . Observe that $\text{dist}_T(r, t') + \text{dist}_T(r, t^*) < \text{dist}_T(r, t) + \text{dist}_T(r, t^*)$. So we may apply induction with $(t, t^*) := (t', t^*)$. We find that $|B_{t^*}| \leq |B_{t'}| \leq k + 1$. Since t' is the parent of t , we also have that $|B_{t'}| \leq |B_t| \leq \max(|B_{t'}|, |J_t|)$. Since $(J_t : t \in V(T))$ has width at most k , this gives $|B_{t^*}| \leq |B_t| \leq k + 1$, as desired. ■

In particular, for each $t \in V(T)$, by applying Claim 1 with $t^* := r$, we find that $|B_t| \leq k + 1$. Using this, we show that the parts of $(P_x : x \in V(H))$ have bounded weak diameter in G .

Claim 2. *For each $x \in V(H)$ and $v \in P_x$, $\text{dist}_G(x, v) \leq (k + 1)\ell$.*

Proof. Since $v \in P_x$, we have $a(v) = x$. Thus, by definition of $a(v) = x$ and $t^*(v)$, we have $v \in C_{t^*(v)}(x)$. Hence, $\text{dist}_G(x, v) \leq (|B_{t^*(v)}| - \sigma_{t^*(v)}(x))\ell$. Since $\sigma_{t^*(v)}(x) \geq 0$ and since $|B_{t^*(v)}| \leq k + 1$ (by Claim 1), this gives $\text{dist}_G(x, v) \leq (k + 1)\ell$, as desired. ■

It follows from Claim 2 that for each $x \in V(H)$, P_x has weak diameter in G at most $2(k+1)\ell$.

We next show that $(B_t : t \in V(T))$ is the desired T -decomposition of H . For each $x \in V(H)$, let T'_x be the subgraph of T induced by the vertices $t \in V(T)$ such that $x \in B_t$. The main difficulty is showing that T'_x is connected. For this, we show that the path from t to $r(x)$ in T is contained in $V(T'_x)$.

Claim 3. *Let $x \in V(H)$ and $t^* \in V(T'_x)$. Then for each vertex t' on the path from t^* to $r(x)$ in T , we have $x \in B_{t'}$.*

Proof. By induction on the distance from t' to t^* . In the base case, $t' = t^*$, and the claim is trivial (as $x \in B_{t^*}$ since $t^* \in V(T'_x)$). So we may proceed to the inductive step. In particular, $t' \neq t^*$.

Since $t^* \in V(T'_x)$, we have $x \in B_{t^*}$. Hence, recall that $r(x)$ is an ancestor of t^* . Thus and since $t' \neq t^*$, there is a child t of t' on the path from t^* to $r(x)$. Observe that t is closer to t^* than t' is, and thus by induction $x \in B_t$.

Presume, for a contradiction, that $x \notin B_{t'}$. So $x \in B_t \setminus B_{t'} = J_t \setminus S_{t'}(B_{t'})$. Since $J_{t'} \subseteq S_{t'}(B_{t'})$, this gives $x \in J_t \setminus J_{t'}$. Hence, we have $t, r(x) \in V(T_x)$, but $t' \notin V(T_x)$. This contradicts the fact that T_x is a subtree of T , as t' is on the unique path from t to $r(x)$ in T (since $r(x)$ is an ancestor of t^* and thus t' , and since t is a child of t'). Hence, we have $x \in B_{t'}$. The claim follows. ■

We can now show that $(B_t : t \in V(T))$ is the desired T -decomposition of H . By definition of H , $B_t \subseteq V(H)$ for each $t \in V(T)$. For each $x \in V(H)$, by definition of H , T'_x is nonempty, and by Claim 3, T'_x is connected (since the path from any $t^* \in V(T'_x)$ to $r(x)$ in T is contained in $V(T'_x)$). For each $uv \in E(H)$, by definition of H , there exists $t \in V(T)$ such that $u, v \in B_t$. Therefore, $(B_t : t \in V(T))$ is a T -decomposition of H . Recalling that $|B_t| \leq k+1$ for each $t \in V(T)$ (by Claim 1), we find that $(B_t : t \in V(T))$ has width at most k , as desired.

For each $x \in V(H)$, let $r'(x)$ be the vertex of T'_x closest to r (which is well-defined and uniquely defined, as T'_x is a nonempty subtree of T). Note that $r'(x)$ is an ancestor of each $t \in V(T'_x)$ (each $t \in V(T)$ with $x \in B_t$).

Claim 4. *For each $x \in V(H)$, $r(x) = r'(x)$.*

Proof. By Claim 3 (and since T'_x is nonempty), $r(x) \in V(T'_x)$. So $r(x)$ is a descendant of $r'(x)$. As $x \in B_{r'(x)}$, recall that $r(x)$ is an ancestor of $r'(x)$. Thus, $r(x) = r'(x)$. ■

We are now left with the hardest step, showing that $(P_x : x \in V(H))$ is ℓ -compressing in G (as an H -indexed partition). We first need the following fact regarding, for $v \in V(G)$, the root of $a(v)$.

Claim 5. *Let $v \in V(G)$ and $t, t^* \in V(T)$ be such that t^* is an ancestor of $r(v)$ and a descendant of t . If $v \in C_{t^*}(B_t \cap B_{t^*})$, then $r(a(v))$ is an ancestor of t .*

Proof. So there exists $x \in B_t \cap B_{t^*}$ such that $v \in C_{t^*}(x)$. As $x \in B_t$, $r(x)$ is an ancestor of t , and thus t^* and $r(v)$. So we have that:

1. $r(x)$ is an ancestor of $r(v)$, and
2. there exists an ancestor t^* of $r(v)$ such that $x \in B_{t^*}$ and $v \in C_{t^*}(x)$.

Hence, by choice of $a(v)$, x does not have higher priority than $a(v)$ in Q . Recall that $r(a(v))$ and $r(x)$ are ancestors of $r(v)$. Thus, $r(a(v))$ is related to $r(x)$. Since x does not have higher priority than $a(v)$ in Q , $r(x)$ is not a strict ancestor of $r(a(v))$. Thus, $r(a(v))$ is an ancestor of $r(x)$. Recalling that $r(x)$ is an ancestor of t , this implies that $r(a(v))$ is an ancestor of t , as desired. ■

We can now relate the roots of $a(u)$ and $a(v)$ whenever $u, v \in V(G)$ are at distance at most ℓ in G .

Claim 6. *If $u, v \in V(G)$ are such that $\text{dist}_G(u, v) \leq \ell$, then $r(a(u))$ and $r(a(v))$ are related, and both are ancestors of both $r(u)$ and $r(v)$.*

Proof. Let $t \in V(T)$ be the (unique) vertex farthest from r that is an ancestor of both $r(u)$ and $r(v)$. Note that J_t separates $J_{r(u)}$ from $J_{r(v)}$. Thus and since $u \in J_{r(u)}$ and $v \in J_{r(v)}$, observe that there exists $z \in J_t$ such that $\text{dist}_G(u, v) = \text{dist}_G(u, z) + \text{dist}_G(z, v)$. In particular, observe that $\text{dist}_G(u, z), \text{dist}_G(v, z) \leq \ell$. Since $z \in J_t \subseteq S_t(B_t)$, we obtain $u, v \in N_G^\ell(S_t(B_t)) \subseteq C_t(B_t)$. By Claim 5 (with $t^* := t$), $r(a(u))$ and $r(a(v))$ are both ancestors of t , and thus both $r(u)$ and $r(v)$. This also implies that $r(a(u))$ and $r(a(v))$ are related, completing the proof of the claim. ■

We also need the following fact about the penalty of a vertex that appears in multiple bags B_t .

Claim 7. *For each $t \in V(T)$, each ancestor t^* of t , and each $x \in B_t \cap B_{t^*}$, we have $\sigma_t(x) \leq \sigma_{t^*}(x)$.*

Proof. By Claims 3 and 4, observe that for each $y \in B_t \setminus B_{t^*}$, $r'(y) = r(y)$ is a strict descendant of t^* , and thus also a strict descendant of $r(x)$ (as $x \in B_{t^*}$). Therefore,

y has lower priority than x in Q . Recall that $\sigma_t(x)$ and $\sigma_{t^*}(x)$ count the number of vertices in B_t and B_{t^*} respectively that have higher priority than x in Q . It follows that $\sigma_t(x) \leq \sigma_{t^*}(x)$, as desired. ■

Finally, we need the following result regarding, for $v \in V(G)$, the priorities of vertices in a bag B_t containing $a(v)$.

Claim 8. *If $v \in V(G)$ and $t \in V(T)$ are such that $a(v) \in B_t$ and t is an ancestor of $r(v)$, then for each $x \in B_t$ with $v \in C_t(x)$, x does not have higher priority than $a(v)$ in Q .*

Proof. As $x \in B_t$, $r(x)$ is an ancestor of t and thus $r(v)$. So we have that:

1. $r(x)$ is an ancestor of $r(v)$, and
2. there exists an ancestor $t^* := t$ of $r(v)$ such that $x \in B_{t^*}$ and $v \in C_{t^*}(x)$.

Hence, by definition of $a(v)$, x does not have higher priority than $a(v)$ in Q . ■

To prove that $(P_x : x \in V(H))$ is ℓ -compressing, for $u, v \in V(G)$ with $\text{dist}_G(u, v) \leq \ell$, we need to find a bag B_t with $a(u), a(v) \in B_t$ (as then $a(u), a(v)$ are adjacent in H).

Claim 9. *If $u, v \in V(G)$ are such that $\text{dist}_G(u, v) \leq \ell$, then there exists $w \in \{u, v\}$ such that $a(u), a(v) \in B_{r(a(w))}$.*

Proof. By Claim 6, $r(a(u))$ and $r(a(v))$ are related and are both ancestors of both $r(u)$ and $r(v)$. Without loss of generality, say that $r(a(v))$ is an ancestor of $r(a(u))$. Presume, for a contradiction, that $a(v) \notin B_{r(a(u))}$. Note that $a(v) \in B_{r'(a(v))} = B_{r(a(v))}$ by Claim 4. Thus and since $r(a(v))$ is an ancestor of $r(a(u))$, there exists a vertex $t' \in V(T)$ and a child t of t' such that:

1. t' is a descendant of $r(a(v))$,
2. t is an ancestor of $r(a(u))$,
3. $a(v) \in B_{t'}$, and
4. $a(v) \notin B_t$.

Note that t' is a strict ancestor of $r(a(u))$, and that t is a strict descendant of $r(a(v))$. Further, since $r(a(u))$ is an ancestor of $r(v)$, both t and t' are ancestors of $r(v)$, and are on the path from $r(v)$ to $r(a(v))$ in T .

Recall that $v \in C_{t^*(v)}(a(v))$, so $\text{dist}_G(v, a(v)) \leq (|B_{t^*(v)}| - \sigma_{t^*(v)}(a(v)))\ell$. Since t is on the path from $r(v)$ to $r(a(v))$ in T , J_t separates $J_{r(v)}$ from $J_{r(a(v))}$ in G . Thus

and since $v \in J_{r(v)}$ and $a(v) \in J_{r(a(v))}$, there exists $z \in J_t$ such that $\text{dist}_G(a(v), v) = \text{dist}_G(a(v), z) + \text{dist}_G(v, z)$. Hence, $\text{dist}_G(v, z) \leq (|B_{t^*(v)}| - \sigma_{t^*(v)}(a(v)))\ell - \text{dist}_G(a(v), z)$.

Recall that $t^*(v)$ is an ancestor of $r(v)$. Thus, $t^*(v)$ is related to both t and t' . Recall also that $a(v) \in B_{t^*(v)}$. Thus, by Claim 3 and since $a(v) \in B_{t'} \setminus B_t$, $t^*(v)$ is a strict ancestor of t and an ancestor of t' . Hence, by Claims 1 and 7 (with $(t, t^*) := (t', t^*(v))$), we have $|B_{t^*(v)}| \leq |B_{t'}|$ and $\sigma_{t'}(a(v)) \leq \sigma_{t^*(v)}(a(v))$. Therefore, we obtain $\text{dist}_G(v, z) \leq (|B_{t'}| - \sigma_{t'}(a(v)))\ell - \text{dist}_G(a(v), z)$. By Claim 8 (with $t := t'$), there does not exist $x \in B_{t'}$ with higher priority than $a(v)$ in Q such that $v \in C_{t'}(x)$. Thus (and since $z \in J_t$), we have $v \in L_{t'}(a(v), t)$.

Since t' is the parent of t and since $a(v) \in B_{t'} \setminus B_t$, we know that $L_{t'}(a(v), t) \subseteq S_t(B_t \cap B_{t'})$. So $v \in S_t(B_t \cap B_{t'})$, and thus $u \in N_G^\ell(v) \subseteq N_G^\ell(S_t(B_t \cap B_{t'})) \subseteq C_t(B_t \cap B_{t'})$. By Claim 5 (with $(v, t, t^*) := (u, t', t)$, and recalling that t is an ancestor of $r(a(u))$ and thus $r(u)$), $r(a(u))$ is an ancestor of t' . But this contradicts the fact that t' is a strict ancestor of $r(a(u))$. Thus, we can conclude that $a(v) \in B_{r(a(u))}$. Noting that $a(u) \in B_{r'(a(u))} = B_{r(a(u))}$ by Claim 4, the claim is satisfied with $w := u$. ■

Let distinct $x, y \in V(H)$ be such that $\text{dist}_G(P_x, P_y) \leq \ell$. So there exists $u \in P_x, v \in P_y$ with $\text{dist}_G(u, v) \leq \ell$. Note that $a(u) = x$ and $a(v) = y$. By Claim 9, there exists $w \in \{u, v\}$ such that $a(u) = x, a(v) = y \in B_{r(a(w))}$. Thus, by definition of H , x and y are adjacent in H . Hence, $(P_x : x \in V(H))$ is ℓ -compressing. This completes the proof of the theorem. □

Theorem 4.4 (and the generalisation to other trees) follows directly from Theorem 4.6.

Chapter 5

Assouad–Nagata Dimension of Minor-Free Graphs

This chapter proves Theorem 2.18, which says that every proper minor-closed class of graphs $\mathcal{G}$ has Assouad–Nagata dimension at most 2, dropping to 1 if and only if the graphs in $\mathcal{G}$ have bounded treewidth. We also prove Theorem 2.19 at the end of this chapter (Section 5.6), which says that a subdivision-closed class has bounded Assouad–Nagata dimension only if it excludes some graph as a minor (in which case, it has Assouad–Nagata dimension at most 2, by Theorem 2.18).

5.1 Weighted Graphs

This chapter considers weighted graphs. A *weighted graph* is a graph G equipped with an (edge) *weighting* function $w : E(G) \mapsto \mathbb{R}^{>0} \cup \{\infty\}$. For each $e \in E(G)$, $w(e)$ is the *weight* of e . By abuse of notation, we usually just refer to a weighted graph by the graph G , rather than the pair (G, w) (leaving the weighting implicit). To avoid confusion, in this chapter, we refer to graphs as *unweighted graphs*.

We consider structures and properties of unweighted graphs that do not relate to a metric (such as tree-decompositions and treewidth) to be defined for weighted graphs by ignoring the edge-weighting (that is, defined with respect to the unweighted graph G). In particular, minors ignore edge-weights. As such, we only discuss unweighted minors of a graph.

Regarding metric-related definitions (those given in Section 1.4), we now consider the length of a path P to be the sum of the edge-weights on edges in P . This changes all

definitions that depend on path-lengths (most notably distances) accordingly.

A *weighted subgraph* of a weighted graph G is a subgraph H of G whose edges have the same weight in H as in G . That is, the weighting for H is the restriction of the weighting for G to $E(H)$.

We consider an unweighted graph to be a weighted graph where every edge has weight 1. This recovers the standard definition of the length of a path. We can also *imbue* an unweighted graph G with a weighting $w : E(G) \mapsto \mathbb{R}^{>0} \cup \{\infty\}$, after which we consider G to be the weighted graph with weighting w . By contrast, we can remove the weighting of a weighted graph G to recover the unweighted graph, which we denote *unweighted*(G).

We also want to consider the following variant of the torso. For a tree-decomposition $(J_t : t \in V(T))$ of a weighted graph G , we say that the *weighted-torso* at $t \in V(T)$ is the weighted graph obtained from $G[J_t]$ by, for each $t' \in V(T)$ adjacent to t and each pair of distinct $x, y \in J_t \cap J_{t'}$, adding the edge xy with weight $\text{dist}_G(x, y)$ if it does not already exist, or changing the weight of xy to $\text{dist}_G(x, y)$ if it does exist. In this chapter, we use $G\langle J_t \rangle$ to refer to the weighted torso, rather than the torso.

5.2 Completions

The primary reason for considering weighted graphs is that it allows edges additions without changing the metric. Specifically, between any two non-adjacent vertices $u, v \in V(G)$ in a (weighted) graph G , we can add the edge uv with weight $\text{dist}_G(u, v)$. The resulting weighted graph G' is isometric in G . This can be repeated for multiple pairs of non-adjacent vertices, giving each new edge weight equal to the distance between the endpoint in G (which is also the distance in G' , and the distance in the final weighted graph).

If an edge uv has weight more than $\text{dist}_G(u, v)$, it is not contributing to the metric. Thus, it could be removed, or more importantly, have its weight updated to $\text{dist}_G(u, v)$, without changing the metric of the graph. An edge uv cannot have weight less than $\text{dist}_G(u, v)$, so we can update each edge to have weight exactly $\text{dist}_G(u, v)$.

A critical use-case for these weighted edges is when there is a separation (A, B) with separator S . For each pair of vertices $u, v \in S$, either add or update the edge uv so that it has weight $\text{dist}_G(u, v)$, as above. Call the resulting graph G' , and let $A' := G'[V(A)]$ and $B' := G'[V(B)]$. Observe that (A', B') is a separation of G' with separator S . Moreover, A' and B' are isometric in G' (and G), because any path that goes to the

other side can be shortcut with the weighted edges. Thus, we now have isometric subgraphs, which will be useful for an inductive argument.

Of course, we can add such a weighted edge between any pair of non-adjacent vertices, and then any subgraph would be isometric. However, this would destroy many structural properties of the graph. So we need to strike a balance, adding as many edges as possible while preserving key structural properties. Motivated by the Graph Minor Structure Theorem (Theorem 1.2 [121]), we seek to preserve the (weighted) torsos of a tree-decomposition. In this case, we can add these weighted edges between any pair of vertices in a common adhesion set without changing the important structure. Because adhesion sets also act as separators in the graph, we can also take advantage of our previous observation about separations giving isometric subgraphs. This motivates the following definition.

For a tree-decomposition $(J_t : t \in V(T))$ of a weighted graph G , the *completion* of G by $(J_t : t \in V(T))$ is the graph obtained from G by, for each edge $tt' \in E(T)$ and each pair of distinct $u, v \in J_t \cap J_{t'}$, adding the edge uv with weight $\text{dist}_G(u, v)$ if it does not already exist, or updating the weight of uv to $\text{dist}_G(u, v)$ if it does exist. We say that $(J_t : t \in V(T))$ is *complete* if the completion of G by $(J_t : t \in V(T))$ is G .

The completion preserves the width, adhesion, and weighted torsos of the tree-decomposition, along with the metric of the graph. In fact, the weighted torsos are just induced subgraphs of the completion. Additionally, once a graph is ‘completed’, it does not need further completing. That is, the completion of the completion is the completion. This is summarised in the following observation.

Observation 5.1. *Let $(J_t : t \in V(T))$ be a tree-decomposition of a weighted graph G . Then G is isometric in the completion $\widehat{G}$ of G by $(J_t : t \in V(T))$, and $(J_t : t \in V(T))$ is a complete tree-decomposition of $\widehat{G}$. Further, for each $t \in V(T)$, $G\langle J_t \rangle = \widehat{G}\langle J_t \rangle = \widehat{G}[J_t]$.*

Proof. Observe for any shortest path in G , each edge uv on the path has weight exactly $\text{dist}_G(u, v)$, and thus this path is also a path in $\widehat{G}$ of the same length. For any shortest path in $\widehat{G}$, since each edge uv on this path has weight exactly $\text{dist}_G(u, v)$, there is an path in G of the same length. It follows that G is isometric in $\widehat{G}$.

Since $V(\widehat{G}) = V(G)$ and since each edge $e \in E(\widehat{G}) \setminus E(G)$ is between two vertices in a common adhesion-set, it is immediate that $(J_t : t \in V(T))$ is a tree-decomposition of $\widehat{G}$. Further, for each $t \in V(T)$ and each pair of distinct $u, v \in J_t$, by definition of $\widehat{G}$, $uv \in E(\widehat{G})$ with weight $\text{dist}_G(u, v) = \text{dist}_{\widehat{G}}(u, v)$ as G is isometric in $\widehat{G}$. Thus, $(J_t : t \in V(T))$ is complete, and for each $t \in V(T)$, $G\langle J_t \rangle = \widehat{G}\langle J_t \rangle = \widehat{G}[J_t]$. $\square$

For a subtree T' of T , let $G' := G[\bigcup_{t \in V(T')} J_t]$. Since adhesion sets give rise to separations, we find that in the completion, G' is an isometric subgraph. Further, we find that $(J_t : t \in V(T'))$ is a complete tree-decomposition of G' . These results are summarised in the following observation.

Observation 5.2. *Let $(J_t : t \in V(T))$ be a complete tree-decomposition of a weighted graph G , let $T' \subseteq T$, and set $G' := G[\bigcup_{t \in V(T')} J_t]$. Then G' is an isometric subgraph of G , and $(J_t : t \in V(T'))$ is a complete tree-decomposition of G' .*

Proof. Consider any shortest path between vertices in G' . If any subpath of this path is in $G - V(G')$, observe that the vertices u, v directly before and after this subpath are contained in a common bag J_t (with $t \in V(T')$). Since $(J_t : t \in V(T))$ is complete, $uv \in E(G)$ with weight $\text{dist}_G(u, v)$, so we can remove this subpath and replace it with the edge uv to get a path of the same length. Repeating this process, we find a path in G' of the same length. It follows that G' is isometric in G .

Since $(J_t : t \in V(T))$ is a tree-decomposition of G , it follows that $(J_t : t \in V(T'))$ is a tree-decomposition of G' . Further, since $(J_t : t \in V(T))$ is complete, for any $t \in V(T')$ and pair of distinct $u, v \in J_t$, we have that $u, v \in V(G')$ and $uv \in E(G)$ (and thus $E(G')$) with weight $\text{dist}_G(u, v) = \text{dist}_{G'}(u, v)$ as G' is isometric in G . Thus, $(J_t : t \in V(T'))$ is complete. $\square$

Finally, while the completion cannot preserve the metric after an arbitrary set of vertices are deleted, it can preserve the weighted torsos (up to an induced subgraph).

Observation 5.3. *Let $(J_t : t \in V(T))$ be a complete tree-decomposition of a weighted graph G , and let $Z \subseteq V(G)$. Then $(J_t \setminus Z : t \in V(T))$ is a complete tree-decomposition of $G - Z$, and for each $t \in V(T)$, $(G - Z)\langle J_t \setminus Z \rangle = G\langle J_t \rangle[J_t \setminus Z] = G[J_t \setminus Z]$.*

Proof. Since $(J_t : t \in V(T))$ is a tree-decomposition of G , it follows that $(J_t \setminus Z : t \in V(T))$ is a tree-decomposition of $G - Z$. Further, since $(J_t : t \in V(T))$ is complete, for any $t \in V(T)$ and pair of distinct $u, v \in J_t \setminus Z$, we have that $u, v \in V(G - Z)$ and $uv \in E(G)$ (and thus $E(G - Z)$) with weight $\text{dist}_G(u, v)$. Since $G - Z$ is a subgraph of G , this gives $\text{dist}_{G-Z}(u, v) = \text{dist}_G(u, v)$. So uv exists in $G - Z$ and has weight $\text{dist}_G(u, v) = \text{dist}_{G-Z}(u, v)$. It follows that $(J_t \setminus Z : t \in V(T))$ is complete. Further, we find that for each $t \in V(T)$, $(G - Z)\langle J_t \setminus Z \rangle = (G - Z)[J_t \setminus Z] = G[J_t \setminus Z]$. Since $(J_t : t \in V(T))$ is complete, by Observation 5.1, $G[J_t] = G\langle J_t \rangle$. Thus, $(G - Z)\langle J_t \setminus Z \rangle = G[J_t \setminus Z] = G\langle J_t \rangle[J_t \setminus Z]$, as desired. $\square$

5.3 Path Rerouting

One of the key ideas of this chapter is the ability to ‘adjust’ or ‘reroute’ particular r -paths to obtain an r' -path with the same endpoints but whose interior is contained within a better part of the weighted graph. This is formalised in Lemma 5.4. Note that this concept of rerouting is not completely new, and exists in some capacity within Lemma 3.2 of Bonamy et al. [22] without being formally identified.

Lemma 5.4. *Let $r \in \mathbb{R}^{\geq 0}$ and $d, \ell \in \mathbb{R}^{\geq 0}$, let G be a weighted graph, let $x, y \in V(G)$, and let P be an r -path in G from x to y . Assume there exist sets $Z, S \subseteq V(G)$ and a map $\iota : Z \mapsto S$ such that:*

- (a) $\text{dist}_G(v, \iota(v)) \leq \ell$ for each $v \in Z$, and
- (b) every r -subpath of $P \setminus Z$ has weak diameter in G at most d .

Let $v_1, \dots, v_n$ be the (possibly empty) subsequence of P that is in Z , including x, y if applicable. Define P' to be the sequence $x, \iota(v_1), \iota(v_2), \dots, \iota(v_n), y$. Then P' is a $(d + 2r + 2\ell)$ -walk from x to y in G whose interior is contained in S .

Proof. Recall that we allow consequent vertices in a walk (and thus a $(d + 2r + 2\ell)$ -walk) to be identical. Thus, it suffices to show that consecutive vertices of P' are at distance at most $d + 2r + 2\ell$. If P does not intersect Z , then P is an r -subpath of $P \setminus Z$, and hence $\text{dist}_G(x, y) \leq d \leq d + 2r + 2\ell$ as desired. So we may assume that P intersects Z , and that $v_1, \dots, v_n$ are defined with $n \in \mathbb{N}^{\geq 1}$.

Consider consecutive vertices $\iota(v_i), \iota(v_{i+1})$ in the interior of P' . We must show that $\text{dist}_G(\iota(v_i), \iota(v_{i+1})) \leq d + 2r + 2\ell$. It suffices to show that $\text{dist}_G(v_i, v_{i+1}) \leq d + 2r$. If v_i and v_{i+1} are consecutive in P , then $\text{dist}_G(v_i, v_{i+1}) \leq r \leq d + 2r$, as desired. Otherwise, let u_i be the vertex directly after v_i in P , and let w_{i+1} be the vertex directly before v_{i+1} in P . It suffices to show that $\text{dist}_G(u_i, w_{i+1}) \leq d$. Consider the r -subpath P_i of P going from u_i to w_{i+1} . $P_i \cap Z = \emptyset$ by definition of the subsequence $v_1, \dots, v_n$, and thus P_i is an r -subpath of $P \setminus Z$. Hence, P_i has weak diameter in G at most d . Therefore, $\text{dist}_G(u_i, w_{i+1}) \leq d$, as desired.

It remains to show that $\text{dist}_G(x, \iota(v_1))$ and $\text{dist}_G(y, \iota(v_n))$ are at most $d + 2r + 2\ell$. We show that $\text{dist}_G(x, \iota(v_1)) \leq d + 2r + 2\ell$; the argument for $\text{dist}_G(y, \iota(v_n))$ is symmetric. If $x \in Z$, then $v_1 = x$, and thus $\text{dist}_G(x, \iota(v_1)) = \text{dist}_G(x, \iota(x)) \leq \ell \leq d + 2r + 2\ell$. Otherwise, let w_1 be the vertex directly before v_1 in P . Noting that $\text{dist}_G(w_1, \iota(v_1)) \leq \ell + r$, it suffices to show that $\text{dist}_G(x, w_1) \leq d + r + \ell$. Let P' be the r -subpath of P from x to w_1 . We have $P' \cap Z = \emptyset$ by definition of v_1 , so P' is an r -subpath of $P \setminus Z$. Thus, P' has weak diameter in G at most d . In particular $\text{dist}_G(x, w_1) \leq d \leq d + r + \ell$, as desired. $\square$

An important application for Lemma 5.4 is when the set S is small, so that Z is a (k, ℓ) -centred-set.

Corollary 5.5. *Let $k \in \mathbb{N}^{\geq 1}$, $r \in \mathbb{R}^{\geq 0}$, and $\ell \in \mathbb{R}^{\geq 0}$, let G be a weighted graph, let Z be a (k, ℓ) -centred set in G , and let $x, y \in Z$ be such that there exists an r -path P from x to y in G that is contained in Z . Then $\text{dist}_G(x, y) \leq (k + 1)(2r + 2\ell)$.*

Proof. Let S be a (k, ℓ) -centre of Z , and define $\iota : Z \mapsto S$ such that $\iota(v) \in N_G^\ell(v) \cap S$ for each $v \in Z$. Since $P \subseteq Z$, it is immediate that every r -subpath of $P \setminus Z$ has weak diameter in G at most 0. We also know that for any $v \in Z$, $\text{dist}_G(v, \iota(v)) \leq \ell$. Thus, we can apply Lemma 5.4 to obtain a $(2r + 2\ell)$ -walk from x to y whose interior is in S . Since $|S| \leq k$, P' can involve at most $k + 2$ distinct vertices, and thus $\text{dist}_G(x, y) \leq (k + 1)(2r + 2\ell)$, as desired. $\square$

The next result is very similar to [22, Lemma 2.1], although this result is not tight enough for our purposes. We instead use Lemma 5.4 for a different proof that avoids this problem.

Corollary 5.6. *Let G be a weighted graph, let $m \in \mathbb{N}^{\geq 1}$, $r \in \mathbb{R}^{\geq 0}$, and $d, \ell \in \mathbb{R}^{\geq 0}$, and let $Z \subseteq V(G)$ be a (k, ℓ) -centred set in G such that $G - Z$ admits an (m, r, d) -colouring c . Then for any m -colouring c' of Z , $c \cup c'$ is an $(m, r, (k + 1)(d + 4r + 2\ell))$ -colouring of G .*

Proof. Let S be a (k, ℓ) -centre of Z , and let $Z' := N_G^{r+\ell}(S)$. For any $u, v \in V(G) \setminus Z'$ at distance at most r in G and any shortest path Q from u to v in G , observe that Q is disjoint from Z , as otherwise we would have $u, v \in N_G^r(Z) \subseteq Z'$. Thus, Q is also a path of length at most r in $G - Z$, and $\text{dist}_{G-Z}(u, v) \leq r$. Therefore, for any monochromatic r -path P in G under $c \cup c'$, any r -subpath P' of $P \setminus Z'$ is a monochromatic r -path in $G - Z$ under c . Thus, P' has weak diameter in $G - Z$ (and G) at most d . Define a map $\iota : Z' \mapsto S$ by selecting $\iota(z) \in N_G^{r+\ell}(z) \cap S$ for each $z \in Z'$. By Lemma 5.4, we obtain a $(d + 2r + 2(r + \ell)) = (d + 4r + 2\ell)$ -walk which has the same endpoints as P but whose interior is contained in S . Since $|S| \leq k$, this $(d + 4r + 2\ell)$ -walk involves at most $k + 2$ distinct vertices, and thus has weak diameter at most $(k + 1)(d + 4r + 2\ell)$ in G , as desired. $\square$

Corollaries 5.5 and 5.6 are also conceptually similar to [64, Observation 11]. Specifically, [64, Observation 11] proves the $r = 1$ cases of Corollaries 5.5 and 5.6 for unweighted graphs and with slightly different bounds.

The main purpose of Lemma 5.4 and Corollaries 5.5 and 5.6 is to give sufficient conditions for showing that a given colouring is an (m, r, D) -colouring. With this in mind, we

introduce one more tool to do this in a non-trivial setting. We first need the following definition.

Given $r \in \mathbb{R}^{\geq 0}$, $m \in \mathbb{N}^{\geq 2}$, a weighted graph G , $S \subseteq V(G)$, and $Z \supseteq N_G^{3r}(S)$, we say that a colouring c of Z has an (S, r) -barrier in G if there exists distinct colours i, j such that:

- (a) for each $v \in N_G^r(S)$, there exists $s \in N_G^r(v) \cap S$ with $c(s) = c(v)$
- (b) $N_G^{2r}(S) \setminus N_G^r(S)$ and $N_G^{3r}(S) \setminus N_G^{2r}(S)$ are i -monochromatic and j -monochromatic respectively under c .

This definition is not entirely new; a similar idea was used in the proof of [22, Lemma 3.2] without being formally identified. Note that for any m -colouring c_S of S , there exists an m -colouring of Z that extends c_S and admits an (S, r) -barrier. Further, note that the presence of an (S, r) -barrier in G is preserved upon extending a colouring.

The choice of name ‘barrier’ comes from the following observation. For any r -path P from $u \in N_G^{2r}(S)$ to $v \in V(G) \setminus N_G^{2r}(S)$, P must contain a vertex in both $N_G^{2r}(S) \setminus N_G^r(S)$ and $N_G^{3r}(S) \setminus N_G^{2r}(S)$. Since these sets are monochromatic of different colours, P cannot be monochromatic. Hence, this colouring contains a barrier near S that blocks monochromatic r -paths. The requirement on how $N_G^r(S)$ is coloured is to prevent a sudden change in colouring, in the sense that vertices close to S still use similar colours to S , and the monochromatic sets only appear further away from S . This is not important when G is the full graph, but we use it when G is one half of a separation of a larger graph, and S is the separator. Property (a) then prevents a sudden transition of colours upon crossing the separator, and property (b) prevents monochromatic r -components from leaking far across the separator. See also Figure 5.1 in the proof of Lemma 5.7.

We are now in position to state the main tool of this section, Lemma 5.7. While the statement of Lemma 5.7 is new, the proof uses similar techniques to later parts of the proof of [22, Lemma 3.2]. The key idea behind this result is that if a weighted graph can be isometrically separated into a central region and disjoint outer regions, and we can (m, r, D) -colour the outer regions with barriers at the separator, and (m, r', d) -colour the central region for sufficiently large r' and small d , then the union of these colourings is an (m, r, D) -colouring of the entire graph. This is because any monochromatic r -component that intersects something far from the central region (in an outer-region outside the barrier) is stuck within one outer region. Any other monochromatic r -path is confined to the central region plus (part of) the barriers. Using the fact that the first r -neighbourhood in a barrier copies the colouring of the separator, we can reroute a monochromatic r -path in this expanded region to a monochromatic r' -path in the

central region (using Lemma 5.4), which then has bounded weak diameter.

The following lemma formalises this idea.

Lemma 5.7. *Let $k \in \mathbb{N}$, $r, \ell \in \mathbb{R}^{\geq 0}$, and set $\ell' := (k+1)(6r+2\ell)$ and $r' := 2r+2\ell'$. Let $m \in \mathbb{N}^{\geq 2}$, and let $d, D \in \mathbb{R}^{\geq 0}$ be such that $D \geq d+2r'$. Let G be a weighted graph, and for some $a \in \mathbb{N}$, let $G_0, G_1, \dots, G_a$ be isometric weighted subgraphs of G such that:*

- (a) $G = \bigcup_{i=0}^a G_i$,
- (b) for each $i \in \{1, \dots, a\}$, if $\tilde{G}_i := \bigcup_{j=0, j \neq i}^a G_j$ and $S_i := V(G_i \cap \tilde{G}_i)$, then:
 - (i) $S_i \subseteq V(G_0)$, and
 - (ii) S_i is a (k, ℓ) -centred set in G ,
- (c) G_0 admits an (m, r', d) -colouring c_0 ,
- (d) for each $i \in \{1, \dots, a\}$, G_i admits an (m, r, D) -colouring such that:
 - (i) $c_i = c_0$ on S_i , and
 - (ii) c_i has an (S_i, r) -barrier.

Then $c := \bigcup_{i=0}^a c_i$ is a (well-defined) (m, r, D) -colouring of G .

Proof. If $a = 0$ the claim is trivially true, as $G = G_0$ by (a) and $c = c_0$ is an (m, r, D) -colouring of $G_0 = G$ by (c) (and since $r' \geq r$ and $d \leq D$). So assume $a \geq 1$.

Observe that c is a well-defined colouring of G by (a), (b)(i) and (d)(i). So it suffices to show that for every $r \in \mathbb{R}^{\geq 0}$ and every monochromatic r -component M of G under c , $V(M)$ has weak diameter in G at most D .

By (a), for each $i \in \{1, \dots, a\}$, $(G_i, \tilde{G}_i)$ is a separation of G with separator S_i , which we use implicitly throughout this proof. Also, note that $\bigcap_{i=1}^a V(\tilde{G}_i) = V(G_0)$ by (b)(i).

For all $i \in \{1, \dots, a\}$ and $j \in \{1, 2, 3\}$, let $N_i^j := N_{G_i}^{jr}(S_i)$ and $S_i^j := N_{G_i}^{jr}(S_i) \setminus N_{G_i}^{(j-1)r}(S_i)$. Note that for each $i \in \{1, \dots, a\}$, since c_i has an (S_i, r) -barrier, S_i^2, S_i^3 are both monochromatic of different colours, and for each $v \in S_i^1$ there is an $s \in S_i \cap N_{G_i}^r(v) \subseteq S_i \cap N_G^r(v)$ with the same colour as v . Let $V' := V(G_0) \cup \bigcup_{i=1}^a N_i^2$.

Claim 1. *For each monochromatic r -component M of G under c , either $V(M) \subseteq V'$, or there exists $i \in \{1, \dots, a\}$ such that $V(M) \subseteq V(G_i)$.*

Proof. For any $i \in \{1, \dots, a\}$, any r -path P from $u \in V(\tilde{G}_i) \cup N_i^2$ to $v \in V(G_i) \setminus N_i^2$ must contain a vertex in S_i^2 and S_i^3 . See Figure 5.1. Since S_i^2 and S_i^3 are monochromatic of different colours, P cannot be monochromatic. It follows that for any monochromatic r -component M of G , $V(M)$ must be contained in either $V(\tilde{G}_i) \cup N_i^2$ or $V(G_i) \setminus N_i^2$. Accounting for the fact that this holds for each $i \in \{1, \dots, a\}$, we can conclude that

$V(M)$ is either contained in $V(G_i) \setminus N_i^2 \subseteq V(G_i)$ for some $i \in \{1, \dots, a\}$, or in $\bigcap_{i=1}^a (V(\tilde{G}_i) \cup N_i^2) = V'$. ■

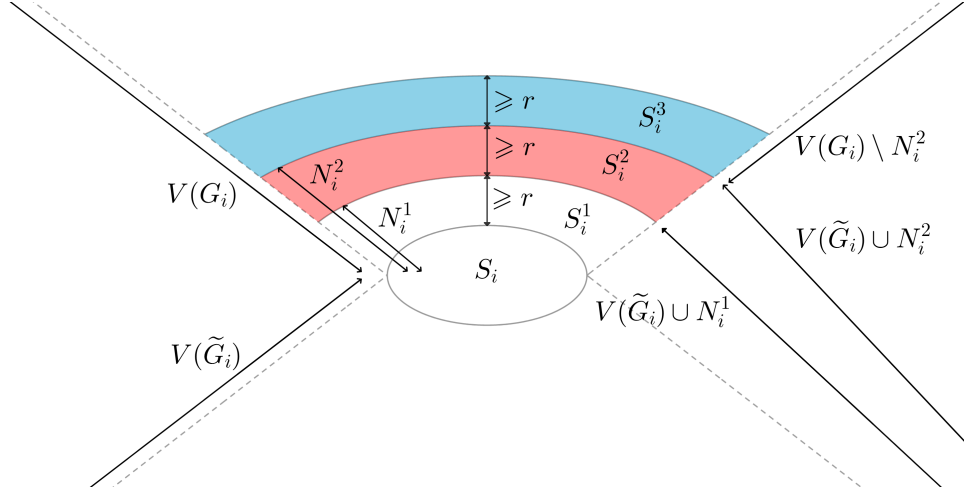


Figure 5.1: A diagram of the separation $(G_i, \tilde{G}_i)$ of G for some $i \in \{1, \dots, a\}$, with several relevant sets of vertices near the separator S_i labelled. Vertices and edges of G are not depicted, instead the dotted line denotes a region in which all vertices and edges are contained. Observe that any r -path P from a vertex in $V(G_i) \setminus N_i^2$ to a vertex in $V(\tilde{G}_i) \cup N_i^2$ must contain a vertex in both S_i^2 and S_i^3 , which are monochromatic of different colours, depicted here as red and blue. Thus, P cannot be monochromatic. Additionally, observe that any r -path from a vertex in S_i^2 to a vertex in $\tilde{G}_i \cup N_i^1$ must contain a vertex in S_i^1 .

By assumption, $G_0, \dots, G_a$ are all isometric weighted subgraphs of G . Thus, any monochromatic r -component M of G under c whose vertices are contained in $V(G_i)$ for some $i \in \{1, \dots, a\}$ is also a monochromatic r -component of G_i under c_i . In this case, $V(M)$ has weak diameter at most D in G_i and G . Similarly, any monochromatic r -component M of G under c whose vertices are contained in $V(G_0)$ is a subgraph of a monochromatic r' -component of G_0 under c_0 as $r \leq r'$, and thus $V(M)$ has weak diameter at most $d \leq D$ in G_0 and G . Thus and by Claim 1, we only need to consider monochromatic r -components whose vertices are contained in V' but are not contained in $V(G_i)$ for any $i \in \{0, \dots, a\}$. Consider such a monochromatic r -component M .

Claim 2. *For each $v \in V(M)$, there exists $s \in V(G_0) \cap N_G^{\ell'}(v)$ with the same colour as v under c .*

Proof. This is trivially true for each $v \in V(G_0)$, and for each $i \in \{1, \dots, a\}$ and $v \in S_i^1$, the claim follows since c_i has an (S_i, r) -barrier and $r \leq \ell'$. Since $V(M) \subseteq V'$, we now only need to consider $v \in S_i^2$ for some $i \in \{1, \dots, a\}$. Observe that $V(M)$ is not contained in S_i^2 , as it is not contained in $V(G_i)$, thus there must exist some

$u \in V(M) \setminus S_i^2$. Since $V(M) \subseteq V'$, this means that $u \in V(\tilde{G}_i) \cup N_i^1$. Because u, v are both in M , there exists a monochromatic r -path P from $v \in S_i^2$ to $u \in V(\tilde{G}_i) \cup N_i^1$ in G of the same colour. Observe that P must contain a vertex in S_i^1 . See Figure 5.1. Let x be the first (starting from v) such vertex in $S_i^1 \cap P$. Observe that every vertex before x must be in S_i^2 . Thus, the r -subpath P' of P from v to x is an r -path contained in $S_i^1 \cup S_i^2 \subseteq N_i^2$. Since c_i has an (S_i, r) -barrier, there must be an $s \in S_i \cap N_G^r(x) \subseteq V(G_0)$ with the same colour as x . Hence v and s have the same colour, and there is an r -path contained in N_i^2 from v to s obtained by appending s to the end of P' . Since S_i is a (k, ℓ) -centred set in G , N_i^2 is a $(k, \ell + 2r)$ -centred set in G . Thus, we can apply Corollary 5.5 to find that $\text{dist}_G(v, s) \leq (k + 1)(2r + 2(2r + \ell)) = \ell'$, as desired. ■

By Claim 2, there exists a map $\iota : V(M) \mapsto S$ that sends each $v \in V(M)$ to some $s \in V(G_0) \cap N_G^{\ell'}(v)$ with the same colour as v . Now, for any two vertices x, y in $V(M)$, there exists a monochromatic r -path P from x to y in G whose vertices are contained in $V(M)$. Thus, every r -subpath of $P \setminus V(M)$ has weak diameter in G at most 0. By applying Lemma 5.4, we obtain a monochromatic $(0 + 2r + 2\ell') = r'$ -path P' in G of the same colour as P but whose interior is contained in $V(G_0)$. As G_0 is an isometric weighted subgraph, the interior of P' is a monochromatic r' -path in G_0 under c_0 , and thus has weak diameter in G_0 and G at most d . Hence, P has weak diameter in G at most $d + 2r'$. In particular, $\text{dist}_G(x, y) \leq d + 2r' \leq D$, as desired. □

5.4 Colourings over Tree-Decompositions

In this section, we (m, r, D) -colour a graph using (m, r, D) -colourings of the weighted torsos. This can then be used to find (m, r, D) -colourings of bounded treewidth graphs, and can also be used in conjunction with the Graph Minor Structure Theorem (Theorem 1.2 [121]). In the latter case, it is important that we can find colourings of almost-embeddable graphs with specific edge-weights, so we need to use the weighted torsos, not torsos. This inspires the following definition.

For $k \in \mathbb{N}$ and a class of weighted graphs $\mathcal{H}$, we say that a tree-decomposition $(B_t : t \in V(T))$ of a weighted graph G is a $(k, \mathcal{H})$ -construction of G if it has adhesion at most k and, for each $t \in V(T)$, $G\langle B_t \rangle \in \mathcal{H}$ (where $G\langle B_t \rangle$ denotes the weighted torso). If G admits a $(k, \mathcal{H})$ -construction, then we say that G is $(k, \mathcal{H})$ -constructable.

We also work in a setting slightly stronger than bounded treewidth. Specifically, we only ask that disjoint subgraphs of bounded radius can be contracted to obtain a bounded treewidth graph. This leads to the following definitions.

For $\ell \in \mathbb{R}^{\geq 0}$, say that a proper partition $\mathcal{P}$ of a graph G is ℓ -*shallow* if, for each part $P \in \mathcal{P}$, $G[P]$ has radius at most ℓ . If, for $k \in \mathbb{N}$, $\mathcal{P}$ also has treewidth at most k , then we say that $\mathcal{P}$ is a (k, ℓ) -*proper-partition* of G .

The main results of this section are as follows.

Theorem 5.8. *Let $k \in \mathbb{N}$, let $\mathcal{H}$ be a hereditary class of weighted graphs, and let $\mathcal{G}$ be a class of $(k, \mathcal{H})$ -constructable weighted graphs. Then $\text{ANdim}(\mathcal{G}) \leq \max(\text{ANdim}(\mathcal{H}), 1)$*

Theorem 5.9. *For every $k \in \mathbb{N}$, there exists a dilation f_k such that for every $\ell \in \mathbb{R}^{\geq 0}$, f_k is an ℓ -almost 1-dimensional control function for every weighted graph that admits a (k, ℓ) -proper-partition.*

(Recall that an ℓ -almost n -dimensional control function f gives $(n+1, r, f(r))$ -colourings only when $r \geq \ell$.)

For any graph G of treewidth at most k , the (proper) partition into singletons is a $(k, 0)$ -proper-partition of G . This gives the following as an easy application of Theorem 5.9.

Theorem 5.10. *Let $\mathcal{G}$ be a class of graphs such that there exists $k \in \mathbb{N}$ such that $\text{tw}(G) \leq k$ for each $G \in \mathcal{G}$. Then $\text{ANdim}(\mathcal{G}) \leq 1$.*

Theorems 5.8 and 5.9 are both special cases of a more general result, Theorem 5.11, which we work towards stating now.

Let G be a weighted graph, $\mathcal{P}$ be a proper partition of G , and $(B_t : t \in V(T))$ be a tree-decomposition of $G/\mathcal{P}$. For each $t \in V(T)$, set $J_t := \bigcup_{P \in B_t} P$. Observe that $(J_t : t \in V(T))$ is a tree-decomposition of G , which we call the tree-decomposition of G *induced* by $(B_t : t \in V(T))$.

For $k \in \mathbb{N}$, $\ell \in \mathbb{R}^{\geq 0}$, and a class of weighted graphs $\mathcal{H}$, we say that a pair $(\mathcal{P}, (B_t : t \in V(T)))$ is a $(k, \ell, \mathcal{H})$ -*strong-construction* for a weighted graph G if:

- (a) $\mathcal{P}$ is an ℓ -shallow proper partition of G ,
- (b) $(B_t : t \in V(T))$ is a tree-decomposition of $G/\mathcal{P}$ of adhesion at most k , and
- (c) if $(J_t : t \in V(T))$ is the tree-decomposition of G induced by $(B_t : t \in V(T))$, then for each $t \in V(T)$ and $S \subseteq B_t$, $G\langle J_t \rangle[\bigcup_{P \in S} P] \in \mathcal{H}$,

Say that $(\mathcal{P}, (B_t : t \in V(T)))$ is *complete* if $(J_t : t \in V(T))$ is complete. Say that G is $(k, \ell, \mathcal{H})$ -*strongly-constructable* if it admits a $(k, \ell, \mathcal{H})$ -strong-construction.

Theorem 5.11. *For each $k \in \mathbb{N}$ and function $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$, there exists a function $f_k : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ such that the following holds. Let $n \in \mathbb{N}$, let $\ell \in \mathbb{R}^{\geq 0}$, let $\mathcal{H}$ be a class*

of weighted graphs that admits f as an ℓ -almost n -dimensional control function, and set $n' := \max(n, 1)$. Then f_k is an ℓ -almost n' -dimensional control function for every $(k, \ell, \mathcal{H})$ -strongly-constructable weighted graph G . Further, if f is a dilation, then for each $k \in \mathbb{N}$, f_k is a dilation.

If G is $(k, \mathcal{H})$ -constructable and $\mathcal{H}$ is hereditary, then G is $(k, 0, \mathcal{H})$ -strongly-constructable using the (proper) partition into singletons. Theorem 5.8 then follows.

If $\mathcal{H}$ is the class of weighted graphs H such that $V(H)$ is a $(k+1, \ell)$ -centred set in H , then observe that any weighted graph G that admits a (k, ℓ) -proper-partition is $(k, \ell, \mathcal{H})$ -strongly-constructable (as for any weighted graph A of radius ℓ , $V(A)$ is $(1, \ell)$ -centred in G). Using Corollary 5.5, we find that, for any $r \geq \ell$ and any $H \in \mathcal{H}$, any r -path in H has weak diameter in H at most $4(k+2)r$. It follows any 1-colouring of H is a $(1, r, 4(k+2)r)$ -colouring. Thus, $r \mapsto 4(k+2)r$ is a dilation that is an ℓ -almost 0-dimensional control function for $\mathcal{H}$. Theorem 5.9 follows.

The remainder of this section is therefore dedicated to the proof of Theorem 5.11. We need the following fact about strong-constructability and completions.

Observation 5.12. *Let $k \in \mathbb{N}$, let $\ell \in \mathbb{R}^{\geq 0}$, and let $\mathcal{H}$ be a class of weighted graphs. Let $(\mathcal{P}, (B_t : t \in V(T)))$ be a $(k, \ell, \mathcal{H})$ -strong-construction for a weighted graph G , let $(J_t : t \in V(T))$ be the tree-decomposition of G induced by $(B_t : t \in V(T))$, and let $\widehat{G}$ be the completion of G by $(J_t : t \in V(T))$. Then $(\mathcal{P}, (B_t : t \in V(T)))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of $\widehat{G}$.*

Proof. Observe that $\mathcal{P}$ is an ℓ -shallow proper partition of $\widehat{G}$, as $V(G) = V(\widehat{G})$, $E(G) \subseteq E(\widehat{G})$, and since each $e \in E(G) \cap E(\widehat{G})$ has weight in $\widehat{G}$ no greater than its weight in G .

For each $PP' \in E(\widehat{G}/\mathcal{P}) \setminus E(G/\mathcal{P})$, there exists $uv \in E(\widehat{G}) \setminus E(G)$ with $u \in P$ and $v \in P'$. By definition of $\widehat{G}$, there exists $t \in V(T)$ such that $u, v \in J_t$, and thus $P, P' \in B_t$. It follows that $(B_t : t \in V(T))$ is a tree-decomposition of $\widehat{G}/\mathcal{P}$ of adhesion at most k . Note that the tree-decomposition of $\widehat{G}$ induced by $(B_t : t \in V(T))$ is $(J_t : t \in V(T))$.

By Observation 5.1, for each $t \in V(T)$, $G\langle J_t \rangle = \widehat{G}\langle J_t \rangle = \widehat{G}[J_t]$. Therefore, for each $S \subseteq B_t$, $\widehat{G}\langle J_t \rangle[\bigcup_{P \in S} P] = G\langle J_t \rangle[\bigcup_{P \in S} P] \in \mathcal{H}$. Hence, $(\mathcal{P}, (B_t : t \in V(T)))$ is a $(k, \ell, \mathcal{H})$ -strong-construction of $\widehat{G}$. By Observation 5.1, $(J_t : t \in V(T))$, and thus $(\mathcal{P}, (B_t : t \in V(T)))$, is complete. This completes the proof. $\square$

Since G is isometric in the completion $\widehat{G}$, by Observation 5.1, any control function for $\widehat{G}$ is a control function for G . Thus, by Observation 5.12, to prove Lemma 5.13, it suffices to consider weighted graphs that admit complete strong-constructions. It is then clear that Theorem 5.11 follows from the following technical lemma.

Lemma 5.13. *For every $k \in \mathbb{N}$ and function $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$, there exists a function $f_k : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ such that the following holds. Let $n \in \mathbb{N}$, let $\ell \in \mathbb{R}^{\geq 0}$, let $\mathcal{H}$ be a class of weighted graphs that admits f as an ℓ -almost n -dimensional control function, and let G be a weighted graph with a complete $(k, \ell, \mathcal{H})$ -strong-construction $(\mathcal{P}, (B_t : t \in V(T)))$. Set $n' := \max(n, 1)$, let $r \in \mathbb{R}^{\geq \ell}$, let $q \in V(T)$, and let $S^{\mathcal{P}} \subseteq B_q$ be of size at most k . Set $S := \bigcup_{P \in S^{\mathcal{P}}} P$, $Z^{\mathcal{P}} := \{P \in \mathcal{P} : P \cap N_G^{3r}(S) \neq \emptyset\}$, and let c_Z be an $(n' + 1)$ -colouring of $Z := \bigcup_{P \in Z^{\mathcal{P}}} P$. Then c_Z can be extended to an $(n' + 1, r, f_k(r))$ -colouring c of G . Further, if f is a dilation, then so is f_k .*

The idea behind the proof of Lemma 5.13 is as follows. We find that the vertices t indexing bags B_t that intersect $Z^{\mathcal{P}}$ induce a subtree T' of T . For each maximal subtree T_e of $T - V(T')$, along with T' , we consider the graph induced by the bags indexed by vertices in that subtree, which we call G_e and G' respectively. Each of these subgraphs also admit a strong-construction (see Observation 5.14).

By the choice of T' , we find that Z is contained in $V(G')$ and disjoint from each $V(G_e)$. $G' - Z$ still admits a strong-construction (see Observation 5.15), and further, the adhesion of this strong-construction drops by 1 (since each adhesion set beforehand intersects $Z^{\mathcal{P}}$). This allows us to inductively colour $G' - Z$ using the control function f_{k-1} . We choose to find an $(n' + 1, r', f_{k-1}(r'))$ -colouring, where r' is sufficiently larger than r .

We then add Z back with the colouring c_Z . Z is a centred set, so this colouring is still an (m, r', d) -colouring, where d is larger than $f_{k-1}(r')$ but smaller than $f_k(r)$. For each other subgraph G_e , we then colour the vertices of G_e near G' to create a barrier. We can take this barrier as the set $Z := Z_e$ and apply induction on the subgraph G_e , with the same adhesion but a smaller subgraph. This then puts us in a situation to apply Lemma 5.7, which shows that each monochromatic component is either contained in some G_e , or in a region close to G' (specifically, G' plus the barriers). In the latter case, the monochromatic r -paths can be rerouted to monochromatic r' -paths in G' .

Before we start the formal proof, we first prove some simple facts regarding strong-constructions and subgraphs.

Observation 5.14. *Let $k \in \mathbb{N}$, let $\ell \in \mathbb{R}^{\geq 0}$, and let $\mathcal{H}$ be a class of weighted graphs. Let $(\mathcal{P}, (B_t : t \in V(T)))$ be a complete $(k, \ell, \mathcal{H})$ -strong-construction for a weighted graph G , let $(J_t : t \in V(T))$ be the tree-decomposition of G induced by $(B_t : t \in V(T))$, and let T' be a subtree of G . Let $G' := G[\bigcup_{t \in V(T')} J_t]$, and let $\mathcal{P}' := \{P \in \mathcal{P} : P \cap V(G') \neq \emptyset\}$. Then $(\mathcal{P}', (B_t : t \in V(T')))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of G' .*

Proof. Observe that $G' = G[\bigcup_{t \in V(T')} \bigcup_{P \in B_t} P]$. Thus, for each $P \in \mathcal{P}'$, either P is

disjoint to $V(G')$, or $P \subseteq V(G')$. In the latter case, $G'[P] = G[P]$ has radius at most ℓ . It follows that $\mathcal{P}'$ is an ℓ -shallow proper-partition of G' . Note also that $G'/\mathcal{P}' = (G/\mathcal{P})[\mathcal{P}']$. Further, for each $P \in \mathcal{P}$, observe that $P \in \mathcal{P}'$ if and only if $P \in B_t$ for some $t \in V(T')$. Thus and since $G'/\mathcal{P}' = (G/\mathcal{P})[\mathcal{P}']$, it follows that $(B_t : t \in V(T'))$ is a tree-decomposition of $G'/\mathcal{P}'$ of adhesion at most k . Note that the tree-decomposition of G' induced by $(B_t : t \in V(T'))$ is $(J_t : t \in V(T'))$.

By Observation 5.2, $(J_t : t \in V(T'))$ is a complete tree-decomposition of G' . Since both $(J_t : t \in V(T))$ and $(J_t : t \in V(T'))$ are complete, by two applications of Observation 5.1, for each $t \in V(T)$, we have $G'\langle J_t \rangle = G'[J_t] = G[J_t] = G\langle J_t \rangle$. Thus, for any $S \subseteq B_t$, we have $G'\langle J_t \rangle[\bigcup_{P \in S} P] = G\langle J_t \rangle[\bigcup_{P \in S} P] \in \mathcal{H}$. Hence, $(\mathcal{P}, (B_t : t \in V(T)))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of G' , as desired. $\square$

Observation 5.15. *Let $k \in \mathbb{N}$, let $\ell \in \mathbb{R}^{\geq 0}$, and let $\mathcal{H}$ be a class of weighted graphs. Let $(\mathcal{P}, (B_t : t \in V(T)))$ be a complete $(k, \ell, \mathcal{H})$ -strong-construction for a weighted graph G , let $S \subseteq \mathcal{P}$, and let $Z := \bigcup_{P \in S} B_t$. Then $(\mathcal{P} \setminus S, (B_t \setminus S : t \in V(T)))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of $G - Z$.*

Proof. Observe that each $P \in S$ is disjoint to $G - Z$, and that for each $P \in \mathcal{P} \setminus S$, $P \subseteq V(G - Z)$, and $(G - Z)[P] = G[P]$ has radius at most ℓ . It follows that $\mathcal{P} \setminus S$ is an ℓ -shallow proper-partition of $G - Z$. Note also that $(G - Z)/(\mathcal{P} \setminus S) = (G/\mathcal{P})[\mathcal{P} \setminus S]$. It follows that $(B_t \setminus S : t \in V(T))$ is a tree-decomposition of $(G - Z)/(\mathcal{P} \setminus S)$ of adhesion at most k . Note that the tree-decomposition of $G - Z$ induced by $(B_t : t \in V(T))$ is $(J_t \setminus Z : t \in V(T))$.

By Observation 5.3, $(J_t \setminus Z : t \in V(T))$ is a complete tree-decomposition of $G - Z$, and for each $t \in V(T)$, we have $(G - Z)\langle J_t \setminus Z \rangle = G\langle J_t \rangle[J_t \setminus Z]$. Thus, for each $Q \subseteq B_t \setminus S$, we have $(G - Z)\langle J_t \setminus Z \rangle[\bigcup_{P \in Q} P] = G\langle J_t \rangle[J_t \setminus Z][\bigcup_{P \in Q} P] = G\langle J_t \rangle[\bigcup_{P \in Q} P]$ as each $P \in B_t \setminus S$ is disjoint to Z . Thus, $(G - Z)\langle J_t \setminus Z \rangle[\bigcup_{P \in Q} P] = G\langle J_t \rangle[\bigcup_{P \in Q} P] \in \mathcal{H}$. Hence, $(\mathcal{P}, (B_t : t \in V(T)))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of $G - Z$, as desired. $\square$

We now prove Lemma 5.13, which we restate for ease of reference.

Lemma 5.13. *For every $k \in \mathbb{N}$ and function $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$, there exists a function $f_k : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$ such that the following holds. Let $n \in \mathbb{N}$, let $\ell \in \mathbb{R}^{\geq 0}$, let $\mathcal{H}$ be a class of weighted graphs that admits f as an ℓ -almost n -dimensional control function, and let G be a weighted graph with a complete $(k, \ell, \mathcal{H})$ -strong-construction $(\mathcal{P}, (B_t : t \in V(T)))$. Set $n' := \max(n, 1)$, let $r \in \mathbb{R}^{\geq \ell}$, let $q \in V(T)$, and let $S^{\mathcal{P}} \subseteq B_q$ be of size at most k . Set $S := \bigcup_{P \in S^{\mathcal{P}}} P$, $Z^{\mathcal{P}} := \{P \in \mathcal{P} : P \cap N_G^{3r}(S) \neq \emptyset\}$, and let c_Z be an $(n' + 1)$ -colouring*

of $Z := \bigcup_{P \in Z^{\mathcal{P}}} P$. Then c_Z can be extended to an $(n' + 1, r, f_k(r))$ -colouring c of G . Further, if f is a dilation, then so is f_k .

Proof. Let $f_0 := f$, and for every $k \in \mathbb{N}^{\geq 1}$ and $r \in \mathbb{R}^{\geq 0}$, let:

$$\begin{aligned} g'_k(r) &:= 8(k+1)r, \\ g_k^*(r) &:= 2g'_k(r) + 2r, \\ f_k^*(r) &:= f_{k-1}(g_k^*(r)), \\ f_k^\#(r) &:= (k+1)(f_k^*(r) + 4g_k^*(r) + 12r), \text{ and} \\ f_k(r) &:= f_k^\#(r) + 2g_k^*(r). \end{aligned}$$

Observe that if f is a dilation, then for every $k \in \mathbb{N}$, g'_k , g_k^* , f_k^* , $f_k^\#$, and f_k are all dilations. Further, note that for every $k \in \mathbb{N}$ and $r \in \mathbb{R}^{\geq \ell}$, $g_k^*(r) \geq r \geq \ell$.

Now, fix $n, \ell, \mathcal{H}, G, \mathcal{P}, (B_t : t \in V(T)), r, q, S^{\mathcal{P}}, c_Z$ as above, with $n', S, Z^{\mathcal{P}}, Z$ defined as above. Note that f is also an ℓ -almost n' -dimensional control function for $\mathcal{H}$. Let $(J_t : t \in V(T))$ be the tree-decomposition of G induced by $(B_t : t \in V(T))$. As $(\mathcal{P}, (B_t : t \in V(T)))$ is complete, $(J_t : t \in V(T))$ is complete. So the completion of G by $(J_t : t \in V(T))$ is G . We use this implicitly from now on.

We show that G admits an $(n' + 1, r, f_k(r))$ -colouring that extends c_Z . We do this via induction, primarily on k and secondarily on $|V(G)|$.

The base case occurs when $k = 0$. So $(B_t : t \in V(T))$ has adhesion 0. In this case, observe that $(J_t : t \in V(T))$ also has adhesion 0, and thus that $G\langle J_t \rangle = G[J_t]$ for each $t \in V(T)$. Since $G[J_t] = G\langle J_t \rangle \in \mathcal{H}$, since $f_0 = f$ is an ℓ -almost n' -dimensional control function for $\mathcal{H}$, and since $r \geq \ell$, $G[J_t]$ admits an $(n' + 1, r, f_0(r))$ -colouring c_t . Since $(J_t : t \in V(T))$ has adhesion 0, $G = \bigsqcup_{t \in V(T)} G[J_t]$. Thus, $\bigcup_{t \in V(T)} c_t$ is well-defined, and is an $(n' + 1, r, f_0(r))$ -colouring of G . Noting that $Z = \emptyset$ as $|S| = 0$, $\bigcup_{t \in V(T)} c_t$ is the desired colouring. So we may now assume that $k \geq 1$, and that the lemma holds for all smaller values of k and when k is the same but $|V(G)|$ is smaller.

We may assume that $S^{\mathcal{P}}$ is nonempty. Otherwise, if G is empty, we are clearly done, and if G is nonempty, pick any part P' of $\mathcal{P}$ and pick some vertex $q' \in V(T)$ with $P' \in B_{q'}$. Let $S^{\mathcal{P}'} := \{P'\}$, let $S' := \bigcup_{P \in S^{\mathcal{P}'}} P = P'$, let $Z^{\mathcal{P}'}$ be the set of parts that intersect $N_G^{3r}(S')$, and let $c_{Z'}$ be an arbitrary $(n' + 1)$ -colouring of $Z' := \bigcup_{P \in Z^{\mathcal{P}'}} V(P)$. Observe that Z is empty, thus any colouring that extends $c_{Z'}$ also extends c_Z . Therefore, since $|S^{\mathcal{P}'}| = 1 \leq k$, we may proceed by setting $S^{\mathcal{P}} := S^{\mathcal{P}'}$, $S := S'$, $Z^{\mathcal{P}} := Z^{\mathcal{P}'}$, $Z := Z'$ and $c_Z := c_{Z'}$. So we can assume that $S^{\mathcal{P}}$ is nonempty. Since parts are nonempty (as $\mathcal{P}$ is proper), this implies that S and hence Z are nonempty.

For each $e = tt' \in E(T)$, let $S_e^{\mathcal{P}} := B_t \cap B_{t'}$ and let $S_e := \bigcup_{P \in S_e^{\mathcal{P}}} P$. Note that $|S_e^{\mathcal{P}}| \leq k$. Thus, S_e is a (k, r) -centred set in G , as $\ell \leq r$ and since $\mathcal{P}$ is ℓ -shallow.

For each $P \in S^{\mathcal{P}}$, let $N_P := \{P \in \mathcal{P} : P \cap N_G^{3r}(P) \neq \emptyset\}$. Since $\mathcal{P}$ is ℓ -shallow, P induces a nonempty connected weighted subgraph (of radius at most ℓ). Thus, $N_G^{3r}(P)$ also induces a nonempty connected weighted subgraph of G , and hence N_P induces a nonempty connected subgraph of $G/\mathcal{P}$. Let T_P be the subgraph of T induced by the vertices $t \in V(T)$ such that $B_t \cap N_P \neq \emptyset$. It follows that T_P is nonempty and connected, and for each $tt' \in E(T_P)$, $B_t \cap B_{t'} \cap N_P \neq \emptyset$. Additionally, T_P contains q as a vertex, as $P \in S^{\mathcal{P}} \subseteq B_q$ (and thus $P \in N_P$).

Let $T' := \bigcup_{P \in S^{\mathcal{P}}} T_P$. Notice that T' is connected, as each T_P , $P \in S^{\mathcal{P}}$, is connected and contains q , and nonempty, as $|S^{\mathcal{P}}| > 0$. Further, notice that $t \in V(T)$ is in $V(T')$ if and only if $B_t \cap Z^{\mathcal{P}} \neq \emptyset$, and that $B_t \cap B_{t'} \cap Z^{\mathcal{P}} \neq \emptyset$ for any $tt' \in E(T')$. Let $G' := G[\bigcup_{t \in V(T')} J_t]$, and let $\mathcal{P}' := \{P \in \mathcal{P} : P \cap V(G') \neq \emptyset\}$. By Observation 5.2, G' is isometric in G , and by Observation 5.14, $(\mathcal{P}', (B_t : t \in V(T')))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of G' . Note also that $Z \subseteq V(G')$.

Next, let E' denote the set of edges between T' and $T - V(T')$. For each $e \in E'$, let T_e be the connected component of $T - V(T')$ incident to e . Note that e is the only edge between T_e and $T - V(T_e)$. Thus, T_e is disjoint from $T_{e'}$, $e' \in E' \setminus \{e\}$, and there is a unique vertex $q_e \in V(T_e)$ adjacent to $T - V(T_e)$. In particular, q_e is adjacent to T' and e is incident with q_e . Additionally, notice that $V(T') \cup \bigcup_{e \in E'} V(T_e) = V(T)$. In particular, for each $e \in E'$, $V(T) \setminus V(T_e) = V(T') \cup \bigcup_{e' \in E' \setminus \{e\}} V(T_{e'})$. Finally, since $V(T_e) \subseteq V(T) \setminus V(T')$, we have that for each $t \in V(T_e)$, $B_t \cap Z^{\mathcal{P}} = \emptyset$.

Now, for each $e \in E'$, let $G_e := G[\bigcup_{t \in V(T_e)} J_t]$, and let $\tilde{G}_e := G[\bigcup_{t \in V(T) \setminus V(T_e)} J_t]$. Since the only edge between T_e and $T - V(T_e)$ was e , whose endpoints are q_e and some vertex of T' , note that $V(G_e \cap \tilde{G}_e) = S_e \subseteq V(G')$ and that $S_e^{\mathcal{P}} \subseteq B_{q_e}$. Set $\mathcal{P}_e := \{P \in \mathcal{P} : P \cap V(G_e) \neq \emptyset\}$. By Observations 5.2 and 5.14, G_e is isometric in G , and $(\mathcal{P}_e, (B_t : t \in V(T_e)))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of G_e . Finally, notice that $Z \cap V(G_e) = \emptyset$, as $Z^{\mathcal{P}} \cap B_t = \emptyset$ for each $t \in V(T_e) \subseteq V(T) \setminus V(T')$.

We now seek to apply Lemma 5.7 on G , using G' as G_0 and $(G_e : e \in E')$ as $G_1, \dots, G_a$. We use r as both r and ℓ , $n' + 1$ for m , k as itself, $f_k^{\#}(r)$ for d , and $f_k(r)$ for D . Notice that ℓ' is $g_k'(r)$ and r' is $g_k^*(r)$, hence $D = f_k(r) \geq d + 2r' = f_k^{\#}(r) + 2g_k^*(r)$, as required.

As $(J_t : t \in V(T))$ is a tree-decomposition of G , and noting that $\{V(T')\} \cup \{V(T_e) : e \in E'\}$ is a (proper) partition of T , it follows that $G = G' \cup \bigcup_{e \in E'} G_e$. So Lemma 5.7 (a) is satisfied.

Next, for each $e \in E'$, recall that $S_e = V(G_e \cap \tilde{G}_e) \subseteq V(G')$ and that S_e is a (k, r) -centred

set in G . Since $V(T) \setminus V(T_e) = V(T') \cup \bigcup_{e' \in E' \setminus \{e\}} V(T_{e'})$, we have $\tilde{G}_e = G' \cup \bigcup_{e' \in E' \setminus \{e\}} G_{e'}$. Thus, the $\tilde{G}_e$ [respectively the S_e], $e \in E'$, are the $\tilde{G}_i$ [respectively the S_i], $i \in \{1, \dots, a\}$, in the statement of Lemma 5.7. So Lemma 5.7 (b) is satisfied.

Now, consider Lemma 5.7 (c). By Observation 5.15 (with $S := Z^{\mathcal{P}}$), $(\mathcal{P}' \setminus Z^{\mathcal{P}}, (B_t \setminus Z^{\mathcal{P}} : t \in V(T')))$ is a $(k, \ell, \mathcal{H})$ -strong-construction for $G' - Z$. In fact, since $B_t \cap B_{t'} \cap Z^{\mathcal{P}} \neq \emptyset$ for each $tt' \in E(T')$, $(B_t \setminus Z^{\mathcal{P}} : t \in V(T'))$ has adhesion at most $k-1$ (as $(B_t : t \in V(T))$ has adhesion at most k). Thus, $(\mathcal{P}' \setminus Z^{\mathcal{P}}, (B_t \setminus Z^{\mathcal{P}} : t \in V(T')))$ is a $(k-1, \ell, \mathcal{H})$ -strong-construction for $G' - Z$. Thus, using $g_k^*(r) \geq \ell$ as r and recalling that $f_{k-1}(g_k^*(r)) = f_k^*(r)$, we may apply the induction hypothesis, keeping q unchanged and letting $S^{\mathcal{P}}$ be empty, to find an $(n' + 1, g_k^*(r), f_k^*(r))$ -colouring c' of $G' - Z$.

Note that $S^{\mathcal{P}} \subseteq \mathcal{P}'$, so each $P \in S^{\mathcal{P}}$ induces a weighted subgraph of G' of radius at most $\ell \leq r$ (as $\mathcal{P}'$ is ℓ -shallow). Thus (and since $|S^{\mathcal{P}}| \leq k$), $S = \bigcup_{P \in S^{\mathcal{P}}} P$ is (k, r) -centred in G' . Since $N_G^{3r}(S) \subseteq Z \subseteq V(G')$, observe that $N_G^{3r}(S) = N_{G'}^{3r}(S)$. Thus, $N_G^{3r}(S)$ is $(k, 4r)$ -centred in G' . Recall that each $P \in Z^{\mathcal{P}}$ intersects $N_G^{3r}(S)$, and observe that P has weak diameter in G' at most $2\ell \leq 2r$ (as $\mathcal{P}'$ is ℓ -shallow). Thus, $Z = \bigcup_{P \in Z^{\mathcal{P}}} P$ is $(k, 6r)$ -centred in G' . Since $(k+1)(f_k^*(r) + 4g_k^*(r) + 12r) = f_k^{\#}(r)$, by Corollary 5.6 (with $\ell := 6r$), $c'' := c' \cup c_Z$ is an $(n' + 1, g_k^*(r), f_k^{\#}(r))$ -colouring of G' . So Lemma 5.7 (c) is satisfied with c'' as c_0 . Note that c'' also extends c_Z , by definition.

Now, for each $e \in E'$, let c_{S_e} be the restriction of c'' to S_e . Let $Z_e^{\mathcal{P}} := \{P \in \mathcal{P}_e : P \cap N_{G_e}^{3r}(S_e) \neq \emptyset\}$ and $Z_e := \bigcup_{P \in Z_e^{\mathcal{P}}} P$. Observe that there exists a colouring c_{Z_e} of Z_e that extends c_{S_e} and has an (S_e, r) -barrier in G_e . Recall that $(\mathcal{P}_e, (B_t : t \in V(T_e)))$ is a complete $(k, \ell, \mathcal{H})$ -strong-construction of G_e . Note that $|V(G_e)| \leq |V(G - Z)| < |V(G)|$, since $Z \cap V(G_e) = \emptyset$ and since Z is nonempty. Hence, we can apply induction on G_e , using r as itself, q_e as q , $S_e^{\mathcal{P}} \subseteq B_{q_e}$ as $S^{\mathcal{P}}$, and c_{Z_e} as c_Z . This allows us to extend c_{Z_e} to an $(n' + 1, r, f_k(r))$ -colouring c_e of G_e . Note that since c_e extends c_{Z_e} , c_e has an (S_e, r) -barrier in G_e , and that $c_e = c''$ on S_i , by definition of c_{S_e} . Hence, Lemma 5.7 (d) is satisfied with the various c_e , $e \in E'$, acting as $c_1, \dots, c_a$.

Thus, we can now apply Lemma 5.7 with the parameters specified earlier. We find that $c := c'' \cup \bigcup_{e \in E'} c_e$ is an $(n' + 1, r, f_k(r))$ -colouring. Since c extends c'' , which extends c_Z , c also extends c_Z . Thus, c is the desired colouring. $\square$

5.5 Colourings of Minor-Free Graphs

Recall that the Graph Minor Structure Theorem (Theorem 1.2 [121]) gives a tree-decomposition of a K_h -minor-free (weighted) graph G of adhesion at most j such that

each torso is (g, p, k, a) -almost-embeddable. So each weighted torso is a (g, p, k, a) -almost-embeddable weighted graph. Thus, if $\mathcal{H}$ is the class of all (g, p, k, a) -almost-embeddable weighted graphs, G is $(j, \mathcal{H})$ -constructable.

To apply Theorem 5.8, we need $\mathcal{H}$ to be hereditary. Thus, we need to also consider all induced weighted subgraphs of almost-embeddable graphs. Unfortunately, it is difficult to convert an almost-embedding of a (weighted) graph to an almost-embedding of an induced (weighted) subgraph. This is because we might delete a boundary vertex, is needed to index a bag of the vortex-decomposition. Fortunately, because we are working in a weighted setting, we can substitute ‘deleting a vertex’ with ‘setting all weights of incident edges to ∞ ’. Therefore, each (induced) weighted subgraph H of an almost-embeddable graph is isometric in an almost-embeddable weighted graph G . We can also embed the missing edges of the underlying cycles around the disc with weight ∞ , so we can further take G to be snugly almost-embeddable. See Lemma 5.22 for the proof.

Therefore, a 2-dimensional control function for snugly almost-embeddable weighted graphs gives a 2-dimensional control function for (induced) weighted subgraphs of almost-embeddable weighted graphs. We can then apply Theorem 5.8 and Theorem 1.2 to show that each proper-minor-closed class $\mathcal{G}$ (whose graphs are K_h -minor-free for some $h \in \mathbb{N}$) has Assouad–Nagata dimension at most 2, proving Theorem 2.18(a). Recall that Theorem 2.18(b) says that the Assouad–Nagata dimension drops to 1 if and only if the graphs in $\mathcal{G}$ have bounded treewidth. The ‘if’ follows from Theorem 5.10, and the ‘only if’ follows from Theorem 2.15(b), since the asymptotic dimension lower bounds the Assouad–Nagata dimension. So this is enough to prove Theorem 2.18.

We now construct a 2-dimensional control function for snugly almost-embeddable weighted graphs. Let $\Gamma = (\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s))$ be a snug (g, p, k, a) -almost-embedding of such a weighted graph G . Recall that $G - A = \bigcup_{i=0}^s G_i = G_0 \cup H$, where $H := H(\Gamma) = \bigcup_{i=1}^s G_i$ is the vortex-union subgraph. Noting that A is an $(a, 0)$ -centred set in G , we can use Corollary 5.6 to convert an (m, r, D) -colouring of $G_0 \cup H$ to an (m, r, D') -colouring of G . So it suffices to focus on the union $G_0 \cup H$.

Recall that Bonamy et al. [22] showed that the class of unweighted graphs of Euler genus at most $g \in \mathbb{N}$ has Assouad–Nagata dimension 2 (Theorem 2.16). In fact, they showed that this holds even in a weighted setting.

Theorem 5.16. *For each $g \in \mathbb{N}$, the class of weighted graphs of Euler genus at most g has Assouad–Nagata dimension 2.*

G_0 has bounded Euler genus, and thus admits a 2-dimensional control function by Theorem 5.16. Recall that H has pathwidth, and thus treewidth, at most $2k + 1$,

since the vortices are disjoint and pairwise have pathwidth at most $2k + 1$. Thus, H admits a 1-dimensional (and thus 2-dimensional) control function by Theorem 5.10. So individually, both G_0 and H admit a 2-dimensional control function.

However, the union $G_0 \cup H$ no longer has bounded treewidth nor Euler genus. Further, trying to colour the two halves of the separation independently fails, as by jumping back and forth between G_0 and H , we can find short paths in the union that exist in neither half individually. However, the problem occurs close to where the two subgraphs meet, the boundary B of G . Far away from the boundary, naively colouring as either a graph of bounded Euler genus or as a graph of bounded treewidth suffices. So we just need to deal with the part of the graph close to the boundary.

Specifically, for any $\ell \in \mathbb{R}^{\geq 0}$ and $r \in \mathbb{R}^{\geq \ell}$, we ask that we can (m, r, D) -colour the graph induced by the ℓ -neighbourhood of the boundary. For $r \in \mathbb{R}^{\geq 0}$, we then colour G as follows. Any vertex not in the r -neighbourhood Z of the boundary lies strictly in either G_0 or H , and receives the colour from the (m, r, D) -colouring of either G_0 or H respectively. We colour the boundary according to the (m, r', D') -colouring of the graph induced by the r' -neighbourhood of B , where $r' \geq r$ is sufficiently large (bigger than D). Finally, we colour the vertices in Z according to the colour of a vertex on the boundary within distance r .

Any monochromatic r -path disjoint from Z is far from the boundary, and is thus an r -path in either G_0 or H . It is also monochromatic under the colouring of G_0 or H respectively. Such a path hence has weak diameter at most D . So any long monochromatic r -path must intersect Z . Using the same colour vertex on the boundary, this path can be translated to a monochromatic r' -path whose internal vertices are on the boundary. The interior is also a (monochromatic) r' -path in the graph induced by the r' -neighbourhood of the boundary, and so has weak diameter at most D' . Accounting for the two endpoints, the entire r' -path has weak diameter at most $D'' := D' + 2r'$. So this gives an (m, r, D'') -colouring of the union $G_0 \cup H$.

The following lemma formalises this idea.

Lemma 5.17. *Let $n \in \mathbb{N}$, let $f, g : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$, let G be a weighted graph, and let (A, B) be a separation of G with separator S such that:*

- (a) *f is an n -dimensional control function for both A and B , and*
- (b) *for any $\ell \in \mathbb{R}^{\geq 0}$, g is an ℓ -almost n -dimensional control function for $G[N_G^\ell(S)]$.*

Then the function $r \mapsto g(f(r) + 4r) + 2(f(r) + 4r)$ is an n -dimensional control function for G .

Proof. Fix $r \in \mathbb{R}^{\geq 0}$, and set $r' := f(r) + 4r$ and $d' := g(r')$. We must show that G admits an $(n + 1, r, d' + 2r')$ -colouring. Let c_A and c_B be $(n + 1, r, f(r))$ -colourings of A, B respectively, set $Z := N_G^r(S)$ and $Z' := N_G^{r'}(S)$, and let $c_{Z'}$ be an $(n + 1, r', d')$ -colouring of $G[Z']$. Let $\iota : Z \mapsto S$ be the identity on S , and for $x \in Z \setminus S$, let $\iota(x)$ be a vertex in $S \cap N_G^r(x)$. Now, define a colouring c of G via, for each $x \in V(G)$:

$$c(x) := \begin{cases} c_A(x) & \text{if } x \in V(A) \setminus Z, \\ c_B(x) & \text{if } x \in V(B) \setminus Z, \\ c_{Z'}(\iota(x)) & \text{otherwise.} \end{cases}$$

Observe that for any $x, y \in V(A) \setminus Z$ at distance at most r in G and any shortest path Q from x to y in G , Q does not intersect S , since otherwise we would have $x, y \in N_G^r(S) = Z$. It follows that Q is also a path of length at most r in A , so $\text{dist}_A(x, y) \leq r$. Consequently, any r -path in G whose vertices are contained in $V(A) \setminus Z$ is also an r -path in A . A symmetric argument shows that any r -path in G whose vertices are contained in $V(B) \setminus Z$ is also an r -path in B .

Now, notice that for any $x \in V(A)$ and $y \in V(B)$ with $\text{dist}_G(x, y) \leq r$, we have that $x, y \in N_G^r(S) = Z$ (since the shortest path from x to y must intersect S). It follows that any r -path in G disjoint to Z is contained either in $V(A) \setminus Z$ or in $V(B) \setminus Z$, and is thus an r -path either in A or in B respectively. Therefore, for any monochromatic r -path P from $u \in V(G)$ to $v \in V(G)$ in G under c , any r -subpath P' of $P \setminus Z$ is an r -path either in A or in B . Since c is c_A on $V(A) \setminus Z$ and c_B on $V(B) \setminus Z$, P' is also monochromatic either in A under c_A or in B under c_B respectively. Thus, P' has weak diameter at most $f(r)$ either in A or in B respectively, and consequently in G as well.

Therefore, we can apply Lemma 5.4 to find that there exists a $f(r) + 2r + 2r = r'$ -walk P'' from u to v in G whose interior consists of vertices of the form $\iota(x)$ with $x \in P \cap Z$. Observe that for $x \in P \cap Z$, $c(x) = c_{Z'}(\iota(x)) = c_{Z'}(\iota(\iota(x))) = c(\iota(x))$, as $\iota(\iota(x)) = \iota(x)$ since ι is the identity on S . Therefore, P'' is also monochromatic under c .

Now, for any $x, y \in S$ at distance at most r' in G and any shortest path Q from x to y in G , observe that Q is contained in $N_G^{r'}(S) = Z'$. Thus, Q is also a path of length at most r' in $G[Z']$, and $\text{dist}_{G[Z']}(x, y) \leq r'$. Since the interior of P'' is contained in S , the interior of P'' is also a (possibly empty) r' -path in $G[Z']$. Further, since c is $c_{Z'}$ on S , as ι is the identity map on S , we also have that the interior of P'' is monochromatic under $c_{Z'}$. Hence, the interior of P'' has weak diameter at most d' in both $G[Z']$ and G . Thus, P'' has weak diameter at most $d' + 2r'$ in G . In particular, $\text{dist}_G(u, v) \leq d' + 2r'$, as desired. $\square$

It remains to find a dilation that is an ℓ -almost 2-dimensional control function for $(G - A)[N_{G-A}^\ell(S)]$. In fact, we find an ℓ -almost 1-dimensional control function for $(G - A)[N_{G-A}^\ell(S)]$. We do this by showing $(G - A)[N_{G-A}^\ell(S)]$ admits a (w, ℓ) -proper-partition (where w is independent of ℓ), across multiple steps. Theorem 5.9 then gives the desired dilation. The first step is to find a natural ℓ -shallow proper partition of $G_0[N_{G_0}^\ell(S)]$. We do this by assigning each $v \in N_{G_0}^\ell(S)$ to the closest $s \in S$ (which is at distance at most ℓ), breaking ties consistently to ensure connectedness.

Lemma 5.18. *Let $r \in \mathbb{R}^{\geq 0}$, let G be a weighted graph, and let $S \subseteq V(G)$. Then $G' := G[N_G^r(S)]$ admits an r -shallow proper partition $\mathcal{P}$ such that each part of $\mathcal{P}$ contains exactly one vertex of S .*

Proof. Let $\preceq$ be an arbitrary ordering of the vertices of S . For $v \in N_G^r(S)$, let $\iota(v)$ be the smallest (with respect to $\preceq$) $s \in S$ such that $\text{dist}_G(v, S) = \text{dist}_G(v, s)$. Then, for each $s \in S$, define $P_s := \{v \in N_G^r(S) : \iota(v) = s\}$, and finally, define $\mathcal{P} := \{P_s : s \in S\}$. We argue that $\mathcal{P}$ is the desired (proper) partition of G' .

Each vertex of G' is in exactly one part of $\mathcal{P}$, since $\preceq$ acts as a tiebreaker. Additionally, for each $s \in S$, we have that $s \in P_s$, since $\text{dist}_G(s, s) = 0$ and $\text{dist}_G(s, s') > 0$ for $s' \in S \setminus \{s\}$. It follows that each part is nonempty and contains exactly one vertex of S . So it remains only to show that $\mathcal{P}$ is r -shallow; that is, for each $s \in S$, P_s induces a (connected) weighted subgraph of radius at most r . It suffices to show that for each $v \in P_s$, there exists a path from v to s in $G'[P_s]$ of length at most r .

Let Q be a shortest path from v to s in G . By definition of P_s , Q must have length $\text{dist}_G(v, S) \leq r$. Thus, $Q \subseteq N_G^r(S) = V(G')$, as each vertex of Q is at least as close to s as v is. Consequently, each vertex of Q lies in some part of $\mathcal{P}$.

Assume, for a contradiction, that some $u \in Q$ is not in P_s , and instead is in $P_{s'}$ for some $s' \in S \setminus \{s\}$. Since Q is a shortest path, $\text{dist}_G(s, v) = \text{dist}_G(s, u) + \text{dist}_G(u, v)$, and since $u \in P_{s'}$ and $v \in P_s$, we have $\text{dist}_G(s, u) \geq \text{dist}_G(s', u)$ and $\text{dist}_G(s', v) \geq \text{dist}_G(s, v)$. If $\text{dist}_G(s, u) > \text{dist}_G(s', u)$, then $\text{dist}_G(s, v) > \text{dist}_G(s', u) + \text{dist}_G(u, v) \geq \text{dist}_G(s', v)$, a contradiction. Thus, we have that $\text{dist}_G(s, u) = \text{dist}_G(s', u)$ and $\text{dist}_G(s, v) = \text{dist}_G(s', u) + \text{dist}_G(u, v) \geq \text{dist}_G(s', v)$. This forces $\text{dist}_G(s, v) = \text{dist}_G(s', v)$. If $s' \prec s$, then since $\text{dist}_G(s', v) = \text{dist}_G(s, v) = \text{dist}_G(v, S)$, $\preceq$ would put v in $P_{s'}$ over P_s , contradicting the fact that $v \in P_s$. By contrast, if $s' \succ s$, then since $\text{dist}_G(s, u) = \text{dist}_G(s', u) = \text{dist}_G(u, S)$, u would have been placed in P_s over $P_{s'}$, another contradiction. Thus, we can conclude that $u \in P_s$.

Therefore, $Q \subseteq P_s$. So Q is also a path from v to s in $G'[P_s]$, and is of the same length in $G'[P_s]$ as in G . Since Q had length at most r in G , Q is therefore the desired path.

Thus, each part induces a weighted subgraph of radius at most r , so $\mathcal{P}$ is r -shallow. This completes the proof of the lemma. $\square$

We next show that the proper partition of $G_0[N_{G_0}^\ell(S)]$ obtained from Lemma 5.18 is a (w, ℓ) -proper-partition. For this, we add handles so that all the discs are in the same face, and then contract each part to a vertex on that face. We find that the quotient is contained in a radius 1 (unweighted) graph of bounded Euler genus, to which we can apply the following lemma of Dujmović, Morin, and Wood [61].

Lemma 5.19. *For all $g, r \in \mathbb{N}$, every (unweighted) graph of radius at most r and Euler genus at most g has treewidth at most $(2g + 3)r$.*

See also [68] for an earlier $O(gr)$ bound and [117] for an earlier proof of the planar case. We now proceed to the proof.

Lemma 5.20. *Let $g, p \in \mathbb{N}$, let $r \in \mathbb{R}^{\geq 0}$, let G be a weighted graph embedded in a surface of Euler genus at most g . Let $s \in \{0, \dots, p\}$, let $D_1, \dots, D_s$ be G -clean discs, and let $B := \bigcup_{i=1}^s B(D_i, G)$ and $G' := G[N_G^r(B)]$. If $\mathcal{P}$ is an r -shallow proper partition of G' such that each part contains at least vertex in B , then $\mathcal{P}$ is a $(2g + 4p, r)$ -proper-partition of G' .*

Proof. Since $\mathcal{P}$ is r -shallow, it remains only to show that $G'/\mathcal{P}$ has treewidth at most $2g + 4p$. If $s = 0$, then B is empty, and thus G' and $\mathcal{P}$ are empty. $G'/\mathcal{P}$ is then the empty graph, which has treewidth at most $0 \leq 2g + 4p$. So we may assume that $s \geq 1$. For each $i \in \{2, \dots, s\}$, add a handle connecting the interior of D_i to the interior of D_1 . These handles can be added pairwise disjointly and disjointly to G . Thus, we obtain an embedding of G (and G') in a surface Σ of Euler genus at most $g + 2(s - 1) \leq g + 2p - 2$. In this embedding, observe that all the vertices of B lie on a common face. Contracting each part of $\mathcal{P}$ down to a vertex in B , we obtain an embedding of $G'/\mathcal{P}$ where all the vertices lie on a common face F (this can be done even if the parts contain multiple vertices in B). Inside F , we can embed a new vertex v and edges vP for each $P \in \mathcal{P}$. We obtain an embedding of an unweighted graph H of radius 1 with $G'/\mathcal{P} \subseteq H$. By Lemma 5.19, H has treewidth at most $2(g + 2p - 2) + 3 \leq 2g + 4p$. Since $G'/\mathcal{P} \subseteq H$, $G'/\mathcal{P}$ also has treewidth at most $2g + 4p$, as desired. $\square$

Finally, we extend the (w, ℓ) -proper-partition of $G_0[N_{G_0}^\ell(S)]$ to a (w', ℓ) -proper-partition of $(G - A)[N_{G-A}^\ell(S)]$. For this, we create singleton parts for every vertex not already in a part (in a vortex but not in the boundary). To obtain the tree-decomposition of the new quotient, we take the tree-decomposition $(B_t : t \in V(T))$ of the original quotient, and for each vertex on the boundary v and each bag B_t containing P_v , the part

containing v , we add all the parts that intersect the bag J_v of the vortex-decomposition to B_t . The fact that the vortex-decomposition is planted and that the disc is snug means that the result is a tree-decomposition (of bounded width).

Recall that an H -decomposition $(J_x : x \in V(H))$ of a graph G (with $V(H) \subseteq V(G)$) is planted if $x \in J_x$ for each $x \in V(H)$.

Lemma 5.21. *Let $k, w \in \mathbb{N}$, let $r \in \mathbb{R}^{\geq 0}$, and let G' be a weighted graph that admits a separation (G, H) with separator S such that:*

- (a) *G admits a (w, r) -proper-partition $\mathcal{P}$ such that each part contains at most one vertex in S , and*
- (b) *H admits a planted $G[S]$ -decomposition of width at most k .*

Then $\mathcal{P}' := \mathcal{P} \cup \{\{v\} : v \in V(H) \setminus S\}$ is a $((k+1)(w+1) - 1, r)$ -proper-partition of G' .

Proof. It is immediate that $\mathcal{P}'$ is an r -shallow proper partition of G' (since $G' = G \cup H$ and $V(G \cap H) = S$), so it remains only to show that the treewidth of $G'/\mathcal{P}'$ is at most $(k+1)(w+1) - 1$. Let $(B_t : t \in V(T))$ be a tree-decomposition of $G/\mathcal{P}$ of width at most w , and let $(J_s : s \in S)$ be a planted $G[S]$ -decomposition of H of width at most k .

For each $s \in S$, let $P_s \in \mathcal{P}$ be the unique part of $\mathcal{P}$ (and $\mathcal{P}'$) that contains s , and let $\mathcal{P}'_s := \{P \in \mathcal{P}' : P \cap J_s \neq \emptyset\}$. Note that $|\mathcal{P}'_s| \leq |J_s| \leq k+1$ (since $(J_s : s \in S)$ has width at most k), and that $P_s \in \mathcal{P}'_s$ (since $(J_s : s \in S)$ is planted). Let $\mathcal{P}_S \subseteq \mathcal{P}$ be the set of parts in $\mathcal{P}$ (and $\mathcal{P}'$) that intersect S . Note that $\mathcal{P}_S = \{P_s : s \in S\}$.

For each $t \in V(T)$, let $S_t := \bigcup_{P \in B_t} P \cap S$, and let $K_t := B_t \cup \bigcup_{s \in S_t} \mathcal{P}'_s$. Note that $|S_t| \leq |B_t \cap \mathcal{P}_S|$ since each $P \in B_t$ contains at most one vertex in S (and since $\mathcal{P}_S$ is the parts of $\mathcal{P}$ that intersect S). Observe that for each $P \in B_t \cap \mathcal{P}_S$, there exists $s \in P \cap S$; we then have $s \in S_t$, and $P = P_s \in \mathcal{P}'_s$. So $B_t \cap \mathcal{P}_S \subseteq \bigcup_{s \in S_t} \mathcal{P}'_s$.

For each $h \in V(H)$, the vertices $s \in S$ such that $h \in J_s$ induce a nonempty connected weighted subgraph C_h of $G[S]$. Let $C_h^{\mathcal{P}}$ be the subgraph of $G/\mathcal{P}$ induced by the parts $P \in \mathcal{P}$ such that $P \cap V(C_h) \neq \emptyset$. Since C_h is nonempty and connected, observe that $C_h^{\mathcal{P}}$ is nonempty and connected (this is true even if $\mathcal{P}$ is not a connected partition). Thus, the vertices $t \in V(T)$ such that $B_t \cap V(C_h^{\mathcal{P}}) \neq \emptyset$ induce a nonempty subtree T_h of T . Also, since $(J_s : s \in S)$ is planted, observe that for each $s \in S$, we have $s \in V(C_s)$ and thus $P_s \in V(C_s^{\mathcal{P}})$.

Fix $P \in \mathcal{P}'$. If $P \in \mathcal{P} \setminus \mathcal{P}_S$, then observe that P is disjoint to $V(H)$, and thus $\bigcup_{s \in S} \mathcal{P}'_s$. It follows that for each $t \in V(T)$, $P \in K_t$ if and only if $P \in B_t$. Thus, the vertices $t \in V(T)$ such that $P \in K_t$ induce a nonempty subtree of T . Now, presume that $P \notin \mathcal{P} \setminus \mathcal{P}_S$. So either $P \in \mathcal{P} \cap \mathcal{P}_S$ or $P \in \mathcal{P}' \setminus \mathcal{P}$. In the former case, observe that

there is exactly one vertex $s \in P \cap S$ (since each part of $\mathcal{P}$ contains at most one vertex of S), and that $P \cap V(H) = \{s\}$. In the latter case, $P = \{v\}$ for some $v \in V(H) \setminus S$. Either way, there exists exactly one vertex h in $P \cap V(H)$.

We argue that for each $t \in V(T)$, $P \in K_t$ if and only if $t \in V(T_h)$. Let $t \in V(T)$ be such that $P \in K_t$. Since $B_t \cap \mathcal{P}_S \subseteq \bigcup_{s \in S_t} \mathcal{P}'_s$ (and since $P \notin \mathcal{P} \setminus \mathcal{P}_S$), we have that $P \in \mathcal{P}'_s$ for some $s \in S_t$. Since $P \cap V(H) = \{h\}$ and $J_s \subseteq V(H)$, this implies that $h \in J_s$. Thus, $s \in V(C_h)$, and $P_s \in V(C_h^\mathcal{P})$. Further, since $s \in P_s \cap S_t$ and since $\mathcal{P}$ is a partition of G , it follows that $P_s \in B_t$. Thus, $P_s \in V(C_h^\mathcal{P}) \cap B_t$. In particular, $V(C_h^\mathcal{P}) \cap B_t \neq \emptyset$, so $t \in V(T_h)$. By contrast, for each $t \in V(T_h)$, there exists $P' \in B_t \cap V(C_h^\mathcal{P})$. This part P' must contain some $s \in V(C_h) \subseteq S$. By definition of C_h , we have $h \in J_s$, so $P \in \mathcal{P}'_s$. However, we also know that $s \in P' \cap S \subseteq S_t$ (since $P' \in B_t$), so $P \in \bigcup_{s \in S_t} \mathcal{P}'_s \subseteq K_t$. Thus, for each $t \in V(T)$, $P \in K_t$ if and only if $t \in V(T_h)$. Since T_h is nonempty and connected, it follows that the vertices $t \in V(T)$ such that $P \in K_t$ induce a nonempty subtree of T .

Let $P_1 P_2 \in E(G'/\mathcal{P}')$. So there exists some $uv \in E(G')$ such that $u \in P_1$, $v \in P_2$. Observe that uv is an edge either in G or in H . In the former case, P_1 and P_2 are parts in $\mathcal{P}$ and $P_1 P_2$ is an edge of $E(G/\mathcal{P})$. Thus, there exists $t \in V(T)$ such that $P_1, P_2 \in B_t \subseteq K_t$. In the latter case, there exists $s \in S$ such that $u, v \in J_s$; note that $P_1, P_2 \in \mathcal{P}'_s$. Observe that for any $t \in V(T)$ with $P_s \in B_t$ (at least one such t must exist), we have $s \in S_t$, and thus $P_1, P_2 \in K_t$. It follows that $(K_t : t \in V(T))$ is a tree-decomposition of $G'/\mathcal{P}'$.

Since $B_t \cap \mathcal{P}_S \subseteq \bigcup_{s \in S_t} \mathcal{P}'_s$, observe that $|K_t| \leq |B_t \setminus \mathcal{P}_S| + |\bigcup_{s \in S_t} \mathcal{P}'_s|$. Recalling that $|S_t| \leq |B_t \cap \mathcal{P}_S|$ and that $|\mathcal{P}'_s| \leq k + 1$ for each $s \in S$, we obtain $|K_t| \leq |B_t \setminus \mathcal{P}_S| + (k + 1)|B_t \cap \mathcal{P}_S| \leq (k + 1)|B_t|$ (since $k \geq 0$). Since $(B_t : t \in V(T))$ has width at most w , this gives $|K_t| \leq (k + 1)(w + 1)$. So $(K_t : t \in V(T))$ has width at most $(k + 1)(w + 1) - 1$. The lemma follows. $\square$

It remains to prove that any (induced) weighted subgraph of a (g, p, k, a) -almost-embeddable weighted graph is isometric in a snugly (g, p, k, a) -almost-embeddable weighted graph.

Lemma 5.22. *Let $g, p, k, a \in \mathbb{N}$, and let H be a weighted subgraph of a (g, p, k, a) -almost-embeddable weighted graph. Then H is an isometric weighted subgraph of a snugly (g, p, k, a) -almost-embeddable weighted graph G .*

Proof. By the choice of H , there exists a graph G' with $V(H) \subseteq V(G')$ that admits a (g, p, k, a) -almost-embedding $\Gamma' = (\Sigma, s, A, (G'_0, G'_1, \dots, G'_s), (D_1, \dots, D_s))$.

For each $i \in \{1, \dots, s\}$, let $U_i := U(D_i, G'_0)$. Let $G_0 := G'_0 \cup \bigcup_{i=1}^s U_i$ and $G := G' \cup \bigcup_{i=1}^s U_i$. Give each $e \in E(H) \subseteq E(G)$ weight in G equal to its weight in H , and weight each $e \in E(G) \setminus E(H)$ by ∞ . Observe that each path in H is a path in G of the same length, and that each path in G not in H has weight ∞ . It follows that H is isometric in G .

For each $i \in \{1, \dots, s\}$, embed the edges of $E(U_i) \setminus E(G'_0)$ around D_i . This can be done such that the interiors of the new edges are embedded disjointedly to each other, G'_0 , and $D_1, \dots, D_s$. We obtain an embedding of G_0 in Σ such that $D_1, \dots, D_s$ are G_0 -clean. Further, for each $i \in \{1, \dots, s\}$, we have $U(D_i, G_0) = U_i \subseteq G_0$. Thus, D_i is snug in G_0 . Note also that $V(G_0) = V(G'_0)$ and $B(D_i, G_0) = B(D_i, G'_0)$, and thus $V(G_0 \cap G_i) = V(G'_0 \cap G_i) = B(D_i, G'_0) = B(D_i, G_0)$. It follows that G_i is a (D_i, G_0) -vortex. From these observations and choice of Γ , it follows that $\Gamma := (\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s))$ is a snug (g, p, k, a) -almost-embedding of G . The lemma follows. $\square$

We are finally in a position to show that the class of all (induced) weighted subgraphs of (g, p, k, a) -almost-embeddable weighted graphs has Assouad–Nagata dimension 2.

Lemma 5.23. *Let $g, p, k, a \in \mathbb{N}$. Then the class $\mathcal{G}$ of all weighted subgraphs of (g, p, k, a) -almost-embeddable weighted graphs is hereditary and has Assouad–Nagata dimension at most 2.*

Proof. The fact that $\mathcal{G}$ is hereditary is immediate, so we only need to show the existence of a dilation f that is a 2-dimensional control function for $\mathcal{G}$.

By Theorem 5.16, there exists a dilation $f_1^\#$ that is a 2-dimensional control function for the class of weighted graphs of Euler genus at most g . By Theorem 5.10, there exists a dilation $f_2^\#$ that is a 1-dimensional control function for the class of weighted graphs of treewidth at most $2k + 1$. Let $f^\# := \max(f_1^\#, f_2^\#)$; note that $f^\#$ is a dilation. By Theorem 5.9, there exists a dilation f' such that for every $\ell \in \mathbb{R}^{\geq 0}$, f' is an ℓ -almost 1-dimensional control function for every weighted graph G that admits a $((2g + 4p + 1)(k + 1) - 1, \ell)$ -proper-partition. Let $f^*(r) := f'(f^\#(r) + 4r) + 2(f^\#(r) + 4r)$ for any $r \in \mathbb{R}^{\geq 0}$. Note that f^* is also a dilation, as both $f^\#$ and f' are dilations. Finally, let $f(r) := (a + 1)(f^*(r) + 4r)$ for any $r \in \mathbb{R}^{\geq 0}$. Note that f is also a dilation. We show that f is a 2-dimensional control function for $\mathcal{G}$.

Consider any $G' \in \mathcal{G}$. By Lemma 5.22, G' is an isometric weighted subgraph of some weighted graph G that admits a snug (g, p, k, a) -almost-embedding $\Gamma = (\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s))$. Note that any 2-dimensional control function for G is a 2-dimensional control function for G' .

Let $H := H(\Gamma)$ and $B := B(\Gamma)$. So $G - A = G_0 \cup H$ and $V(G_0 \cap H) = B$. Since Γ is snug, recall that H admits a planted $G_0[B]$ -decomposition of width at most k , obtained from the union of the vortex-decompositions. Here, we use that discs have pairwise disjoint boundary vertices, as the discs are disjoint. Observe also that (G_0, H) is a separation of $G - A$ with separator B , and that H has treewidth at most $2k - 1$.

We seek to apply Lemma 5.17 using $G - A$ as G , (G_0, H) as (A, B) (and $B = B(\Gamma)$ as S), $f^\#$ as f and f' as g . Lemma 5.17 (a) is immediately satisfied by definition of $f^\#$, as G_0 has Euler genus at most g and H has treewidth at most $2k + 1$. Therefore, we only need to show Lemma 5.17 (b), that for any $\ell \in \mathbb{R}^{\geq 0}$, f' is an ℓ -almost 2-dimensional control function for $(G - A)[N_{G-A}^\ell(B)]$. We show that $(G - A)[N_{G-A}^\ell(B)]$ admits a $((2g + 4p + 1)(k + 1) - 1, \ell)$ -proper-partition; the desired result then follows by definition of f' .

Let $\ell \in \mathbb{R}^{\geq 0}$, and set $B^\ell := N_{G-A}^\ell(B)$, $B_{G_0}^\ell := B^\ell \cap V(G_0)$ and $B_H^\ell := B^\ell \cap V(H)$. Note that $B_{G_0}^\ell = N_{G_0}^\ell(B)$, and that $B_H^\ell = N_H^\ell(B)$. We use these facts implicitly in the remainder of this proof.

By Lemma 5.18, $G_0[B_{G_0}^\ell]$ admits an ℓ -shallow proper partition $\mathcal{P}$ such that each part contains exactly one vertex of B . Observe that the discs $D_1, \dots, D_s$ are $G_0[B_{G_0}^\ell]$ -clean and that for each $i \in \{1, \dots, s\}$, $B(D_i, G_0[B_{G_0}^\ell]) = B(D_i, G_0)$ (as $B \subseteq B_{G_0}^\ell$). Thus, by Lemma 5.20 (noting that each part contains exactly one vertex of B), $\mathcal{P}$ is $(2g + 4p, \ell)$ -proper-partition of $G_0[B_{G_0}^\ell]$.

Now, notice that $(G_0[B_{G_0}^\ell], H[B_H^\ell])$ is a separation of $(G - A)[B^\ell]$ with separator B . Further, from the planted $G_0[B]$ -decomposition of width at most k of H , we obtain a planted $G_0[B] = (G_0[B_{G_0}^\ell])[B]$ -decomposition of width at most k of $H[B_H^\ell]$ (as $B \subseteq B_H^\ell$). Thus, by Lemma 5.21 (with $w := 2g + 4p$, and noting that each part contains exactly one vertex of B), $\mathcal{P}' := \mathcal{P} \cup \bigcup_{v \in S_H^\ell \setminus S} \{\{v\}\}$ is a $((2g + 4p + 1)(k + 1) - 1, \ell)$ -partition of $(G - A)[B^\ell]$, as desired.

Thus, we can apply Lemma 5.17 with the parameters specified above to find that $f'(f^\#(r) + 4r) + 2(f^\#(r) + 4r) = f^*(r)$ is a 2-dimensional control function for $G - A$. Observing that A is an $(a, 0)$ -centred set in G , by Corollary 5.6, $(a + 1)(f^*(r) + 4r) = f(r)$ is a 2-dimensional control function for G . Since G' is isometric in G , f is also a 2-dimensional control function for G' . Hence, f is a 2-dimensional control function for $\mathcal{G}$. Since f is a dilation, we obtain $\text{ANdim}(\mathcal{G}) \leq 2$, completing the proof. $\square$

We can now bound the Assouad–Nagata dimension of any minor-free class of weighted graphs.

Theorem 5.24. *For any $h \in \mathbb{N}$ and any K_h -minor-free class of weighted graphs $\mathcal{G}$,*

- (a) $\text{ANdim}(\mathcal{G}) \leq 2$, and
- (b) $\text{ANdim}(\mathcal{G}) \leq 1$ if there exists $k \in \mathbb{N}$ such that $\text{tw}(G) \leq k$ for each $G \in \mathcal{G}$.

Proof. Theorem 5.24(b) follows from Theorem 5.10, so we focus on Theorem 5.24(a).

Let $\mathcal{H}$ be the class of weighted graphs that are a weighted subgraph of a (g, p, k, a) -almost-embeddable weighted graph. By Lemma 5.23 $\mathcal{H}$ is hereditary and $\text{ANdim}(\mathcal{H}) \leq 2$.

By Theorem 1.2, there exists $g, p, k, a, j \in \mathbb{N}$ such that every unweighted K_h -minor-free graph has a tree-decomposition of adhesion at most j such that every torso is (g, p, k, a) -almost-embeddable. Thus, every weighted K_h -minor-free graph (including each $G \in \mathcal{G}$) has a tree-decomposition of adhesion at most j such that every weighted torso is (g, p, k, a) -almost-embeddable. Hence, each $G \in \mathcal{G}$ is $(j, \mathcal{H})$ -constructable. By Theorem 5.8, $\text{ANdim}(\mathcal{G}) \leq \max(\text{ANdim}(\mathcal{H}), 1) \leq 2$. $\square$

Theorem 2.18 then follows due to Theorem 5.24 and Theorem 2.15(b), as the Assouad–Nagata dimension is at least the asymptotic dimension (and since each proper minor-closed class excludes K_h as a minor for some $h \in \mathbb{N}$). We remark that the converse of Theorem 5.24(b) is not true. For example, take the class $\mathcal{G}$ of unweighted 2-dimensional grids (which is K_5 and $K_{3,3}$ -minor-free by the Kuratowski–Wagner Theorem, Theorem 1.1 [96, 127], since all 2-dimensional grids are planar), and for each $G \in \mathcal{G}$, fix $r \in V(G)$, and weight each $uv \in E(G)$ by $2^{\text{dist}_G(r, \{u, v\})}$. It can be seen that any ℓ -path in the weighted graph is either contained in 2ℓ -neighbourhood of r , or consists of only a single vertex. It follows that $\ell \mapsto 4\ell$ is a 0-dimensional control function for the corresponding class of weighted graphs $\mathcal{G}'$, despite these weighted graphs having unbounded treewidth.

5.6 Assouad–Nagata Dimension of Subdivision-Closed Classes

We close this chapter by looking at classes of weighted graphs that admit every graph as a minor. In particular, we establish some necessary conditions for such a class to have bounded asymptotic or Assouad–Nagata dimension, which leads to a proof of Theorem 2.19 (Assouad–Nagata dimension of subdivision-closed classes). We achieve this by migrating a control function for a graph G in the class to a control function for any minor of G , achieved by reweighting edges appropriately.

We first show that for any unweighted graphs G, H with H connected and $H \leq G$, there is a weighted graph G' obtained from imbuing G with rational edge weights whose

control functions translate to control functions for H . This is obtained by taking a proper model of H in G , and then giving each edge not in the proper model very large weight and each contracted edge very small weight. The resulting weighted graph G' has a very similar metric to H .

We require the following observation, which is a slight variation of Observation 2.20 (which says that quasi-isometries to a class with a known control function give rise to a control function).

Observation 5.25. *Let $n \in \mathbb{N}$, let G, H be weighted graphs, and let f be an n -dimensional control function for G . If there exist $\alpha, \beta \in \mathbb{R}^{>0}$ and a map $\iota : V(H) \mapsto V(G)$ such that for all $u, v \in V(H)$, $\beta \operatorname{dist}_H(u, v) \leq \operatorname{dist}_G(\iota(u), \iota(v)) \leq \alpha \operatorname{dist}_H(u, v)$, then $r \mapsto \frac{1}{\beta} f(\alpha r)$ is an n -dimensional control function for H .*

Proof. Fix $r \in \mathbb{R}^{>0}$, and let c_G be an $(n+1, \alpha r, f(\alpha r))$ -colouring of G . Define a colouring c_H of H via $c_H(v) := c_G(\iota(v))$ for each $v \in V(H)$. Observe that for any monochromatic r -path $v_0, v_1, \dots, v_m$ in H under c_H , $\iota(v_0), \iota(v_1), \dots, \iota(v_m)$ is a monochromatic αr -walk in G under c_G , and thus $\operatorname{dist}_G(\iota(v_0), \iota(v_n)) \leq f(\alpha r)$. Since $\beta \operatorname{dist}_H(v_0, v_n) \leq \operatorname{dist}_G(\iota(v_0), \iota(v_n))$, we have that $\operatorname{dist}_H(v_0, v_n) \leq \frac{1}{\beta} f(\alpha r)$. Thus, c_H is an $(n+1, r, \frac{1}{\beta} f(\alpha r))$ -colouring of H . The observation follows. $\square$

We remark that the fact that control functions are preserved upon taking an isometric subgraph (or any other isometric graph) is a special case of Observation 5.25, using $\alpha := \beta := 1$ and the identity map for ι .

Lemma 5.26. *Let $\epsilon \in \mathbb{R}^{>0}$, let G be a unweighted graph, and let H be a connected unweighted graph that is a minor of G . Then there exists a function $w : E(G) \mapsto \mathbb{R}^{>0}$ such that for any $n \in \mathbb{N}$ and $f : \mathbb{R}^{>0} \mapsto \mathbb{R}^{>0}$, if f is an n -dimensional control function for the weighted graph G' obtained from imbuing G with w , then the function $r \mapsto f((1+\epsilon)r)$ is an n -dimensional control function for H . Furthermore, if ϵ is rational, then w is a rational-valued function.*

Proof. Let $\mathcal{P} := (P_x : x \in V(H))$ be a proper model of H in G , which exists as $H \leq G$. Observe that there exists a subgraph H' of G (consisting of the branch paths and the subgraphs induced by the branch sets) such that $\mathcal{P}$ is a partition of H' and $H'/\mathcal{P}$ is isomorphic to H .

Let E be the set of edges $uv \in E(H')$ such that u and v belong to different parts of $\mathcal{P}$, let E' be the set of edges $uv \in E(H')$ such that u and v belong to the same part of $\mathcal{P}$, and let $E'' := E(G) \setminus E(H')$. Define $\iota : V(H) \mapsto V(G)$ via setting, for each $x \in V(H)$,

$\iota(x) \in P_x$. Set $p := \max(|E'|, 1)$ and let d be the diameter of H . Note that $d < \infty$ since H is connected. Define w via

$$w(e) := \begin{cases} 1 & \text{if } v \in E, \\ \frac{\epsilon}{p} & \text{if } v \in E', \\ (1 + \epsilon)d + 1 & \text{otherwise.} \end{cases}$$

Observe that w maps to $\mathbb{R}^{>0}$ since $\epsilon \in \mathbb{R}^{>0}$. Further, note that if ϵ is rational, then w is rational-valued, since $p \in \mathbb{N}^{\geq 1}$ and $d \in \mathbb{N}$ (as H is unweighted). We show that w is the desired weighting via showing that $\text{dist}_H(u, v) \leq \text{dist}_{G'}(\iota(u), \iota(v)) \leq (1 + \epsilon) \text{dist}_H(u, v)$ for any $u, v \in V(H)$; the desired result then follows from Observation 5.25. This is clearly true when $u = v$, so assume that u and v are distinct. Set $\ell := \text{dist}_H(u, v)$ and $\ell' := \text{dist}_{G'}(\iota(u), \iota(v))$. Note that $\ell \geq 1$ as H is unweighted.

Let P be a shortest path from u to v in H . Observe that P corresponds to a path P^* in G' from $\iota(u)$ to $\iota(v)$ that uses exactly ℓ edges in E and no edges in E'' . Thus, the length of P^* is at most $\ell + |E'| \frac{\epsilon}{p} \leq \ell + p \frac{\epsilon}{p} \leq (1 + \epsilon)\ell$ as $\ell \geq 1$. Thus, $\ell' \leq (1 + \epsilon)\ell$. Noting that $\ell \leq d$ (as d is the diameter of H), we obtain $\ell' \leq (1 + \epsilon)d$. Therefore, if P' is a shortest path from $\iota(u)$ to $\iota(v)$ in G' , then P' does not use any edges in E'' , since these edges all have weight larger than ℓ' . Thus, P' corresponds to a path in H of length $|P' \cap E|$. Note that $|P' \cap E| \leq \ell'$, since each edge in E has weight 1 in G' . This gives $\ell \leq |P' \cap E| \leq \ell'$, completing the proof of the lemma. $\square$

We now construct a weighted graph G'' obtained from imbuing G with integer edge weights such that dilations that are control functions for G'' translate to dilations that are control functions for G' . This is just obtained by rescaling all the rational-valued edge weights by a (fixed) sufficiently large integer so that they become integer-valued. This works because the rescaling factor cancels out if the control function is a dilation (and not in general).

Lemma 5.27. *For every dilation f , there exists a dilation f' such that the following holds. Let G be a graph, and let H be the result of imbuing G with some rational-valued weighting function w . Then there exists a function $w' : E(G) \mapsto \mathbb{N}^{\geq 1}$ such that for any $n \in \mathbb{N}$, if f is an n -dimensional control function for the weighted graph G' that is the result of imbuing G with w' , then f' is an n -dimensional control function for H .*

Proof. Since f is a dilation, there exists $\alpha \in \mathbb{R}^{\geq 0}$ such that $f(r) \leq \alpha r$ for every $r \in \mathbb{R}^{\geq 0}$. Set $f'(r) := \alpha r$ for every $r \in \mathbb{R}^{\geq 0}$. Note that f' is a dilation. We claim that f' is the desired function.

Since w is rational-valued and $|E(G)| < \infty$, there exists $k \in \mathbb{N}^{\geq 1}$ such that the function $w' : E(G) \mapsto \mathbb{R}^{>0} \cup \{\infty\}$ defined by $w'(e) := kw(e)$ for each $e \in E(G)$ is integer-valued. Let G' be the result of imbuing G with w . Observe that for every $u, v \in V(H) = V(G')$, $k \operatorname{dist}_H(u, v) = \operatorname{dist}_{G'}(u, v)$. Thus, if f is an n -dimensional control function for G (for some $n \in \mathbb{N}$), then by Observation 5.25 the function $r \mapsto \frac{1}{k}f(kr)$ is an n -dimensional control function for H . Since $f(r) \leq \alpha r$ for every $r \in \mathbb{R}^{\geq 0}$, $\frac{1}{k}f(kr) \leq \frac{1}{k}\alpha kr = \alpha r = f'(r)$. Thus, f' is an n -dimensional control function for H , as desired. $\square$

The last step is to find a graph G''' in the subdivision-closed class whose control functions translate to control functions for G'' . This is achieved by subdividing G according to the weights of G'' . Before we proceed to the proof, we introduce one last piece of notation. Given a set $S \subseteq \mathbb{R}^{>0}$ and a class of unweighted graphs $\mathcal{G}$, let $\mathcal{G}_S$ denote the class of all weighted graphs G such that all edge weights of G are in S and $\operatorname{unweighted}(G) \in \mathcal{G}$.

Theorem 5.28. *Let $\mathcal{G}$ be a class of unweighted graphs that does not exclude any unweighted graph as a minor. Then:*

- (a) $\operatorname{asdim}(\mathcal{G}_{\mathbb{R}^{>0}}) = \infty$,
- (b) $\operatorname{ANdim}(\mathcal{G}_{\mathbb{N}^{\geq 1}}) = \infty$, and
- (c) if $\mathcal{G}$ is closed under subdivision, then $\operatorname{ANdim}(\mathcal{G}) = \infty$.

Proof. Let $\mathcal{H}$ be any class of connected unweighted graphs with infinite asymptotic (and Assouad–Nagata) dimension, such as the class of all grids of all dimensions [80].

Fix any $n \in \mathbb{N}$ and any function $f : \mathbb{R}^{\geq 0} \mapsto \mathbb{R}^{\geq 0}$. We show that if f is an n -dimensional control function for $\mathcal{G}_{\mathbb{R}^{>0}}$, then there is an n -dimensional control function for $\mathcal{H}$, a contradiction by the choice of $\mathcal{H}$. Theorem 5.28(a) follows. Additionally, we show that if f is a dilation and an n -dimensional control function for $\mathcal{G}_{\mathbb{N}^{\geq 1}}$, then we again have that there is an n -dimensional control function for $\mathcal{H}$, another contradiction. Theorem 5.28(b) follows. Finally, we show that if $\mathcal{G}$ is closed under subdivision, and f is a dilation and an n -dimensional control function for $\mathcal{G}$, then f is a dilation that is an n -dimensional control function for $\mathcal{G}_{\mathbb{N}^{\geq 1}}$, contradicting Theorem 5.28(b). Theorem 5.28(c) follows.

We start with the first claim. Presume that f is an n -dimensional control function for $\mathcal{G}_{\mathbb{R}^{>0}}$, and fix a rational number $\epsilon > 0$. Now, for every $H \in \mathcal{H}$, there exists $G \in \mathcal{G}$ that contains H as minor. Noting that H is connected, by Lemma 5.26, there exists a rational-valued function $w : E(G) \mapsto \mathbb{R}^{>0}$ such that if f is an n -dimensional control function for the weighted graph G' obtained by imbuing G with w , then the function $r \mapsto f((1+\epsilon)r)$ is an n -dimensional control function for H . Observe that $G' \in \mathcal{G}_{\mathbb{R}^{>0}}$, thus f is an n -dimensional function for G' and $r \mapsto f((1+\epsilon)r)$ is an n -dimensional control

function for $\mathcal{H}$. It follows that $r \mapsto f((1 + \epsilon)r)$ is an n -dimensional control function for $\mathcal{H}$, giving the desired contradiction. This completes the proof of Theorem 5.28(a).

Next, we tackle the second claim. Presume that f is a dilation that is an n -dimensional control function for $\mathcal{G}_{\mathbb{N} \geq 1}$. Since f is a dilation, we can apply Lemma 5.27; let f' be the function we obtain. For every $H \in \mathcal{H}$, let G , G' , and w be defined as in the previous paragraph. Since G' is the result of imbuing G with the rational-valued weighting function w , by Lemma 5.27 (with $H := G'$), there exists a function $w' : E(G) \mapsto \mathbb{N}^{\geq 1}$ such that if f is an n -dimensional control function for the weighted graph G'' obtained from imbuing G with w' , then f' is an n -dimensional control function for G' . Observe that $G'' \in \mathcal{G}_{\mathbb{N} \geq 1}$, thus f is an n -dimensional control function for G'' , f' is an n -dimensional control function for G' , and $r \mapsto f'((1 + \epsilon)r)$ is an n -dimensional control function for H (by Lemma 5.26). It follows that $r \mapsto f'((1 + \epsilon)r)$ is an n -dimensional control function for $\mathcal{H}$, giving the desired contradiction. This completes the proof of Theorem 5.28(b).

We now proceed to the final claim. Presume that $\mathcal{G}$ is closed under subdivision, and that f is a dilation that is an n -dimensional control function for $\mathcal{G}$. For each $G \in \mathcal{G}_{\mathbb{N} \geq 1}$, let w be the weighting of G , and let G' be the graph obtained from $\text{unweighted}(G)$ by subdividing each $e \in E(\text{unweighted}(G)) = E(G)$ exactly $w(e) - 1$ times. Since $\mathcal{G}$ is closed under subdivision, observe that $G' \in \mathcal{G}$, and thus f is an n -dimensional control function for G' . Since G' is a subdivision of $\text{unweighted}(G)$ where each $e \in E(G)$ has been split into a path of length exactly $w(e)$, observe that G is isometric in G' . Thus, f is also an n -dimensional control function for G . It follows that f is an n -dimensional control function for $\mathcal{G}_{\mathbb{N}}$, giving the desired contradiction. This completes the proof of Theorem 5.28(c), and the overall proof of Theorem 5.28. $\square$

Theorem 2.19 (the Assouad–Nagata dimension of subdivision-closed classes of unweighted graphs) follows directly from Theorem 5.24 and Theorem 5.28(c).

Chapter 6

Blowups over Tree-Decompositions

This chapter proves Theorem 2.11, which says that each n -vertex K_h -minor-free graph is a $O_h(\sqrt{n} \log(n)^2)$ -blowup of a fan. We also start the proof of Theorem 2.9, which says that these graphs are a $O_h(\sqrt{n})$ -blowup of a treewidth 3 graph. We complete the proof of Theorem 2.9 in Chapter 7.

6.1 Almost Partitions

Recall that Distel et al. [45] showed $\text{tpw}_4(G) \in O_h(\sqrt{n})$ for each n -vertex K_h -minor-free graph G . In fact, they showed something stronger.

Theorem 6.1. *Let $h, n \in \mathbb{N}$. Then each n -vertex K_h -minor-free graph G is a $O_h(\sqrt{n})$ -blowup of a graph H which contains a vertex v such that $\text{tw}(H - v) \leq 3$.*

This implies $\text{tpw}_4(G) \in O_h(\sqrt{n})$, since adding a vertex increases the treewidth by at most 1.

The existence of this vertex v is natural and required for large h . One can take any arbitrary $(n - 1)$ -vertex K_{h-1} -free graph, and add a single vertex x adjacent to every other vertex to obtain an n -vertex K_h -minor-free graph G' of radius 1. Since $\text{tpw}_4(G) \in O_h(\sqrt{n})$, G' (and thus G) is a $O_h(\sqrt{n})$ -blowup of a graph H of treewidth 4. However, up to deleting empty parts, this graph H must also have radius 1; whichever part P that contains x must be adjacent to every other nonempty part in H . Deleting P then decreases the treewidth of H by 1.

To help capture this detail, we introduce some new terminology. Say that the *apex-treewidth* of a graph G , denoted $\text{atw}(G)$, is the minimum treewidth of $G - v$ across

all $v \in V(G)$. Note that $\text{tw}(G) - 1 \leq \text{atw}(G) \leq \text{tw}(G)$. For $k \in \mathbb{N}$, say that the *apex- k -tree-partition-width*, denoted $\text{atpw}_k(G)$, is the minimum $c \in \mathbb{N}$ such that G is a c -blowup of a graph of apex-treewidth at most k .

Theorem 6.1 can be rephrased as ‘ $\text{atpw}_3(G) \in O_h(\sqrt{n})$ ’. This thesis shows the following.

Theorem 6.2. *For each $h, n \in \mathbb{N}$ and each n -vertex K_h -minor-free graph G , $\text{atpw}_2(G) \in O_h(\sqrt{n})$.*

Theorem 6.2 directly implies Theorem 2.9, which says that n -vertex K_h -minor-free graphs are $O_h(\sqrt{n})$ -blowups of treewidth 3 graphs.

For the purposes of a proof, it is easier to use partitions to describe blowups. To capture the idea that there is a vertex of the base graph that will be deleted, we introduce the following variant of a partition.

An *almost-partition* of a graph G is a pair $(X, \mathcal{P})$, where $X \subseteq V(G)$ and $\mathcal{P}$ is a partition of $G - X$. We call X the *loss set*. It is immediate that $(X) \sqcup \mathcal{P}$ is a partition of G . So an almost-partition is just a special type of partition. However, we favour this notation, because it better allows us to isolate and focus on key properties.

Say that the *width* and *treewidth* of $(X, \mathcal{P})$ are the width and treewidth of $\mathcal{P}$ (as a partition of $G - X$) respectively, and that the *loss* of $(X, \mathcal{P})$ is $|X|$. The *quotient* of G by $(X, \mathcal{P})$ is $(G - X)/\mathcal{P}$.

So if $(X, \mathcal{P})$ has treewidth at most k , then $G/\mathcal{P}$ has apex-treewidth at most k . Also, if $(X, \mathcal{P})$ has width and loss at most w , then $(X) \sqcup \mathcal{P}$ has width at most w . However, we prefer to decouple these parameters, because we typically want the loss to be substantially smaller than the width. This is because, at a later stage, we will merge multiple almost-partitions together by taking the (disjoint) union of the partitions (which does not increase the width) and the union of the loss sets (which does increase the loss).

Observe that a graph G is a b -blowup of a graph of apex-treewidth at most k if and only if G admits an almost-partition of width and loss at most b and treewidth at most k . Thus, Theorems 6.1 and 6.2 can be restated as ‘Every n -vertex K_h -minor-free graph G admits an almost-partition of width and loss at most $O_h(\sqrt{n})$ and treewidth at most $3/2$ ’ respectively. We actually show a slightly stronger result.

Theorem 6.3. *For each $h \in \mathbb{N}$, there exists $f \in O(n)$ and $g \in O(\sqrt{n})$ such that for each $n \in \mathbb{N}$, and every $d \in \mathbb{R}^{>0}$ with $d \geq g(n)$, each n -vertex K_h -minor-free graph G admits an almost-partition of treewidth at most 2, width at most d , and loss at most $f(n)/d$.*

Picking $d := \sqrt{n}$ (or $d := 1$ if $n = 0$) then implies Theorem 6.2. So we focus on proving Theorem 6.3.

We have a similar situation occurring with Theorem 2.11 (which says that n -vertex K_h -minor-free graphs are $O_h(\sqrt{n} \log(n)^2)$ -blowups of fans), in that after deleting a centre of a fan, we obtain a path. This leads to the following definition.

Say that an almost-partition $(X, \mathcal{P})$ of a graph G is an *almost-layering* if $(G - X)/\mathcal{P}$ is contained in an path (equivalently, $\mathcal{P}$ is a layering, as a partition of $G - X$). We can view a partition whose quotient is contained in a fan as an almost-layering. Thus, a graph is a b -blowup of a fan if and only if it admits an almost-layering of width and loss at most b . So Theorem 2.11 can be restated as ‘Every n -vertex K_h -minor-free graph G admits an almost-layering of width and loss at most $O_h(\sqrt{n} \log(n)^2)$ ’. We again show a slightly stronger result.

Theorem 6.4. *For each $h \in \mathbb{N}$, there exists $f \in O(n \log(n))$ and $g \in O(\log(n)^3)$ such that for each $n \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, every n -vertex K_h -minor-free graph admits an almost-layering of width at most $dg(n)$ and loss at most $f(n)/d$.*

Theorem 2.11 follows from Theorem 6.4 with $d := \sqrt{n}/\log(n)$ (or $d := 1$ if $n \leq 1$).

Theorem 6.3 and Theorem 6.4 (and Theorem 6.1 [45]) have a large overlap in their proof strategy. To avoid repeating the same proof, we identify a common structure.

For a class of graphs $\mathcal{H}$, say that an almost-partition $(X, \mathcal{P})$ of a graph G is an *almost- $\mathcal{H}$ -partition* if $(G - X)/\mathcal{P}$ is contained in some $H \in \mathcal{H}$.

If $\mathcal{H}$ is the class of graphs of treewidth at most $k \in \mathbb{N}$, then the almost-partitions of treewidth at most k are exactly the almost- $\mathcal{H}$ -partitions. If $\mathcal{H}$ is the class of graphs whose components are paths, then almost-layerings are exactly the almost- $\mathcal{H}$ -partitions. (This is true if we take $\mathcal{H}$ to be the class of paths, but for reasons we will see later, it is important that $\mathcal{H}$ is closed under disjoint unions.) So this notation allows us to describe both structures we are interested in.

6.2 Carving Tree-Decompositions

Theorems 6.3 and 6.4 (and Theorem 6.1 [45]) have the same starting point. Starting from a tree-decomposition $(J_t : t \in V(T))$ from the Graph Minor Structure Theorem (Theorem 1.2 [121]), we attempt to carve up the tree-decomposition into manageable chunks by deleting only $O(\sqrt{n})$ vertices. Specifically, each chunk consists of a torso and many small sections disjointedly hanging off the torso.

We find a small set $Z \subseteq V(T)$ such that for each component of T' of $T - Z$, the total number of vertices in bags indexed by $V(T')$ is small. See Lemma 6.5. We then consider T to be rooted at some $r \in V(T)$, and delete the parent adhesion sets of $z \in Z$. Since $(J_t : t \in V(T))$ has bounded adhesion, we delete only a small number of vertices. These vertices X will go into the loss set.

We find that, at worst, each component of $G - X$ is the torso at some $z \in Z \cup \{r\}$ plus small sections disjointedly hanging off this torso, which correspond to the subtrees T' of $T - Z$ whose parent (vertex closest to r adjacent to a vertex in $V(T')$) is z . See Figure 6.1 in the proof of Lemma 6.6. We can capture this structure as a star-decomposition (of bounded adhesion), where the torso of the centre is contained in one of our original torsos, and all other bags are small ($O(\sqrt{n})$).

If our graph has n vertices and a given component has N vertices, we set $d := \sqrt{n}$, and ask that the component admits an almost- $\mathcal{H}$ -partition of width at most d and loss at most N/d . Achieving this is the main challenge, and it is here that these three proofs deviate. So we skip this step for now, and discuss how to finish the proof. Handling this step will be the subject of Sections 6.3 and 6.4.

We combine the loss sets of the almost-partitions of the components, along with the vertices we deleted, into the loss set X' for our graph. Since the loss sets for the components had size $N/\sqrt{n}$ (rather than $\sqrt{n}$) and since the components are disjoint, the total loss is still $O(\sqrt{n})$. The partition of $G - X'$ is just the disjoint union of the partitions of the components. Assuming that $\mathcal{H}$ is closed under disjoint union (which is the case for our purposes), the resulting almost-partition is an almost- $\mathcal{H}$ -partition of width $d = \sqrt{n}$, as desired.

We now give the proofs for these two steps. To find the set Z , we make use of the following technical lemma by Distel et al. [45].

Lemma 6.5. *For every $q \in \mathbb{N}$ and $W \in \mathbb{R}^{\geq 0}$, every vertex-weighted tree T with weight at most W has a set Z of at most q vertices such that each component of $T - Z$ has total weight at most $\frac{W}{q+1}$.*

Here, a *vertex-weighting* of a graph G is a map $w : V(G) \mapsto \mathbb{R}^{\geq 0}$, and the *total weight* of subgraph H [respectively $S \subseteq V(G)$] is $\sum_{v \in V(H)} w(v)$ [respectively $\sum_{v \in S} w(v)$].

We first prove the carving up step. Here, we aim to not just preserve the torsos, but also the adhesion sets. This will be relevant for the proof of Theorem 6.3. We also need to view the components as being contained in a larger partition. This is because almost-embeddings do not behave nicely under taking subgraphs, so we need to mirror the structure of the original torsos as closely as possible.

Lemma 6.6. *Let $n, j \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, let G be an n -vertex graph, let $(J_t : t \in V(T))$ be a tree-decomposition of G of adhesion at most j , and let $r \in V(T)$ be a root for T . Then there exists a partition $\mathcal{P}$ of G and $X \subseteq V(G)$ with $|X| \leq jn/d$ such that:*

1. *for each connected component C of $G - X$, $V(C)$ intersects exactly one part of $\mathcal{P}$*
2. *for each $P \in \mathcal{P}$, there exists $r_P \in V(T)$ with parent-adhesion set K_P and a star-decomposition $(B_s^P : s \in V(S_P))$ of $G[P]$ such that:*
 - (a) *$(B_s^P : s \in V(S_P))$ has adhesion at most j ,*
 - (b) *$r_P \in V(S_P)$, is a centre of S_P , and $B_{r_P}^P = J_{r_P} \setminus K_P$,*
 - (c) *$(G[P]) \langle B_{r_P}^P \rangle$ is a spanning subgraph of $G \langle J_{r_P} \rangle - K_P$, and*
 - (d) *for each $s \in V(S_P) \setminus \{r_P\}$:*
 - i. *s is a child of r_P in T ,*
 - ii. *$|B_s^P \setminus B_{r_P}^P| \leq d$, and*
 - iii. *$B_s^P \cap B_{r_P}^P = J_s \cap J_{r_P} \setminus K_P$.*

Proof. For each $t \in V(T)$, let K_t be the parent adhesion set at t . Note that $|K_t| \leq j$.

Weight each $t \in V(T)$ by $|J_t \setminus K_t|$. Observe that $(J_t \setminus K_t : t \in V(T))$ is a partition of G . Thus, the total vertex-weight of T is exactly n . By Lemma 6.5 with $q := \lfloor n/d \rfloor$, there is a set Z with $|Z| \leq q \leq n/d$ such that the total weight of each connected component of $T - Z$ is at most $n/(q+1) \leq d$ (as $n/d \leq q+1$).

Let $X := \bigcup_{z \in Z} K_z$. Note that $|X| \leq j|Z| \leq jn/d$. Let F be the forest obtained from T by, for each $z \in Z \setminus \{r\}$, deleting the edge $p(z)z$. See Figure 6.1.

Let $\mathcal{T}$ be the connected components of F . For each $Q \in \mathcal{T}$, let $r_Q \in V(Q)$ be the (unique) vertex closest to r , and let $V_Q := \bigcup_{t \in V(Q)} J_t \setminus K_t = \bigcup_{t \in V(Q)} J_t \setminus K_{r_Q}$. Note that $r_Q \in Z \cup \{r\}$, and thus $K_{r_Q} \subseteq X$ (as $K_r = \emptyset$).

Noting that $\bigcup_{Q \in \mathcal{T}} V(Q) = V(T)$ and since $(J_t \setminus K_t : t \in V(T))$ is a partition of G , it follows that $\mathcal{P} := (V_Q : Q \in \mathcal{T})$ is a partition of G . For each $P \in \mathcal{P}$, let Q_P be such that $P = V_{Q_P}$ (which must exist), and let $r_P := r_{Q_P}$. Thus, $K_P = K_{r_P}$.

Let C be a connected component of $G - X$, and let T_C be the (nonempty) subtree of T induced by the vertices $t \in V(T)$ such that $J_t \cap V(C) \neq \emptyset$. Since $K_z \subseteq X$ for each $z \in Z \setminus \{r\}$, observe that $T_C \subseteq F$. Hence, $T_C \subseteq Q$ for some $Q \in \mathcal{T}$, and $V(C) \subseteq V_Q \cup K_{r_Q}$. Since $K_{r_Q} \subseteq X$, it follows that $V(C) \subseteq V_Q$. Since $\mathcal{P}$ is a partition of G (and since C is nonempty), it follows that $V(C)$ intersects exactly one part of $\mathcal{P}$.

Fix $P \in \mathcal{P}$, and set $Q := Q_P$. So $P = V_Q$, $r_P = r_Q$, and $K_P = K_{r_Q}$. Let $\mathcal{T}_P$ denote the set of connected components of $Q - r_Q$. Note that $Q \cap (Z \cup \{r\}) = r_Q$, and thus each

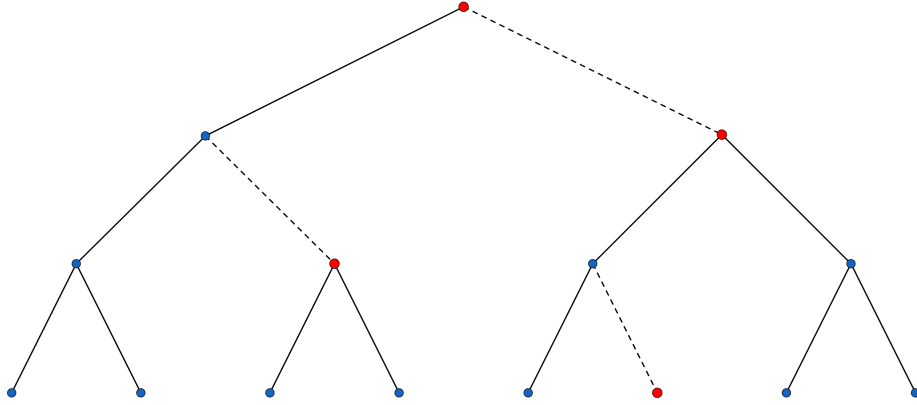


Figure 6.1: A diagram of the tree T , with the vertices in $Z \cup \{r\}$ coloured red and the edges removed to make F represented as dashed lines. The parts P of $\mathcal{P}$ correspond to the components Q of F , and the leaves of the star S_P correspond to the components Y of Q after the unique (red) vertex in $Z \cup \{r\}$ is deleted.

$Y \in \mathcal{T}_P$ is a subtree of $T - Z$.

For each $Y \in \mathcal{T}_P$, let $r_Y \in V(Y)$ be the (unique) vertex closest to r . Note that r_Y is a child of r_Q . Let $V_Y := \bigcup_{t \in V(Y)} J_t \setminus K_{r_Q}$. Note that $V_Y \subseteq V_Q$.

Let c_Q denote the children of r_Q not in Z . Observe that the map $Y \mapsto r_Y$ is a bijection between $\mathcal{T}_P$ and c_Q . For each $s \in c_Q$, let $Y_s \in \mathcal{T}_P$ be such that $r_{Y_s} = s$ (which exists, and is uniquely defined).

Let S_P be the star with centre $r_P = r_Q$ and leaves c_Q . Let $B_{r_P}^P := J_{r_Q} \setminus K_{r_Q} = J_{r_P} \setminus K_P$, and for each $s \in c_Q$, let $B_s^P := V_{Y_s}$. Observe that $(B_s^P : s \in V(S_P))$ is a star-decomposition of $G[V_Q] = G[P]$.

For each $s \in c_Q = V(S) \setminus \{r_P\}$, observe that $B_s^P \cap B_{r_P}^P = J_s \cap J_{r_Q} \setminus K_{r_Q} = J_s \cap J_{r_P} \setminus K_P$. In particular, since $(J_t : t \in V(T))$ has adhesion at most j , so does $(B_s^P : s \in V(S_P))$. Note also that $J_s \setminus J_{r_P} = \bigcup_{t \in V(Y_s)} J_t \setminus K_t$ (as r_P is the parent of r_{Y_s}). Thus, $|J_s \setminus J_{r_P}| \leq \sum_{t \in V(T)} |J_t \setminus K_t|$, which is the total weight on Y_s . Since $Y_s \subseteq T - Z$, the total weight on Y_s is at most d . Hence, $|J_s \setminus J_{r_P}| \leq d$.

It remains only to show that $(G[P]) \langle B_{r_P}^P \rangle$ is a spanning subgraph $G \langle J_{r_P} \rangle - K_P$. Note that $V((G[P])[B_{r_P}^P]) = J_{r_Q} \setminus K_{r_Q} = V(G \langle J_{r_P} \rangle - K_P)$. Thus, $(G[P])[B_{r_P}^P] = G \langle J_{r_P} \rangle - K_P$.

Now, let $uv \in E((G[P]) \langle B_{r_P}^P \rangle) \setminus E((G[P])[B_{r_P}^P])$. So there exists $s \in c_Q$ such that $u, v \in B_s^P \cap B_{r_P}^P = J_s \cap J_{r_P} \setminus K_P$. As s is a child of r_P , $sr_P \in E(T)$. Hence, $uv \in G \langle J_{r_P} \rangle - K_P$. Since $G \langle J_{r_P} \rangle - K_P$ is a spanning subgraph of $G \langle J_{r_P} \rangle - K_P$, it

follows that $(G[P])\langle B_{r_P}^P \rangle$ is a spanning subgraph of $G\langle J_{r_P} \rangle - K_P$, as desired. (We remark, without proof, that equality is achieved whenever r_P has no children in Z .) This completes the proof. $\square$

We now prove the recombining step, to be applied once each component of $G - X$ has been processed individually. As a guide to the reader, the following lemma will be applied with $f(n) \in O(\log(n)/d)$ (so that the loss is $O(n \log(n)/d)$) and $g(n) \in O(d \log(n)^3)$ to prove Theorem 6.4, and $f(n) \in O(1/d)$ (so that the loss is $O(n/d)$) and $g(n) \in O(d)$ to prove Theorem 6.3. (We bound the loss using the function $nf(n)$, rather than the function $f'(n) := nf(n)$, as $nf(n)$ is superadditive whenever f is increasing. We prefer this to adding the condition that f' is superadditive.)

Lemma 6.7. *Let $n \in \mathbb{N}$ and $q \in \mathbb{R}^{\geq 0}$, let $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ be non-decreasing functions, let $\mathcal{H}$ be a class of graphs closed under disjoint union, and let G be an n -vertex graph. If there exists a set $X' \subseteq V(G)$ and a partition $\mathcal{P}'$ of G such that:*

1. $|X'| \leq q$,
2. *each component of $G - X'$ intersects exactly one part of $\mathcal{P}'$, and*
3. *for each $P \in \mathcal{P}'$, $G[P]$ admits an almost- $\mathcal{H}$ -partition of width at most $g(|P|)$ and loss at most $|P|f(|P|)$,*

then G admits an almost- $\mathcal{H}$ -partition $(X, \mathcal{P})$ of width at most $g(n)$ and loss at most $q + nf(n)$.

Proof. For each $P \in \mathcal{P}'$, let $n_P := |P|$. By assumption, there exists a triple $(H_P, X_P, \mathcal{P}_P)$ such that $(X_P, \mathcal{P}_P)$ is an almost-partition of $G[P]$ of width at most $g(n_P)$ and loss at most $n_P f(n_P)$, and $H_P \in \mathcal{H}_P$ contains $(G[P] - X_P)/\mathcal{P}_P$. Let $\mathcal{P}'_P := (P' \setminus X' : P' \in \mathcal{P}_P)$, and let $X'_P := X_P \cup (X' \cap P)$. Observe that $(X'_P, \mathcal{P}'_P)$ is an almost-partition of $G[P]$ of width at most $g(n_P)$ and whose quotient is contained in H_P .

Let $X := X' \cup \bigcup_{P \in \mathcal{P}'} X_P$. Note that for each $P \in \mathcal{P}'$, $P \cap X = X'_P$. Let $\mathcal{P} := \bigcup_{P \in \mathcal{P}'} \mathcal{P}'_P$. Since $\mathcal{P}'$ is a partition of G , and since $(X'_P, \mathcal{P}'_P)$ is an almost-partition of $G[P]$ of width at most $g(n_P)$, it follows that $(X, \mathcal{P})$ is an almost-partition of G of width at most $g(n)$. Here, we use that g is non-decreasing.

Let $P_1, P_2 \in \mathcal{P}$ be distinct and adjacent in G . Observe that there exists (unique) $P'_1, P'_2 \in \mathcal{P}'$ such that $P_1 \subseteq P'_1$ and $P_2 \subseteq P'_2$. Since each connected component of $G - X$ intersects at most one part of $\mathcal{P}'$, observe that $P'_1 = P'_2$. Setting $Q := P'_1 = P'_2$, it follows that $P_1, P_2 \in \mathcal{P}'_Q$. Note also that P'_1, P'_2 are adjacent in $(G[Q] - X'_Q)/\mathcal{P}'_Q$. It follows that $(G - X)/\mathcal{P}$ is a subgraph of $\bigsqcup_{Q \in \mathcal{P}'} (G[Q] - X'_Q)/\mathcal{P}'_Q$.

Let $H := \bigsqcup_{P \in \mathcal{P}'} H_P$. Note that $H \in \mathcal{H}$, as $\mathcal{H}$ is closed under disjoint union. Further, observe that $\bigsqcup_{Q \in \mathcal{P}'} (G[Q] - X'_Q)/\mathcal{P}'_Q$ is contained in H . Thus, $(G - X)/\mathcal{P}$ is also contained in H . Since $H \in \mathcal{H}$, this means that $(X, \mathcal{P})$ is an almost- $\mathcal{H}$ -partition of G .

Since f is non-decreasing (and since $n_Q \geq 0$ for each $Q \in \mathcal{P}'$), observe that $\sum_{Q \in \mathcal{P}'} n_Q f(n_Q) \leq (\sum_{Q \in \mathcal{P}'} n_Q) f(\sum_{Q \in \mathcal{P}'} n_Q)$. Noting that $\sum_{Q \in \mathcal{P}'} n_Q = n$ (as $\mathcal{P}'$ is a partition of G), this gives $\sum_{Q \in \mathcal{P}'} n_Q f(n_Q) \leq n f(n)$.

For each $Q \in \mathcal{P}'$, recall that $|X_Q| \leq n_Q f(n_Q)$. Thus, $|X| \leq |X'| + \sum_{Q \in \mathcal{P}'} |X_Q| \leq q + \sum_{Q \in \mathcal{P}'} n_Q f(n_Q) \leq q + n f(n)$. So $(X, \mathcal{P})$ has loss at most $q + n f(n)$. This completes the proof. $\square$

6.3 Fan Partitions

The next two sections are devoted to handling the components of $G - X$, completing the proof. Here, the proofs of Theorems 6.3 and 6.4 (and Theorem 6.1 [45]) diverge. This section is devoted to the proof of Theorem 6.4, which is the easier of the two proofs. Section 6.4 (and Chapter 7) is devoted to the proof of Theorem 6.3.

The proof of Theorem 6.4 makes use of two key facts: (1) we know how to find good almost-layerings of graphs of bounded row-treewidth (Theorem 2.12 [59]), and (2) layerings of a graph give rise to layerings of the graph powers (Lemma 6.10).

We utilise Theorem 2.7 [58], which allows us to describe the torsos in the Graph Minor Structure Theorem as only graph of bounded row-treewidth plus a bounded number of extra vertices (the apices). By putting the apices in the loss-set, Theorem 2.12 gives us a way of finding almost-layerings of small width and loss. However, recall that we generally need the loss to be much smaller than the width. Thankfully, Dujmović et al. [59] showed the following strengthening of Theorem 2.12.

Theorem 6.8. *For each $b \in \mathbb{N}$, there exists $f \in O(n \log(n))$ and $g \in O(\log(n)^3)$ such that for every $n \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, every n -vertex graph of row-treewidth at most b admits an almost-layering of width at most $d g(n)$ and loss at most $f(n)/d$.*

By adding the apices to the loss set, we can handle the torsos on their own. However, we still need to attach the vertices in the smaller bags B_s , $s \neq r$ (where r is the centre of the star-decomposition). For this, we create an auxiliary graph G' , obtained from the central torso by attaching degree 1 vertices. Specifically, for each $s \neq r$, each vertex v in $B_s \setminus B_r$, and each vertex x in $B_s \cap B_r$, we add an auxiliary vertex v_x adjacent to x . See Figure 6.2 in the proof of Lemma 6.11. We can delete the same apex vertices

from the resulting graph to get a graph of the same row-treewidth (see Lemma 6.9). Thus, we can still process G' using Theorem 6.8. By adding the apices to the loss set, we obtain an almost-layering $(X', \mathcal{P}')$ of G' .

For each B_s , $s \neq r$, one of two things must happen. One possibility is that $B_s \cap B_r \subseteq X'$, in which case $B_s \setminus B_r$ can exist as an isolated part in the quotient. The second possibility is that there exists some $x \in (B_s \cap B_r) \setminus X'$. In this case, for each vertex $v \in B_s \setminus B_r$, one of two outcomes must have occurred. Either $v_x \in X'$, in which case v is added to our loss-set, or v_x is at distance at most 2 in $G' - X'$ of every vertex in $(B_s \cap B_r) \setminus X'$ and every vertex u_x with $u \in B_s \setminus B_r$ and $u_x \notin X'$. In particular, these vertices become adjacent in $(G' - X')^2$. In this case, we place v in the part of v_x .

By merging pairs of consecutive layers, we can convert our layering of $G' - X'$ to a layering of $(G' - X')^2$ of similar width (at most double). See Lemma 6.10. Converting this almost-layering to an almost-layering of G as above, we find that this almost-partition is an almost-layering, since all possible adjacencies were already handled by $(G' - X')^2$.

We now formalise these ideas.

Lemma 6.9. *Let $b \in \mathbb{N}^{\geq 1}$, and let G be a graph for which there exists a set $Z \subseteq V(G)$ such that:*

1. *each $z \in Z$ has degree at most 1, and*
2. *$G - Z$ has row-treewidth at most b .*

Then G has row-treewidth at most b .

Proof. Since $G - Z$ has row-treewidth at most b , there exists a graph H of treewidth at most b and a path P such that $G - Z$ is contained in $H \boxtimes P$. So there exists an isomorphism ψ from G to a subgraph of $H \boxtimes P$. Further, we may assume without loss of generality that $V(H)$ is disjoint to $V(Z)$. For each $v \in V(G - Z)$, we have $\psi(v) = (h, p) \in V(H \boxtimes P)$; let $\psi_H(v) := h$ and $\psi_P(v) := p$.

Let E_z be the set of edges of G with an endpoint in Z , and let H' be the graph with vertex set $V(H) \cup Z$ and edge set $E(H) \cup \{\psi_H(v)z : vz \in E_z\}$. Observe that H' is obtained from H by adding some number of degree 1 vertices. Thus, $\text{tw}(H') \leq \max(\text{tw}(H), 1) \leq b$. Next, notice that for each $vz \in E_z$, we have $\psi_H(v)z \in E(H')$ and thus $(\psi_H(v), \psi_P(v))(z, \psi_P(v)) \in E(H' \boxtimes P)$. Therefore, we find that G is isomorphic to a subgraph of $H' \boxtimes P$, via the mapping $v \mapsto \psi(v) = (\psi_H(v), \psi_P(v))$ for $v \in V(G - Z)$, and, for each $z \in Z$, $z \mapsto (z, \psi_P(v))$ for the unique $v \in V(G - Z)$ such that $vz \in E_z$. This gives the desired result. $\square$

Lemma 6.10. *For each $d \in \mathbb{N}$ and every $w \in \mathbb{R}^{\geq 0}$, if a graph G admits a layering of width at most w , then G^d admits a layering of width at most wd .*

Proof. Let $\mathcal{P} = \{P_1, P_2, \dots, P_m\}$ be a layering of G of width at most w , with the layers ordered by their position on the path. Set $P_i := \emptyset$ for each $i \in \mathbb{Z} \setminus \{1, \dots, m\}$. Now, for each $i \in \{0, \dots, \lceil \frac{m}{d} \rceil - 1\}$, set $P'_i := \bigcup_{j=1}^d P_{id+j}$, and then let $\mathcal{P}' := (P'_i : i \in \{0, \dots, \lceil \frac{m}{d} \rceil - 1\})$. Observe that for each $k \in \{1, \dots, m\}$, $k \in \{id + 1, \dots, (i+1)d\}$ for exactly one $i \in \{0, \dots, \lceil \frac{m}{d} \rceil - 1\}$. It follows that $\mathcal{P}'$ is a partition of G (and G^d). As each $P' \in \mathcal{P}'$ is a union of at most d parts in $\mathcal{P}$, it is immediate that $\mathcal{P}'$ has width at most wd . So it remains only to show that $\mathcal{P}'$ is a layering of G^d .

As $\mathcal{P}$ is a layering, it follows that for each $vw \in E(G^d)$, if $k, j \in \{1, \dots, m\}$ are such that $v \in P_k$ and $w \in P_j$, then $j \in \{k - d, \dots, k + d\}$. Thus, if $i \in \{0, \dots, \lceil \frac{m}{d} \rceil - 1\}$ is such that $v \in P'_i$, then $k \in \{id + 1, \dots, (i+1)d\}$, and $j \in \{(i-1)d + 1, \dots, (i+2)d\}$. But $\bigcup_{\ell=(i-1)d+1}^{(i+2)d} P_\ell \subseteq P'_{i-1} \cup P'_i \cup P'_{i+1}$, so $w \in P'_{i-1} \cup P'_i \cup P'_{i+1}$. It follows that $\mathcal{P}'$ is a layering for G^d , as desired. $\square$

Lemma 6.11. *Let $a, n \in \mathbb{N}$, $b, j \in \mathbb{N}^{\geq 1}$, and $w, w', q \in \mathbb{R}^{\geq 0}$ be such that each graph of row-treewidth at most b and with at most jn vertices admits an almost-layering of width at most w and loss at most q . Let G be an n -vertex graph that admits a star-decomposition $(B_s : s \in V(S))$ of adhesion at most j with centre r such that:*

1. *there exists $A \subseteq B_r$ with $|A| \leq a$ such that $G\langle B_r \rangle - A$ has row-treewidth at most b , and*
2. *for each $s \in V(S) \setminus \{r\}$, $|B_s \setminus B_r| \leq w'$.*

Then G admits an almost-layering of width at most $\max(2w, w')$ and loss at most $q + a$.

Proof. For each $s \in V(S) \setminus \{r\}$, let $K_s := B_s \cap B_r \setminus A$ and $B'_s := B_s \setminus B_r$. Note that $|K_s| \leq j$ and $|B'_s| \leq w'$. Observe also that B_s is disjoint and non-adjacent in G to $B_{s'}$ whenever $s, s' \in V(S) \setminus \{r\}$ are distinct.

Let G' be the graph obtained from $G\langle B_r \rangle - A$ by adding, for each $s \in V(S) \setminus \{r\}$, each $v \in B'_s$, and each $x \in K_s$, a new vertex v_x adjacent only to x . See Figure 6.2. Observe that $|V(G')| \leq |B_r| + j|V(G) \setminus B_r| \leq j|V(G)| = jn$ (since $j \geq 1$). By Lemma 6.9 (since $b \geq 1$), G' has row-treewidth at most b . Thus, by choice of w and q , G' admits an almost-layering $(X', \mathcal{P}'')$ of width at most w and loss at most q . By Lemma 6.10, $(G' - X')^2$ admits a layering $\mathcal{P}'$ of width at most $2w$. Note that $(X', \mathcal{P}')$ is also an almost-layering of G' (of width at most $2w$ and loss at most q).

For each $v \in V(G) \setminus B_r$, there exists exactly one $s \in V(S) \setminus \{r\}$ such that $v \in B'_s$; let $M_v := \{v_x : x \in K_s\}$. Let $X_1 := X' \cap B_r$, let $X_2 := \{v \in V(G) \setminus B_r : M_v \cap X' \neq \emptyset\}$,

and let $X := X_1 \cup X_2 \cup A$. Observe that $|X_2| \leq |X' \setminus B_r|$, and thus $|X_1 \cup X_2| \leq |X'|$. Hence, $|X| \leq |X'| + |A| \leq q + a$. Let $U := \{s \in V(S) \setminus \{r\} : K_s \setminus X' = \emptyset\}$ and $W := \{s \in V(S) \setminus \{r\} : K_s \setminus X' \neq \emptyset\}$. Let $V_U := (\bigcup_{s \in U} B'_s) \setminus X$ and $V_W := (\bigcup_{s \in W} B'_s) \setminus X$. For each $s \in U$, define $P_s := B'_s \setminus X$. Note that $|P_s| \leq |B'_s| \leq w'$. For each $s \in W$, pick some $x \in K_s \setminus X'$ and set $x_s := x$. Then, for each $v \in B'_s \setminus X$, set $a_v := v_x = v_{x_s}$. Note that since $v \notin X \supseteq X_2$, $M_v \cap X' = \emptyset$. In particular, $a_v \notin X'$.

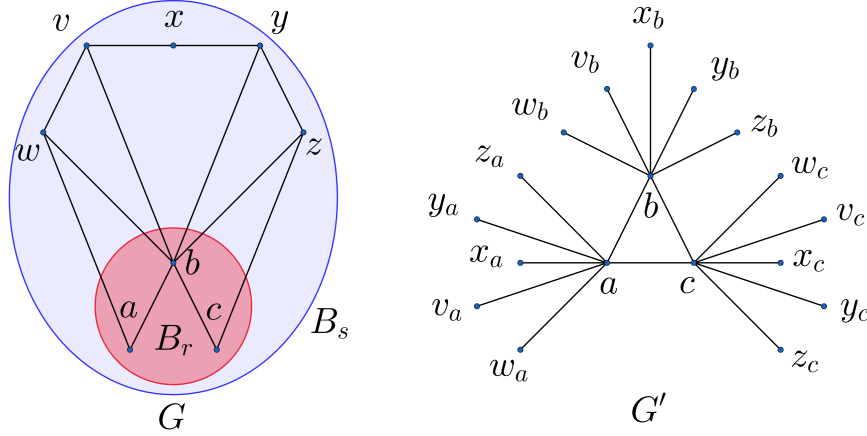


Figure 6.2: A diagram of how G is turned into G' . The left graph is G , the right graph is G' . On the left, the vertices in the red section are B_r , where r is the centre. The vertices in the blue section (all vertices) are B_s . Note that, on the right, each vertex in $B_s \setminus B_r = B'_s$ has been replaced by three new vertices, each adjacent to a different vertex of $B_r = \{a, b, c\}$, and that B_r has become a clique. In this case, $M_v = \{v_a, v_b, v_c\}$ (and similar for M_w, M_x, M_y, M_z).

For $P \in \mathcal{P}'$, define $P' := (P \cap B_r) \cup \{v \in V_W : a_v \in P\}$. Note that $|\{v \in V_W : a_v \in P\}| \leq |P \setminus B_r|$, thus $|P'| \leq |P| \leq 2w$. Let $\mathcal{P} := (P' : P \in \mathcal{P}') \sqcup (P_s : s \in U)$. We claim that $(X, \mathcal{P})$ is the desired almost-layering of G . We first show that $\mathcal{P}$ is a partition of $G - X$.

Observe that $\{B_r \setminus X, V_U, V_W\}$ is a partition of $G - X$, and that $\{P_s : s \in U\}$ is a partition of $G[V_U]$. Note that $B_r \setminus X = V(G' - X') \cap B_r$ (since $X \cap B_r = X_1 \cup A = (X' \cap B_r) \cup A$ and since $V(G) \cap V(G') = B_r \setminus A$), and that $P' \cap B_r = P \cap B_r$ for each $P \in \mathcal{P}'$. It follows that $\{P' \cap B_r : P \in \mathcal{P}'\}$ is a partition for $G[B_r \setminus X]$. Next, for each $v \in V_W$, since $v \notin X$, we have $a_v \notin X'$. Thus, $a_v \in P$ for exactly one $P \in \mathcal{P}'$. Hence, for each $Q \in \mathcal{P}$, we have $v \in Q$ if and only if $Q = P'$. It follows that $\mathcal{P}$ is a partition of $G - X$.

Next, we show that $\mathcal{P}$ is a layering of $G - X$. Observe that for each $s \in U$, $K_s \subseteq X$ (as $K_s \subseteq X' \cap B_r \subseteq X$). Therefore, P_s is an isolated vertex in $(G - X)/\mathcal{P}$. Noting that a graph is contained in a path if and only if each connected component is, it suffices to show that the subgraph of $(G - X)/\mathcal{P}$ induced by $\{P' : P \in \mathcal{P}'\}$ is contained in a path.

As $\mathcal{P}'$ is a layering of $(G' - X')^2$, it further suffices to show that for distinct $P_1, P_2 \in \mathcal{P}$, P'_1, P'_2 are adjacent in $G - X$ only if P_1, P_2 are adjacent in $(G' - X')^2$.

Presume that P'_1, P'_2 are adjacent in G . So there exists $u \in P'_1$ and $v \in P'_2$ such that $uv \in E(G)$. Note that either $u \in P_1 \cap B_r$, or $u \in V_W$ and $a_u \in P_1$. Likewise, either $v \in P_2 \cap B_r$, or $v \in V_W$ and $a_v \in P_2$. Further, $u \notin X$ and $v \notin X$. As $uv \in E(G)$, there exists $s \in V(S)$ such that $u, v \in B_s$. If $s = r$ then observe that $u \notin V_W$ and $v \notin V_W$. Thus, we obtain $u \in P_1$ and $v \in P_2$, so P_1, P_2 are adjacent in $G' - X' \subseteq (G' - X')^2$, as desired. So we may assume that $s \neq r$, and at least one of u, v is not in B_r . Without loss of generality, say $v \notin B_r$. It follows that $v \in V_W$, and thus $s \in W$.

Recall that x_s is a vertex in $K_s \setminus X'$, and that for each $z \in B'_s$, a_z is adjacent to x_s in G' . Note also that for each $z \in B'_s \setminus X$, $a_z \notin X'$. Thus, if $u \in B'_s$ (which occurs if and only if $u \notin B_r$), then since neither u nor v is in X , we have that a_u, a_v are at distance at most 2 in $G' - X'$ via the common neighbour x_s . Noting that $a_u \in P_1$ and $a_v \in P_2$ (as neither u nor v are in B_r), we find that P_1, P_2 are adjacent in $(G' - X')^2$, as desired.

It remains only to consider the case when $u \in B_r$ (and $v \notin B_r$). Since $u \in B_s$ and since $u \notin X$, we have $u \in K_s$ (as $A \subseteq X$). Since $X' \cap B_r \subseteq X$, we obtain $u \notin X'$. Also, since $v \notin X$, we have $a_v \notin X'$. Noting that K_s is a clique in $G' \supseteq G\langle B_r \rangle - A$ and recalling that $a_v \in K_s$, we find that u is adjacent or equal to x_s in $G' - X'$. By the same argument as before, x_s is also adjacent to a_v in $G' - X'$. Thus, u is at distance at most 2 to a_v in $G' - X'$, so u and a_v are adjacent in $(G' - X')^2$. Noting that $u \in P_1$ and $a_v \in P_2$, this gives that P_1 and P_2 are adjacent in $(G' - X')^2$, as desired.

Hence, $\mathcal{P}$ is a layering of $G - X$. So $(X, \mathcal{P})$ is an almost-layering of G . As $|P'| \leq 2w$ for each $P \in \mathcal{P}'$ and $|P_s| \leq w'$ for each $s \in U$, $(X, \mathcal{P})$ has width at most $\max(2w, w')$. Since $|X| \leq q + a$, $(X, \mathcal{P})$ has loss at most $q + a$. This completes the proof. $\square$

We now prove Theorem 6.4. In fact, because we only use Theorem 6.8 as a black box, and otherwise do not create a dependence on n larger than $\sqrt{n}$, we can prove the following stronger result (in which, the key case is $d = \sqrt{n}$).

Theorem 6.12. *Let $d \in \mathbb{R}^{>0}$, $h \in \mathbb{N}$, and let a, b, j be from Theorem 2.7. Let $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ be non-decreasing functions such that for each $m \in \mathbb{N}$, every graph of row-treewidth at most $\max(b, 1)$ and with at most $\max(j, 1)m$ vertices admits an almost-layering of width at most $dg(m)$ and loss at most $mf(m)/d$. Then for each $n \in \mathbb{N}$, each n -vertex K_h -minor-free graph G admits an almost-layering of width at most $\max(2g(n), 1)d$ and loss at most $(f(n) + j + a)n/d$.*

Proof. For each $m \in \mathbb{N}$, let $g'(m) := \max(2g(m), 1)d$ and let $f'(m) := (f(m) + a)/d$.

Note that f', g' are non-decreasing. Set $b' := \max(b, 1)$ and $j' := \max(j, 1)$.

Let $(J_t : t \in V(T))$ be the tree-decomposition of G given by Theorem 2.7. So $(J_t : t \in V(T))$ has adhesion at most $j \leq j'$, and for each $t \in V(T)$, there exists $A_t \subseteq J_t$ with $|A_t| \leq a$ such that $G\langle J_t \rangle - A_t$ has row-treewidth at most $b \leq b'$.

Fixing $r \in V(T)$, by Lemma 6.6, there exists a partition $\mathcal{P}$ of G and $X' \subseteq V(G)$ with $|X'| \leq jn/d$ such that:

1. for each connected component C of $G - X'$, $V(C)$ intersects exactly one part of $\mathcal{P}$
2. for each $P \in \mathcal{P}$, there exists $r_P \in V(T)$ with parent-adhesion set K_P and a star-decomposition $(B_s^P : s \in V(S_P))$ of $G[P]$ such that:
 - (a) $(B_s^P : s \in V(S_P))$ has adhesion at most j ,
 - (b) $r_P \in V(S_P)$, is a centre of S_P , and $B_{r_P} = J_{r_P} \setminus K_P$,
 - (c) $(G[P])\langle B_{r_P}^P \rangle$ is a spanning subgraph of $G\langle J_{r_P} \rangle - K_P$, and
 - (d) for each $s \in V(S_P) \setminus \{r_P\}$:
 - i. s is a child of r_P in T ,
 - ii. $|B_s^P \setminus B_{r_P}^P| \leq d$, and
 - iii. $B_s^P \cap B_{r_P}^P = J_s \cap J_{r_P} \setminus K_P$.

Fix $P \in \mathcal{P}$. Let $A_P := A_{r_P} \cap P$. Note that $|A_P| \leq a$. Since row-treewidth does not increase under taking subgraphs, $(G[P])\langle B_{r_P}^P \rangle - A_P \subseteq G\langle J_{r_P} \rangle - A_{r_P}$ has row-treewidth at most b . If $|P| \leq d$, then $G[P]$ trivially admits an almost-layering (consisting of one part) of width at most $d \leq g'(|P|)$ and loss $0 \leq |P|f'(|P|)$. Otherwise, by Lemma 6.11 (with $(b, j, w, w', q) := (b', j', dg(|P|), d, |P|f(|P|)/d)$, and noting that $b', j' \geq 1$) and choice of f, g , $G[P]$ admits an almost-layering of width at most $\max(2dg(|P|), d) = g'(|P|)$ and loss at most $|P|f(|P|)/d + a \leq |P|(f(|P|) + a)/d = |P|f'(|P|)$ (as $|P|/d \geq 1$).

Let $\mathcal{H}$ be the class of graphs whose components are paths. Note that $\mathcal{H}$ is closed under disjoint union, and that each $H \in \mathcal{H}$ is contained in a path. So any almost- $\mathcal{H}$ -partition is an almost-layering. Thus and by Lemma 6.7 (with $(f, g, q) := (f', g', jn/d)$), G admits an almost-layering of width at most $g'(n) = \max(2g(n), 1)d$ and loss at most $f'(n) + jn/d = (f(n) + a + j)/d$, as desired. $\square$

We now prove Theorem 6.4, which we restate for convenience.

Theorem 6.4. *For each $h \in \mathbb{N}$, there exists $f \in O(n \log(n))$ and $g \in O(\log(n)^3)$ such that for each $n \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, every n -vertex K_h -minor-free graph admits an almost-layering of width at most $dg(n)$ and loss at most $f(n)/d$.*

Proof. Let a, b, j be from Theorem 2.7. Set $b' := \max(b, 1)$ and $j' := \max(j, 1)$. By

Theorem 6.8, there exists $f' \in O(m \log(m))$ and $g' \in O(\log(m)^3)$ such that for each $m \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, every m -vertex graph of row-treewidth at most b' admits an almost-layering of width at most $d g'(m)$ and loss at most $f'(m)/d$.

As $\log(j'm) \in O(\log(m))$, there exists non-decreasing functions $f'', g'' : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ with $f'' \in O(\log(m))$ and $g'' \in O(\log(m)^3)$ such that $m f''(m) \geq f'(j'm)$ and $g''(m) \geq g'(j'm)$ for all $m \in \mathbb{N}^{\geq 1}$. Thus, since f'', g'' are non-decreasing, for each $m \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, every graph of row-treewidth at most $\max(b, 1) = b'$ and with at most $\max(j, 1)m = j'm$ vertices admits an almost-layering of width at most $d g''(m)$ and loss at most $m f''(m)/d$. (This includes the case $m = 0$, since any graph on $j'm = 0$ vertices admits an almost-layering of width $0 \leq d g''(m)$ and loss at most $0 \leq m f''(m)/d$.)

By Theorem 6.12, for each $n \in \mathbb{N}$ and $d \in \mathbb{R}^{>0}$, every K_h -minor-free graph admits an almost-layering of width at most $\max(2g''(n), 1)d$ and loss at most $(f''(n) + a + j)n/d$. Note that $(f''(n) + a + j)n \in O(n \log(n))$ and $\max(2g''(n), 1) \in O(\log(n)^3)$. Thus, the theorem is satisfied with $f(n) := (f''(n) + j)n$ and $g(n) := \max(2g''(n), 1)$. $\square$

6.4 Concentrated Partitions

This section is devoted to the proof of Theorem 6.3, and is the counterpart to Section 6.3. Once again, the goal is to find a way to almost-partition the connected components (of $G - X$) whose structures are torsos plus small sections attached to adhesion sets. This time, the goal is to bound the treewidth of the almost-partition, rather than to find an almost-layering. Specifically, we aim for treewidth 2.

For this section, we assume that we have a way of constructing almost-partitions of treewidth $b(= 2)$, and width and loss $O(\sqrt{n})$ for almost-embeddable graphs. Actually finding these almost-partitions is the subject of Chapter 7. So the goal of this section is to extend the almost-partitions of the torsos (which are almost-embeddable) to almost-partitions of the connected components.

It is first worth noting why we need a different proof strategy. Lemma 6.10 implies that powers of paths are blowups of paths. We exploited this to prove Theorem 6.4. However, this is not true for graphs of bounded treewidth (when the degree is unbounded). So there is no obvious way to recycle our proof, and we instead need a new proof strategy.

Observe that for each clique in the torso, the parts that intersect that clique form a clique in the quotient of the torso. Since the almost-partition has treewidth at most b , this clique has size at most $b + 1$. (Here, we use that each graph of treewidth b has clique number at most $b + 1$.) Since each small section attaches to a clique (adhesion

set) in the torso, we could make each of these a part. These parts then therefore attach (disjointedly) to cliques of size at most $b + 1$, so the resulting almost-partition has treewidth at most $b + 1$.

However, this thesis only finds almost-partitions of almost-embeddable graphs of treewidth 2. This method then increases the treewidth to 3, which is not an improvement over Theorem 6.1. So we must avoid this increase by 1 in the treewidth. This increase occurs only if a small section is attached to a clique in the quotient of size exactly $b + 1$. So if we only attach to cliques in the quotient of size at most b , the resulting almost-partition would have treewidth at most b , as desired.

Here, we take some inspiration from the proof of Theorem 6.1. Distel et al. [45] used a variant of the Graph Minor Structure Theorem, which we cover later (see Theorem 7.1) to precisely describe which cliques in the torso could be an adhesion set. They then showed that every such clique only intersected at most three parts in the almost-partition. (This used some special properties of the almost-partition, and would not be true in general.) Since their almost-partition already had treewidth 3, this meant that they could avoid increasing the treewidth by 1.

We use the same theorem (Theorem 7.1) to control the location of the adhesion sets. However, in our case, it is too much to ask that each adhesion set intersects only $b = 2$ parts. However, the number of ways this can fail is, in a sense, limited. The idea is that given a (large) clique of parts, the adhesion sets usually only intersect a few (at most b) of those parts. All adhesion sets that are exceptions to this intersect a small ‘concentrated’ region of the graph. The deletion of this region then ensures that no adhesion set attaches to this large clique.

The following definition formalises this idea. Let $(X, \mathcal{P})$ be an almost-partition of a graph G , let $b, m \in \mathbb{N}$, and let $\mathcal{K} \subseteq 2^{V(G)}$. We say that $(X, \mathcal{P})$ is *$(b, m, \mathcal{K})$ -concentrated* if, for each clique Q in $(G - X)/\mathcal{P}$, there exists $S_Q \subseteq V(G)$ with $|S_Q| \leq m$ such that, for each $K \in \mathcal{K}$, $K \setminus S_Q$ intersects at most b parts in Q . See Figure 6.3.

As a guide to the reader, the set $\mathcal{K}$ will be the adhesion sets. Each adhesion set is a clique (in the torso), and thus the parts that intersect that adhesion set form a clique Q in the quotient. This is why we limit Q to being a clique. Also, b will be the treewidth of the almost-partition. So the parts that intersect $K \setminus S_Q$ form a clique of size at most b , strictly smaller than the maximum of $b + 1$.

If we were to delete the set S_Q for each clique Q in the quotient of size $b + 1$, then each small section would attach to a clique of size at most b , as desired. Unfortunately, this is still too much, as there can be more than $O(\sqrt{n})$ such cliques (if the parts have size

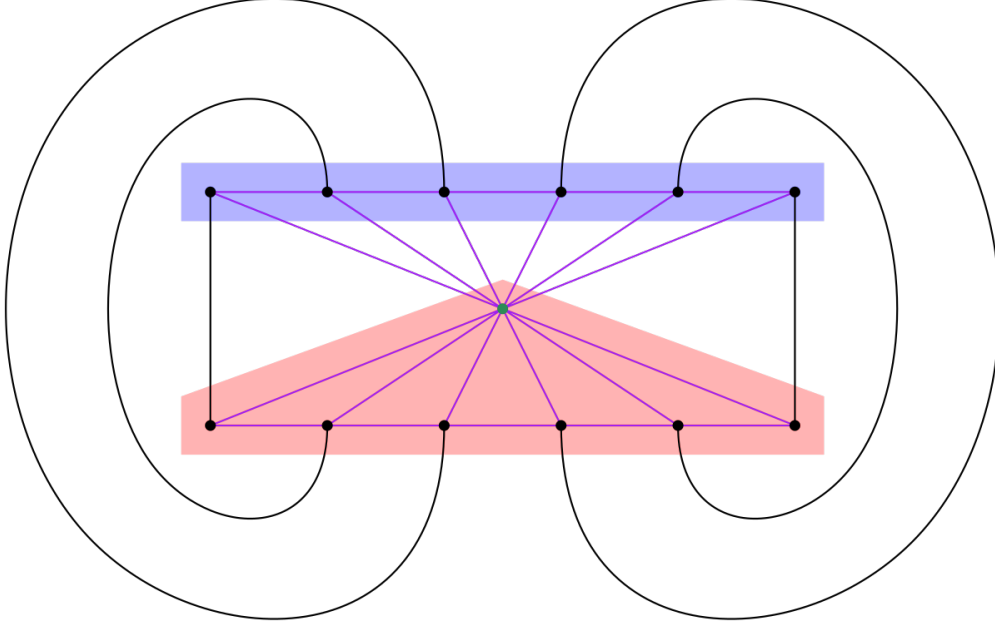


Figure 6.3: An example of a $(1,1,\mathcal{K})$ -concentrated (almost-)partition, where $\mathcal{K}$ is the set of cliques of size 2 corresponding to the purple edges. The partition consists of two parts, shown in red and blue. While there are many purple edges between the two parts, all of them share a common endpoint, namely the green vertex. After this vertex is deleted, each clique in $\mathcal{K}$ either consists only of a single vertex, or is an edge between two vertices in the same part. The black edges that go between the parts are not relevant, as they do not correspond to cliques in $\mathcal{K}$.

$o(\sqrt{n})$). (There is also no easy way to avoid this issue just by merging adjacent small parts.) So we only delete S_Q for a carefully chosen set of cliques Q in the quotient.

Recall that each component C of the remainder attaches to an adhesion set K_C , and that the parts that intersect that adhesion set form a clique Q_C in the quotient. We think of the component (and its vertices) as attaching to that clique in the quotient. For each clique Q in the quotient, we ask ‘how many vertices v attach to exactly that clique’ (that is, $v \in V(C)$ for some C with $Q_C = Q$). Only $O(\sqrt{n})$ cliques Q can have $\Omega(\sqrt{n})$ attaching vertices. Call these the large cliques. For each large clique Q , we delete (put in the loss-set) the set S_Q from the definition of ‘concentrated’. Since the sets S_Q have bounded size, and there are few large cliques, only a few vertices are deleted.

Now, for each component C of the remainder, only a subset K'_C of the adhesion set K_C it attaches to remains in the graph post-deletion. If Q'_C is the set of parts that K'_C intersects, then Q'_C is contained in the clique Q_C in the quotient that C attaches to. We can think of C as attaching to Q'_C post-deletion. We now ask ‘how big is Q'_C ’. If $|Q'_C| \leq b$ (recalling that $b = 2$ in our case), we can add $V(C)$ as a part as discussed earlier. So presume that $|Q'_C| \geq b + 1$. In particular, we have $|Q'_C| = b + 1$ (as the

quotient has clique number at most $b + 1$). Note that for every large clique Q , since we deleted S_Q , every adhesion set post-deletion intersects at most b parts in Q . It follows that Q'_C is not a large clique. Thus, if we take all the vertices attaching onto Q'_C (post-deletion) and put them into a single pre-existing part, that part would still be small afterwards.

Using the degeneracy of bounded treewidth graphs, we can ensure that each part only receives the vertices attaching to one clique Q (post-deletion). So no part is made large ($\Omega(\sqrt{n})$). We also find that adding these extra vertices do not create any new edges in the quotient. Thus, the final quotient still has treewidth at most b , as desired.

We now proceed to the proof.

Lemma 6.13. *Let $b, m, n \in \mathbb{N}$, $w, w', q \in \mathbb{R}^{\geq 0}$, and $d \in \mathbb{R}^{> 0}$. Let G be an n -vertex graph, let $\mathcal{K} \subseteq 2^{V(G)}$, and let $(J_s : s \in V(S))$ be a star-decomposition of G with centre r such that:*

1. $\mathcal{K} \subseteq 2^{J_r}$,
2. $G\langle J_r \rangle$ admits a $(b, m, \mathcal{K})$ -concentrated almost partition $(X_r, \mathcal{P}_r)$ of treewidth at most b , width at most w , and loss at most q , and
3. for each $s \in V(S) \setminus \{r\}$:
 - (a) $J_s \cap J_r \in \mathcal{K}$, and
 - (b) $|J_s \setminus J_r| \leq w'$.

Then G admits an almost-partition $(X, \mathcal{P})$ of treewidth at most b , width at most $\max(w + d, w')$, and loss at most $q + mn/d$.

Proof. Let $G_r := G\langle J_r \rangle$, and set $Y := V(S) \setminus \{r\}$. For each $s \in Y$, let $K_s := J_s \cap J_r$ and $J'_s := J_s \setminus J_r$. So $K_s \in \mathcal{K}$ and is a clique in G_r , and $|J'_s| \leq w'$. Observe also that J'_s is disjoint and non-adjacent in G to $J'_{s'}$ whenever $s, s' \in Y$ are distinct, and that $(J'_s : s \in Y)$ is a partition of $G - J_r$.

Let $\mathcal{Q}$ be the set of cliques in $(G_r - X_r)/\mathcal{P}_r$. Since $(X, \mathcal{P})$ has treewidth at most b , $(G_r - X_r)/\mathcal{P}_r$ has treewidth at most b , and thus each $Q \in \mathcal{Q}$ has size at most $b + 1$. Since $(X_r, \mathcal{P}_r)$ is $(b, m, \mathcal{K})$ -concentrated, for each $Q \in \mathcal{Q}$, there exists $X_Q \subseteq V(G_r) = J_r$ with $|X_Q| \leq m$ such that for each $K \in \mathcal{K}$, $K \setminus X_Q$ intersects at most b parts in Q .

For each $s \in Y$, let Q_s be the set of parts in $\mathcal{P}_r$ that intersect K_s . Since K_s is a clique in G_r , observe that $Q_s \in \mathcal{Q}$. Thus, $|Q_s| \leq b + 1$. Let $X_s := X_{Q_s}$. As $K_s \in \mathcal{K}$, $K_s \setminus X_s$ intersects at most b parts in Q_s .

For each $Q \in \mathcal{Q}$, let $Y_Q := \{s \in Y : Q_s = Q\}$, $V_Q := \bigcup_{s \in Y_Q} J'_s$, and let $n_Q := |V_Q|$.

Observe that for each pair of distinct $Q, Q' \in \mathcal{Q}$, Y_Q is disjoint to $Y_{Q'}$, and thus V_Q is disjoint to $V_{Q'}$. Therefore, $\sum_{Q \in \mathcal{Q}} n_Q \leq n$. Thus, if $\mathcal{Q}^+ \subseteq \mathcal{Q}$ consists of the $Q \in \mathcal{Q}$ with $n_Q \geq d$, then $|\mathcal{Q}^+| \leq n/d$.

Let $Y^+ := \{s \in Y : Q_s \in \mathcal{Q}^+\}$, and let $X' := \bigcup_{s \in Y^+} X_s$. Note that $X' \subseteq \bigcup_{Q \in \mathcal{Q}^+} X_Q$, and thus $|X'| \leq m|\mathcal{Q}^+| \leq mn/d$. Note also that $X' \subseteq V(G_r)$. Let $X := X_r \cup X'$. Observe that $|X| \leq |X_r| + |X'| \leq q + mn/d$. Note also that $X \subseteq V(G_r)$.

Let $\mathcal{P}'_r := (P \setminus X : P \in \mathcal{P}_r)$. Observe that $(X, \mathcal{P}'_r)$ is an almost-partition of G_r of treewidth at most b , width at most w , and loss at most $q + mn/d$. Let $\psi : \mathcal{P}'_r \mapsto \mathcal{P}_r$ be the obvious bijection that satisfies $\psi^{-1}(P) = P \setminus X$ for each $P \in \mathcal{P}_r$, and let $\mathcal{Q}'$ be the set of cliques of $(G_r - X)/\mathcal{P}'_r$. Note that each $Q \in \mathcal{Q}'$ has size at most $b + 1$ as $(G_r - X)/\mathcal{P}'_r$ has treewidth at most b .

For each $s \in Y$, let Q'_s be the set of parts in $\mathcal{P}'_r$ that intersect K_s . Since the parts of $\mathcal{P}'_r$ are disjoint to X , observe that each part in Q'_s intersects $K_s \setminus X$. Since K_s is a clique in G_r , observe that $Q'_s \in \mathcal{Q}'$. Thus, $|Q'_s| \leq b + 1$. Also, observe that $\psi(Q'_s) \subseteq Q_s$. Since $|\psi(Q'_s)| = |Q'_s|$, it follows that $|Q'_s| \leq |Q_s|$, with equality if and only if $\psi(Q'_s) = Q_s$. In particular, since $|Q_s| \leq b + 1$, if $|Q'_s| = b + 1$, then $\psi(Q'_s) = Q_s$.

Let $U := \{s \in Y : |Q'_s| \leq b\}$ and $W := \{s \in Y : |Q'_s| = b + 1\}$. Observe that Y is the disjoint union of U and W (as $b \in \mathbb{N}$). For each $s \in W$, since $|Q'_s| = b + 1$, recall that $\psi(Q'_s) = Q_s$. Also, recall that $K_s \setminus X_s$ intersects at most b parts in $Q_s = \psi(Q'_s)$. Thus, $K_s \setminus X_s$ intersects at most b parts in Q'_s , as $P \subseteq \psi(P)$ for each $P \in \mathcal{P}'_r$. However, since $|Q'_s| = b + 1$ and since $K_s \setminus X$ intersects each part in Q'_s , we obtain $X_s \not\subseteq X$. Observe that if $s \in Y^+$, then we would have $X_s \subseteq X' \subseteq X$, a contradiction. Thus, $s \notin Y^+$. Hence, $Y^+ \cap W = \emptyset$.

As $(G_r - X)/\mathcal{P}'_r$ has treewidth at most b , it has degeneracy at most b . Thus, we can find an ordering $\preceq$ of $\mathcal{P}'_r$ such that each $P \in \mathcal{P}'_r$ is adjacent in $(G_r - X)/\mathcal{P}'_r$ to at most b parts P' with $P' \prec P$. For each $s \in W$, let P_s be the part in Q'_s that is preceded by every other part in Q'_s (which exists since $|Q'_s| = b + 1 \geq 1$). Since $P_s \in Q'_s$, observe that P_s intersects K_s . In particular, P_s is nonempty. Since Q'_s is a clique in $(G_r - X)/\mathcal{P}'_r$ of size exactly $b + 1$, observe that $Q'_s \setminus \{P_s\}$ is exactly the set of parts P' of $\mathcal{P}'_r$ with $P' \prec P$ that are adjacent to P_s in $(G_r - X)/\mathcal{P}'_r$. Thus, if $s, s' \in W$ are such that $P_s = P_{s'}$, then $Q'_s = Q'_{s'}$, and thus $Q_s = \phi(Q'_s) = \phi(Q'_{s'}) = Q_{s'}$.

For each $P \in \mathcal{P}'_r$, let $Y_P := \{s \in W : P_s = P\}$, and let $V_P := P \cup \bigcup_{s \in Y_P} J'_s$. If $Y_P = \emptyset$, then $V_P = P$, and thus $|V_P| = |P| \leq w$ as $\mathcal{P}'_r$ has width at most w . Otherwise, fix $s \in Y_P$. Recalling that $Y_P \subseteq W$, which is disjoint to Y^+ , we obtain $Q_s \notin \mathcal{Q}^+$. So $n_{Q_s} \leq d$. Observe that for each $s' \in Y_P$, $P_{s'} = P = P_s$, and thus $Q_{s'} = Q_s$. It follows that

$Y_P \subseteq Y_{Q_s}$. Thus, $\bigcup_{t \in Y_P} J'_t \subseteq \bigcup_{t \in Y_{Q_s}} J'_t = V_{Q_P}$. So $|V_P| \leq |P| + |V_{Q_s}| \leq w + n_{Q_P} \leq w + d$ (as $\mathcal{P}'_r$ has width at most w).

Let $\mathcal{P} := \{V_P : P \in \mathcal{P}'_r\} \cup \{J'_s : s \in U\}$. We show that $(X, \mathcal{P})$ is the desired almost-partition. Recall that for each pair of distinct $s, s' \in Y$, J_s is disjoint to $J_{s'}$. Recall also that Y is the disjoint union of U and W , and note that $W = \bigsqcup_{P \in \mathcal{P}'_r} Y_P$. Further, observe that for distinct $P, P' \in \mathcal{P}'_r$, Y_P is disjoint to $Y_{P'}$, since the existence of some $s \in Y_P \cap Y_{P'}$ would give $P_s = P = P'$, a contradiction (as P_s is nonempty). Thus, $(J'_s : s \in U) \sqcup (\bigcup_{s \in Y_P} J'_s : P \in \mathcal{P}'_r)$ is a partition of $G - B_r$ (as $(J'_s : s \in Y)$ is a partition of $G - B_r$). Since $X \subseteq B_r$ and since $\mathcal{P}'_r$ is a partition of $G_r - X$ (and thus $G[B_r] - X$), it follows that $(X, \mathcal{P})$ is an almost-partition of G (as $V_P = P \cup \bigcup_{s \in Y_P} J'_s$ for each $P \in \mathcal{P}'_r$). Recall that $|X| \leq q + mn/d$, that $|J'_s| \leq w'$ for each $s \in Y \supseteq U$, and that $|V_P| \leq w + d$ for each $P \in \mathcal{P}'_r$. Thus, $(X, \mathcal{P})$ has width at most $\max(w + d, w')$ and loss at most $q + mn/d$. It remains only to show that $(X, \mathcal{P})$ has treewidth at most b .

Recall that for distinct $s, s' \in Y$, J'_s is disjoint and non-adjacent in G to $J_{s'}$. Further, observe that for each $s \in Y$ and $P \in \mathcal{P}'_r$, J'_s is disjoint to P , and is adjacent to P in G only if P intersects K_s . In this case, we have $P \in Q'_s$. Thus, since Q'_s is a clique (in $(G_r - X)/\mathcal{P}'_r$) and since $P_s \in Q'_s$, P is either equal to P_s , or adjacent in $G_r - X$ to P_s .

Thus, for each $s \in U$, the parts of $\mathcal{P}$ that J'_s is adjacent in G (and thus $G - X$) to is contained in $(V_P : P \in Q'_s)$. Note that Q'_s is a clique of size at most b in $(G_r - X)/\mathcal{P}'_r$ (as $s \in U$). Further, for each pair of distinct $P, P' \in \mathcal{P}'_r$, recalling that $P_s = P$ for each $s \in Y_P$, we find that $\bigcup_{s \in Y_P} J'_s$ is adjacent in $G - X$ to P' only if $P' \in Q'_s$ for some $s \in Y_P$, and thus is adjacent to $P = P_s$ in $G_r - X$. Applying a symmetric argument with P' and since $G[B_r] - X \subseteq G_r - X$, it follows that V_P is adjacent to $V_{P'}$ in $G - X$ only if P is adjacent to P' in $G_r - X$.

Therefore, we find that $(G - X)/\mathcal{P}$ can be obtained from $(G_r - X)/\mathcal{P}'_r$ by attaching new vertices (J'_s with $s \in U$) to cliques of size at most b , then deleting edges and relabelling vertices. The treewidth after the first step is at most $\max(\text{tw}((G_r - X)/\mathcal{P}'_r), b) = b$ (as $(X, \mathcal{P}'_r)$ has treewidth at most b), and the second step does not increase the treewidth. Thus, $(X, \mathcal{P})$ has treewidth at most b . This completes the proof. $\square$

To complete the proof of Theorem 6.3, we need the following theorem, which loosely says that we can almost-partition the torsos in a concentrated way. Its proof is the subject of Chapter 7.

Theorem 6.14. *For each $h \in \mathbb{N}$, there exists $j, m \in \mathbb{N}$ and maps $f \in O(n)$ and $g \in O(\sqrt{n})$ such that the following holds. For each $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq g(n)$, and n -vertex K_h -minor-free graph G , there exists a tree T , a root $r \in V(T)$, and a collection*

of pairs $((J_t, \mathcal{K}_t) : t \in V(T))$ such that:

1. $(J_t : t \in V(T))$ is a tree-decomposition of G of adhesion at most j , and
2. for each $t \in V(T)$, if K_t is the parent-adhesion set at t , then:
 - (a) $\mathcal{K}_t \subseteq 2^{J_t \setminus K_t}$,
 - (b) $G \langle J_t \rangle - K_t$ admits a $(2, m, \mathcal{K}_t)$ -concentrated almost-partition of treewidth at most 2, width at most d , and loss at most $f(|J_t \setminus K_t|)/d$, and
 - (c) for each child t' of t , $J_t \cap J_{t'} \setminus K_t \in \mathcal{K}_t$.

We can now prove Theorem 6.3, which we restate for ease of reference.

Theorem 6.3. *For each $h \in \mathbb{N}$, there exists $f \in O(n)$ and $g \in O(\sqrt{n})$ such that for each $n \in \mathbb{N}$, and every $d \in \mathbb{R}^{>0}$ with $d \geq g(n)$, each n -vertex K_h -minor-free graph G admits an almost-partition of treewidth at most 2, width at most d , and loss at most $f(n)/d$.*

Proof of Theorem 6.3. Let $(j, m, f', g') := (j, m, f, g)$ be from Theorem 6.14. Let $g(n) := 2g'(n)$ for each $n \in \mathbb{N}$. Note that $g \in O(\sqrt{n})$. Further, observe that we can find a non-decreasing map $f'' : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ with $f'' \in O(1)$ such that for each $n \in \mathbb{N}^{\geq 1}$, $nf''(n) \geq 2f'(n) + mn$. Then, for each $n \in \mathbb{N}$, let $f(n) := jn + nf''(n)$. Observe that $f \in O(n)$.

Fix $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq g(n)$, and an n -vertex K_h -minor-free graph G . Noting that $d/2 \geq g'(n)$, there exists a tree T , a root $r \in V(T)$, and a collection of pairs $((J_t, \mathcal{K}_t) : t \in V(T))$ such that:

1. $(J_t : t \in V(T))$ is a tree-decomposition of G of adhesion at most j , and
2. for each $t \in V(T)$, if K_t is the parent-adhesion set at t , then:
 - (a) $\mathcal{K}_t \subseteq 2^{J_t \setminus K_t}$,
 - (b) $G \langle J_t \rangle - K_t$ admits a $(2, m, \mathcal{K}_t)$ -concentrated almost-partition of treewidth at most 2, width at most d , and loss at most $f(|J_t \setminus K_t|)/d$, and
 - (c) for each child t' of t , $J_t \cap J_{t'} \setminus K_t \in \mathcal{K}_t$.

By Lemma 6.6, there exists a partition $\mathcal{P}'$ of G and $X' \subseteq V(G)$ with $|X'| \leq jn/d$ such that:

1. for each connected component C of $G - X'$, $V(C)$ intersects exactly one part of $\mathcal{P}'$
2. for each $P \in \mathcal{P}'$, there exists $r_P \in V(T)$ with parent-adhesion set K_P and a star-decomposition $(B_s^P : s \in V(S_P))$ of $G[P]$ such that:
 - (a) $(B_s^P : s \in V(S_P))$ has adhesion at most j ,

- (b) $r_P \in V(S_P)$, is a centre of S_P , and $B_{r_P} = J_{r_P} \setminus K_P$,
- (c) each $s \in V(S_P) \setminus \{r_P\}$ is a child of r_P in T ,
- (d) $(G[P])\langle B_{r_P}^P \rangle$ is a spanning subgraph of $G\langle J_{r_P} \rangle - K_P$, and
- (e) for each $s \in V(S_C) \setminus \{r_C\}$:
 - i. $|B_s^P \setminus B_{r_P}^P| \leq d$, and
 - ii. $B_s^P \cap B_{r_P}^P = J_s \cap J_{r_P} \setminus K_P$.

Fix $P \in \mathcal{P}'$. We show that $G[P]$ admits an almost-partition of treewidth at most 2, width at most d , and loss at most $|P|f''(|P|)/d$. If $|P| = 0$, this is trivial as $G[P]$ admits an almost-partition of treewidth $0 \leq 2$, width $0 \leq d$, and loss $0 \leq |P|f''(|P|)/d$. So we may assume that $|P| \geq 1$.

Since $(G[P])\langle B_{r_P}^P \rangle$ is a spanning subgraph of $G\langle J_{r_P} \rangle - K_P$, note that $(X_{r_P}, \mathcal{P}_{r_P})$ is also a $(2, m, \mathcal{K}_{r_P})$ -concentrated almost-partition of $(G[P])\langle B_{r_P}^P \rangle$ of treewidth at most 2, width at most $d/2$, and loss at most $2f'(|J_t \setminus K_t|)/d = 2f'(|B_{r_P}^P|) \leq 2f'(|P|)$ (as f' is non-decreasing). For each $s \in V(S_P) \setminus \{r_P\}$, we have that $B_s^P \cap B_{r_P}^P = J_s \cap J_{r_P} \setminus K_P \in \mathcal{K}_{r_P}$ as s is a child of r_P in T (and since $K_{r_P} = K_P$). Thus, by Lemma 6.13 (with $(b, w, w', q, d, n) := (2, d/2, d, 2f'(|P|)/d, d/2, |P|)$, $G[P]$ admits an almost-partition of treewidth at most 2, width at most $\max(d/2 + d/2, d) = d$, and loss at most $2f'(|P|)/d + m|P|/d \leq |P|f''(|P|)/d$ (as $|P| \geq 1$). Hence, in all cases, $G[P]$ admits an almost-partition of treewidth at most 2, width at most d , and loss at most $|P|f''(|P|)/d$.

For each $i \in \mathbb{N}$, let $g^*(i) := d$ and $f^*(i) := f''(i)/d$. Note that f^*, g^* are non-decreasing. By Lemma 6.7 (with $(f, g, q) := (f^*, g^*, jn/d)$, and $\mathcal{H}$ being the class of graphs of treewidth at most 2), G admits an almost-partition of treewidth at most 2, width at most d , and loss at most $jn/d + nf''(n)/d = f(n)/d$. This completes the proof. $\square$

This completes the proof of Theorem 2.9, conditional on the proof of Theorem 6.14.

Chapter 7

Treewidth 3 Partitions of Minor-Free Graphs

This chapter completes the proof of Theorem 2.9, which says that n -vertex K_h -minor-free graphs are $O_h(\sqrt{n})$ -blowups of treewidth 3 graphs.

Chapter 6 proved Theorem 2.9 conditionally on the unproven Theorem 6.14. This theorem roughly says that n -vertex K_h -minor-free graphs have a tree-decomposition in which each torso admits a concentrated almost-partition of treewidth 2 and appropriate width and loss (based on a variable parameter d).

The proof of Theorem 6.14 combines the Graph Minor Structure Theorem and a new result about almost-embeddable graphs. Specifically, we show that almost-embeddable graphs admit concentrated almost-partitions for a specific set of cliques $\mathcal{K}$ that depend on the almost-embedding (see Theorem 7.2). This set will include all the adhesion sets. However, the standard version of the Graph Minor Structure Theorem (Theorem 1.2 [121]) is not strong enough, as it does not tell us where the adhesion sets lie. We will instead employ a variant shown by Diestel, Kawarabayashi, Müller, and Wollan [37]. To state this variant, we will need to establish some more background and terminology. We will also need to work in the setting of pseudographs. See Section 7.1.1.

We warn the reader that we do not give a complete proof of Theorem 6.14 (and thus Theorem 2.9) in this thesis. We have excluded the proof of a few key lemmas, namely Theorem 7.9 and Lemma 7.5. The proof of both of these results can be found in [42].

Theorem 7.9 gives a way of reducing a (g, p, k, a) -almost-embedding to something close to a $(0, 1, k, 0)$ -almost-embedding. See Section 7.1.6 for the details. We exclude its proof because it is extremely long (roughly 70 pages), technical, and strays too far from the main themes of this thesis. We do, however, explain the key proof ideas in Section 7.1.5.

Lemma 7.5 loosely says that for any plane pseudograph, we can add edges and then find a connected treewidth 2 partition of the spanning super-pseudograph of an appropriate width that satisfies some technical conditions. We exclude this proof because it is essentially a repeat of the proof of Theorem 2.10 [45], along with some technical work to reprove a known result in a slightly modified setting. See Section 7.1.8 for details.

We have also excluded one other lemma, Lemma 7.14, which relates to how we can tweak a particular construction. We exclude this proof because (1) it is tedious and technical, and (2) within the scope of this thesis, we only use a special case with a much easier proof. As such, we have elected to only include the proof of this special case (Observation 7.15). The proof of the general case can again be found in [42].

7.1 Proof Overview

7.1.1 Adhesion Sets in the Graph Minor Structure Theorem

For this chapter, we consider pseudographs. A *pseudograph* is defined similarly to a graph, except that it allows loops and parallel edges. We do not consider edges to be labelled. To avoid confusion, in this chapter, we often refer to a graph as a *simple graph*. From a pseudograph, we can delete all parallel edges and loops to obtain a simple graph. This simple graph is also uniquely defined (as we ignore edge labels).

Most definitions on graphs translate naturally to pseudographs. However, we clarify a few important definitions and details. A *sub-pseudograph* H of a pseudograph G is a pseudograph with $V(H) \subseteq V(G)$ and, for each pair $u, v \in V(G)$, the number uv -edges (v -loops, if $u = v$) in H is at most the number of uv -edges in G . This uses the fact that we do not consider edges to be labelled. We define a *super-pseudograph* analogously. We consider a simple graph H to be *minor* of a pseudograph G if H is a minor of the simple graph obtained from G by deleting all parallel edges and loops. For $b \in \mathbb{N}$, we consider G to be a *b -blowup* of H if the simple graph obtained from G after deleting all parallel edges and loops is a b -blowup of H .

We consider the quotient of a pseudograph by a partition to be a simple graph, not a pseudograph. For an embedded graph G and a G -clean disc D , we consider the underlying cycle $U(D, G)$ to be a simple graph. This is only relevant when $|B(D, G)| \leq 2$. So if $|B(D, G)| = 1$ we consider $U(D, G)$ to be a single vertex with no edges (rather than a loop), and if $|B(D, G)| = 2$, we consider $U(D, G)$ to be a single edge (rather than a pair of parallel edges).

We also want to consider the following variant of the torso. For a tree-decomposition $(J_t : t \in V(T))$, we say that the *pseudo-torso* at $t \in V(T)$ is pseudograph obtained from $G[J_t]$ by, for each $t' \in V(T)$ adjacent to t and each pair of distinct $x, y \in J_t \cap J_{t'}$, adding a new edge with endpoints x, y . So this can result in many parallel edges being added. In this chapter, we use $G\langle J_t \rangle$ to refer to the pseudo-torso, rather the torso.

We now work towards stating the variant of the Graph Minor Structure Theorem proven by Diestel et al. [37].

Say that a face F of an embedded pseudograph is a *triangle-face* if F is 2-cell (homeomorphic to a disc) and the boundary walk of F is a cycle of length exactly 3. Observe that the boundary vertices in a triangle-face form a clique of size exactly 3 in G . Say that a clique K of size 3 in a pseudograph G is a *facial triangle* if there is a triangle-face of G whose boundary vertices are exactly K . We use $F_3(G)$ to refer to the set of facial triangles of G .

Let $(\Sigma, A, s, (G_0, \dots, G_s), (D_1, \dots, D_s))$ be an almost-embedding of a pseudograph G . We define the *facial triangles* of Γ , denoted $F_3(\Gamma)$, to be $F_3(G_0)$. Define the *attachable embedded cliques* of Γ , denoted $\mathcal{EC}(\Gamma)$, to be the set of cliques in G_0 that are either in $F_3(G_0) = F_3(\Gamma)$ or have size at most 2. Define the *vortex cliques* of Γ , denoted $\mathcal{VC}(\Gamma)$, to be the set of cliques in $H(\Gamma)$ (the vortex-union subgraph). Define the *attachable cliques* of Γ , denoted $\mathcal{AC}(\Gamma)$, to be the set of cliques K in G such that $K \setminus A \in \mathcal{EC}(\Gamma) \cup \mathcal{VC}(\Gamma)$. Observe that $\mathcal{VC}(\Gamma)$, $\mathcal{EC}(\Gamma)$, and $\mathcal{AC}(\Gamma)$ are all closed under subsets.

We can now state the result of Diestel et al. [37, Theorem 4].

Theorem 7.1. *For each $h \in \mathbb{N}$, there exists $g, p, k, a \in \mathbb{N}$ such that for each K_h -minor-free pseudograph G , there exists a tree T , a root $r \in V(T)$ of T , and a collection $((J_t, \Gamma_t) : t \in V(T))$ such that:*

1. $(J_t : t \in V(T))$ is a tree-decomposition of G ,
2. for each $t \in V(T)$, Γ_t is a (g, p, k, a) -almost-embedding of $G\langle J_t \rangle$, and
3. for each edge $tt' \in E(T)$, if t' is the parent of t and A_t are the set of apices of Γ_t , then:
 - (a) $J_t \cap J_{t'} \subseteq A_t$, and
 - (b) $J_t \cap J_{t'} \in \mathcal{AC}(\Gamma_{t'})$.

We remark that Diestel et al. [37, Theorem 4] uses torsos, rather than pseudo-torsos. We believe that their proof would be more rigorously written in terms of pseudo-torsos. However, even if torsos suffice, the almost-embedding of the torsos can easily be extended to almost-embeddings of the pseudo-torsos satisfying the same properties. Notably, we can add parallel edges to the embedded subgraph in a manner that preserves clean

discs and facial triangles. The reverse is not true, because deleting a parallel edge might destroy a facial triangle. So we prefer to err on the side of caution and use the pseudo-torso (although it makes little difference to our proof).

7.1.2 Concentrated Partitions of Almost-Embeddable Pseudographs

As explained before, Theorem 6.14 is essentially a combination of Theorem 7.1 and a result about concentrated almost-partitions of almost-embeddable (pseudo)graphs. We can now state the latter.

Theorem 7.2. *For each $g, p, k, a \in \mathbb{N}$, there exists $m \in \mathbb{N}$ and $c, \alpha \in \mathbb{R}^{\geq 0}$ such that the following holds. For each $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq \alpha\sqrt{n}$, and n -vertex pseudograph G , if G admits a (g, p, k, a) -almost-embedding Γ , then G admits a $(2, m, \mathcal{AC}(\Gamma))$ -concentrated almost-partition of treewidth at most 2, width at most d , and loss at most cn/d .*

Before proving Theorem 7.2, we show how it implies Theorem 6.14. For this, we need the following observation.

Observation 7.3. *Let $g, p, k, a \in \mathbb{N}$, let $\Gamma = (\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s))$ be a (g, p, k, a) -almost-embedding of a pseudograph G , and let $S \subseteq A$. Then $\Gamma' := (\Sigma, s, A \setminus S, (G_0, \dots, G_s), (D_1, \dots, D_s))$ is a (g, p, k, a) -almost-embedding of $G - S$, and for each $K \in \mathcal{AC}(\Gamma)$, $K \setminus S \in \mathcal{AC}(\Gamma')$.*

Proof. Since $S \subseteq A \subseteq V(G)$, $G - S$ is a well-defined pseudograph. Observe that $(G - S) - (A \setminus S) = G - A$. The fact that Γ' is an almost-embedding of $G - S$ then follows directly from the fact that Γ is an almost-embedding of G . Further, since Γ is a (g, p, k, a) -almost-embedding of G and since $|A \setminus S| \leq |A|$, Γ' is a (g, p, k, a) -almost-embedding of $G - S$.

Observe that $H(\Gamma') = H(\Gamma)$. Thus, we find that $\mathcal{EC}(\Gamma) = \mathcal{EC}(\Gamma')$ and $\mathcal{VC}(\Gamma) = \mathcal{VC}(\Gamma')$, as these depend only on G_0 and $H(\Gamma') = H(\Gamma)$ respectively. For each $K \in \mathcal{AC}(\Gamma)$, recall that $K \setminus A \in \mathcal{EC}(\Gamma) \cup \mathcal{VC}(\Gamma) = \mathcal{EC}(\Gamma') \cup \mathcal{VC}(\Gamma')$. Observe also that $K \setminus S$ is a clique in $G - S$, and that $K \setminus A = (K \setminus S) \setminus (A \setminus S)$. Hence, $(K \setminus S) \setminus (A \setminus S) \in \mathcal{EC}(\Gamma') \cup \mathcal{VC}(\Gamma')$. Thus, by definition of $\mathcal{AC}(\Gamma')$, we have $K \setminus S \in \mathcal{AC}(\Gamma')$. This completes the proof. $\square$

We now prove Theorem 6.14, which we recall for convenience. (In the original statement of this theorem, $G\langle J_t \rangle$ refers to the torso, not pseudo-torso. However, it makes little difference, since the absence or presence of parallel edges/loops does not prevent an

almost-partition from being concentrated, nor does it change the treewidth, width, or loss.)

Theorem 6.14. *For each $h \in \mathbb{N}$, there exists $j, m \in \mathbb{N}$ and maps $f \in O(n)$ and $g \in O(\sqrt{n})$ such that the following holds. For each $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq g(n)$, and n -vertex K_h -minor-free graph G , there exists a tree T , a root $r \in V(T)$, and a collection of pairs $((J_t, \mathcal{K}_t) : t \in V(T))$ such that:*

1. $(J_t : t \in V(T))$ is a tree-decomposition of G of adhesion at most j , and
2. for each $t \in V(T)$, if K_t is the parent-adhesion set at t , then:
 - (a) $\mathcal{K}_t \subseteq 2^{J_t \setminus K_t}$,
 - (b) $G\langle J_t \rangle - K_t$ admits a $(2, m, \mathcal{K}_t)$ -concentrated almost-partition of treewidth at most 2, width at most d , and loss at most $f(|J_t \setminus K_t|)/d$, and
 - (c) for each child t' of t , $J_t \cap J_{t'} \setminus K_t \in \mathcal{K}_t$.

Proof. Let $g, p, k, a \in \mathbb{N}$ be from Theorem 7.1, and let c, m, α be from Theorem 7.2. Set $j := a$. For each $n \in \mathbb{N}$, set $f(n) := cn$ and $g(n) := \alpha\sqrt{n}$. So $f \in O(n)$ and $g \in O(\sqrt{n})$.

Fix $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq g(n)$, and n -vertex K_h -minor-free graph G . So $d \geq \alpha\sqrt{n}$.

By Theorem 7.1, there exists a tree T , a root $r \in V(T)$ of T , and a collection $((J_t, \Gamma_t) : t \in V(T))$ such that:

1. $(J_t : t \in V(T))$ is a tree-decomposition of G ,
2. for each $t \in V(T)$, Γ_t is a (g, p, k, a) -almost-embedding of $G\langle J_t \rangle$, and
3. for each edge $tt' \in E(T)$, if t' is the parent of t and A_t are the set of apices of Γ_t , then:
 - (a) $J_t \cap J_{t'} \subseteq A_t$, and
 - (b) $J_t \cap J_{t'} \in \mathcal{AC}(\Gamma_{t'})$.

For each $t \in V(T)$, let A_t be the set of apices of Γ_t , and let K_t be the parent-adhesion set at t . So $K_t \subseteq A_t$ (even if $t = r$, as then $K_t = \emptyset$), and if $t \neq r$ and has parent t' , $K_t \in \mathcal{AC}(\Gamma_{t'})$. Note also that $|K_t| \leq |A_t| \leq a = j$, and thus $(J_t : t \in V(T))$ is a tree-decomposition of G of adhesion at most j .

Let Γ'_t be obtained from Γ_t by replacing the set of apices A_t with $A_t \setminus K_t$. By Observation 7.3 (with $S := K_t$), Γ'_t is a (g, p, k, a) -almost-embedding of $G\langle J_t \rangle - K_t$, and for each $K \in \mathcal{AC}(\Gamma_t)$, $K \setminus K_t \in \mathcal{AC}(\Gamma'_t)$.

Set $\mathcal{K}_t := \mathcal{AC}(\Gamma'_t)$. Note that $\mathcal{K}_t \subseteq 2^{V(G\langle J_t \rangle - K_t)} = 2^{J_t \setminus K_t}$. Further, for each child t' of t , since $J_t \cap J_{t'} \in \mathcal{AC}(\Gamma_t)$, we have $J_t \cap J_{t'} \setminus K_t \in \mathcal{K}_t$. By Theorem 7.2 (with $n := |V(G\langle J_t \rangle - K_t)| = |J_t \setminus K_t|$), $G\langle J_t \rangle - K_t$ admits a $(2, m, \mathcal{K}_t)$ -concentrated almost-

partition $(X_t, \mathcal{P}_t)$ of treewidth at most 2, width at most d , and loss at most $c|J_t \setminus K_t|/d = f(|J_t \setminus K_t|)/d$. This completes the proof. $\square$

It remains to prove Theorem 7.2. We want to emphasise that our method for finding these almost-partitions is not particularly specific to the treewidth 2 case. Instead, if a technical condition relating to the existence of good almost-partitions of plane pseudographs is satisfied, we show that (g, p, k, a) -almost-embeddable pseudographs have good almost-partitions (with respect to an almost-embedding). Specifically, for some $b \in \mathbb{N}$, we ask that a pair of functions (f, g) satisfy a technical condition referred to as ‘generate adjustments of treewidth b ’. Very loosely, this condition says that, given a plane pseudograph G , there is a plane pseudograph G' obtained from G by deleting vertices and adding edges such that G' admits a connected partition of treewidth at most b . f controls the number of deleted vertices, and g controls the width of the partition. See Section 7.1.7 for the definition. For now, we just state the theorem.

Theorem 7.4. *For each $b, g, p, k, a \in \mathbb{N}$, there exists $m_1, m_2 \in \mathbb{N}$ and $c_1, c_2, \beta \in \mathbb{R}^{\geq 0}$ such that the following holds. If $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ are such that (f, g) generate adjustments of treewidth b , then for each $n \in \mathbb{N}$, every $d \in \mathbb{R}^{> 0}$ with $d \geq \beta g(n)$, and each (g, p, k, a) -almost-embedding Γ of an n -vertex pseudograph G , G admits an almost-partition of treewidth at most b , width at most d , and loss at most $(c_1 f(n) + c_2 n)/d$ that is $(1, m_1, \mathcal{VC}(\Gamma))$ -concentrated and $(2, m_2, \mathcal{AC}(\Gamma))$ -concentrated.*

We do not make use of the fact that the almost-partition is $(1, m_1, \mathcal{VC}(\Gamma))$ -concentrated in this thesis, but we believe that it could be useful for potentially showing that K_h -minor-free pseudographs admit treewidth 1 almost-partitions of small width/loss, and it is not much extra effort to show.

Theorem 7.4 is then combined with the following lemma.

Lemma 7.5. *Let $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ be defined by $f(n) := 0$ and $g(n) := 12\sqrt{3n}$ for each $n \in \mathbb{N}$. Then (f, g) generate adjustments of treewidth 2.*

Since f is the 0 map, no vertices are deleted to get the plane pseudograph G' from the plane pseudograph G (only edges are added). So within the scope of this thesis, the f term in generating adjustments is redundant. We include it for the sake of generalisation, as it could be relevant for the treewidth 1 almost-partitions case (and to be consistent with [42]).

We use these results to prove Theorem 7.2.

Theorem 7.2. *For each $g, p, k, a \in \mathbb{N}$, there exists $m \in \mathbb{N}$ and $c, \alpha \in \mathbb{R}^{\geq 0}$ such that the following holds. For each $n \in \mathbb{N}$, $d \in \mathbb{R}^{> 0}$ with $d \geq \alpha\sqrt{n}$, and n -vertex pseudograph G , if*

G admits a (g, p, k, a) -almost-embedding Γ , then G admits a $(2, m, \mathcal{AC}(\Gamma))$ -concentrated almost-partition of treewidth at most 2, width at most d , and loss at most cn/d .

Proof of Theorem 7.2. Let $m_1, m_2, c_1, c_2, \beta$ be from Theorem 7.4 with $b := 2$. Set $m := m_2$, $c := c_2$, and $\alpha := 12\beta\sqrt{3}$.

Let $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ be defined by $f(n) := 0$ and $g(n) := 12\sqrt{3n}$ for each $n \in \mathbb{N}$. By Lemma 7.5, (f, g) generate adjustments of treewidth 2.

Fix $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq \alpha\sqrt{n} = \beta g(n)$, and a (g, p, k, a) -almost-embedding Γ of an n -vertex pseudograph G . By Theorem 7.4, G admits a $(2, m, \mathcal{AC}(\Gamma))$ -concentrated almost-partition of treewidth at most 2, width at most d , and loss at most $(c_1 f(n) + c_2 n)/d = cn/d$ (as $f(n) = 0$), as desired. $\square$

For the reasons explained at the beginning of this chapter, we omit the proof of Lemma 7.5. See [42] for the proof. So the remainder of this chapter is devoted to proving Theorem 7.4.

7.1.3 Partitioning a Vortex

We now discuss how to almost-partition a pseudograph that admits an almost-embedding. We assume that an almost-partition of the embedded pseudograph is given. The question is ‘how can we use the almost-partition of the embedded pseudograph to find an almost-partition of the almost-embedding’? We can safely delete (put in the loss set) the apices, as there are only a few of them. So this is really a question of how to almost-partition the vortex.

One idea (which is loosely what is done by Distel et al. [45] to obtain the treewidth 4 blowup result) is as follows. Add edges along the boundary of each disc so that they are snug. Then, find a few boundary vertices along the disc such that each interval of the underlying cycle between these points is small. We refer to these points as the breakpoints. All vertices in bags indexed by breakpoints will be deleted. The remaining intervals get contracted, and we find a partition of the resulting embedded pseudograph. We then uncontract the intervals to get the final partition.

This method works for Distel et al. [45] as (using a result from Distel et al. [49]) they find a treewidth 3 partition of (simple) embedded graphs of width $O(\sqrt{n})$ for which each part only contains a fixed number of the contracted vertices. Thus, when the contraction is undone, the total number of vertices in each part is small. However, we cannot use this partition, since we need a treewidth 2 partition.

Recall that Distel et al. [45] showed that each graph of Euler genus g is a $O_g(\sqrt{n})$ -blowup of a treewidth 2 graph (Theorem 2.10). Can we use this partition instead? The immediate answer is no, because unlike before, this partition cannot control how many contracted vertices are in a single part.

What if, instead of contracting at the start, we instead find the partition first, choose the breakpoints with this partition in mind, and then merge parts according to the intervals? The difficulty now is ensuring that each part only meets one interval. Some parts might go between vertices far apart in the cycle, or even touch the cycle many times. However, if the part is connected and goes between two intervals, then these intervals can be contracted together, even if they are far apart in the cycle. So, provided that we can work with connected partitions, we can instead focus on contracting groups of intervals, connected by common parts that they intersect.

Another annoyance is the fact that, even if the parts are connected, they can cross in the cycle. That is, we can find four vertices in cyclic order such that the first and third are contained in one part, and the second and fourth are contained in a different part. However, this cannot occur if the embedded pseudograph was a plane pseudograph (and the partition is connected). In a plane pseudograph, these parts would block each other, separating the pseudograph. See Figure 7.1. We prove this.

Lemma 7.6. *Let G be a plane pseudograph, let D be a G -clean disc, set $B := B(D, G)$, and let $\preceq_U$ denote the cyclic ordering of B induced by D . Then for any pair of disjoint connected sub-pseudographs H_1, H_2 of G , there does not exist pairwise distinct $x, y, a, b \in B$ with $x \prec_U a \prec_U y \prec_U b$ and $x, y \in V(H_1)$, $a, b \in V(H_2)$.*

Proof. Presume that such vertices $x, y, a, b \in B$ exist. Set $U := U(D, G)$. Since x, y, a, b are all distinct, $|V(U)| \geq 4$, and thus U is a cycle. Observe that we can add (possibly parallel) edges along the boundary of D to find a spanning plane super-pseudograph G' of G such that D is G' -clean, snug in G' , and such that $U(D, G') = U$. So $U \subseteq G'$.

Let G^* be the plane pseudograph obtained from G' by embedding a new vertex q inside D , and, for each $v \in V(U)$, embedding a new edge with endpoints q and v inside D (this can be done so that the interiors of the edges are disjoint). We will construct a K_5 -minor in G^* , contradicting Theorem 1.1. (Note that Theorem 1.1 still holds for pseudographs, as the absence or presence of parallel edges and loops does not change the Euler genus or excluded minors of a pseudograph.)

As H_1 is connected, there exists a path P_1 from x to y in H_1 . Let y' be the first vertex on this path such that $y' \in B$ and $a \prec_U y' \prec_U b$, and let x' be the last vertex on this path before y' such that $b \prec_U x' \prec_U a$. Since $x \prec_U a \prec_U y \prec_U b$, note that x' and y'

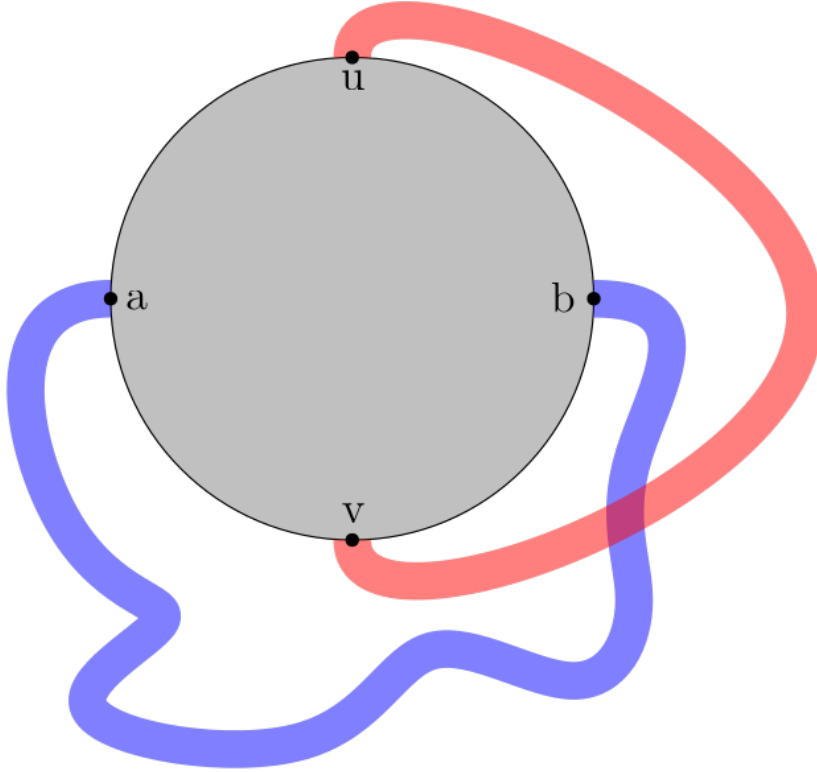


Figure 7.1: A diagram of a plane graph with a clean disc in grey. Any connected subgraph (in red) containing the points u and v must intersect any connected subgraph (in blue) containing the points a and b . So the red and blue subgraphs cannot correspond to parts of a connected partition.

are well-defined and distinct. Notice also $x', y' \in V(H_1)$, and that $x' \prec_U a \prec_U y' \prec_U b$. Let $P_{x'y'}$ be the subpath of P_1 from x' to y' (inclusive). By choice of x' and y' (and since $a, b \in V(H_2)$, which is disjoint to $V(H_1) \supseteq P_1$), observe that $P_{x'y'}$ is internally disjoint to B (since for each $z \in B \setminus \{a, b\}$, either $a \prec_U z \prec_U b$ or $b \prec_U z \prec_U a$).

By a symmetric argument, we can find vertices $a', b' \in V(H_2)$ such that $x' \prec_U a' \prec_U y' \prec_U b'$, and a path $P_{a'b'}$ from a' to b' in H_2 whose interior is disjoint to $B = V(U)$. Since $x' \prec_U a' \prec_U y' \prec_U b'$ and since U is a cycle, we can find internally disjoint paths $P_{x'a'}, P_{a'y'}, P_{y'b'}, P_{b'x'}$ in U such that, for each $P_{uv} \in \{P_{x'a'}, P_{a'y'}, P_{y'b'}, P_{b'x'}\}$, the endpoints of P_{uv} are u and v . Thus and since H_1 and H_2 are disjoint, observe that the paths $P_{x'a'}, P_{a'y'}, P_{y'b'}, P_{b'x'}, P_{x'y'}, P_{a'b'}$ are pairwise internally disjoint, contained in $G^* - \{q\}$, and that each path P_{uv} in this list has endpoints u and v . Thus and since q is adjacent in G^* to each vertex in B , which includes x', y', a' , and b' , we can find a K_5 minor in G^* (with $\{x'\}, \{y'\}, \{a'\}, \{b'\}$, and $\{q\}$ being the branch sets), a contradiction. The lemma follows. $\square$

While Lemma 7.6 might seem like a relatively mundane result, it is absolutely critical

to the proof in multiple places. As such, we name it the *non-crossing property*. Because of the non-crossing property, in the case when the embedded pseudograph is plane, we can make progress on finding these groups of intervals. We withhold on explaining the details until Section 7.3, when we tackle that part of the proof.

Conveniently (and for unrelated reasons) in a plane-triangulation-like setting (see Section 7.1.8), the algorithm of Distel et al. [45] gives a connected partition. This provides the structure needed to make use of the non-crossing property.

Now, to employ Lemma 7.6, we need the embedded pseudograph to be plane. Of course, this is not always the case. However, via a tedious and technical process, we can reduce the (g, p, k, a) -almost-embeddable graph to a very similar pseudograph that admits something that is very close to a $(0, 1, k, 0)$ -almost-embedding (with the same k). Note that we have also reduced the number of discs to 1. This is very useful, since only one vortex needs to be almost-partitioned, rather than multiple vortices at the same time.

7.1.4 Plane+Quasi-Vortex Embeddings

First, we describe what we reduce to. We get close to a $(0, 1, k, 0)$ -almost-embedding, except there is a small sub-pseudograph in the plane acting as an obstruction. Specifically, the vertices of this sub-pseudograph lie within the interior of the disc we would like to attach the vortex to. Deleting the vertices of this sub-pseudograph gives a $(0, 1, k, 0)$ -almost-embedding.

This motivates the following definitions, which parallel the definitions of clean-discs and $(0, 1, k, 0)$ -almost-embeddings. Let G be a pseudograph embedded in a surface Σ , and let $S \subseteq V(G)$. Say that a closed disc in Σ is *(G, S) -clean* if:

1. D is $(G - S)$ -clean, and
2. the interior of D contains S .

Note that a G -clean disc is also $(G, \emptyset)$ -clean. A *(G, S) -clean disc* for some embedded pseudograph G and $S \subseteq V(G)$ means a closed disc in the same surface that G is embedded in that is (G, S) -clean.

We define $B(D, G, S) := B(D, G - S)$ and $U(D, G, S) := U(D, G, S)$. So $V(U(D, G, S)) = B(D, G, S)$. D is *S -snug* in G if D is snug in $G - S$.

A pseudograph H is a *(D, G, S) -quasi-vortex* if $V(H)$ is disjoint to S and H is a $(D, G - S)$ -vortex. Note that $V(H) \cap V(G) = V(H) \cap V(G - S) = B(D, G - S) = B(D, G, S)$. A *(D, G, S) -quasi-vortex-decomposition* of H is a $(D, G - S)$ -vortex-decomposition of

H and the (D, G, S) -quasi-vortex-width of H is the $(D, G - S)$ -vortex-width of H .

A *plane+quasi-vortex embedding* Λ of a pseudograph G is a tuple $(G_0^+, W, D, H, \mathcal{J})$ such that:

1. G_0^+, H are sub-pseudographs of G , and $G = G_0^+ \cup H$,
2. G_0^+ is a plane pseudograph,
3. W is a connected and nonempty plane sub-pseudograph of G_0^+ ,
4. D is a $(G_0^+, V(W))$ -clean disc, and
5. H is a $(D, G_0^+, V(W))$ -quasi-vortex, and $\mathcal{J}$ is a $(D, G_0^+, V(W))$ -quasi-vortex-decomposition of H .

Call G the *pseudograph* (of Λ), G_0^+ the *plane sub-pseudograph*, W the *obstruction sub-pseudograph*, D the *disc*, H the *quasi-vortex*, and $\mathcal{J}$ the *quasi-vortex-decomposition*. Say that the *quasi-vortex-width* of Λ is the width of $\mathcal{J}$. Let $G_0 := G_0(\Lambda)$ be the plane pseudograph $G_0^+ - V(W)$. Since D is $(G_0^+, V(W))$ -clean, D is G_0 -clean. Set $B(\Lambda) := B(D, G_0^+, V(W)) = B(D, G_0) = V(G_0^+ \cap H)$, and $U(\Lambda) := U(D, G_0^+, V(W)) = U(D, G_0)$. So $V(U(\Lambda)) = B(\Lambda) = V(G_0^+ \cap H)$.

Note that the quasi-vortex-decomposition $\mathcal{J}$ explicit in the definition of a plane+quasi-vortex embedding (rather than implicitly choosing one of the correct width). This is because there are future definitions that depend on the choice of $\mathcal{J}$.

As mentioned before, we can delete the obstruction sub-pseudograph to obtain a $(0, 1, k, 0)$ -almost-embedding. Specifically, we have the following observation, which we do not use but include for intuition.

Observation 7.7. *Let $k \in \mathbb{N}$, and let $\Lambda = (G_0^+, W, D, H, \mathcal{J})$ be a plane+quasi-vortex embedding with quasi-vortex-width at most k of a pseudograph G . Then $(\mathbb{R}^2, 1, \emptyset, (G_0(\Lambda), H), (D))$ is a $(0, 1, k, 0)$ -almost-embedding of $G - V(W)$.*

Proof. Note that $W \subseteq G_0^+ \subseteq G$, so $G - V(W)$ is well-defined.

Recall that $G_0 := G_0(\Lambda)$ is the plane pseudograph $G_0^+ - V(W)$. As H is a $(D, G_0^+, V(W))$ -vortex, recall that $V(H)$ is disjoint to $V(W)$. Thus, $G - V(W) = (G_0^+ \cup H) - V(W) = (G_0^+ - V(W)) \cup H = G_0 \cup H$. By definition of a $(G_0^+, V(W))$ -clean disc and a $(D, G_0^+, V(W))$ -vortex (and the $(D, G_0^+, V(W))$ -vortex-width), D is G_0 -clean and H is a (D, G_0) -vortex with (D, G_0) -vortex-width at most k . The observation follows. $\square$

There are a few extra properties that our plane+quasi-vortex embeddings should satisfy. Firstly, D should be $V(W)$ -snug. This is because we want to be able to contract along $U(D, V(W))$. Secondly, the boundary vertices of the disc should be adjacent to a vertex

in $V(W)$. This is because we perform modifications to parts that intersect $B(D, V(W))$. Specifically, we add and remove vertices in $B(D, V(W))$ to these parts. This condition ensures that if a part was adjacent to $V(W)$ beforehand, it will also be afterwards. This is relevant, as $V(W)$ will be a part of the partition (both before and after modification).

This leads to the following definition. Say that a plane+quasi-vortex embedding $\Lambda = (G_0^+, W, D, H, \mathcal{J})$ is *standardised* if:

1. D is $V(W)$ -snug in G_0^+ , and
2. $B(D, G_0^+, V(W)) \subseteq N_{G_0^+}(V(W))$.

The former condition is equivalent to requiring that D is snug in $G_0(\Lambda)$. This condition is slightly different from the one that appears in [42], which essentially only requires that specific subgraphs of the underlying cycle are contained in G_0^+ . However, the missing edges can be added (possibly as parallel edges) along the boundary of the disc, to find a spanning sub-pseudograph that does have this stronger property. For our purposes, taking a spanning super-pseudograph is not an obstacle, as the reduction is already quite imprecise (see Section 7.1.6). It also makes little difference to the proof; this change is made only to simplify and re-use terminology.

7.1.5 Finding a Reduction

Our goal is to reduce to a plane+quasi-vortex embedding. As mentioned at the beginning of this chapter, we do not include the proof of this reduction in this thesis (see [42] instead). However, it is worth briefly explaining the key ideas, even if only to motivate the definitions. See also Figure 7.2 for a (partial) example.

To reduce the genus of the surface, we use an algorithm from Dujmović et al. [58] (which was then repeated by Distel et al. [49]). Buried within the proof of key results in these papers, the authors describe a way to cut open a specific sub-pseudograph of an embedded pseudograph to obtain a plane pseudograph. This can be extended to an algorithm to cut open an almost-embedding.

There are a few key features of this algorithm. Firstly, it takes a sub-pseudograph in the embedded pseudograph (the ‘cutting sub-pseudograph’), and splits its vertices and edges. Let Y be the vertices that were split up, and let W be the sub-pseudograph consisting of all the post-split vertices and edges (the ‘cut sub-pseudograph’). W will always be connected (in fact, it is a cycle bounding a face; see Distel [42, Lemma A.2.1]).

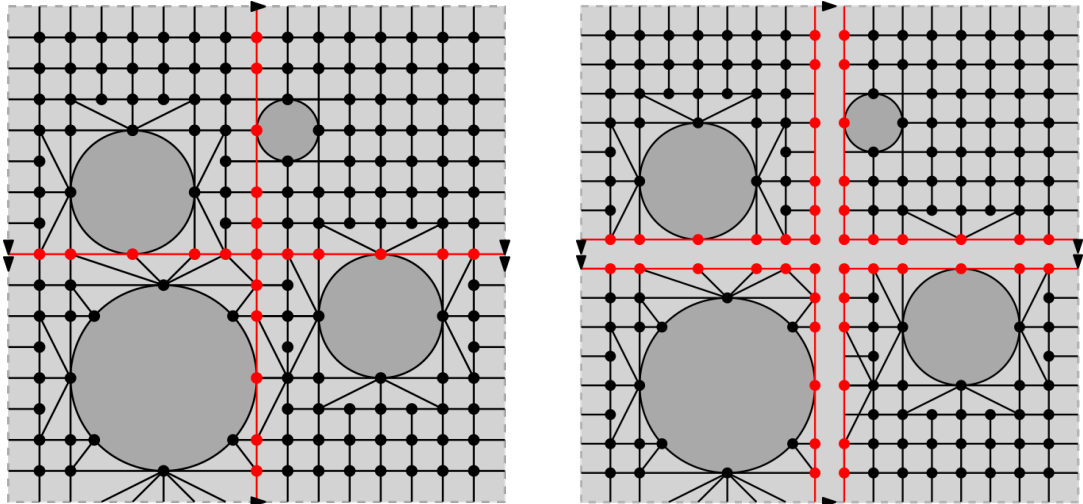
The algorithm leaves the vortices mostly untouched. The exception is vertices lying in a bag J_v of a vertex $v \in Y$. v corresponds to multiple vertices in $V(W)$, and these

may be far apart. As such, we cannot naturally translate our vortex-decomposition. However, once we delete the vertices in (all such bags) J_v from the vortex, the remainder of the vortex translates to a vortex of the plane pseudograph.

The algorithm chooses the cutting sub-pseudograph so that these vortices attach near W . Specifically, after deleting $V(W)$, all the discs lie in a common face. Using this, the algorithm finds a big disc that captures the union of all the vortices as a single (quasi-)vortex. We omit several details here that enable this, however some key ideas include (1) adding handles so that all the original discs are contained in a single face and (2) removing at least one bag from each original vortex to split the underlying cycle into a path.

Since we are deleting several vertices in the vortex already, we may as well delete the apices as well. Call the set of deleted vertices X . Thus, the final pseudograph G' consists of the union of a plane pseudograph G_0^+ , and a quasi-vortex H . After deleting the plane sub-pseudograph W , H is an actual vortex.

Undoing the split allows us to recover the embedded pseudograph from G_0^+ . We can extend this to recover G (except for X) by treating the ‘undoing the split’ operation as the identity elsewhere. This defines a map $\phi : V(G') \mapsto V(G)$ that is the identity outside $V(W)$ and maps $V(W)$ to Y . (ϕ can map to X , as G' contains some vertices of X . These are boundary vertices x that were contained in a bag J_v that was deleted, but their bag J_x was not deleted. So x is needed in G' to index a bag of the quasi-vortex-decomposition, and is instead made isolated in G' .) A key feature of this map is that adjacent vertices in $G - X$ are the images of adjacent vertices in G' . Further, the cutting algorithm preserves facial triangles, in the sense that ϕ -image of facial triangles of G_0^+ gives all facial triangles in the original embedded sub-pseudograph G_0 , except for those that intersect X (which are not a concern, as X will be contained in the loss-set of the almost-partition).



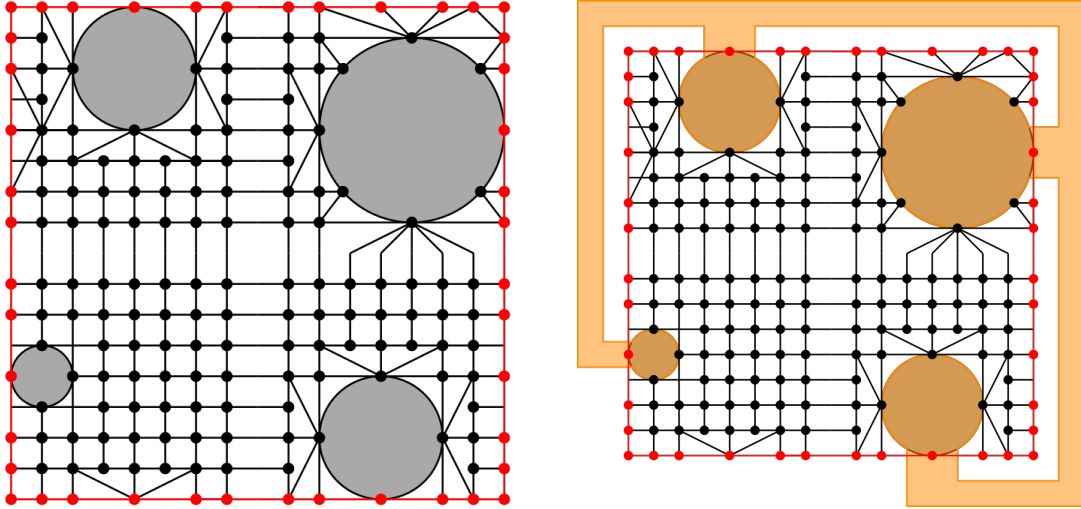


Figure 7.2: A diagram of how to cut open an almost-embedding. In this example, the embedded subgraph is embedded in the torus. The algorithm finds a special subgraph (red), which each disc (in grey) touches (after some modifications are performed). The red vertices of this subgraph are the split-vertices Y . This subgraph is cut open, splitting each edge into two and each vertex into several vertices (depending on the degree within the red subgraph), and creating a new face. The boundary of this face is a cycle, so it can be replaced with a disc. The result is a plane graph. In this example, the embedding into the plane is obtained by diagonally swapping the four quadrants. The split subgraph (the red cycle along the outside in the plane) is W . All the discs touch W , so ignoring W , they can be merged into a single disc (in orange).

7.1.6 Reductions

Let $\Gamma = (\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s))$ be an almost-embedding of a pseudograph G , and set $H := H(\Gamma)$. A *reduction* $\mathcal{R}$ for Γ is a tuple $(X, Y, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ such that:

1. G' is a pseudograph, and $(G_0^+, W, D, H', \mathcal{J}')$ is a standardised plane+quasi-vortex embedding of G' ,
2. $Y \subseteq V(G_0)$, and $A \subseteq X \subseteq V(G)$,
3. Y is disjoint to $V(G')$,
4. $H - (X \cap V(H)) \subseteq H' \subseteq H$, and
5. ϕ is a map from $V(G')$ to $V(G)$ such that:
 - (a) for each $v \in V(G - X)$, $N_{G-X}(v) \subseteq \phi(N_{G'}(\phi^{-1}(v)))$,
 - (b) ϕ is the identity on $V(G') \setminus V(W)$,
 - (c) $\phi(V(W)) = Y$, and
 - (d) for each $K \in F_3(\Gamma) = F_3(G_0)$ disjoint to X , there exists $K' \in F_3(G_0^+)$ with $\phi(K') = K$.

Call Y the *split-vertices*, X the *loss-set*, and ϕ the *projection*. G' is called the *modified pseudograph*, D' the *modified disc*, H' the *quasi-vortex*, and $\mathcal{J}'$ the *quasi-vortex-decomposition*. G_0^+ and W are called the *plane sub-pseudograph* and *obstruction sub-pseudograph* respectively, as before. Say that the *loss* of $\mathcal{R}$ is $|X|$, the *quasi-vortex-width* of $\mathcal{R}$ is the width of $\mathcal{J}'$.

Set $\Lambda(\mathcal{R}) := (G_0^+, W, D, H', \mathcal{J}')$, let $G'_0(\mathcal{R}) := G_0(\Lambda(\mathcal{R})) = G_0^+ - V(W)$, let $B'(\mathcal{R}) := B(\Lambda(\mathcal{R})) = B(D, G_0^+, V(W)) = B(D, G_0^+ - V(W)) = V(G_0^+ \cap H')$, and let $U'(\mathcal{R}) := U(\Lambda(\mathcal{R})) = U(D, G_0^+, V(W)) = U(D, G_0^+ - V(W))$. So $\mathcal{J}'$ is a planted $U'(\mathcal{R})$ -decomposition of H' , and D is $G'_0(\mathcal{R})$ -clean. We call $B'(\mathcal{R})$ the *modified boundary* and $U'(\mathcal{R})$ the *modified underlying cycle*.

The definition of $\mathcal{R}$ is so that the structure of G' loosely matches the structure of G . Particularly, most vertices are preserved, most adjacencies are preserved, most facial triangles are preserved, and most of the vortex is preserved. ϕ^{-1} is also well-behaved, as given by the following observation.

Observation 7.8. *Let Γ be an almost-embedding of a pseudograph G , and let $\mathcal{R} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ be a reduction of Γ . Then:*

1. $V(G) \setminus (Y \cup X) \subseteq V(G') \setminus V(W) \subseteq V(G) \setminus Y$,
2. $\phi^{-1}(Y) = V(W)$,
3. for each $v \in V(G - X)$, $\phi^{-1}(v)$ is nonempty, and
4. for each $Z \subseteq V(G) \setminus Y$, $\phi^{-1}(Z) \subseteq Z$. Further, if $Z \subseteq V(G') \setminus V(W)$, then $\phi^{-1}(Z) = Z$.

Proof. Since ϕ is the identity on $V(G') \setminus V(W)$, we have $V(G') \setminus V(W) \subseteq V(G)$. Since Y is disjoint to $V(G')$, it follows that $V(G') \setminus V(W) \subseteq V(G) \setminus Y$.

Since $\phi(V(W)) = Y$, observe that $V(W) \subseteq \phi^{-1}(Y)$. Presume, for a contradiction, that there exists $x \in \phi^{-1}(Y) \setminus V(W)$. So $\phi(x) \in Y$ and $x \in V(G') \setminus V(W)$. Thus, $x = \phi(x)$. We obtain $x = \phi(x) \in V(G') \cap Y$, which contradicts the fact that $V(G')$ and Y are disjoint. So $\phi^{-1}(Y) = V(W)$.

Fix $v \in V(G - X)$. Presume, for a contradiction, that $\phi^{-1}(v)$ is empty. Then $\phi(N_{G'}(\phi^{-1}(v)))$ is empty. However, observe that $v \in N_{G-X}(v)$. So $N_{G-X}(v) \not\subseteq \phi(N_{G'}(\phi^{-1}(v)))$, a contradiction. Thus, $\phi^{-1}(v)$ is nonempty.

Let $Z \subseteq V(G) \setminus Y$. Presume, for a contradiction, that there exists $x \in \phi^{-1}(Z) \setminus Z$. So $\phi(x) \in Z$ and $x \notin Z$. If $x \in V(G') \setminus V(W)$, we would have $x = \phi(x) \in Z$, contradicting the fact that $x \notin Z$. So $x \in V(W)$, and thus $\phi(x) \subseteq \phi(V(W)) = Y$. We obtain $\phi(x) \in Y \cap Z$, contradicting the choice of Z . So $\phi^{-1}(Z) \subseteq Z$, as desired.

If $Z \subseteq V(G') \setminus V(W) \subseteq V(G) \setminus Y$, note that $\phi^{-1}(Z)$ is well-defined. Since $Z \subseteq V(G') \setminus V(W)$, observe that $\phi(Z)$ is well-defined and $\phi(Z) = Z$. Thus, $Z \subseteq \phi^{-1}(Z)$. Combined with the previous paragraph, this gives $\phi^{-1}(Z) = Z$, as desired.

Fix $v \in V(G) \setminus (Y \cup X)$. Since $v \in V(G - X)$, there exists $x \in \phi^{-1}(v)$. Since $\{v\} \subseteq V(G) \setminus Y$, $\phi^{-1}(v) \subseteq \{v\}$. Thus, $\{v\} = \{x\} = \phi^{-1}(v)$. So $v \in \phi^{-1}(v) \subseteq V(G')$. Since $\phi(V(W)) = Y$ and since $\phi(v) = v \notin Y$, we obtain $v \notin V(W)$. So $v \in V(G') \setminus V(W)$. Thus, $V(G) \setminus (Y \cup X) \subseteq V(G') \setminus V(W)$. Combining this with our earlier result, we obtain $V(G) \setminus (Y \cup X) \subseteq V(G') \setminus V(W) \subseteq V(G) \setminus Y$, as desired. This completes the proof of the lemma. $\square$

There is one more important parameter of a reduction $\mathcal{R}$, but it requires some explanation. We want to partition G according to a partition of G' . However, it is not clear as to how to assign the vertices of Y to parts, as they correspond to multiple vertices of W . The easiest solution is to just force all the vertices of the W into a single part. However, this creates a new problem, because $V(W)$ might be large. The solution is to find vertices whose deletion breaks G into components that each only contain a small number of split-vertices. We include these vertices in X . We then partition each component according to how we would have partitioned the entire pseudograph. Now, the part corresponding to the split-vertices is small for each component.

This leads to the following definition. Let Γ be an almost-embedding of a pseudograph G , and let $\mathcal{R} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ be a reduction of Γ . Let $\mathcal{C}(\mathcal{R})$ be the set of connected components of $G - X$. Define the *remainder-width* to be the maximum intersection between $V(C)$ and Y for some $C \in \mathcal{C}(\mathcal{R})$.

It is worth noting that the loss-set X serves two roles. Firstly, it contains any vertices that cannot be described using the plane+quasi-vortex embedding. Second, it controls the remainder-width. We do not distinguish between these uses because X will be the loss set of the almost-partition.

For $k \in \mathbb{N} \cup \{\infty\}$ and $w, q \in \mathbb{R}^{\geq 0} \cup \{\infty\}$, say that a reduction $\mathcal{R}$ of an almost-embedding Γ (of a pseudograph G) is a (k, w, q) -*reduction* of Γ if it has quasi-vortex-width at most k , remainder-width at most w , and loss at most q .

The following theorem establishes the existence of a well-behaved reduction. See [42] for the proof.

Theorem 7.9. *For all $g, p, k, a \in \mathbb{N}$, there exists $c, q \in \mathbb{N}$ such that the following holds. For each nontrivial (g, p, k, a) -almost-embedding of a pseudograph G , there exists a layering $\mathcal{L}$ of G and a (k, ∞, q) -reduction of Γ whose split vertices Y satisfy $|Y \cap L| \leq c$ for each $L \in \mathcal{L}$.*

Recall that an almost-embedding is nontrivial if it has vertices other than the apex vertices. The nontrivial condition in Theorem 7.9 exists to avoid some annoying and uninteresting edge cases (and because we want the obstruction subgraph W to be nonempty).

We believe that Theorem 7.9 may be of independent interest, as a general strategy to simplify an almost-embedding.

To bound the remainder-width, we employ the following observation.

Observation 7.10. *Let $b, n \in \mathbb{N}$, let $\mathcal{L}$ be a layering of an n -vertex pseudograph G , and let $S \subseteq V(G)$ be such that for each $L \in \mathcal{L}$, $|L \cap S| \leq c$. Then for any $d \in \mathbb{R}^{>0}$, there exists $X \subseteq V(G)$ with $|X| \leq cn/d$ such that each connected component of $G - X$ contains at most d vertices in S .*

Proof. Let $\mathcal{L} = (L_0, L_1, \dots, L_m)$ be the layers in the order they appear in the path, and let $q := \lfloor d/c \rfloor + 1$. Note that $n/q \leq cn/d$ and that $c(q-1) \leq d$. Set also $L_i := \emptyset$ for each $i \in \mathbb{Z} \setminus \{0, \dots, m\}$. For each $i \in \{0, \dots, q-1\}$, $L'_i := \bigcup_{\{j \in \{0, \dots, m\} : j \equiv i \pmod{q}\}} L_j$. Observe that $(L'_0, \dots, L'_{q-1})$ is a partition of G into exactly q parts. Thus, there exists $i \in \{0, \dots, q-1\}$ such that $|L'_i| \leq n/q \leq cn/d$. Set $X := L'_i$. For each connected component C of $G - X$, observe that there exists $j \in \mathbb{Z}$ such that $V(C) \subseteq L_{qj+i+1} \cup \dots \cup L_{q(j+1)+i-1}$. In particular, C intersects at most $q-1$ layers of $\mathcal{L}$. Thus, $|V(C) \cap S| \leq c(q-1) \leq d$, as desired. $\square$

More vertices can be added to the loss-set of a reduction, at the cost of increasing the loss.

Observation 7.11. *Let $k \in \mathbb{N}$, $w \in \mathbb{R}^{\geq 0} \cup \{\infty\}$, and $q, q' \in \mathbb{R}^{\geq 0}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{R} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ be a (k, w, q) -reduction of Γ , and let $X' \subseteq V(G)$ with $|X'| \leq q'$. Then $\mathcal{R}' := (Y, X \cup X', G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is a $(k, w, q + q')$ -reduction of Γ .*

Proof. It is easy to check that $\mathcal{R}'$ is a reduction of Γ of the same quasi-vortex-width (which is at most k) and at most the same remainder-width (which is at most w). The loss of $\mathcal{R}'$ is $|X \cup X'| \leq |X| + |X'| \leq q + q'$. Thus, $\mathcal{R}'$ is a $(k, w, q + q')$ -reduction of Γ , as desired. $\square$

We employ Observations 7.10 and 7.11 to convert the reduction from Theorem 7.9 to one with bounded remainder-width.

Corollary 7.12. *For all $g, p, k, a \in \mathbb{N}$, there exists $c, q \in \mathbb{N}$ such that the following holds. For each $n \in \mathbb{N}$, every nontrivial (g, p, k, a) -almost-embedding Γ of an n -vertex pseudograph G , and every $d \in \mathbb{R}^{>0}$, there exists a $(k, d, cn/d + q)$ -reduction of Γ .*

Proof. Let c, q be from Theorem 7.9. By Theorem 7.9, there exists a layering $\mathcal{L}$ of G and a (k, ∞, q) -reduction $\mathcal{R} = (Y, X \cup X', G', G_0^+, W, D, H', \mathcal{J}', \phi)$ of Γ such $|Y \cap L| \leq c$ for each $L \in \mathcal{L}$. By Observation 7.10 (with $S := Y$), there exists $X' \subseteq V(G)$ with $|X'| \leq cn/d$ such that each connected component of $G - X'$ contains at most d vertices in Y . By Observation 7.11, $\mathcal{R}' := (Y, X \cup X', G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is a $(k, \infty, cn/d + q)$ -reduction of Γ . Further, each connected component of $G - (X \cup X')$ is contained in a connected component of $G - X'$, and thus has at most d vertices of Y . So $\mathcal{R}'$ has remainder width at most d . Thus, $\mathcal{R}'$ is a $(k, d, cn/d + q)$ -reduction of Γ , as desired. $\square$

7.1.7 Adjustments

We need a good almost-partition of the plane sub-pseudograph G_0^+ , since this will be the starting point for finding an almost-partition for the pseudograph G . However, we only find these good almost-partitions for very specific plane pseudographs (resembling a triangulation). So we adjust G_0^+ so that it becomes one of these specific plane pseudographs. This adjustment needs to be mindful of the disc D and obstruction sub-pseudograph W , but we can strip away all other information from the reduction. This leads to the following definition. An *adjustable triple* is a triple (G, W, D) such that:

1. G is a plane pseudograph,
2. W is a connected and nonempty plane sub-pseudograph of G , and
3. D is a $(G, V(W))$ -clean disc.

Observation 7.13. *Let Γ be an almost-embedding, and let $\mathcal{R} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ be a reduction of Γ . Then (G_0^+, W, D) is an adjustable triple.*

Proof. Follows directly from the definition of a plane+quasi-vortex embedding. $\square$

We use Observation 7.13 implicitly from now on.

To adjust G_0^+ , we embed new edges and delete vertices. The main restrictions are: (1) we do not delete vertices from the obstruction sub-pseudograph W (as we need to keep it connected), and (2) the edges should not destroy facial triangles or prevent the disc from being clean (after deleting $V(W)$). (We are not worried about facial triangles

lost by deleting vertices, as these become cliques of size at most 2.) This leads to the following definition.

An *adjustment* for an adjustable triple (G, W, D) is a pair $\mathcal{A} = (G', S)$ such that:

1. $S \subseteq V(G) \setminus V(W)$,
2. G' is a spanning plane super-pseudograph of $G - S$,
3. for each $K \in \mathcal{F}_3(G)$ disjoint to S , $K \in \mathcal{F}_3(G')$, and
4. D is $(G', V(W))$ -clean.

The first two conditions ensure that W is a (connected and nonempty) sub-pseudograph of G' , so the fourth condition is well-defined. Say that the *loss* of $\mathcal{A}$ is $|S|$.

For our purposes, S will be empty, and G' will be a triangulation-like super-pseudograph. However, we allow for S to be nonempty to maximise the chance that our lemmas can be reused in the future (specifically, towards a proof that $\text{tpw}_2(G) \in O_h(\sqrt{n})$ for n -vertex K_h -minor-free G).

An *adjustment* for a reduction $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is an adjustment for (G_0^+, W, D) . We have a method of applying an adjustment to a reduction to obtain a similar reduction with many of the same properties.

Lemma 7.14. *Let $k \in \mathbb{N}$, $w \in \mathbb{R}^{\geq 0} \cup \{\infty\}$, and $q, q' \in \mathbb{R}^{\geq 0}$, let $\mathcal{R}$ be a (k, w, q) -reduction of an almost-embedding Γ (of a pseudograph G), and let $\mathcal{A} = (G_0^-, S)$ be an adjustment for $\mathcal{R}$ of loss at most q' . Then there exists a $(k, w, (k+1)q' + q)$ -reduction $\mathcal{R}'$ for Γ such that:*

1. *the plane sub-pseudograph of $\mathcal{R}'$ is G_0^- , and*
2. *the split vertices Y , obstruction sub-pseudograph W , and disc D of $\mathcal{R}'$ are the split vertices, obstruction sub-pseudograph, and disc of $\mathcal{R}$ respectively.*

The proof of Lemma 7.14 is rather tedious and technical. However, the technical detail relates to dealing with modified boundary vertices in S , as this requires changing the quasi-vortex. In our case, S is empty ($q' = 0$). In this case, the proof is easy, as the new reduction only replaces the plane sub-pseudograph (with G_0^-). All desired properties follow from the definition of reductions and adjustments. We do not include the proof of the general case, but for completeness of this thesis, we include the proof of this special case. See [44] for the proof of the general case (Lemma 7.14).

Observation 7.15. *Let $k \in \mathbb{N}$ and $w, q \in \mathbb{R}^{\geq 0} \cup \{\infty\}$, let $\mathcal{R} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ be a (k, w, q) -reduction of an almost-embedding Γ (of a pseudograph G), and let $\mathcal{A} = (G_0^-, \emptyset)$ be an adjustment for $\mathcal{R}$ (of loss 0). Then $(Y, X, G_0^- \cup H', G_0^-, W, D, H', \mathcal{J}', \phi)$ is a (k, w, q) -reduction for Γ .*

Proof. Since $(G_0^-, \emptyset)$ is an adjustment for $\mathcal{R}$, G_0^- is a spanning plane super-pseudograph of G_0^+ , D is $(G_0^-, V(W))$ -clean, and $F_3(G_0^+) \subseteq F_3(G_0^-)$. Observe that $U(D, G_0^-, V(W)) = U(D, G_0^+, V(W))$ (as G_0^+ is spanning). From this, it is easy to check that $(Y, X, G_0^- \cup H', G_0^-, W, D, H', \mathcal{J}', \phi)$ is a (k, w, q) -reduction of Γ . $\square$

We mentioned that we wanted a good partition of the plane sub-pseudograph G_0^+ . What exactly does this mean? Firstly, and most importantly, the partition should be connected, so that the non-crossing property (Lemma 7.6) can be exploited. Secondly, $V(W)$ should be a part of the partition. This is because we want to handle it separately to all other parts (using the remainder-width). Thirdly, we want the partition to have small treewidth. Finally, it would be ideal if the partition had bounded width. However, the requirement that $V(W)$ is a part will normally make this impossible, as W has unbounded size. The compromise is that $V(W)$ is the only large part.

We need to ensure that for any possible reduction, there exists an adjustment whose new plane sub-pseudograph admits a partition with all of these properties. This leads to the following definition. Fix $b \in \mathbb{N}$. Say that a pair of functions $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ *generate adjustments of treewidth b* if for each $n \in \mathbb{N}$, every $d \in \mathbb{R}^{\geq 0}$ with $d \geq g(n)$, and each adjustable triple (G, W, D) with $|V(G)| - |V(W)| + 1 \leq n$, there exists a triple $(G', S, \mathcal{P})$ such that:

1. (G', S) is an adjustment of (G, W, D) of loss at most $f(n)/d$, and
2. $\mathcal{P}$ is a connected partition of G' of treewidth at most b with $V(W) \in \mathcal{P}$ such that each $P \in \mathcal{P}$ distinct from $V(W)$ has size at most d .

Since W is nonempty, a part is distinct from $V(W)$ if and only if it is not equal to $V(W)$. We only allow partitions of G' , rather than almost-partitions, as any would-be loss vertices can instead be added to S (although this is not needed in this thesis). The condition on the number of vertices of G should be thought of as ‘after contracting $V(W)$ into a single vertex, the result should have at most n vertices’. Since W is nonempty, this reduces the number of vertices by $|V(W)| - 1$. We are not concerned with the size of $V(W)$ (as it is already large, and will be split in a later step), so it is effectively equivalent to finding a partition of the pseudograph when $V(W)$ is contracted. This reduces the number of vertices in the pseudograph, which is ideal, given that we expect the sizes of the parts to depend on the number of vertices. In particular, this is useful because G_0^+ can have more vertices than G , but only because of $V(W)$ being larger than Y . ($V(W)$ actually contains a bounded number of vertices more than Y ; see Distel [42, Lemma A.2.1].) This detail allows us to bound the size of the parts in terms of $|V(G)|$, rather than $|V(G')|$.

7.1.8 Generating Adjustments

Recall that Lemma 7.5 says that if f is the zero map and $g = 12\sqrt{3n}$, then (f, g) generate treewidth 2 adjustments. Since f is the zero map, this means that there is a spanning super-pseudograph G' that (1) contains W , (2) preserves every facial triangle of G , and (3) admits a connected treewidth 2 partition that contains $V(W)$ as a part. As mentioned at the beginning of this chapter, we do not give the full proof of Lemma 7.5 in this thesis. However, we explain the key ideas here, to motivate the definition of adjustments.

Edges are added until a triangulation-like setting called a *plane quasi-triangulation* is reached. This is a plane pseudograph where for each face F , the boundary vertices B_F of F induce a clique. This clique can be of size smaller than 3 due to parallel edges or loops, and can include edges that do not bound the face. We remark, without proof, that for any simple plane graph G with at least 3 vertices, if G is a plane quasi-triangulation then G is a plane triangulation (every face is a triangle-face). However, deleting all parallel edges and loops from a plane quasi-triangulation may result in a simple plane graph that is not a triangulation, so quasi-triangulations are more general than triangulations. By adding edges inside large faces (those with more than 3 boundary vertices), it is easy to turn an arbitrary plane pseudograph into a plane quasi-triangulation (without destroying any facial triangles). See [42] for the proof.

In a plane quasi-triangulation, any minimal separator between connected and disjoint subgraphs induces a connected subgraph. This also holds if we first delete a connected subgraph. We believe that this is a known fact in the case of plane triangulations (see [83] for a similar lemma), but since plane quasi-triangulations are a non-standard setting, it must be proved from scratch, which is rather tedious.

The algorithm of Distel et al. [45] that constructs treewidth 2 partitions of embedded graphs works by finding a (minimal) separator between two existing parts and then making that separator plus several paths between those parts into a new part. The properties of a quasi-triangulation then guarantee that the partition is connected, because the separator induces a connected subgraph and the paths intersect the separator. Additionally, the algorithm allows for one part to be fixed in advance, which we chose to be $V(W)$. Thus, the proof of Lemma 7.5 largely just repeats the original proof of Distel et al. [45] with these observations in mind, along with the proofs regarding quasi-triangulations. It is also not short, hence the decision to omit the proof from this thesis.

7.1.9 Partitioned-Reductions

It is convenient to come up with a term for a reduction for which we have found a desired partition of the plane sub-pseudograph. This leads to the following definition.

A *partitioned-reduction* of an almost-embedding Γ (of a pseudograph G) is a collection $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ such that:

1. $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is a reduction of Γ , and
2. $\mathcal{P}_0^+$ is a connected partition of G_0^+ containing $V(W)$.

As before, we call $Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi$ the *split-vertices*, *loss-set*, *modified pseudograph*, *plane sub-pseudograph*, *obstruction sub-pseudograph*, *quasi-vortex*, *quasi-vortex-decomposition*, and *projection* of $\mathcal{PR}$ respectively. We call $\mathcal{P}_0^+$ the *plane partition* of $\mathcal{PR}$. We set $\mathcal{R} := \mathcal{R}(\mathcal{PR}) := (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$, $B'(\mathcal{PR}) := B'(\mathcal{R})$, $U'(\mathcal{PR}) := U'(\mathcal{R})$, $G'_0(\mathcal{PR}) := G'_0(\mathcal{R})$, $\mathcal{C}(\mathcal{PR}) := \mathcal{C}(\mathcal{R})$, and $\Lambda(\mathcal{PR}) := \Lambda(\mathcal{R})$. Say that the *quasi-vortex-width*, *remainder-width*, and *loss* of $\mathcal{PR}$ are the quasi-vortex-width, remainder-width, and loss of $\mathcal{R}$, the *treewidth* of $\mathcal{PR}$ is the treewidth of $\mathcal{P}_0^+$, and that *near-width* of $\mathcal{PR}$ is the maximum size of a part $P \in \mathcal{P}_0^+$ distinct from $V(W)$.

For $b, k \in \mathbb{N} \cup \{\infty\}$ and $w, w', q \in \mathbb{R}^{\geq 0} \cup \{\infty\}$, say that a partitioned-reduction $\mathcal{PR}$ of an almost-embedding Γ (of a pseudograph G) is a *(b, k, w, w', q) -partitioned-reduction* of Γ if it has treewidth at most b , quasi-vortex-width at most k , near-width at most w , remainder-width at most w' , and loss at most q .

Given a pair of functions (f, g) that generate adjustments, we use Lemma 7.14 to turn reductions into partitioned-reductions. (Within the scope of this thesis, Observation 7.15 suffices, since $f = 0$ in Lemma 7.5.)

Lemma 7.16. *Let $b, k, n \in \mathbb{N}$ and $w, q \in \mathbb{R}^{\geq 0}$, let $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ be such that (f, g) generate adjustments of treewidth b , and let $d \in \mathbb{R}^{\geq 0}$ be such that $d \geq g(n)$. If an almost-embedding Γ of an n -vertex pseudograph G admits a (k, w, q) -reduction, then Γ admits a $(b, k, d, w, (k+1)f(n)/d + q)$ -partitioned-reduction.*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi) := \mathcal{R}$. By Observation 7.8, $V(G') \setminus V(W) \subseteq V(G) \setminus Y$. So $|V(G')| - |V(W)| + 1 \leq |V(G)| - |Y| + 1$. Since W is nonempty and since $\phi(V(W)) = Y$, Y is nonempty. Thus, $|V(G)| - |Y| + 1 \leq |V(G)| = n$. So $|V(G')| - |V(W)| + 1 \leq n$. Recalling that $G_0^+ \subseteq G'$, this gives $|V(G_0^+)| - |V(W)| + 1 \leq n$.

By Observation 7.13, (G_0^+, W, D) is an adjustable triple. By definition of f, g (and since $d \geq g(n)$ and $|V(G_0^+)| - |V(W)| + 1 \leq n$), there exists a triple $(G_0^-, S, \mathcal{P}_0^+)$ such that:

1. (G_0^-, S) is an adjustment of G_0^+ of loss at most $f(n)/d$, and

2. $\mathcal{P}_0^+$ is a connected partition of G_0^- of treewidth at most b with $V(W) \in \mathcal{P}_0^+$ such that each $P \in \mathcal{P}_0^+$ distinct from $V(W)$ has size at most d .

By Lemma 7.14 (with $q' := f(n)/d$), there exists a $(k, w, (k+1)f(n)/d + q)$ -reduction $\mathcal{R}'$ for Γ such that:

1. the plane sub-pseudograph of $\mathcal{R}'$ is G_0^- , and
2. the split vertices Y , obstruction sub-pseudograph W , and disc D of $\mathcal{R}'$ are the split vertices, obstruction sub-pseudograph, and disc of $\mathcal{R}$ respectively.

So $\mathcal{R}'$ is of the form $(Y, X^+, G^-, G_0^-, W, D, H^-, \mathcal{J}^-, \phi^-)$ for some unknown (and unimportant) $X^+, G^-, H^-, \mathcal{J}^-, \phi^-$. (If $f(n) = 0$, we can instead apply Observation 7.15, and we have $(Y, X^+, G^-, G_0^-, W, D, H^-, \mathcal{J}^-, \phi^-) := (Y, X, G', G_0^-, W, D, H', \mathcal{J}', \phi)$.)

Let $\mathcal{PR} := (Y, X^+, G^-, G_0^-, W, D, H^-, \mathcal{J}^-, \phi^-, \mathcal{P}_0^+)$. It follows directly from the definitions of $\mathcal{R}'$ and $\mathcal{P}_0^+$ that $\mathcal{PR}$ is a $(b, k, d, w, (k+1)f(n)/d + q)$ -partitioned-reduction of Γ , as desired. $\square$

7.1.10 Raises

As stated earlier, the key difficulty is finding an appropriate breakpoints and an appropriate way to merge parts, so that we can then almost-partition the vortex. Specifically, it would be ideal to ensure that each interval that avoids the breakpoints only intersects one part. Then, we can put all quasi-vortex vertices that lie in a bag along that interval into the corresponding part. However, we do not achieve this. In fact, we consider this goal not realistic, since one part could intersect many boundary vertices of the disc, creating many small intervals between intersections. We would then have to merge all these small intervals into the part, or add breakpoints for each of these intervals, or some combination of these two methods. However, regardless of the method, either the width or the loss would be high.

Thus, instead of requiring that each interval only intersects one part, we instead search for a clear way of assigning vertices v of the quasi-vortex to a part. The assignment method will be completely determined by the interval induced by the vertices x whose bags J_x contain v . The part we assign v to will contain a modified boundary vertex x whose bag J_x contains v . Think of v as using x to decide its part. We allow modified boundary vertices to be assigned to a part different from the part that originally contained them. Think of the other vertices as being assigned to the parts that originally contained them.

This assignment describes a partition of the modified pseudograph. We then use

the inverse of the projection, along with the loss-set, to almost-partition the original pseudograph. We want to ensure that this creates no new edges in the quotient. We achieve this by ensuring that adjacent vertices in the modified pseudograph (that do not map to vertices in the loss set) are either assigned to the same part or adjacent parts. This leads to the following definition.

Let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of an almost-embedding Γ of a pseudograph G . Set $B' := B'(\mathcal{R})$ and set $(J'_x : x \in B') := \mathcal{J}'$. A *raise* for $\mathcal{PR}$ is a map $\tau : V(G') \mapsto \mathcal{P}_0^+$ such that:

1. for each $v \in V(G_0^+) \setminus B'$, $v \in \tau(v)$,
2. for each $v \in V(H')$, there exists $x \in B' \cap \tau(v)$ such that $v \in J'_x$, and
3. for each pair of adjacent vertices u, v in G' with $\phi(u), \phi(v) \in V(G - X)$, $N_{G_0^+}(\tau(u))$ intersects $\tau(v)$.

We remark that the final condition is equivalent to requiring that either $\tau(u) = \tau(v)$ or $\tau(u)$ is adjacent to $\tau(v)$ in $G_0^+/\mathcal{P}_0^+$.

For each $C \in \mathcal{C}(\mathcal{PR})$, let $\mathcal{P}_C := \mathcal{P}_{C,\tau} := (\phi(\tau^{-1}(P)) \cap V(C) : P \in \mathcal{P}_0^+)$, and let $\psi_C := \psi_{C,\tau}$ be the obvious bijection from $\mathcal{P}_C$ to $\mathcal{P}_0^+$ such that $\phi(\tau^{-1}(\psi_C(P))) \cap V(C) = P$ for each $P \in \mathcal{P}_C$. Let $\mathcal{P}_\tau := \bigsqcup_{C \in \mathcal{C}} \mathcal{P}_C$. As a guide to the reader, the pair $(X, \mathcal{P}_\tau)$ will be the almost-partition of G .

We have two important results regarding raises. The first uses the raise to find the almost-partition, and the second is an existence result, in which we merge parts and delete vertices (add to the loss-set) until we can find a raise. Specifically, we have the following results (where we consider $\binom{m}{n} := 0$ when $m < n$).

Lemma 7.17. *Let $b, k \in \mathbb{N}$ and $w, w', q \in \mathbb{R}^{\geq 0}$, let Γ be an almost-embedding of a pseudograph G , and let $\mathcal{PR}$ be a (b, k, w, w', q) -partitioned-reduction of Γ . If $\mathcal{PR}$ admits a raise τ , then G admits an almost-partition of treewidth at most b , width at most $\max((k+1)w, w')$, and loss at most q that is $(1, b(b+1)(k+1), \mathcal{VC}(\Gamma))$ -concentrated and $(2, (6k+8)\binom{b+1}{3}, \mathcal{AC}(\Gamma))$ -concentrated.*

Lemma 7.18. *Let $b, k, n \in \mathbb{N}$, $w, w', q \in \mathbb{R}^{\geq 0}$, and $d \in \mathbb{R}^{>0}$ with $d \geq w$, and let Γ be an almost-embedding of an n -vertex pseudograph G . If Γ admits a (b, k, w, w', q) -partitioned-reduction $\mathcal{PR}$, then there exists a $(b, k, d, w', 28(k+1)n/d + q)$ -partitioned-reduction $\mathcal{PR}'$ of Γ that admits a raise.*

Lemma 7.17 and Lemma 7.18 are the main results we prove in this chapter. Using all the results we have listed thus far, we can prove Theorem 7.4.

Theorem 7.4. *For each $b, g, p, k, a \in \mathbb{N}$, there exists $m_1, m_2 \in \mathbb{N}$ and $c_1, c_2, \beta \in \mathbb{R}^{\geq 0}$ such that the following holds. If $f, g : \mathbb{N} \mapsto \mathbb{R}^{\geq 0}$ are such that (f, g) generate adjustments of treewidth b , then for each $n \in \mathbb{N}$, every $d \in \mathbb{R}^{>0}$ with $d \geq \beta g(n)$, and each (g, p, k, a) -almost-embedding Γ of an n -vertex pseudograph G , G admits an almost-partition of treewidth at most b , width at most d , and loss at most $(c_1 f(n) + c_2 n)/d$ that is $(1, m_1, \mathcal{VC}(\Gamma))$ -concentrated and $(2, m_2, \mathcal{AC}(\Gamma))$ -concentrated.*

Proof. Let c, q be from Corollary 7.12. Set $\beta := c_1 := k + 1$, $m_1 := b(b + 1)(k + 1)$, $m_2 := (6k + 8) \binom{b+1}{3}$, and $c_2 := \max(c + q + 28(k + 1)^2, a)$.

Fix $n \in \mathbb{N}$, $d \in \mathbb{R}^{>0}$ with $d \geq \beta g(n)$, and a (g, p, k, a) -almost-embedding Γ of an n -vertex pseudograph G .

If $n \leq d$, then $(\emptyset, \{V(G)\})$ is trivially an almost-partition of G of treewidth $0 \leq b$, width $n \leq d$, and loss $0 \leq (c_1 f(n) + c_2 n)/d$. Further, since there is only one part, this almost-partition is trivially $(1, m_1, \mathcal{VC}(\Gamma))$ -concentrated and $(2, m_2, \mathcal{AC}(\Gamma))$ -concentrated (this would be true even if m_1 and m_2 were 0). So we may assume that $d \leq n$.

If Γ is trivial, we have that $V(G) = A$, where A is the set of apices of Γ . Thus, $|V(G)| = |A| \leq a$, and $(V(G), \emptyset)$ is trivially an almost-partition of G of treewidth at most $0 \leq b$, width $0 \leq d$, and loss $a \leq c_2 \leq (c_1 f(n) + c_2 n)/d$ (as $d \leq n$). Further, since there are no parts, this almost-partition is trivially $(1, m_1, \mathcal{VC}(\Gamma))$ -concentrated and $(2, m_2, \mathcal{AC}(\Gamma))$ -concentrated (this would be true even if m_1 and m_2 were 0). So we may assume that Γ is nontrivial. By Corollary 7.12, there exists a $(k, d, cn/d + q)$ -reduction $\mathcal{R}$ for Γ . Noting that $cn/d + q \leq (c + q)n/d$ (as $n \geq d$), $\mathcal{R}$ is also a $(k, d, (c + q)n/d)$ -reduction.

Set $d' := d/(k + 1) = d/\beta$. Note that $d' \geq g(n)$. Thus, by Lemma 7.16 (with $(d, w, q) := (d', d, (c + q)n/d)$), there exists a $(b, k, d', d, ((k + 1)f(n) + (c + q)n)/d)$ -partitioned-reduction $\mathcal{PR}$ for Γ . By Lemma 7.18 (with $(w, w', d, q) := (d', d, d', ((k + 1)f(n) + (c + q)n)/d)$), and noting that $28(k + 1)n/d' = 28(k + 1)^2 n/d$, there exists a $(b, k, d', d, ((k + 1)f(n) + (c + q + 28(k + 1)^2)n)/d)$ -partitioned-reduction of Γ that admits a raise. By Lemma 7.17 (with $(w, w', q) := (d', d, ((k + 1)f(n) + (c + q + 28(k + 1)^2)n)/d)$), G admits an almost-partition of treewidth at most b , width at most $\max((k + 1)d', d) = \max(d, d) = d$, and loss at most $((k + 1)f(n) + (c + q + 28(k + 1)^2)n)/d \leq (c_1 f(n) + c_2 n)/d$ that is $(1, b(b + 1)(k + 1), \mathcal{VC}(\Gamma))$ -concentrated and $(2, (6k + 8) \binom{b+1}{3}, \mathcal{AC}(\Gamma))$ -concentrated. Recalling that $m_1 := b(b + 1)(k + 1)$ and $m_2 := (6k + 8) \binom{b+1}{3}$, this completes the proof. $\square$

We prove Lemma 7.17 in Section 7.2, and Lemma 7.18 in Section 7.3. We consider Lemma 7.18 to be the heart of the proof of Theorem 2.9 (K_h -minor-free graphs are $O_h(\sqrt{n})$ -blowups of treewidth 3 graphs).

7.2 Partitioning using a Raise

This section proves Lemma 7.17, which uses a raise τ to construct the almost-partition. As stated before, the pair $(X, \mathcal{P}_\tau)$ will be the almost-partition.

7.2.1 Confirming the Partition

We first verify that the pair $(X, \mathcal{P}_\tau)$ is an almost-partition of the pseudograph G .

Lemma 7.19. *Let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR}$ be a partitioned-reduction of Γ , let X be the loss set of $\mathcal{PR}$, and let τ be a raise for $\mathcal{PR}$. Then $(X, \mathcal{P}_\tau)$ is an almost-partition of G .*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$, $B' := B'(\mathcal{PR})$, and $\mathcal{C} := \mathcal{C}(\mathcal{PR})$. For each $C \in \mathcal{C}$, let $\psi_C := \psi_{C, \tau}$ and $\mathcal{P}_C := \mathcal{P}_{C, \tau} = (\phi(\tau^{-1}(P)) \cap V(C) : P \in \mathcal{P}_0^+)$. Recall that $\mathcal{P} := \mathcal{P}_\tau = \bigsqcup_{C \in \mathcal{C}} \mathcal{P}_C$. Also, for each $C \in \mathcal{C}$, recall that ψ_C is a bijection from $\mathcal{P}_C$ to $\mathcal{P}_0^+$ such that $\phi(\tau^{-1}(\psi_C(P))) \cap V(C) = P$ for each $P \in \mathcal{P}_C$.

Since each $C \in \mathcal{C}$ is a sub-pseudograph of $G - X$ and since each $P \in \mathcal{P}_C$ is contained in $V(C)$, it is immediate that $\mathcal{P} \subseteq 2^{V(G-X)}$.

Consider any $v \in V(G - X)$. Since $\mathcal{C}$ is the connected components of $G - X$, observe that $(V(C) : C \in \mathcal{C})$ is a partition of $G - X$. Thus, there exists $C \in \mathcal{C}$ such that $v \in V(C)$. By Observation 7.8, there exists $x \in \phi^{-1}(v)$. So $\phi(x) = v$. Let $P' := \tau(x)$ and $P := \psi_C^{-1}(P') = \phi(\tau^{-1}(P')) \cap V(C)$. Note that $P \in \mathcal{P}_C \subseteq \mathcal{P}$. Observe that $x \in \tau^{-1}(P')$, and thus $v \in \phi(\tau^{-1}(P'))$. Since $v \in V(C)$, we obtain $v \in P \in \mathcal{P}$. So each vertex of $G - X$ is contained in at least one element of $\mathcal{P}$.

It remains to show that the elements of $\mathcal{P}$ are pairwise disjoint. Fix distinct $P, P' \in \mathcal{P}$. There exists $C, C' \in \mathcal{C}$ such that $P \in \mathcal{P}_C$ and $P' \in \mathcal{P}_{C'}$. Recall that $P \subseteq V(C)$ and that $P' \subseteq V(C')$. Note also that $(V(C^*) : C^* \in \mathcal{C})$ is a partition of $G - X$. Thus, if $C \neq C'$, then $P \subseteq V(C)$ is disjoint to $P' \subseteq V(C')$. So we may assume that $C = C'$. Then since P, P' are distinct, P and P' are distinct elements of $\mathcal{P}_C$. Since ψ_C is a bijection, $\psi_C(P)$ and $\psi_C(P')$ are distinct elements of $\mathcal{P}_0^+$. Thus, $\psi_C(P)$ is disjoint to $\psi_C(P')$. Hence, $\tau^{-1}(\psi_C(P))$ is disjoint to $\tau^{-1}(\psi_C(P'))$.

Presume, for a contradiction, that P intersects P' . Since $P \subseteq \phi(\tau^{-1}(\psi_C(P)))$ and $P' \subseteq \phi(\tau^{-1}(\psi_C(P')))$, we find that $\phi(\tau^{-1}(\psi_C(P)))$ intersects $\phi(\tau^{-1}(\psi_C(P')))$. So there exists $u \in \tau^{-1}(\psi_C(P))$ and $v \in \tau^{-1}(\psi_C(P'))$ such that $\phi(u) = \phi(v)$. Since $\tau^{-1}(\psi_C(P))$ is disjoint to $\tau^{-1}(\psi_C(P'))$, $u \neq v$. So $\phi(x) \neq x$ for at least one $x \in \{u, v\}$. Without loss

of generality, say $x = u$.

Since $\phi(u) \neq u$, $u \in V(W)$. Thus, $\phi(u) \in Y$. Since Y is disjoint to G' (and since $v \in G'$), $v \neq \phi(u) = \phi(v)$. So $v \in V(W)$. So $u, v \in V(W) \in \mathcal{P}_0^+$. Recall that $B' = V(G_0^+ \cap H) \subseteq V(H)$. Thus, since W is disjoint to H (and since $W \subseteq V(G_0^+)$), $u, v \in V(G_0^+) \setminus B'$. So $u \in \tau(u) \in \mathcal{P}_0^+$ and $v \in \tau(v) \in \mathcal{P}_0^+$. Since $\mathcal{P}_0^+$ is a partition of G_0^+ containing $V(W)$ (and since $u, v \in V(W)$), $\tau(u)$ and $\tau(v)$ are the same part, namely $V(W)$. Further, since $u \in \tau^{-1}(\psi_C(P))$ and $v \in \tau^{-1}(\psi_C(P))$, we find that $\psi_C(P), \tau(u), \tau(v)$, and $\psi_C(P')$ are all the same part, namely $V(W)$. However, we recall that $\psi_C(P)$ and $\psi_C(P')$ are the distinct elements of $\mathcal{P}_0^+$, as P, P' are distinct elements of $\mathcal{P}_C$ and since ψ_C is a bijection. So we have a contradiction. Thus, P is disjoint to P' , as desired. This completes the proof. $\square$

We use Lemma 7.19 implicitly from now on.

7.2.2 Bounding the Treewidth

Next, we bound the treewidth. We first show a relationship between ϕ, τ , and $\psi_{C,\tau}$ for any $C \in \mathcal{C}$.

Lemma 7.20. *Let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ , and let τ be a raise of $\mathcal{PR}$. Then for each $P \in \mathcal{P}_\tau$, if $C \in \mathcal{C}(\mathcal{PR})$ is such that $P \in \mathcal{P}_{C,\tau}$, then $\phi^{-1}(P) \subseteq \tau^{-1}(\psi_{C,\tau}(P))$.*

Proof. Set $\mathcal{P}_C := \mathcal{P}_{C,\tau}$ and $\psi_C := \psi_{C,\tau}$. Presume, for a contradiction, that for some $u \in \phi^{-1}(P)$, $\tau(u) \neq \psi_C(P)$. Set $P^* := \psi_C^{-1}(\tau(u))$. So P^* is distinct from P , and $\phi(u) \in P \subseteq V(C)$. Since $\mathcal{P}$ is a partition of $G - X$ (by Lemma 7.19) and since $P, P^* \in \mathcal{P}_C \subseteq \mathcal{P}$, P is disjoint to P^* . Recall that $P^* = \phi(\tau^{-1}(\tau(u))) \cap V(C)$. Observe that $u \in \tau^{-1}(\tau(u))$, and thus $\phi(u) \in \phi(\tau^{-1}(\tau(u)))$. Since $\phi(u) \in P \subseteq V(C)$, this gives $\phi(u) \in P^*$. Thus, $\phi(u) \in P \cap P^*$, contradicting the fact that P and P^* are disjoint. The lemma follows. $\square$

We now show that the raise does not create new edges in the quotient.

Lemma 7.21. *Let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ and let τ be a raise for $\mathcal{PR}$. If distinct $P, P' \in \mathcal{P}_\tau$ are adjacent in $(G - X)/\mathcal{P}_\tau$, then there exists $C \in \mathcal{C}(\mathcal{PR})$ such that $P, P' \in \mathcal{P}_{C,\tau}$. Further, $\psi_{C,\tau}(P)$ is adjacent to $\psi_{C,\tau}(P')$ in $G_0^+/\mathcal{P}_0^+$.*

Proof. Let $\mathcal{C} := \mathcal{C}(\mathcal{PR})$. For each $C \in \mathcal{C}$, let $\psi_C := \psi_{C,\tau}$ and $\mathcal{P}_C := \mathcal{P}_{C,\tau} = (\phi(\tau^{-1}(P)) \cap V(C) : P \in \mathcal{P}_0^+)$. Recall that $\mathcal{P} := \mathcal{P}_\tau = \bigsqcup_{C \in \mathcal{C}} \mathcal{P}_C$. Also, for each $C \in \mathcal{C}$, recall that ψ_C is a bijection from $\mathcal{P}_C$ to $\mathcal{P}_0^+$ such that $\phi(\tau^{-1}(\psi_C(P))) \cap V(C) = P$ for each $P \in \mathcal{P}_C$. Let $P, P' \in \mathcal{P}$ be adjacent in $(G - X)/\mathcal{P}$. So $N_{G-X}(P)$ intersects P' .

There exists $C, C' \in \mathcal{C}$ such that $P \in \mathcal{P}_C$ and $P' \in \mathcal{P}_{C'}$. Since $\mathcal{C}$ is the connected components of $G - X$, observe that $N_{G-X}(P) \subseteq V(C)$. Since P' intersects $N_{G-X}(P)$, we find that $C = C'$. Thus, observe that $P, P' \subseteq V(C)$. Let $u \in P$ and $v \in P'$ be adjacent in $G - X$. So $v \in N_{G-X}(u) \subseteq \phi(N_{C'}(\phi^{-1}(u)))$ (as $v \in V(G - X)$). So there exists adjacent $u', v' \in V(G')$ such that $\phi(u') = u$ and $\phi(v') = v$. Since $\phi(u') = u, \phi(v') = v \in V(G - X)$ and by definition of a raise, $N_{G_0^+}(\tau(u'))$ intersects $\tau(v')$. By Lemma 7.20, $\tau(u') = \psi_C(P)$ and $\tau(v') = \psi_C(P')$. Thus, $N_{G_0^+}(\psi_C(P))$ intersects $\psi_C(P')$, so $\psi_C(P)$ is adjacent to $\psi_C(P')$ in $G_0^+/\mathcal{P}_0^+$ (as they are distinct), as desired. $\square$

Lemma 7.21 allows us to bound the treewidth.

Corollary 7.22. *Let $b \in \mathbb{N}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR}$ be a partitioned-reduction of Γ of treewidth at most b , and let τ be a raise for $\mathcal{PR}$. Then $\mathcal{P}_\tau$ has treewidth at most b .*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$, and let $\mathcal{C} := \mathcal{C}(\mathcal{PR})$. Set $\mathcal{P} := \mathcal{P}_\tau$, and for each $C \in \mathcal{C}$, let $\psi_C := \psi_{C,\tau}$.

By Lemma 7.21, $(G - X)/\mathcal{P}$ can be obtained as follows. For each $C \in \mathcal{C}$, take a disjoint copy of $G_0^+/\mathcal{P}_0^+$, relabel each $P \in \mathcal{P}_0^+$ to $\psi_C^{-1}(P)$, delete some edges, and then take the disjoint union of the resulting (simple) graphs. Since $\mathcal{P}_0^+$ has treewidth at most b , $G_0^+/\mathcal{P}_0^+$ has treewidth at most b . Relabelling and deleting edges does not increase this bound, nor does taking the disjoint union. So $(G - X)/\mathcal{P}$ has treewidth at most b . Thus, $\mathcal{P}$ has treewidth at most b , as desired. $\square$

7.2.3 Bounding the Width

Next, we bound the width of the partition. This amounts to proving two things. One is that τ and ϕ map the part $V(W)$ to parts of the form $Y \cap V(C)$, where C is a connected component of $G - X$. The remainder-width controls the size of these parts. The second is that τ^{-1} increases the size of the other parts by at most a factor of $(k + 1)$. This is because τ is either the identity, or, in the case of quasi-vortex vertices, sends the vertex v to a vertex x on the modified boundary indexing a bag J'_x containing v .

For the formal proof, we require a strong understanding of how τ^{-1} acts. So we take a detour to prove some related results.

Lemma 7.23. *Let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ , let $B' := B'(\mathcal{PR})$, and let $(J'_x : x \in B') := \mathcal{J}'$. If τ is a raise for $\mathcal{PR}$, then for each $P \in \mathcal{P}_0^+$, $P \setminus B' \subseteq \tau^{-1}(P) \subseteq (P \setminus B') \cup \bigcup_{x \in P \cap B'} J'_x$.*

Proof. Fix $P \in \mathcal{P}_0^+$. Recall that $P \subseteq V(G_0^+)$. Thus, $P \setminus B' \subseteq V(G_0^+) \setminus B$. Hence, for each $v \in P \setminus B'$, $v \in \tau(v)$. Since $P, \tau(v) \in \mathcal{P}$ and since P intersects $\tau(v)$ (as $v \in P \cap \tau(v)$), $\tau(v)$ is P . It follows that $P \setminus B' \subseteq \tau^{-1}(P)$.

Consider any $v \in V(G_0^+) \setminus B'$ such that $v \in \tau^{-1}(P)$. Since $v \in V(G_0^+) \setminus B'$, $v \in \tau(v) = P$. Since $v \notin B'$, $v \in P \setminus B'$. Thus, $\tau^{-1}(P) \cap (V(G_0^+) \setminus B') \subseteq P \setminus B'$. Next, consider any $v \in V(H') \cap \tau^{-1}(P)$. There exists $x \in \tau(v) \cap B' = P \cap B'$ such that $v \in J'_x$. Thus, $\tau^{-1}(P) \cap V(H') \subseteq \bigcup_{x \in P \cap B'} J'_x$. Recall that $V(G') = V(G_0^+ \cup H') = (V(G_0^+) \setminus B') \cup V(H')$. Thus, $\tau^{-1}(P) \subseteq (P \setminus B') \cup \bigcup_{x \in P \cap B'} J'_x$. So $P \setminus B' \subseteq \tau^{-1}(P) \subseteq (P \setminus B') \cup \bigcup_{x \in P \cap B'} J'_x$, as desired. $\square$

Corollary 7.24. *Let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ , and let τ be a raise for $\mathcal{PR}$. Then $V(W) \in \mathcal{P}_0^+$ and $\tau^{-1}(V(W)) = V(W)$.*

Proof. Let $B' := V(G_0^+ \cap H') = B'(\mathcal{PR})$. By definition of $\mathcal{P}_0^+$, $V(W) \in \mathcal{P}_0^+$. Recall that W is disjoint to H' , and thus $V(W)$ is disjoint to B' . Thus, $V(W) \setminus B' = V(W)$ and $V(W) \cap B' = \emptyset$. So by Lemma 7.23, $V(W) \subseteq \tau^{-1}(V(W)) \subseteq V(W) \cup \emptyset$. Thus, $\tau^{-1}(V(W)) = V(W)$, as desired. $\square$

Corollary 7.25. *Let $k \in \mathbb{N}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ of quasi-vortex-width at most k , and let τ be a raise for $\mathcal{PR}$. Then for each $P \in \mathcal{P}_0^+$, $|\tau^{-1}(P)| \leq (k+1)|P|$.*

Proof. Let $B' := V(G_0^+ \cap H') = B'(\mathcal{PR})$, and let $(J'_x : x \in B') := \mathcal{J}'$.

Since $\mathcal{PR}$ has quasi-vortex-width at most k , $\mathcal{J}'$ has width at most k . Thus, for each $x \in B'$, $|J'_x| \leq k+1$.

Fix $P \in \mathcal{P}_0^+$. By Lemma 7.23, $\tau^{-1}(P) \subseteq (P \setminus B') \cup \bigcup_{x \in P \cap B'} J'_x$. Thus, $|\tau^{-1}(P)| \leq |P \setminus B'| + (k+1)|P \cap B'| \leq (k+1)|P|$ (since $k \geq 0$), as desired. $\square$

We now bound the width.

Lemma 7.26. *Let $k \in \mathbb{N}$, $w, w', q \in \mathbb{R}^{\geq 0}$, let $\mathcal{PR}$ be an $(\infty, k, w, w', \infty)$ -partitioned-reduction of Γ , and let τ be a raise for $\mathcal{PR}$. Then $\mathcal{P}_\tau$ has width at most $\max((k+1)w, w')$.*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$, let $B' := V(G_0^+ \cap H') = B(G_0^+ - V(W), D)$, and let $\mathcal{C} := \mathcal{C}(\mathcal{PR})$. For each $C \in \mathcal{C}$, let $\psi_C := \psi_{C, \tau}$ and $\mathcal{P}_C := \mathcal{P}_{C, \tau} = (\phi(\tau^{-1}(P)) \cap V(C) : P \in \mathcal{P}_0^+)$. Let $\mathcal{P} := \mathcal{P}_\tau = \bigsqcup_{C \in \mathcal{C}} \mathcal{P}_C$.

For each $P \in \mathcal{P}$, there exists $C \in \mathcal{C}$ such that $P \in \mathcal{P}_C$. Let $P' := \psi_C(P)$. Recall $P = \phi(\tau^{-1}(P')) \cap V(C)$. We consider two cases.

Case 1: $P' = V(W)$.

By Corollary 7.24, $\tau^{-1}(P') = \tau^{-1}(V(W)) = V(W)$. Recall that $\phi(V(W)) = Y$. So $P = Y \cap V(C)$. As $\mathcal{PR}$ has remainder-width at most w' , $|P| = |Y \cap V(C)| \leq w' \leq \max(w', (k+1)w)$.

Case 2: $P' \neq V(W)$.

Since $\mathcal{PR}$ has near-width at most w , $|P| \leq w$. By Corollary 7.25, $|\tau^{-1}(P')| \leq (k+1)|P| \leq (k+1)w$. So $|P| \leq |\phi(\tau^{-1}(P'))| \leq |\tau^{-1}(P')| \leq (k+1)w \leq \max(w', (k+1)w)$, as desired.

This completes the proof of the lemma. $\square$

7.2.4 Showing that the Partition is Concentrated

It remains to show that the almost-partition is concentrated. This is the most difficult step, and requires some more explaining.

Recall that we start with a connected partition of a plane pseudograph. We use the raise to partition $V(G')$ (note that $(\tau^{-1}(P) : P \in \mathcal{P}_0^+)$ is a partition of $V(G')$). We then transform these parts with ϕ and divide them based on the connected components of $G - X$. The reduction (in particular, the projection) ensures that there is a strong relationship between $\mathcal{VC}(\Gamma)/\mathcal{AC}(\Gamma)$ and the cliques in the quasi-vortex H' , the cliques of size at most 2 in G_0^+ , and the facial triangles of G_0^+ . So, loosely speaking, this problem converts into a problem of ensuring that $(\tau^{-1}(P) : P \in \mathcal{P}_0^+)$ is concentrated. For this, we show that (1) the initial partition of G_0^+ is concentrated with respect to the facial triangles, and (2) the raise does not ruin the concentration too much. The following lemma addresses the first point.

Lemma 7.27. *Let G be a plane pseudograph, and let $\mathcal{P}$ be a connected partition for G . Then for each clique Q in $G/\mathcal{P}$ of size 3, there is at most two cliques in $F_3(G)$ that*

intersect each part in Q .

Proof. Presume otherwise. So there exists a triangle $Q = \{P_1, P_2, P_3\}$ in $G/\mathcal{P}$ and three distinct facial triangles K_1, K_2, K_3 that each intersect all of P_1, P_2, P_3 . So there are distinct faces F_1, F_2, F_3 of G whose boundary vertices are K_1, K_2 , and K_3 respectively. Let G' be the plane pseudograph obtained from G by embedding, for each $i \in \{1, 2, 3\}$, a vertex v_i inside F_i , along with edges inside F_i to each vertex in K_i . (This is possible since the faces F_1, F_2, F_3 are distinct.) Since $\mathcal{P}$ is connected partition of $G \subseteq G'$, we obtain a proper model of $K_{3,3}$ in G' with $\{P_1, P_2, P_3\}$ as branch sets on one side, and $\{\{v_1\}, \{v_2\}, \{v_3\}\}$ as branch sets on the other side, contradicting Theorem 1.1. (Note that Theorem 1.1 still holds for pseudographs, as the absence or presence of parallel edges and loops does not change the Euler genus or excluded minors of a pseudograph.) $\square$

An easy corollary/application of Lemma 7.27 is that $(\emptyset, \mathcal{P}_0^+)$ is a $(2, 2, F_3(G_0^+))$ -concentrated almost-partition of G_0^+ . So the goal is to show that after taking the inverse of the raise, the partition remains concentrated, in terms of cliques in the quasi-vortex, cliques of size at most 2 in G_0^+ , and facial triangles of G_0^+ .

For any pair of parts P_1, P_2 in $\mathcal{P}_0^+$, there are two concerns. The first are edges in the quasi-vortex H' that go between $\tau^{-1}(P_1)$ and $\tau^{-1}(P_2)$. The second are edges in G_0^+ that go between $\tau^{-1}(P_1)$ and $\tau^{-1}(P_2)$ that do not have their endpoints in P_1 and P_2 . The former could prevent the partition from being concentrated with respect to the vortex-cliques, and the latter can occur only near the modified boundary but causes problems both for the vortex-cliques and facial triangles. We show that these problematic edges have at least one endpoint in a small set of vertices.

The proof idea is relatively simple. For the second type of problematic edge described above, assume that the edge does not have an endpoint in P_1 (we can handle the other case symmetrically). In G_0^+ , the non-crossing property (Lemma 7.6) ensures that all the intersection points of P_2 with B' lie between two points a, b of $P_1 \cap B'$ in the cyclic ordering of B' induced by D . We find that for any problematic edge as described above, at least one endpoint lies in $J'_a \cup J'_b$ (where $(J'_v : v \in B')$ is the quasi-vortex-decomposition). There is a decent amount of case analysis required to reach this conclusion, but it fundamentally comes down to the fact that the raise requires $v \in V(H')$ to be mapped to $x \in B'$ with $v \in J'_x$. This immediately takes care of all the edges in the quasi-vortex, and problematic edges in G_0^+ get blocked by P_1 and P_2 using the non-crossing property. See Figure 7.3.

The only exception is that we do not have any control over the number of edges coming

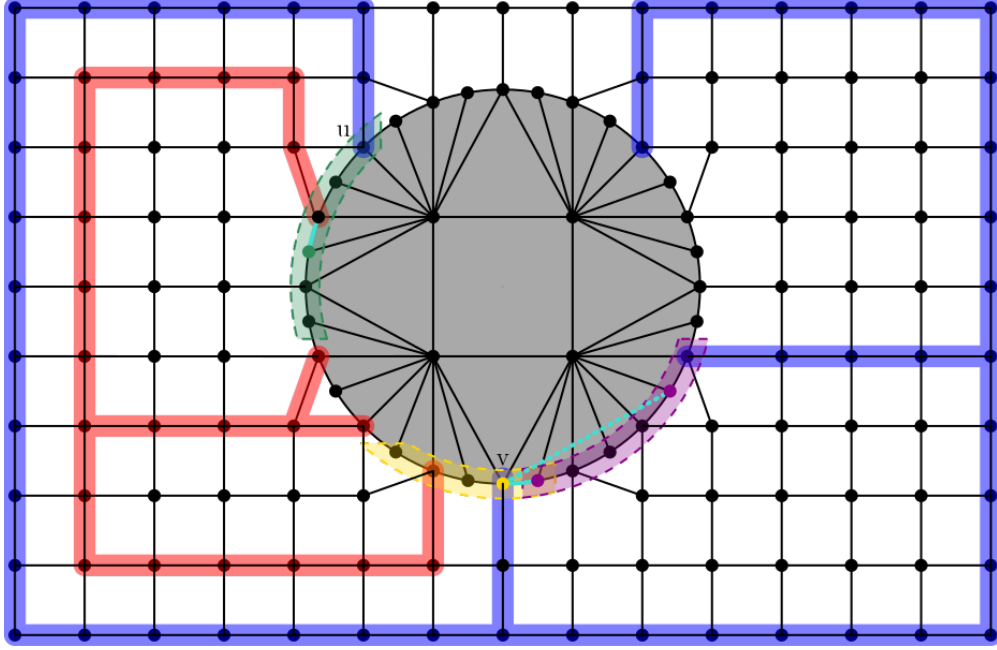


Figure 7.3: An example of some bad edges going between two parts (of the original partition). The blue part is P_1 , and the red part is P_2 (the roles are not symmetric, but the argument can be repeated). The light blue edges represent (some) bad edges, with the dotted light blue edge existing in the quasi-vortex. The vertices $u, v \in P_1$ (the blue part) are such that all the intersections of P_2 (red part) with the boundary of the disc are contained between u and v , and no other vertices of P_1 are contained in this interval. There are three types of bad edges. The first can be seen with the green vertex, whose corresponding interval (in green) causes it to be assigned to the P_1 , creating a bad edge to any neighbours assigned to P_1 . The second and third can be seen with the yellow and purple vertices. The yellow vertices interval (in yellow) causes it to be re-assigned to the red part, whereas the purple vertices (shared) interval (in purple) causes them to be reassigned to the blue part, creating a bad edge to the yellow vertex. These bad edges can occur in the plane, or can be in the quasi-vortex if the intervals overlap. However, in all cases, the bad edges have at least one endpoint in either the bag J'_u indexed by u , or the bag J'_v indexed by v (the yellow vertex). Thus, only $2(k+1)$ vertices need to be deleted to eliminate all bad edges.

from $V(W)$ to B' , so we cannot control the behaviour of bad edges between $V(W)$ and some other part P_1 . Fortunately, there is no need to control these edges (see the paragraph before Lemma 7.30 for the reason). So we ignore the case of $V(W)$ for now. We have the following technical lemma.

Lemma 7.28. *Let $k \in \mathbb{N}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ of quasi-vortex-width at most k , let τ be a raise for $\mathcal{PR}$, and let $P_1, P_2 \in \mathcal{P}_0^+$ be distinct. Then there*

exists $S \subseteq V(G')$ with $|S| \leq 2(k+1)$ such that if $u \in \tau^{-1}(P_1)$ and $v \in \tau^{-1}(P_2)$ are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin P_1$ and $v \notin V(W)$.

then (u, v) intersects S .

Proof. Let $G'_0 := G_0(\Lambda(\mathcal{PR})) = G_0^+ - V(W)$, $B' := B'(\mathcal{PR})$, and $U' := U'(\mathcal{PR})$. Recall that D is G'_0 -clean with $B(D, G'_0) = B'$ and $U(D, G'_0) = U'$, and that U' is a cycle or a path. Let $\preceq_{U'}$ denote the cyclic ordering of B' , and let $(J'_x : x \in B') := \mathcal{J}'$.

Observe that if either P_1 or P_2 is empty, then taking $S := \emptyset$ suffices. So we may assume that neither P_1 nor P_2 is empty. Next, presume that for some $i \in \{1, 2\}$, $P_i = V(W)$. By Corollary 7.24, $\tau^{-1}(P_i) = P_i = V(W)$. Recall that $V(W)$ is disjoint to H' . So for any $z \in \tau^{-1}(P_i) = P_i = V(W)$, we have:

1. $z \notin V(H')$, and
2. $z \in P_i = V(W)$.

So there is no pair $u \in \tau^{-1}(P_1)$ and $v \in \tau^{-1}(P_2)$ that satisfy either:

1. $u, v \in V(H')$, or,
2. $u \notin P_1$ and $v \notin V(W)$.

So we can take $S := \emptyset$ in this case. Thus, we can assume that neither P_1 nor P_2 is $V(W)$ from now on. Since $\mathcal{P}_0^+$ is a connected partition of $V(G_0^+)$ containing $V(W)$, it follows that P_1 and P_2 are contained in $V(G'_0) = V(G_0^+ - V(W))$, and that $H_1 := G'_0[P_1]$ and $G'_0[P_2]$ are connected and disjoint. Let $H_2 := G'_0[N_{G'_0}(P_2) \setminus P_1]$. Since $G'_0[P_2]$ is connected and disjoint to $G'_0[P_1] = H_1$, H_2 is connected and disjoint to H_1 . Note that $P_2 \subseteq V(H_2)$.

By Lemma 7.23, for each $i \in \{1, 2\}$, $\tau^{-1}(P_i) \subseteq (P_i \setminus B') \cup \bigcup_{x \in P_i \cap B'} J'_x$. Thus, if $z \in \tau^{-1}(P_i)$, then either $z \in P_i \setminus B'$ or $z \in \bigcup_{x \in P_i \cap B'} J'_x$. Note that $P_i \setminus B' \subseteq V(G_0^+) \setminus B'$ is disjoint to $V(H') \supseteq \bigcup_{x \in P_i \cap B'} J'_x$. So either:

1. $z \in P_i \setminus B' \subseteq P_i$ and $z \notin V(H')$, or,
2. $z \in \bigcup_{x \in P_i \cap B'} J'_x$.

In particular, if $u \in \tau^{-1}(P_1)$ satisfies either:

1. $u \in V(H')$, or
2. $u \notin P_1$,

then $u \in \bigcup_{x \in P_1 \cap B'} J'_x$. Since $\mathcal{PR}$ has quasi-vortex-width at most k , $|\bigcup_{x \in P_1 \cap B'} J'_x| \leq (k+1)|P_1 \cap B'|$. So if $|P_1 \cap B'| \leq 2$, we can take $S := \bigcup_{x \in P_1 \cap B'} J'_x$. Thus, we may assume that $|P_1 \cap B'| \geq 3$.

By Lemma 7.6, there does not exist $x, y \in V(H_2) \cap B'$ and $a, b \in V(H_1) \cap B'$ such that $x \prec_{U'} a \prec_{U'} y \prec_{U'} b$. Thus and since $|V(H_1) \cap B'| = |P_1 \cap B'| \geq 3$, there exists distinct $a, b \in V(H_1) \cap B' = P_1 \cap B'$ such that for each $y \in V(H_2) \cap B'$ and $z \in P_1 \cap B'$, $z \preceq_{U'} a \prec_{U'} y \prec_{U'} b$. (If $V(H_2) \cap B'$ is empty, we pick a, b such that $z \preceq_{U'} a \prec_{U'} b$ for each $z \in P_1 \cap B'$.) Let $S := J_a \cup J_b$. Observe that $|S| \leq 2(k+1)$ as $\mathcal{J}'$ has width at most k .

Claim 1. *Let $y \in V(H_2) \cap B'$, $z \in P_1 \cap B'$, and let $I \subseteq U'$ be connected with $y, z \in V(I)$. Then $V(I)$ intersects $\{a, b\}$.*

Proof. Recall that $z \preceq_{U'} a \prec_{U'} y \prec_{U'} b$. If $z \in \{a, b\}$, then $V(I)$ intersects $z \in \{a, b\}$, as desired. So we may assume that $z \prec_{U'} a \prec_{U'} y \prec_{U'} b$. Since U is a cycle or path, observe that y and z are in different connected components of $U - a - b$. Thus, since I is connected and contains y, z , $V(I)$ intersects $\{a, b\}$, as desired. ■

Fix $u \in \tau^{-1}(P_1)$ and $v \in \tau^{-1}(P_2)$ that are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin P_1$ and $v \notin V(W)$.

We show that (u, v) intersects S .

If $u, v \in V(H')$ and are adjacent in H' , then recall that $u \in \bigcup_{x \in P_1 \cap B'} J'_x$ and $v \in \bigcup_{x \in P_2 \cap B'} J'_x$. So there exists $z \in P_1 \cap B'$ such that $u \in J'_z$, and $y \in P_2 \cap B'$ such that $v \in J'_y$. For each $m \in \{u, v\}$, let I_m be the subgraph of U' induced by the vertices $x \in B'$ such that $m \in J'_x$. Recall that I_u and I_v are nonempty and connected. Further, since u and v are adjacent in H' , I_u and I_v intersect. Also, note that $z \in V(I_u)$ and $y \in V(I_v)$. So $I_u \cup I_v$ is connected and contains both y and z . Thus, by Claim 1, $I_u \cup I_v$ intersects $\{a, b\}$. So there exists $m \in \{u, v\}$ and $n \in \{a, b\}$ such that $n \in I_m$. So $m \in J'_n \subseteq S$, as desired.

If $u \notin P_1$, recall that $u \in \bigcup_{x \in P_1 \cap B'} J'_x \subseteq V(H')$. So there exists $z \in P_1 \cap B'$ such that $u \in J'_z$. Let I_u be the subgraph of U' induced by the vertices $x \in B'$ such that $u \in J'_x$. Note that I_u is nonempty and connected, and that $u \in V(I_u)$ (since $\mathcal{J}'$ is planted). Observe also that $z \in V(I_u)$ and that $z \preceq_{U'} a \prec_{U'} b$.

By the above case, we may assume that either $v \notin V(H')$, or u and v are not adjacent in H' . Recall that $G' = G_0^+ \cup H$, and that u, v are adjacent in G' . Thus (and since

$u \in V(H')$), we may assume that $u \in B'$, $v \in V(G_0^+)$, and that u, v are adjacent in G_0^+ . Since $u \in P_1$, $v \in P_2$, and since P_1, P_2 are disjoint to $V(W)$, we find that $u, v \in V(G'_0)$ and that u, v are adjacent in G'_0 .

If $v \in P_2$, then observe that $u \in N_{G'_0}(P_2)$. Since $u \notin P_1$, we have $u \in V(H_2)$. So $u \in V(H_2) \cap B'$. Recall that $z \in P_1 \cap B$ and $u, z \in V(I_u)$. Thus, by Claim 1 (with $y := u$), $V(I_u)$ intersects $\{a, b\}$. So there exists $n \in \{a, b\}$ such that $u \in J'_n \subseteq S$, as desired. So we may assume that $v \notin P_2$. Thus, $v \in \bigcup_{x \in P_2 \cap B'} J'_x \subseteq V(H')$. So $v \in B'$ (since $v \in V(G_0^+)$), and there exists $y \in P_2 \cap B'$ such that $v \in J'_y$. Note that $y \in P_2 \subseteq V(H_2)$, and thus $a \prec_{U'} y \prec_{U'} b$. Let I_v be the subgraph of U' induced by the vertices $x \in B'$ such that $v \in J'_x$. Note that I_v is nonempty and connected, and that $v, y \in V(I_v)$.

If $v \in P_1$, then by Claim 1 (with $z := v$), $V(I_v)$ intersects $\{a, b\}$. So there exists $n \in \{a, b\}$ such that $v \in J'_n \subseteq S$, as desired. Thus, we may assume that $v \notin P_1$. Hence, neither u nor v is in P_1 . Since u, v are adjacent in $V(G'_0)$, $G'_0[\{u, v\}]$ and H_1 are connected and disjoint sub-pseudographs of G'_0 . Thus, by Lemma 7.6, there does not exist $m, n \in P_1 \cap B'$ such that $u \prec_{U'} m \prec_{U'} v \prec_{U'} n$. In particular, neither $u \prec_{U'} a \prec_{U'} v \prec_{U'} b$ nor $u \prec_{U'} b \prec_{U'} v \prec_{U'} a$. Since $a, b \in P_1$ and neither u nor v is in P_1 , a, b, u and v are pairwise distinct. Thus, either $u, v \prec_{U'} a \prec_{U'} b$, or $a \prec_{U'} u, v \prec_{U'} b$.

If $u, v \prec_{U'} a \prec_{U'} b$, observe that $v \prec_{U'} a \prec_{U'} y \prec_{U'} b$. Recall that $v, y \in V(I_v)$. By the same argument as that used in Claim 1, $V(I_v)$ intersects $\{a, b\}$, which gives $v \in S$. Otherwise, if $a \prec_{U'} u, v \prec_{U'} b$, observe that $z \preceq_{U'} a \prec_{U'} u \prec_{U'} b$. Recall that $u, z \in V(I_u)$. It follows that $V(I_u)$ intersects $\{a, b\}$, and thus $u \in S$.

So in all cases, (u, v) intersects S . Since $|S| \leq 2(k+1)$, this completes the proof. $\square$

For the purposes of the final proof, it is easier to work with $(X, \mathcal{P}_\tau)$ rather than converting everything to the partition $(\tau^{-1}(P) : P \in \mathcal{P}_0^+)$. So we want to convert Lemma 7.28 to a result about the parts in $\mathcal{P}_\tau$.

Corollary 7.29. *Let $k \in \mathbb{N}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ of quasi-vortex-width at most k , let τ be a raise for $\mathcal{PR}$, and let $P_1, P_2 \in \mathcal{P}_\tau$ be distinct and such that there exists $C \in \mathcal{C}(\mathcal{PR})$ with $P_1, P_2 \in \mathcal{P}_{C, \tau}$. Then there exists $S \subseteq V(G')$ with $|S| \leq 2(k+1)$ such that if $u \in \phi^{-1}(P_1)$ and $v \in \phi^{-1}(P_2)$ are adjacent in G' and satisfy either:*

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin \psi_{C, \tau}(P_1)$ and $\psi_{C, \tau}(P_2)$ is not $V(W)$.

then (u, v) intersects S .

Proof. Set $\psi_C := \psi_{C, \tau}$. By Lemma 7.28 with $(P_1, P_2) := (\psi_C(P_1), \psi_C(P_2))$, there exists $S \subseteq V(G')$ with $|S| \leq 2(k+1)$ such that if $u \in \tau^{-1}(\psi_C(P_1))$ and $v \in \tau^{-1}(\psi_C(P_2))$ are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin \psi_C(P_1)$ and $v \notin V(W)$,

then (u, v) intersects S .

By Lemma 7.20, if $u \in \phi^{-1}(P_1)$ and $v \in \phi^{-1}(P_2)$, then $u \in \tau^{-1}(\psi_C(P_1))$ and $v \in \tau^{-1}(\psi_C(P_2))$. By Corollary 7.24, $\tau^{-1}(V(W)) = V(W)$. Hence, if $\psi_C(P_2)$ is not $V(W)$ (and is thus disjoint to $V(W)$), then $\phi^{-1}(P_2) \subseteq \tau^{-1}(\psi_C(P_2))$ is disjoint to $V(W)$. In particular, $v \notin V(W)$.

So if $u \in \phi^{-1}(P_1)$ and $v \in \phi^{-1}(P_2)$ are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin \psi_C(P_1)$ and $\psi_C(P_2)$ is not $V(W)$,

then $u \in \tau^{-1}(\psi_C(P_1))$, $v \in \tau^{-1}(\psi_C(P_2))$, and u and v are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin \psi_C(P_1)$ and $v \notin V(W)$.

Thus, (u, v) intersects S . Since $|S| \leq 2(k+1)$, this completes the proof. $\square$

We can now prove that the almost-partition is concentrated. This amounts to applying Corollary 7.29 on each pair of parts in a given clique in the quotient. Since the treewidth is bounded (by Corollary 7.22), only a finite number of pairs need to be considered. This, in combination with the fact that $\mathcal{P}_0^+$ is concentrated (with respect to facial triangles), mostly shows that $\mathcal{P}_\tau$ is concentrated.

The only difficulty is dealing with $V(W)$, since the bad edges coming from $V(W)$ cannot be controlled. However, $V(W)$ does not affect the cliques in the vortex, so this is not an issue for showing that the almost-partition is $(1, b(b+1)(k+1), \mathcal{VC}(\Gamma))$ -concentrated. Further, any counterexample to the almost-partition being $(2, (6k+8)\binom{b+1}{3}, \mathcal{AC}(\Gamma))$ -concentrated needs to intersect at least three parts, of which two do not correspond (via $\psi_{C, \tau}$) to $V(W)$. Throwing away the part corresponding to $V(W)$ (if it exists) and controlling the bad edges between the remaining two parts suffices.

We can now complete the proof.

Lemma 7.30. *Let $k, b \in \mathbb{N}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR}$ be a $(k, b, \infty, \infty, \infty)$ -partitioned-reduction of Γ , and let τ be a raise of $\mathcal{PR}$. Then $(X, \mathcal{P}_\tau)$ is $(1, b(b+1)(k+1), \mathcal{VC}(\Gamma))$ -concentrated and $(2, (6k+8)\binom{m}{3}, \mathcal{AC}(\Gamma))$ -concentrated.*

Proof. Let $(\Sigma, s, A, (G_0, \dots, G_s), (D_1, \dots, D_s)) := \Gamma$, let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$, and let $\mathcal{C} := \mathcal{C}(\mathcal{PR})$. For each $C \in \mathcal{C}$, let $\psi_C := \psi_{C, \tau}$ and $\mathcal{P}_C := \mathcal{P}_{C, \tau} = (\phi(\tau^{-1}(P)) \cap V(C) : P \in \mathcal{P}_0^+)$. Recall that $\mathcal{P} := \mathcal{P}_\tau = \bigsqcup_{C \in \mathcal{C}} \mathcal{P}_C$. Also, for each $C \in \mathcal{C}$, recall that ψ_C is a bijection from $\mathcal{P}_C$ to $\mathcal{P}_0^+$ such that $\phi(\tau^{-1}(\psi_C(P))) \cap V(C) = P$ for each $P \in \mathcal{P}_C$.

By Lemma 7.19, $(X, \mathcal{P})$ is an almost-partition of G . By Corollary 7.22, $\mathcal{P}$ has treewidth at most b . Fix a clique Q of $(G - X)/\mathcal{P}$. Since $\mathcal{P}$ has treewidth at most b , note that $|Q| \leq b + 1$. By (repeated application of) Lemma 7.21, there exists $C \in \mathcal{C}$ such that $Q \subseteq \mathcal{P}_C$, and $Q' := \psi_C(Q)$ is a clique in $G_0^+/\mathcal{P}_0^+$. Since ψ_C is a bijection, $|Q'| = |Q| \leq k + 1$. Let $\preceq$ be an ordering of Q such that if $V(W) \in Q'$, then $P \preceq \psi_C^{-1}(V(W))$ for each $P \in Q$. (Other than this, $\preceq$ can be arbitrary.) For each $i \in \{2, 3\}$, let $\mathcal{Q}_i = \binom{Q}{i}$. Observe that $|\mathcal{Q}_i| = \binom{|Q|}{i} \leq \binom{b+1}{i}$.

Fix $E = \{P_1, P_2\} \in \mathcal{Q}_2$ with $P_1 \prec P_2$. Let $P'_1 := \psi_C(P_1)$ and $P'_2 := \psi_C(P_2)$. For each $i \in \{1, 2\}$, by Corollary 7.29 (with $(P_1, P_2) := (P_i, P_{3-i})$), there exists a set $S'_{E,i} \subseteq V(G')$ with $|S'_{E,i}| \leq 2(k+1)$ such that if $u \in \phi^{-1}(P_1)$ and $v \in \phi^{-1}(P_2)$ are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin P'_i$ and $P'_{3-i} \neq V(W)$,

then (u, v) intersects $S'_{E,i}$. Let $S_{E,i} := \phi(S'_{E,i})$. Note that $|S_{E,i}| \leq |S'_{E,i}| \leq 2(k+1)$.

Fix $T = \{P_1, P_2, P_3\} \in \mathcal{Q}_3$ with $P_1 \prec P_2 \prec P_3$. For each $i \in \{1, \dots, 3\}$, let $P'_i := \psi_C(P_i)$. By choice of $\preceq$, $V(W) \notin \{P'_1, P'_2\}$. Let $T' := \psi_C(T) = \{P'_1, P'_2, P'_3\}$. Observe that T' is a clique of size 3 in $G_0^+/\mathcal{P}_0^+$. By Lemma 7.27 (with $(G, \mathcal{P}) := (G_0^+, \mathcal{P}_0^+)$, noting that $\mathcal{P}_0^+$ is a connected partition of the plane pseudograph G_0^+), there exists at most two cliques in $F_3(G_0^+)$ that intersect all three parts in T' . Thus, there exists $S'_{T,0} \subseteq V(G_0^+) \subseteq V(G')$ of size at most two such that no clique in $F_3(G_0^+)$ disjoint to $S'_{T,0}$ intersects all three parts in T' .

Let $E_{12} := \{P_1, P_2\}$ and $E_{13} := \{P_1, P_3\}$. Note that $E_{12}, E_{13} \in \mathcal{Q}_2$. Let $S'_{T,1} := S'_{E_{12},1}$, $S'_{T,2} := S'_{E_{12},2}$, $S'_{T,3} := S'_{E_{13},2}$, and $S'_T := \bigcup_{i=0}^3 S'_{T,i}$. Observe that $|S'_T| \leq 3(2(k+1)) + 2 = 6k + 8$. Set also $j_1 := 2$ and $j_2 := j_3 := 1$. So for each $i \in \{1, 2, 3\}$, if $j := j_i$, then P'_j is not $V(W)$. Thus, if $u \in \phi^{-1}(P_i)$ and $v \in \phi^{-1}(P_j)$ are adjacent in G' and satisfy either:

1. $u, v \in V(H')$ and are adjacent in H' , or,
2. $u \notin P'_i$,

then (u, v) intersects $S'_{T,i} \subseteq S'_T$. Set $S_T := \phi(S'_T)$. So $|S_T| \leq |S'_T| \leq 6k + 8$.

Let $S_1 := \bigcup_{E \in \mathcal{Q}_2} S_{E,1}$, and let $S_2 := \bigcup_{T \in \mathcal{Q}_3} S_T$. So $S_1, S_2 \subseteq V(G)$. Observe that $|S_1| \leq 2(k+1)|\mathcal{Q}_2| \leq 2(k+1)\binom{b+1}{2} = b(b+1)(k+1)$, and that $|S_2| \leq (6k+8)|\mathcal{Q}_{Q,3}| \leq (6k+8)\binom{b+1}{3}$.

Consider any $K \in \mathcal{VC}(\Gamma)$. Note that K is a clique in G . Presume, for a contradiction, that $K \setminus S_1$ intersects at least two distinct parts $P_1, P_2 \in \mathcal{Q}$, where $P_1 \prec P_2$. So there exists (distinct) $x \in P_1 \cap K \setminus S_1$ and $y \in P_2 \cap K \setminus S_1$. Thus, $\{x, y\}$ is disjoint to S_1 . Since $\mathcal{VC}(\Gamma)$ is closed under subsets, observe that $\{x, y\} \in \mathcal{VC}(\Gamma)$. Thus, $x, y \in V(H)$, and are adjacent in H . Noting that neither $x \in X$ nor $y \in X$ (since P_1, P_2 are parts of $\mathcal{P}$, which is a partition of $G - X$) and by the properties of a reduction, we obtain $x, y \in V(H) \setminus X \subseteq V(H')$, and x, y are adjacent in $H - (X \cap V(H)) \subseteq H'$. In particular, $x, y \in V(G')$ and are adjacent in G' .

Recall that H' is disjoint to W . Thus, $x, y \in V(H - X) \subseteq V(H') \subseteq V(G') \setminus V(W)$. So $x, y \in V(G')$, and by Observation 7.8, $\phi^{-1}(x) = \{x\}$ and $\phi^{-1}(y) = \{y\}$. So $\phi(x) = x$ and $\phi(y) = y$. Recall that $x \in P_1$ and $y \in P_2$, thus $\phi^{-1}(x) \subseteq \phi^{-1}(P_1)$ and $\phi^{-1}(y) \subseteq \phi^{-1}(P_2)$. So $x \in \phi^{-1}(P_1)$ and $y \in \phi^{-1}(P_1)$ are adjacent in G' and satisfy $x, y \in V(H')$ and x, y are adjacent in H' .

Let $E = (P_1, P_2)$. Note that $E \in \mathcal{Q}_2$, so $S_{E,1} \subseteq S_1$. By definition of $S'_{E,1}$ (and $S'_{E,2}$), (x, y) intersects $S'_{E,1}$ (and $S'_{E,2}$). Thus, $(\phi(x), \phi(y)) = (x, y)$ intersects $S_{E,1} \subseteq S_1$ (and $S_{E,2}$). But this contradicts the fact that (x, y) is disjoint to S_1 . Thus, $K \setminus S_1$ intersects at most one part in $\mathcal{Q}$. Since $|S_1| \leq b(b+1)(k+1)$ (and recalling that $\mathcal{VC}(\Gamma) \subseteq 2^{V(G)}$), we find that $\mathcal{P}$ is $(1, b(b+1)(k+1), \mathcal{VC}(\Gamma))$ -concentrated.

Consider any $K \in \mathcal{AC}(\Gamma)$. Note that K is a clique in G . Presume, for a contradiction, that $K \setminus S_2$ intersects at least three pairwise distinct parts P_1, P_2, P_3 in $\mathcal{Q}$, where $P_1 \prec P_2 \prec P_3$. So for each $i \in \{1, 2, 3\}$, there exists (distinct) $x_i \in P_i \cap K \setminus S_2$. Note that $\{x_1, x_2, x_3\}$ is disjoint to $S_2 \cup X$. Since $\mathcal{AC}(\Gamma)$ is closed under subsets, $\{x_1, x_2, x_3\} \in \mathcal{AC}(\Gamma)$. Thus, either $\{x_1, x_2, x_3\} \setminus A \in \mathcal{VC}(\Gamma)$ or $\{x_1, x_2, x_3\} \setminus A \in \mathcal{EC}(\Gamma)$. Since $\{x_1, x_2, x_3\}$ is disjoint to $X \supseteq A$, $\{x_1, x_2, x_3\} \setminus A = \{x_1, x_2, x_3\}$, so either $\{x_1, x_2, x_3\} \in \mathcal{VC}(\Gamma)$ or $\{x_1, x_2, x_3\} \in \mathcal{EC}(\Gamma)$. In the former case, set $E := E_{12} = \{P_1, P_2\}$. By repeating the previous argument (replacing x with x_1 and y with x_2), we find that $x_1, x_2 \in V(G')$, that $\phi(x_1) = x_1$ and $\phi(x_2) = x_2$, and that $\{x_1, x_2\}$ intersects $S'_{E,1} = S'_{T,1} \subseteq S'_T$. So $\phi(\{x_1, x_2\}) = \{x_1, x_2\}$ intersects $\phi(S'_T) = S_T \subseteq S_2$, a contradiction. So we may assume that $\{x_1, x_2, x_3\} \in \mathcal{EC}(\Gamma)$.

Recall that $\mathcal{EC}(\Gamma) \setminus F_3(\Gamma) = \mathcal{EC}(\Gamma) \setminus F_3(G_0)$ are the cliques of G_0 of size at most 2. Since $|\{x_1, x_2, x_3\}| = 3$, $\{x_1, x_2, x_3\} \in F_3(G_0)$. Since $\{x_1, x_2, x_3\}$ is disjoint to X , there exists $K' \in F_3(G_0^+)$ with $\phi(K') = \{x_1, x_2, x_3\}$. Since each facial triangle of G_0^+ has size exactly 3, we find that $K' = \{x'_1, x'_2, x'_3\}$ with $\phi(x'_i) = x_i$ for each $i \in \{1, 2, 3\}$. Note that $x'_i \in \phi^{-1}(x_i) \subseteq \phi^{-1}(P_i)$ for each $i \in \{1, 2, 3\}$. Set $T := \{P_1, P_2, P_3\}$ and $T' := \psi_C(T) = \{P'_1, P'_2, P'_3\}$, where $P'_i := \psi_C(P_i)$ for each $i \in \{1, 2, 3\}$. Note that $T \in \mathcal{Q}_3$, so $S_T \subseteq S_2$.

Since $\{x_1, x_2, x_3\}$ is disjoint to $S_2 \supseteq S_T = \phi(S'_T)$, $\{x'_1, x'_2, x'_3\}$ is disjoint to $S'_T \supseteq S'_{T,0}$. Thus, $\{x'_1, x'_2, x'_3\}$ does not intersect all three parts of T' . So there exists $i \in \{1, 2, 3\}$ such that $x'_i \notin P'_i$. Let $j := j_i$. (So $j := 1$ if $i \in \{2, 3\}$ and $j := 2$ if $i = 1$.) Since K' is a clique in G' (as $K' \in F_3(G_0^+)$), x'_i is adjacent to x'_j in G' . So $x'_i \in \phi^{-1}(P_i)$ is adjacent to $x'_j \in \phi^{-1}(P_j)$ in G' and satisfies $x'_i \notin P'_i$. Thus, (x'_i, x'_j) intersects $S'_{T,i} \subseteq S'_T$. Thus, $(x_i, x_j) = (\phi(x'_i), \phi(x'_j))$ intersects $\phi(S'_T) = S_T \subseteq S_2$. But this contradicts the fact that $\{x_i, x_j\} \subseteq \{x_1, x_2, x_3\}$ is disjoint to S_2 . So we conclude that $K \setminus S_2$ intersects at most two parts of Q . Since $|S_2| \leq (6k+8)\binom{b+1}{3}$ (and recalling that $\mathcal{AC}(\Gamma) \subseteq 2^{V(G)}$), we find that $(X, \mathcal{P})$ is $(2, (6k+8)\binom{b+1}{3}, \mathcal{AC}(\Gamma))$ -concentrated. This completes the proof. $\square$

We can now complete the proof of Lemma 7.17, which we now recall.

Lemma 7.17. *Let $b, k \in \mathbb{N}$ and $w, w', q \in \mathbb{R}^{\geq 0}$, let Γ be an almost-embedding of a pseudograph G , and let $\mathcal{PR}$ be a (b, k, w, w', q) -partitioned-reduction of Γ . If $\mathcal{PR}$ admits a raise τ , then G admits an almost-partition of treewidth at most b , width at most $\max((k+1)w, w')$, and loss at most q that is $(1, b(b+1)(k+1), \mathcal{VC}(\Gamma))$ -concentrated and $(2, (6k+8)\binom{b+1}{3}, \mathcal{AC}(\Gamma))$ -concentrated.*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$, and let $\mathcal{P} := \mathcal{P}_\tau$.

By Lemma 7.19, $(X, \mathcal{P})$ is an almost-partition of G . By Corollary 7.22, $(X, \mathcal{P})$ has treewidth at most b . By Lemma 7.26, $(X, \mathcal{P})$ has width at most $\max((k+1)w, w')$. Since $\mathcal{PR}$ has loss at most q , $|X| \leq q$, and thus $(X, \mathcal{P})$ has loss at most q . By Lemma 7.30, $(X, \mathcal{P})$ is $(1, b(b+1)(k+1), \mathcal{VC}(\Gamma))$ -concentrated and $(2, (6k+8)\binom{b+1}{3}, \mathcal{AC}(\Gamma))$ -concentrated. This completes the proof. $\square$

7.3 Finding a Raise

This section is devoted to the proof of Lemma 7.18, which takes a partitioned reduction and returns a new partitioned reduction that admits a raise.

7.3.1 Manipulating a Partitioned-Reduction

We finally reach the heart of the problem, finding a raise. Recall that the goal is not to find a raise of any partitioned-reduction, but rather to manipulate the partitioned-reduction until we can find a raise. Specifically, we merge some parts together, and add some more vertices to the loss set. The former leads to the following definition.

Let $\mathcal{P}, \mathcal{P}'$ be partitions of a pseudograph G . We say that $\mathcal{P}'$ is a *merging* of $\mathcal{P}$ if, for each $P \in \mathcal{P}$, there exists $P' \in \mathcal{P}'$ such that $P \subseteq P'$. In the literature, this is often phrased in reverse and called a ‘refinement’. That is, with the same condition, we would say that $\mathcal{P}$ is a refinement of $\mathcal{P}'$. Given that we are enlarging parts, we prefer to use phrasing that compares the larger to the smaller. Observe that if $\mathcal{P}'$ is a merging of $\mathcal{P}$, then $P \in \mathcal{P}$ intersects $P' \in \mathcal{P}'$ only if $P \subseteq P'$.

Observation 7.31. *Let $\mathcal{P}, \mathcal{P}'$ be proper partitions of a pseudograph G such that $\mathcal{P}'$ is connected and a merging of $\mathcal{P}$. Then $G/\mathcal{P}'$ is a minor of $G/\mathcal{P}$.*

Proof. For each $P \in \mathcal{P}'$, let $\mathcal{P}_P$ be the set of parts of $\mathcal{P}$ contained in P . Since $\mathcal{P}$ and $\mathcal{P}'$ are proper and by definition of a merging, observe that $\mathcal{P}^* := (\mathcal{P}_P : P \in \mathcal{P}')$ is a proper partition of $G/\mathcal{P}$.

For any $P' \in \mathcal{P}'$ and distinct $P_1, P_2 \in \mathcal{P}_{P'}$, since $\mathcal{P}$ is proper, there exists $u \in P_1$ and $v \in P_2$. Since $P_1, P_2 \in \mathcal{P}_{P'}$, we have $u, v \in P'$. Since $\mathcal{P}'$ is connected, there exists a path $u = x_0, \dots, x_n = v$ in $G[P']$. Since $\mathcal{P}$ is a partition of G , there exists a unique sequence $Q_0, \dots, Q_n$ such that for each $i \in \{0, \dots, n\}$, $x_i \in Q_i \in \mathcal{P}$. Note that $Q_0 = P_1$ and $Q_n = P_2$. Further, for each $i \in \{0, \dots, n\}$, since $x_i \in Q_i \cap P'$ and since $\mathcal{P}'$ is a merging of $\mathcal{P}$, observe that $Q_i \subseteq P'$ and thus $Q_i \in \mathcal{P}_{P'}$. Finally, for each $i \in \{0, \dots, n-1\}$, observe that either $Q_i = Q_{i+1}$, or that Q_i is adjacent to Q_{i+1} in $G/\mathcal{P}$. It follows that within $Q_0, \dots, Q_n$ we can find a path from P_1 to P_2 in $G/\mathcal{P}[\mathcal{P}_{P'}]$. Thus, $\mathcal{P}^*$ is a connected partition of $G/\mathcal{P}$, and hence $(G/\mathcal{P})/\mathcal{P}^*$ is a minor of $G/\mathcal{P}$.

Let $\phi : G/\mathcal{P}' \mapsto (G/\mathcal{P})/\mathcal{P}^*$ be the bijection that sends each $P \in \mathcal{P}'$ to $\mathcal{P}_P$. We show that ϕ is an isomorphism. Let distinct $P, P' \in \mathcal{P}'$ be adjacent in $G/\mathcal{P}'$. So there exists $u \in P$ and $v \in P'$ that are adjacent in G . There exists unique $Q, Q' \in \mathcal{P}$ such that $u \in Q$ and $v \in Q'$. Since $\mathcal{P}'$ is a merging of $\mathcal{P}$, note that $Q \in \mathcal{P}_P$ and $Q' \in \mathcal{P}_{P'}$. In particular, since $P \neq P'$, this gives $Q \neq Q'$, and thus Q is adjacent to Q' in $G/\mathcal{P}$. Thus and since $Q \in \mathcal{P}_P$ and $Q' \in \mathcal{P}_{P'}$, we have that $\mathcal{P}_P$ is adjacent to $\mathcal{P}_{P'}$ in $(G/\mathcal{P})/\mathcal{P}^*$.

Let distinct $P, P' \in \mathcal{P}'$ be such that $\mathcal{P}_P$ and $\mathcal{P}_{P'}$ are adjacent in $(G/\mathcal{P})/\mathcal{P}^*$ (and thus $G/\mathcal{P}$). There exists $Q \in \mathcal{P}_P$ and $Q' \in \mathcal{P}_{P'}$ such that Q, Q' are adjacent in $G/\mathcal{P}$. Hence, there exists $u \in Q$ and $v \in Q'$ such that u, v are adjacent in G . Since $Q \in \mathcal{P}_P$ and

$Q' \in \mathcal{P}_{P'}$, we have $u \in P$ and $v \in P'$, and thus P, P' are adjacent in $G/\mathcal{P}'$. Therefore, ϕ is an isomorphism from $G/\mathcal{P}'$ to $(G/\mathcal{P})/\mathcal{P}^*$. Since $(G/\mathcal{P})/\mathcal{P}^* \leq G/\mathcal{P}$, this gives $G/\mathcal{P}' \leq G/\mathcal{P}$, as desired. $\square$

Observation 7.31 leads to the following.

Observation 7.32. *Let $b, k \in \mathbb{N}$ and $w, w', q \in \mathbb{R}^{\geq 0}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a (b, k, ∞, w', q) -partitioned-reduction for Γ , and let $\mathcal{P}'_0$ be a connected partition of G_0^+ that:*

1. *contains $V(W)$,*
2. *is a merging of $\mathcal{P}_0^+$, and*
3. *satisfies $|P| \leq w$ for each $P \in \mathcal{P}'_0$ distinct from $V(W)$.*

Then $\mathcal{PR}' := (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}'_0)$ is a (b, k, w, w', q) -partitioned-reduction of Γ .

Proof. As $\mathcal{PR}$ is a partitioned-reduction of Γ , $\mathcal{R} := (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is a reduction of Γ . Thus and since $\mathcal{P}'_0$ is a connected partition of G_0^+ that contains $V(W)$, $\mathcal{PR}'$ is a partitioned-reduction of Γ . Since the quasi-vortex-width, remainder-width and loss of $\mathcal{PR}'$ and $\mathcal{PR}$ are determined completely by $\mathcal{R}$, the quasi-vortex-width, remainder-width and loss of $\mathcal{PR}'$ are the quasi-vortex-width, remainder-width and loss of $\mathcal{PR}$ respectively. Further, the near-width of $\mathcal{PR}'$ is the maximum size of a part of $\mathcal{P}'_0$ distinct from $V(W)$, which is at most w . So $\mathcal{PR}'$ is an (∞, k, w, w', q) -partitioned-reduction for Γ .

Let $\mathcal{P}_1^+, \mathcal{P}_1^*$ be obtained from $\mathcal{P}_0^+, \mathcal{P}'_0$ respectively by deleting all empty parts. Observe that $\mathcal{P}_1^+, \mathcal{P}_1^*$ are connected proper partitions of G , and that $\mathcal{P}_1^*$ is a merging of $\mathcal{P}_1^+$. Thus, by Observation 7.31, $G/\mathcal{P}_1^+$ is a minor of $G/\mathcal{P}_1^*$. Hence, observe that the treewidth of $G/\mathcal{P}_1^+$ is at most the treewidth of $G/\mathcal{P}_1^*$.

Observe that $G/\mathcal{P}_0^+, G/\mathcal{P}'_0$ are obtained from $G/\mathcal{P}_1^+, G/\mathcal{P}_1^*$ respectively by adding isolated vertices. Noting that neither $G/\mathcal{P}_1^+$ nor $G/\mathcal{P}_1^*$ are empty (as $V(W)$ is a part in both partitions), this does not change the treewidth. Hence, the treewidth of $G/\mathcal{P}'_0$ is at most the treewidth of $G/\mathcal{P}_0^+$, which is at most b . So $\mathcal{PR}'$ is a (b, k, w, w', q) -partitioned-reduction for Γ , as desired. $\square$

Observation 7.32 covers merging parts. As for adding vertices to the loss-set, specifically, we want to take a set of vertices on the modified boundary, and add all vertices in their bags to the loss set. This leads to the following definition.

Let Γ be an almost-embedding of a pseudograph G , and let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of Γ , and let $(J'_x : x \in B'(\mathcal{PR})) := \mathcal{J}'$. For $S \subseteq B'(\mathcal{PR})$, let $X_S := \bigcup_{s \in S} J'_s$, and let $\mathcal{PR} - S := (Y, X \cup X_S, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$.

Observation 7.33. *Let $b, k \in \mathbb{N}$ and $w, w', q \in \mathbb{R}^{\geq 0}$, let Γ be an almost-embedding of a pseudograph G , let $\mathcal{PR}$ be a (b, k, w, w', q) -partitioned-reduction for Γ , and let $S \subseteq B'(\mathcal{PR})$. Then $\mathcal{PR} - S$ is a $(b, k, w, w', q + (k + 1)|S|)$ -partitioned-reduction of Γ .*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$ with $\mathcal{J}' = (J'_x : x \in B'(\mathcal{PR}))$. So $\mathcal{PR} - S = (Y, X \cup X_S, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$, where $X_S := \bigcup_{s \in S} J'_s$. Since $\mathcal{PR}$ has quasi-vortex-width at most k , $|X_S| \leq (k + 1)|S|$. Note that $\mathcal{R} := \mathcal{R}(\mathcal{PR}) = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is a (k, w', q) -reduction of Γ . Observe that $X_S \subseteq V(H') \subseteq V(H) \subseteq V(G)$. Thus, by Observation 7.11, $\mathcal{R}' := (Y, X \cup X_S, G', G_0^+, W, D, H', \mathcal{J}', \phi)$ is a $(k, w', q + (k + 1)|S|)$ -reduction of Γ . Since $\mathcal{R}'$ has the same plane sub-pseudograph and obstruction sub-pseudograph as $\mathcal{R}$, it follows that $(Y, X \cup X_S, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ is a $(b, k, w, w', q + (k + 1)|S|)$ -partitioned-reduction of Γ . $\square$

7.3.2 Proof Idea

As hinted at in Section 7.1.3, the goal is to merge, and then find some breakpoints S in the modified boundary B' such that $\mathcal{PR} - S$ admits a raise. To decide on which part to assign a vertex v of the quasi-vortex to, we look only at the interval I_v of the modified underlying cycle U' induced by the vertices $x \in B'$ whose bag contains v . If $V(I_v)$ intersects S , then v will be in the loss-set, and is thus of no concern. Otherwise, ideally, $V(I_v)$ is contained entirely in one part P , in which case we assign v to P . However, after performing our algorithm, we find that there is exactly one scenario where this is false. In this case, we have one part P that intersects $V(I_v)$ and locally dominates the area around I_v . Specifically, each other part P' that intersects or is adjacent to $V(I_v)$ is adjacent to P . Further, P is the only such part in the area, meaning that if $v \in B'$ and is adjacent to another $u \in B'$, then either $V(I_u)$ (where I_u is the interval of U' induced by the vertices $x \in B'$ whose bag contains u) is contained in a single part, or $V(I_u)$ intersects P . In this case, we instead assign v to P , possibly moving it out of a part that already contained it (if $v \in B'$).

The first question is ‘how to merge the parts’? The idea is to use a variant of Lemma 6.5 [45] to split up the plane sub-pseudograph G_0^+ . Specifically, we want to approximate G_0^+ with a pseudograph of bounded treewidth. For this, we recall Lemma 5.19 [61].

Lemma 5.19. *For all $g, r \in \mathbb{N}$, every (unweighted) graph of radius at most r and Euler genus at most g has treewidth at most $(2g + 3)r$.*

Lemma 5.19 extends trivially to pseudographs, since deleting parallel edges and loops changes neither the treewidth nor the radius.

We utilise an easy corollary of Lemma 6.5 [45], which we recall for convenience.

Lemma 6.5. *For every $q \in \mathbb{N}$ and $W \in \mathbb{R}^{\geq 0}$, every vertex-weighted tree T with weight at most W has a set Z of at most q vertices such that each component of $T - Z$ has total weight at most $\frac{W}{q+1}$.*

Corollary 7.34. *Let $b \in \mathbb{N}$, $W \in \mathbb{R}^{\geq 0}$, and $d \in \mathbb{R}^{> 0}$, and let G be a vertex-weighted pseudograph of treewidth at most b and total weight at most W . Then there exists $S \subseteq V(G)$ with $|S| \leq (b+1)W/d$ such that each connected component of $G - S$ has weight at most d .*

Proof. Let $\mathcal{J} = (J_t : t \in V(T))$ be a tree-decomposition of G of width at most b . For each $t \in V(T)$, let K_t be the parent-adhesion set at t , and let $J'_t := J_t \setminus K_t$. Weight t by the total weight of J'_t . ($J'_t : t \in V(T)$) is a partition of G , so the total weight on T is at most W . By Lemma 6.5 with $q := \lfloor W/d \rfloor$, there exists $Z \subseteq V(T)$ with $|Z| \leq q \leq W/d$ such that each connected component of $T - Z$ has total weight at most $W/(q+1) \leq d$ (as $W/d \leq q+1$).

Let $S := \bigcup_{z \in Z} J_z$. Since $\mathcal{J}$ has width at most b , $|S| \leq (b+1)|Z| \leq (b+1)W/d$. Observe that for each connected component C of $G - S$, there exists a component T_C of $T - Z$ such that $V(C) \subseteq \bigcup_{t \in V(T_C)} J'_t$. Thus, the total weight of C is at most the total weight of T_C , which is at most d , as desired. $\square$

The plan is to use Lemma 5.19 and Corollary 7.34 in combination to split up G_0^+ . The issue is that G_0^+ does not have bounded radius. However, we are only interested in the behaviour of G_0^+ near B' , and in the behaviour of parts rather than individual vertices. So we contract $V(W)$ and each part that intersects B' to a single vertex, and each connected component of the pseudograph after deleting these parts into a single vertex. As $\Lambda(\mathcal{PR})$ is standardised, $B' \subseteq N_G(V(W))$, and thus the resulting plane (pseudo)graph G_0^* has radius 2 after deleting isolated vertices (connected components of G_0^+ that do not intersect B'). So Lemma 5.19 and Corollary 7.34 can be applied. Specifically, we weight each vertex in G_0^* according to how many vertices were contracted into that vertex. Note also that we can view G_0^* as the quotient of a connected partition $\mathcal{P}_0^*$ of G_0^+ (up to deleting parallel edges and loops, which are not important here).

Backtracking, we find that Corollary 7.34 would delete a small number of parts/components S^* , and that every remaining component D of $G_0^* - S^*$ would have small weight. D corresponds to a set of vertices P_D of G_0^+ which induces a connected subgraph

and is small because D has small weight. S^* corresponds to a set of vertices S' of G . If S' is also small, then could make each set P_D an (isolated) part in $G_0^+ - S'$, and assign each interval disjoint to S' to the unique part it intersects (this uses the fact that the disc is snug). However, there is no reason to expect this, as while S^* consists of a small number of subsets (vertices of components/parts), the total number of vertices in these subsets might be large. Fortunately, we do not need to separate the entire pseudograph, because, once again, we are only interested in the area around B' . Even more specifically, we are interested in U' , and where the vertices in quasi-vortex lie above U' . We find that it suffices to only break up U' (by introducing breakpoints), as this controls the behaviour of the quasi-vortex's vertices.

This is where we can make some headway. Because the partition was connected, each of these parts/components are connected, and thus by the non-crossing property (Lemma 7.6), their intersections with U' block off parts of the underlying cycle from the rest of $G'_0 := G'_0(\mathcal{PR}) = G_0^+ - V(W)$. We pick an arbitrary starting point for U' to form a linear ordering of B' , and for each part P we wanted to delete (is in S^*), we remove (add to the breakpoints S) just the first and last vertices of $P \cap B'$ under this linear ordering. This has the effect of restricting any interval of $U' - S$ to a small region of the pseudograph that contains at most one of these parts we wanted to delete. We also need to control the neighbourhoods around intervals, so we repeat the small process with the neighbourhood (with some exclusions) around each part/component we wanted to delete. The result is that every interval of $U' - S$ and its neighbourhood lies in a very restricted region of G_0^+ . Specifically, we find that a part in S^* is locally dominating within this region, as described earlier.

The desired partition is obtained from merging all the parts of $\mathcal{P}_0^*$ contained in a component of $G_0^* - S^*$, and leaving all the parts of $\mathcal{P}_0^+$ contained in part or component in S^* untouched. The parts in S^* are the locally dominating parts mentioned at the start of this explanation.

In terms of a proof, the set S^* inspires the following definition.

Let $\mathcal{PR} = (Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ be a partitioned-reduction of an almost-embedding Γ . Let $G'_0 := G'_0(\mathcal{PR}) = G_0^+ - V(W)$. A *break* for $\mathcal{PR}$ is a set $\mathcal{B}$ of pairwise disjoint connected sub-pseudographs of G'_0 such that:

1. for each $M \in \mathcal{B}$ and $P \in \mathcal{P}_0^+$, if P intersects $V(M)$, then $P \subseteq V(M)$,
2. for each $M \in \mathcal{B}$, at least one of the following is true:
 - (a) $M = G'_0[P]$ for some $P \in \mathcal{P}_0^+$ distinct from $V(W)$, or,
 - (b) $V(M)$ is disjoint to $B'(\mathcal{PR})$, and
3. for each connected sub-pseudograph C of G'_0 , at least one of the following is true:

- (a) C intersects some $M \in \mathcal{B}$, or,
- (b) $V(C) \subseteq P$ for some $P \in \mathcal{P}_0^+$.

Since $V(W) \in \mathcal{P}_0^+$, for each $P \in \mathcal{P}_0^+$ distinct from $V(W)$, $P \subseteq V(G'_0)$. So $G'_0[P]$ is well-defined in 2(a).

The components and (subgraphs induced by) parts in S^* will be the elements of the break. See Figure 7.4 for a step by step visual explanation of the algorithm. The proof is divided into two steps: merging parts to find a small break, and then using the small break to find a small set of breakpoints, which is then used to find a raise. We start with the former step, as it is by far the easier.

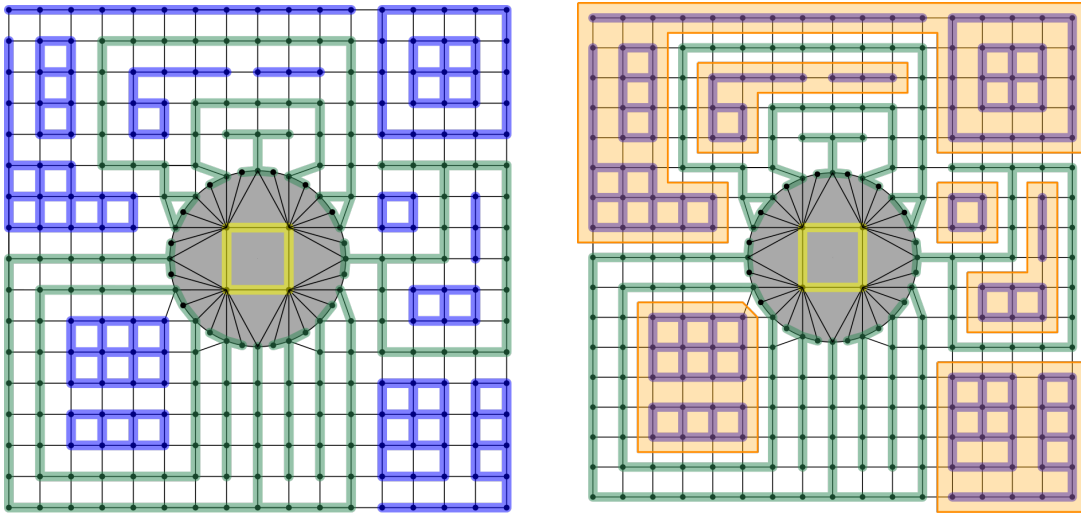


Figure 7.4: A step by step visual guide to how the partition is modified so that a raise can be found. Start with plane pseudograph G_0^+ , a sub-pseudograph W (in yellow), a disc that is (G_0^+, W) -clean and snug (in grey) and whose boundary vertices are contained in the neighbourhood of $V(W)$, and an initial partition $\mathcal{P}_0^+$ containing $V(W)$ as a part. The other parts intersecting the disc have been coloured green, and the parts not intersecting the disc have been coloured blue. These blue parts are then grouped together (in orange) according the components after the green parts (and $V(W)$) are deleted.

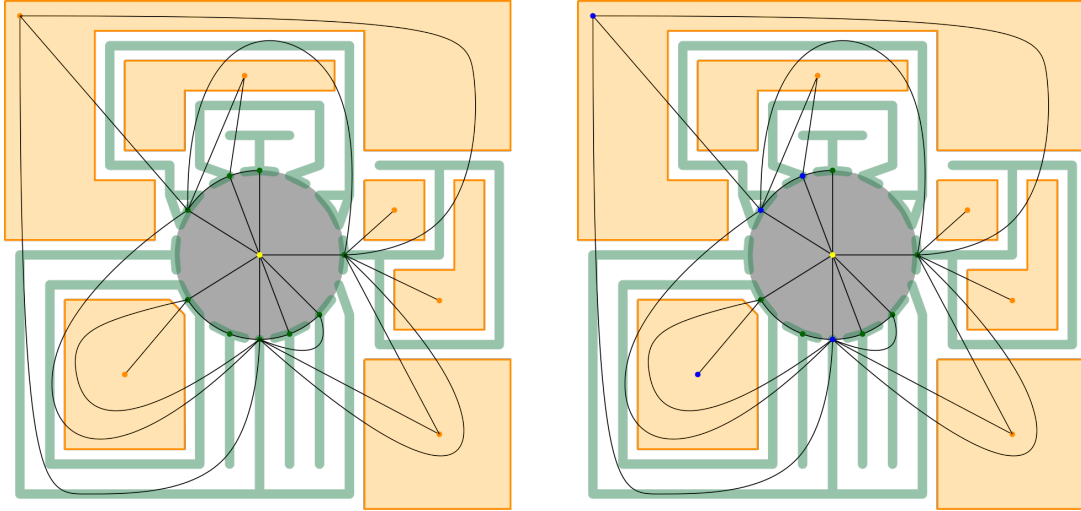


Figure 7.4 continued: The green parts, orange components, and $V(W)$ are then contracted to form an auxiliary graph G_0^* . Note that this graph is planar and has radius 2 (excluding connected components that disjoint to the disc), achieved using $V(W)$ (in yellow). The contracted parts (green) are each adjacent to $V(W)$, and the contracted components (orange) are pairwise non-adjacent nor adjacent to $V(W)$. A set $S' \subseteq V(G_0^*)$ (in blue) is chosen to break up $G_0^* - V(W)$.

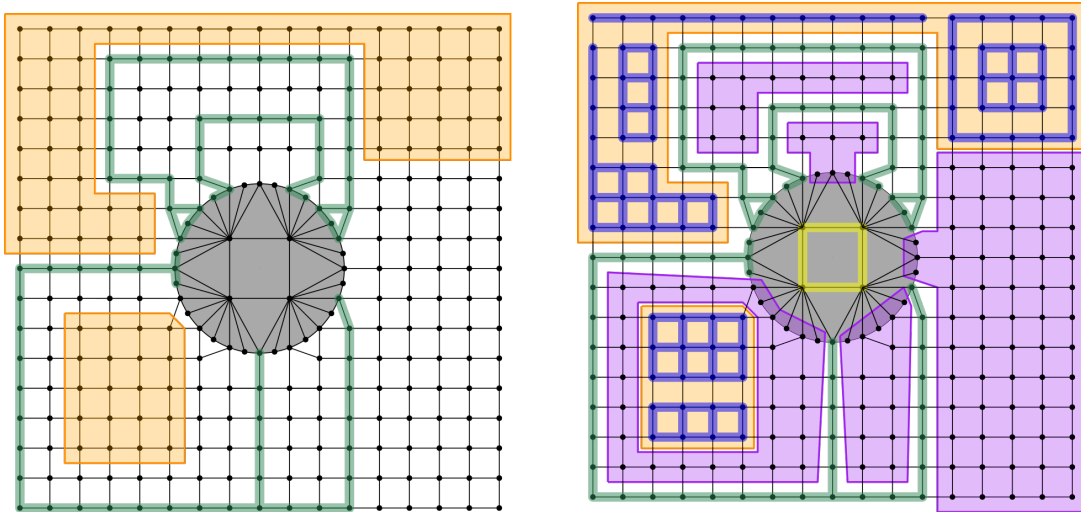


Figure 7.4 continued: S' corresponds to a set of green parts and orange components, which are the elements of the break. The components of the graph after the elements of the break (and $V(W)$) are deleted are merged into parts (in purple). The green parts in the break and the blue parts contained in an orange component in the break stay as parts.

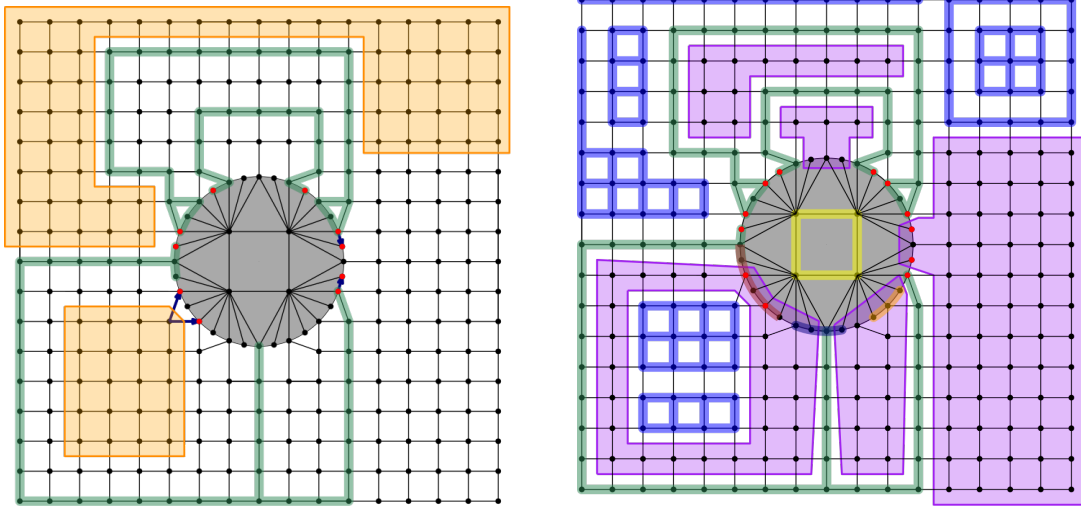


Figure 7.4 continued: A set of breakpoints (in red) is then chosen, with each member of the break contributing up to four. Specifically, each member of the break adds its first and last intersections as breakpoints, and the same for its neighbourhood excluding vertices in other elements of the break (these breakpoints are shown using navy arrows). The raise maps vertices v in the quasi-vortex according to interval I_v indexed by vertices whose bags contain v . If I_v intersects a breakpoint, such as how the brown interval does, then v is added to the loss-set. Otherwise, if I_v intersects a member of the break, such as how the navy interval does, then it intersects at most one member, and this member is a (green) part in both the original and final partition; v is sent to this part. Finally, if I_v intersects neither a breakpoint nor a member of the break, such as how the orange interval does, then I_v is contained in a purple part of the final partition; v is sent to this part.

7.3.3 Finding a Break

Lemma 7.35. *Let $b, k, n \in \mathbb{N}$ and $w, w', q \in \mathbb{R}^{\geq 0}$, let Γ be an almost-embedding of an n -vertex pseudograph G , and let $\mathcal{PR}$ be a (b, k, w, w', q) -partitioned-reduction of Γ . Then for any $d \in \mathbb{R}^{\geq 0}$ with $d \geq w$, there exists a (b, k, d, w', q) -partitioned-reduction $\mathcal{PR}'$ of Γ that admits a break of size at most $7n/d$.*

Proof. Let $\mathcal{PR} = (Y, X', G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$ and $G'_0 := G'_0(\mathcal{PR}) = G_0(\Lambda(\mathcal{PR})) = G_0^+ - V(W)$. Recall that D is G'_0 -clean. Let $B' := B'(\mathcal{PR}) = B(D, G'_0) = V(G_0^+ \cap H')$. Recall that $B' \subseteq V(H')$ is disjoint to $V(W)$. Set $\Lambda := \Lambda(\mathcal{R}) = (G', G_0^+, W, D, H', \mathcal{J}')$. So Λ is a standardised plane+quasi-vortex embedding.

Let $\mathcal{P}_B := (P \in \mathcal{P}_0^+ : P \cap B' \neq \emptyset)$, and let $V_B := \bigcup_{P \in \mathcal{P}_B} P$. Since $V(W) \in \mathcal{P}_0^+$ and since $V(W)$ is disjoint to B' , observe that V_B is disjoint to $V(W)$. So $V_B \subseteq V(G'_0)$. Note also that $B' \subseteq V_B$. Let $\mathcal{C}$ be the connected components of $G'_0 - V_B$. Since $\mathcal{P}_0^+$ is a connected partition of G_0^+ , observe that $\mathcal{P}_0^* := \mathcal{P}_B \sqcup (V(C) : C \in \mathcal{C}) \sqcup (V(W))$ is a

connected (proper) partition of G_0^+ that is a merging of $\mathcal{P}_0^+$.

Let $G_0^* := G_0^+ / \mathcal{P}_0^*$. Since $\mathcal{P}_0^*$ is a connected partition of G_0^+ , G_0^* is a minor of G_0^+ . Thus, G_0^* is planar. Since Λ is standardised, $B' \subseteq N_{G_0^+}(V(W))$. It follows that each $P \in \mathcal{P}_B$ is adjacent to $V(W)$ in G_0^* . Further, observe that for each $C \in \mathcal{C}$, $V(C)$ is either adjacent to some $P' \in \mathcal{P}_B$ in G_0^* , or is isolated in G_0^* . Thus, observe that G_0^* has one component containing $V(W)$ with radius at most 2, and every other component is an isolated vertex. By Lemma 5.19, the component of G_0^* containing $V(W)$ has treewidth at most 6. Thus G_0^* has treewidth at most 6, as adding isolated vertices does not increase this bound. Hence, $G_0^* - V(W)$ also has treewidth at most 6.

Fix $P \in V(G_0^* - V(W))$. Note that $P \in \mathcal{P}_0^*$. In $G_0^* - V(W)$, weight P by $|P|$. Since $\mathcal{P}_0^*$ is a partition of G_0^+ , the total weight on $G_0^* - V(W)$ is exactly $|V(G_0^*)| \leq |V(G)| = n$. By Corollary 7.34, there exists a set $S^* \subseteq V(G_0^*)$ with $|S^*| \leq 7n/d$ such that each connected component of $G_0^* - S^* - V(W)$ has total weight at most d . Note that $V(W) \notin S^*$.

Let $\mathcal{D}$ denote the connected components of $G_0^* - S^* - V(W)$. So each $D \in \mathcal{D}$ has total weight at most d . Let $P_D := \bigcup_{P \in V(D)} P$. Observe that $|P_D|$ is exactly the total weight of D , which is at most d . So $|P_D| \leq d$. Since $\mathcal{P}_0^*$ is a connected partition of G_0^+ and since D is connected, $G_0^+[P_D]$ is connected.

Let $\mathcal{P}_S$ be the set of parts $P \in \mathcal{P}_0^+$ such that $P \subseteq Q$ for some $Q \in S^*$. Note that $V(W) \notin \mathcal{P}_S$ since $V(W) \notin S^*$. Thus and since $\mathcal{P}_0^*$ is a merging of $\mathcal{P}_0^+$, observe that $\mathcal{P}'_0 := (P_D : D \in \mathcal{D}) \sqcup \mathcal{P}_S \sqcup (V(W))$ is a connected partition of G_0^+ containing $V(W)$ that is a merging of $\mathcal{P}_0^+$. Since $\mathcal{PR}$ has near-width at most $w \leq d$, each part of $\mathcal{P}_0^+$ other than $V(W)$ has size at most d . Thus, and since $|P_D| \leq d$ for each $D \in \mathcal{D}$ (and since $\mathcal{P}_S \sqcup (V(W)) \subseteq \mathcal{P}_0^+$), each part of $\mathcal{P}'_0$ other than $V(W)$ has size at most d . Hence, by Observation 7.32 (with $w := d$), $\mathcal{PR}' := (Y, X', G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}'_0)$ is a (b, k, d, w', q) -partitioned-reduction of Γ .

It remains only to find a break of size at most $7n/d$ for $\mathcal{PR}'$. Since $\mathcal{P}_0^*$ is a connected partition of G_0^+ and since $V(W) \notin S^*$, observe that $\mathcal{B} := \{G'_0[Q] : Q \in S^*\}$ is a well-defined collection of pairwise disjoint connected sub-pseudographs of G'_0 . Note that $|\mathcal{B}| = |S^*| \leq 7n/d$. We show that $\mathcal{B}$ is a break for $\mathcal{PR}'$.

Fix $P \in \mathcal{P}_0^+$ and $M \in \mathcal{B}$. So $M = G'_0[Q]$ for some $Q \in S^*$. If $P = P_D$ for some $D \in \mathcal{D}$, observe that P is disjoint to $Q = V(M)$ (as D is a connected component of $G_0^* - S^* - V(W)$). If $P = V(W)$, observe that P is disjoint to $Q = V(M)$ as $Q \subseteq V(G'_0) = V(G_0^+ - V(W))$. Otherwise, observe that $P \in \mathcal{P}_S$, and thus $P \subseteq Q'$ for some $Q' \in S^*$. Since $S^* \subseteq \mathcal{P}_0^*$, either $Q = Q'$, in which case $P \subseteq Q$, or $Q = V(M)$ is disjoint from Q' and thus P . Thus, if P intersects $V(M) = Q$, then $P \subseteq V(M)$ (and

$P \in \mathcal{P}_S$).

Fix $M \in \mathcal{B}$. So $M = G'_0[Q]$ for some $Q \in S^*$. Recall that $S^* \subseteq V(G'_0 - V(W)) = \mathcal{P}_B \sqcup (V(C) : C \in \mathcal{C})$. If $Q \in \mathcal{P}_B \subseteq \mathcal{P}_0^+$, observe that $Q \in \mathcal{P}_S \subseteq \mathcal{P}'_0$ (as $Q \subseteq Q$). So $M = G'_0[P]$ for $P = Q \in \mathcal{P}'_0$. (Note also that Q is distinct from $V(W)$.) Otherwise, if $Q = V(C)$ for some $C \in \mathcal{C}$, then by definition of $\mathcal{C}$ (and since $B' \subseteq V_B$), $Q = V(M)$ is disjoint from B' .

Consider any connected sub-pseudograph C' of G'_0 . Observe that either $V(C')$ intersects some $Q \in S^*$, or $V(C') \subseteq P_D$ for some $D \in \mathcal{D}$. In the former case, C' intersects $G'_0[Q] \in \mathcal{B}$, and in the latter case, $V(C') \subseteq P_D \in \mathcal{P}'_0$. Thus, $\mathcal{B}$ is a break for $\mathcal{PR}'$ of size $|S^*| \leq 7n/d$. This completes the proof. $\square$

7.3.4 Using the Break

We now move to the latter step, using the break to find breakpoints, and then a raise. Really, this step is two steps. The first takes a set of connected sub-pseudographs (related to the elements of a break) and, according to the intersections between the underlying cycle and these sub-pseudographs, finds a set of breakpoints.

Specifically, fixing a starting point of the underlying cycle to create a linear ordering, for each sub-pseudograph, the first and last intersection (with respect to the linear ordering) of that sub-pseudograph with the underlying cycle become breakpoints. This ensures that if we have two disjoint sub-pseudographs H_1, H_2 , each interval disjoint from the breakpoints intersects at most one of H_1 and H_2 (by the non-crossing property, Lemma 7.6). This is used to assign intervals to parts (either an element of the break that the interval intersects, or a part containing the interval). It also ensures that if we have two intersecting or adjacent (in G_0^+) intervals I_0, I'_0 disjoint to H_1 and H_2 (and the breakpoints), and then grow these intervals (while maintaining that they are disjoint from the breakpoints), the union of the grown intervals I, I' intersects at most one of H_1 and H_2 . This is used to ensure that intersecting or adjacent (in G_0^+) intervals I, I' are assigned to the same part or to adjacent parts.

Lemma 7.36. *Let G be a plane pseudograph, let $\mathcal{H}$ be a set of connected sub-pseudographs of G , and let D be a G -clean disc in the plane. Then there exists a set $S \subseteq B(D, G)$ with $|S| \leq 2|\mathcal{H}|$ such that if $\mathcal{I}$ denotes the set of connected and nonempty subgraphs of $U(D, G) - S$, and $H_1, H_2 \in \mathcal{H}$ are disjoint, then the following hold:*

1. *each $I \in \mathcal{I}$ intersects at most one of H_1, H_2 , and*
2. *if $I, I_0, I', I'_0 \in \mathcal{I}$ are such that:*
 - (a) $I_0 \subseteq I$,

- (b) $I'_0 \subseteq I'$,
 - (c) $I_0 \cup I'_0$ is disjoint to $H_1 \cup H_2$, and
 - (d) $N_G(V(I_0))$ intersects $V(I'_0)$,
- then $I \cup I'$ intersects at most one of H_1, H_2 .

Proof. Let $(B, U) := (B(D, G), U(D, G))$. Recall that $B = V(U) \subseteq V(G)$. If $|B| = 0$, then the lemma is trivially true with $S = \emptyset$. So we may assume that $|B| \geq 1$. Fix any $r \in B$. Recall that U is a cycle or a path on at most 2 vertices. Let $\preceq_U$ denote the cyclic ordering of B induced by D , and let $\preceq$ be the total ordering of B obtained from $\preceq_U$ by starting at r . Note that if $a, b, x, y \in B$ are such that $x \prec a \prec y \prec b$, then $x \prec_U a \prec_U y \prec_U b$ also.

For each $H \in \mathcal{H}$, if $|V(H) \cap B| \leq 2$, let $S_H := V(H) \cap B$. Otherwise, observe that there exists distinct $x_H, y_H \in V(H) \cap B$ such that $x_H \prec y_H$ and, for each $z \in V(H) \cap B$, $x_H \preceq z \preceq y_H$. Let $S_H := \{x_H, y_H\}$ in this case. Observe that in either scenario, $|S_H| \leq 2$, and that $S_H \subseteq V(H) \cap B$.

Let $S := \bigcup_{H \in \mathcal{H}} S_H$. Note that $|S| \leq 2|\mathcal{H}|$, and that $S \subseteq B$. Let $\mathcal{I}$ be the set of connected and nonempty subgraphs of $U - S$.

We start by proving the first property. We need the following claim.

Claim 1. *Let $I \in \mathcal{I}$ and $H \in \mathcal{H}$ intersect. Then there exists distinct $x, y \in V(H) \cap S$ such that for each $z \in V(I)$, $x \prec z \prec y$.*

Proof. Since I and H intersect, there exists $u \in V(I \cap H)$.

By definition of $I \in \mathcal{I}$, $V(I) \subseteq V(U - S) = B \setminus S$. Thus, $V(H)$ intersects $B \setminus S$ (in particular, $u \in V(H) \cap (B \setminus S)$).

First, assume, for a contradiction, that $|V(H) \cap B| \leq 2$. Recall then that $V(H) \cap B = S_H \subseteq S$. So $V(H) \cap B$ is contained in S , contradicting the fact that $V(H)$ intersects $B \setminus S$. So we can conclude that $|V(H) \cap B| \geq 3$.

Thus, recall that $S_H = \{x_H, y_H\}$, where $x_H, y_H \in V(H) \cap B$ are distinct and satisfy $x_H \preceq z \preceq y_H$ for each $z \in V(H) \cap B$. Further, recall that $\{x_H, y_H\} = S_H \subseteq S$. So $x_H, y_H \in V(H) \cap S$. Since I is disjoint to S , we find that $\{x_H, y_H\}$ is disjoint to $V(I)$. Thus and since $u \in V(I \cap H) \subseteq V(H) \cap B$, we have $x_H \prec u \prec y_H$.

Since U is a cycle or a path (and since $|V(H) \cap B| \geq 3$), observe that there is a connected component C of $U - x_H - y_H$ such that $V(C)$ is exactly the vertices $z \in B$ for which $x_H \prec z \prec y_H$. In particular, $u \in V(C)$. Since I is connected and contains u but neither

x_H nor y_H , it follows that $x_H \prec z \prec y_H$ for each $z \in V(I)$. The claim follows with $x := x_H$ and $y := y_H$. $\blacksquare$

We can now prove the first property. Presume otherwise. So $H_1, H_2 \in \mathcal{H}$ are disjoint and $I \in \mathcal{I}$ intersects both H_1 and H_2 . Thus, there exists $u, v \in V(I)$ with $u \in V(H_1)$ and $v \in V(H_2)$. Note that since H_1 and H_2 are disjoint, $u \neq v$.

By Claim 1 with $H := H_1$, there exists distinct $x, y \in V(H_1) \cap S$ such that for each $z \in V(I)$, $x \prec z \prec y$. In particular, $x \prec u, v \prec y$. Similarly (by Claim 1 with $H := H_2$), there exists distinct $a, b \in V(H_2) \cap S$ such that $a \prec u, v \prec b$. So $x, a \prec u, v \prec b, y$. Since H_1 and H_2 are disjoint, observe that a, b, u, v, x, y are all pairwise distinct.

Recall that $x, y, u \in V(H_1) \cap B$, and that $a, b, v \in V(H_2) \cap B$. Since H_1, H_2 are connected and disjoint sub-pseudographs of G , by Lemma 7.6, we have neither $x \prec_U a \prec_U u \prec_U b$ nor $x \prec_U v \prec_U y \prec_U a$. Since $a \neq x$, either $x \prec a$ or $a \prec x$. If $x \prec a$, then we obtain $x \prec a \prec u \prec b$, and thus $x \prec_U a \prec_U u \prec_U b$, contradiction. If instead $a \prec x$, then we obtain $a \prec x \prec v \prec y$ and thus $a \prec_U x \prec_U v \prec_U y$. This gives $x \prec_U v \prec_U y \prec_U a$ (as $\prec_U$ is cyclic), another contradiction. So in either scenario, we have a contradiction. It follows that I intersects at most one of H_1, H_2 . This proves the first property.

We now prove the second property. Presume otherwise. So $H_1, H_2 \in \mathcal{H}$ are disjoint, and $I, I_0, I', I'_0 \in \mathcal{I}$ are such that $I_0 \subseteq I, I'_0 \subseteq I', I_0 \cup I'_0$ is disjoint to $H_1 \cup H_2$, $N_G(V(I_0))$ intersects $V(I'_0)$, and $I \cup I'$ intersects both H_1, H_2 . Observe that if I intersects I' , then $I \cup I'$ is connected, and $I \cup I' \in \mathcal{I}$. As $I \cup I'$ intersects both H_1 and H_2 , this contradicts the first property. So we may assume that I and I' are disjoint. Thus, since $I_0 \subseteq I$ and $I'_0 \subseteq I'$, we have that I_0 and I'_0 are disjoint. Since $N_G(V(I_0))$ intersects $V(I'_0)$ and since I_0 is disjoint to I'_0 , there exists distinct $u \in V(I_0)$ and $v \in V(I'_0)$ such that u and v are adjacent in G .

By the first property, neither I nor I' intersect both H_1 and H_2 . Thus and since $I \cup I'$ intersects both H_1 and H_2 , without loss of generality (up to swapping H_1 and H_2), I intersects H_1 and I' intersects H_2 . By Claim 1 with $H := H_1$ and $I := I$, there exists distinct $x, y \in V(H_1) \cap S$ such that for each $z \in V(I)$, $x \prec z \prec y$. In particular, $x \prec u \prec y$. Similarly (by Claim 1 with $H := H_2$ and $I := I'$), there exists distinct $a, b \in V(H_2) \cap S$ such that $a \prec v \prec b$. Since H_1, H_2 , and $I_0 \cup I'_0$ are pairwise disjoint, observe that a, b, u, v, x, y are all pairwise disjoint. Further, observe that $H_1, H_2, G[\{u, v\}]$ are pairwise disjoint connected sub-pseudographs of G .

Since $x \neq a$, observe that either $x \prec a$ or $a \prec x$. Without loss of generality (up to swapping (H_1, I, I_0, u, x, y) and (H_2, I', I'_0, v, a, b)), presume $x \prec a$. Since $u \neq a$, either $u \prec a$ or $a \prec u$. If $u \prec a$, then $u \prec a \prec v \prec b$ and thus $u \prec_U a \prec_U v \prec_U b$. This

contradicts Lemma 7.6 with $u, v \in G[\{u, v\}]$ and $a, b \in V(H_2)$. So we can assume $a \prec u$. Since $u \prec y$, this implies $a \prec y$. Since $u \neq b$, either $u \prec b$ or $b \prec u$. If $b \prec u$, then $a \prec v \prec b \prec u$ and thus $a \prec_U v \prec_U b \prec_U u$. This again contradicts Lemma 7.6 with $u, v \in V(G[\{u, v\}])$ and $a, b \in V(H_2)$. So we can assume $u \prec b$.

Thus, we obtain $x \prec a \prec u \prec b$. Recall that I intersects H_1 . So there exists $w \in V(I \cap H_1)$. Since $a, b \in S$ and since U is a cycle or a path, observe that the component of $U - S$ containing u (which exists since $u \in V(I_0) \subseteq V(U - S)$) consists only of vertices $z \in V(U - S)$ satisfying $a \prec z \prec b$. Since $I \supseteq I_0$ is a connected subgraph of $U - S$ containing w and u , it follows that $a \prec w \prec b$. So $x \prec a \prec w \prec b$, and thus $x \prec_U a \prec_U w \prec_U b$. This contradicts Lemma 7.6 with $x, w \in V(H_1)$ and $a, b \in V(H_2)$. Thus, we have reached a contradiction in all scenarios. So we can conclude that $I \cup I'$ intersects at most one of H_1, H_2 . This completes the proof of the second property, and the lemma. $\square$

The sub-pseudographs $\mathcal{H}$ provided to Lemma 7.36 will be the sub-pseudographs in the break, along with their neighbourhoods (minus any overlap with other elements of the break). We now use the break to find a raise.

Lemma 7.37. *Let $b, k \in \mathbb{N}$ and $w, w', q, q' \in \mathbb{R}^{\geq 0}$, and let $\mathcal{PR}$ be a (b, k, w, w', q) -partitioned-reduction of an almost-embedding Γ of a pseudograph G that admits a break $\mathcal{B}$ of size at most q' . Then Γ admits a $(b, k, w, w', q + 4(k + 1)q')$ -partitioned-reduction $\mathcal{PR}'$ that admits a raise τ .*

Proof. Let $(Y, X, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+) := \mathcal{PR}$, $B' := B'(\mathcal{PR}) = V(G_0^+ \cap H')$, $U' := U'(\mathcal{PR})$, $G'_0 := G'_0(\mathcal{PR}) = G_0^+ - V(W)$, $\Lambda := \Lambda(\mathcal{PR}) = (G', G_0^+, W, D, H', \mathcal{J}')$, and $(J'_x : x \in B') := \mathcal{J}'$. So Λ is a standardised plane+quasi-vortex embedding.

Let $\mathcal{P}_{\mathcal{B}}$ be the set of $P \in \mathcal{P}_0^+$ distinct from $V(W)$ such that $G'_0[P] \in \mathcal{B}$ (recall that $P \subseteq V(G'_0)$ as $V(W) \in \mathcal{P}_0^+$). So by definition of $\mathcal{B}$, for each $M \in \mathcal{B}$, either $V(M)$ is disjoint to B' or $M = G'_0[P]$ for some $P \in \mathcal{P}_{\mathcal{B}}$.

For each $M \in \mathcal{B}$, recall that M is a connected sub-pseudograph of G'_0 . Let $M' := G'_0[N_{G'_0}(V(M)) \setminus \bigcup_{M^* \in \mathcal{B} \setminus \{M\}} V(M^*)]$. Note that $V(M) \subseteq V(M')$ (since the sub-pseudographs in $\mathcal{B}$ are pairwise disjoint), and thus M' is a connected sub-pseudograph of G'_0 .

Let $\mathcal{H} = \bigcup_{M \in \mathcal{B}} \{M\} \cup \{M'\}$. Observe that $|\mathcal{H}| \leq 2|\mathcal{B}| \leq 2q'$, and that $\mathcal{B} \subseteq \mathcal{H}$. Thus, by Lemma 7.36 (with $G := G'_0$), there exists a set $S \subseteq V(U') = B'$ with $|S| \leq 2|\mathcal{H}| \leq 4q'$ such that if $\mathcal{I}$ is the set of connected and nonempty subgraphs of $U' - S$, then for each pair of disjoint $Q_1, Q_2 \in \mathcal{H}$, the following hold:

1. each $I \in \mathcal{I}$ intersects at most one of Q_1, Q_2 , and
2. if $I, I', I_0, I'_0 \in \mathcal{I}$ are such that:
 - (a) $I_0 \subseteq I$,
 - (b) $I'_0 \subseteq I'$,
 - (c) $I_0 \cup I'_0$ is disjoint to $Q_1 \cup Q_2$, and
 - (d) $N_{G'_0}(V(I'_0))$ intersects $V(I_0)$,
 then $I \cup I'$ intersects at most one of Q_1, Q_2 .

Since $\mathcal{R}$ is a reduction of Γ , Λ is standardised, and thus D is snug in G'_0 . So $U' \subseteq G'_0$. Thus, for each $I \in \mathcal{I}$, $I \subseteq G'_0$. In particular, I is a connected and nonempty subpseudograph of G'_0 .

Recall that $S \subseteq B'$ and that $|S| \leq 4q'$. Let $\mathcal{PR}' := \mathcal{PR} - S$. By Observation 7.33 (with $q' := 4q'$), $\mathcal{PR}'$ is a $(b, k, w, w', q + 4(k + 1)q')$ -partitioned-reduction of Γ . It remains to find a raise for $\mathcal{PR}'$. The next claim assigns to each $I \in \mathcal{I}$ a part $P_I \in \mathcal{P}_0^+$ obeying some specific properties, which is used to construct the raise.

Claim 1. *For each $I \in \mathcal{I}$, there exists exactly one $P_I \in \mathcal{P}_0^+$ for which at least one of the following is true.*

1. $V(I) \subseteq P_I$, or,
2. $P_I \in \mathcal{P}_B$ and $V(I)$ intersects P_I .

Proof. We first show that such a P_I exists. Recall that $I \subseteq G'_0$ is connected. Thus, by definition of $\mathcal{B}$, either I intersects some $M \in \mathcal{B}$, or $V(I) \subseteq P$ for some $P \in \mathcal{P}_0^+$. In the latter case, set $P_I := P$. In the former case, $V(I) \subseteq B'$ intersects $V(M)$, so by definition of $\mathcal{B}$ (and $\mathcal{P}_B$), $M = G'_0[P]$ for some $P \in \mathcal{P}_B$. Set $P_I := P$ in this case.

Next, we prove uniqueness. Presume otherwise. So there exists distinct $P, P' \in \mathcal{P}_0^+$ satisfying at least one of the two conditions. Note that since I is nonempty (by definition of $\mathcal{I}$), if $V(I)$ is contained in either P or P' , then it intersects P or P' respectively. So in either scenario, $V(I)$ intersects both P and P' . Thus, if $V(I)$ is contained in either P or P' , then P intersects P' , contradicting the fact that P, P' are distinct parts of $\mathcal{P}_0^+$. Thus, we find that $P, P' \in \mathcal{P}_B$ and that $V(I)$ intersects both P and P' . By definition of $\mathcal{P}_B$, we find that $G'_0[P], G'_0[P'] \in \mathcal{B} \subseteq \mathcal{H}$. Noting that P, P' are disjoint, we have found disjoint elements of $\mathcal{H}$ that both intersect an element of $\mathcal{I}$ (namely I), a contradiction (by choice of S). Thus, P_I is unique, as desired. ■

For each $I \in \mathcal{I}$, we let P_I be from Claim 1. Since I is nonempty, observe that P_I intersects $V(I)$. The goal of the next few claims is to control how the neighbourhoods around $I \in \mathcal{I}$ and the corresponding $P_I \in \mathcal{P}_0^+$ interact.

Claim 2. *For each $I \in \mathcal{I}$, if $P \in \mathcal{P}_0^+$ is not $V(W)$ and is disjoint to $V(I)$, then P is disjoint to $N_{G_0^+}(V(I) \setminus P_I)$.*

Proof. Presume otherwise. Since $V(W) \in \mathcal{P}_0^+$, observe that $P \subseteq V(G'_0)$ and P intersects $N_{G'_0}(V(I) \setminus P_I)$. Thus, there exists a (nonempty) connected component I' of $I - (P_I \cap V(I))$ such that P intersects $N_{G'_0}(V(I'))$. Note that $I' \in \mathcal{I}$, and $I' \subseteq G'_0$. Since $\mathcal{P}_0^+$ is a connected partition of G_0^+ , $G_0^+[P] = G'_0[P]$ is connected. Thus and since P intersects $N_{G'_0}(V(I'))$, $G'_0[V(I') \cup P]$ is a connected sub-pseudograph of G'_0 . By definition of $\mathcal{B}$, either $V(I') \cup P \subseteq P'$ for some $P' \in \mathcal{P}_0^+$, or $V(I') \cup P$ intersects M for some $M \in \mathcal{B}$.

Presume, for a contradiction, that the former holds. Since P is nonempty (as P intersects $N_{G'_0}(V(I'))$) and since $P, P' \in \mathcal{P}_0^+$, we have $P' = P$. Thus, $V(I') \subseteq P$. However, recall that I' is a nonempty subgraph of I . Thus, P intersects $V(I)$, a contradiction (by choice of P). Hence, we may assume that the latter scenario holds. So $G'_0[V(I') \cup P]$ intersects M for some $M \in \mathcal{B}$. Thus, either I' intersects M , or $G'_0[P]$ intersects M .

Presume, for a contradiction, that the former holds. Since $I' \subseteq I$ is nonempty, $V(I) \not\subseteq P_I$. Thus, by definition of P_I , we have $P_I \in \mathcal{P}_\mathcal{B}$ and P_I intersects $V(I)$. So $G'_0[P_I] \in \mathcal{B}$, and $G'_0[P_I]$ intersects I . Since $V(M)$ intersects $V(I') \subseteq B'$, by definition of $\mathcal{B}$ (and $\mathcal{P}_\mathcal{B}$), $M = G'_0[P']$ for some $P' \in \mathcal{P}_\mathcal{B}$. Since $I' \subseteq I - (P_I \cap V(I))$ (and since I' intersects M), $P' \neq P_I$. Thus and since $P', P_I \in \mathcal{P}_\mathcal{B} \subseteq \mathcal{P}_0^+$, P' and P_I are disjoint. So $G'_0[P']$ is disjoint to $G'_0[P_I]$. Thus, I intersects $G'_0[P_I]$ and $G'_0[P']$ (as $I' \subseteq I$), and $G'_0[P'], G'_0[P_I] \in \mathcal{B} \subseteq \mathcal{H}$ are disjoint. This contradicts the choice of S .

Hence, we may assume that the latter scenario holds (and the former does not, for any $M \in \mathcal{B}$). So P intersects $V(M)$. By definition of $\mathcal{B}$, we have $P \subseteq V(M)$. Thus, $N_{G'_0}(P) \setminus \bigcup_{M^* \in \mathcal{B} \setminus \{M\}} V(M^*) \subseteq V(M')$. Since P intersects $N_{G'_0}(V(I'))$, $N_{G'_0}(P)$ intersects $V(I')$. Since the former scenario was false, I' does not intersect any $M^* \in \mathcal{B}$. Thus, M' intersects $I' \subseteq I$.

By the same argument as in the previous scenario, P_I intersects $V(I)$, $P_I \in \mathcal{P}_\mathcal{B} \subseteq \mathcal{P}_0^+$, and $G'_0[P_I] \in \mathcal{B} \subseteq \mathcal{H}$. Since P is disjoint to $V(I)$, P is distinct to P_I , and thus disjoint to P_I . Since P intersects M , it follows that M is not $G'_0[P_I]$. Since distinct elements of a break are pairwise disjoint, M is disjoint to $G'_0[P_I]$. Further, since $P_I \subseteq \bigcup_{M^* \in \mathcal{B} \setminus \{M\}} V(M^*)$, we find that M' is disjoint to $G'_0[P_I]$. Recall that $V(I)$ intersects P_I , we conclude that $I \in \mathcal{I}$ intersects $G'_0[P_I] \in \mathcal{B} \subseteq \mathcal{H}$ and $M' \in \mathcal{H}$, where $G'_0[P_I]$ and M' are disjoint. This contradicts the choice of S . Hence, we arrive at a contradiction in all cases. This completes the proof of the claim. $\blacksquare$

Claim 3. *For each $I \in \mathcal{I}$ and each $P \in \mathcal{P}_0^+$ that intersects $N_{G_0^+}(V(I))$, P intersects $N_{G_0^+}(P_I)$.*

Proof. Recall that $V(I) \subseteq B' \subseteq V(G_0^+)$. So $N_{G_0^+}(V(I))$ is well-defined. If $P = P_I$, then observe that $P = P_I$ is nonempty (since P intersects $N_{G_0^+}(V(I))$), so $P = P_I$ intersects $N_{G_0^+}(P_I)$. Thus, we may assume that P is distinct from P_I .

We first consider the case when P intersects $V(I)$. Since P is distinct from P_I and since P_I is uniquely defined (by Claim 1), observe that $P \notin \mathcal{P}_B$ and $V(I) \not\subseteq P$. Thus (and since I is connected), there exists a connected and nonempty subgraph I' of I such that $|V(I')| \geq 2$ and $|V(I') \setminus P| = 1$. Fix $v \in V(I') \setminus P$. So $V(I') \cap P = V(I' - v)$ is nonempty. Note that $I' \in \mathcal{I}$, and thus $I' \subseteq G'_0$.

Since P_I is distinct from P and since $V(I')$ intersects P , we obtain $V(I') \not\subseteq P_{I'}$. Thus, $P_{I'} \in \mathcal{P}_B$ and $P_{I'}$ intersects $I' \subseteq I$. Thus, by uniqueness of P_I (by Claim 1), we have $P_I = P_{I'}$. So P_I intersects I' . Since P_I is distinct from (and thus disjoint to) P and since $V(I' - v) \subseteq P$, we obtain $v \in P_I$. Since I' is a connected sub-pseudograph of G'_0 and since $V(I' - v)$ is a nonempty subset of P , it follows that P intersects $N_{G'_0}(P_I) \subseteq N_{G_0^+}(P_I)$, as desired.

Thus, we may assume that P does not intersect $V(I)$. So P intersects $N_{G_0^+}(V(I)) \setminus V(I)$. If P is $V(W)$, recall that $B' = B(\Lambda) \subseteq N_{G_0^+}(V(W)) = N_{G_0^+}(P)$ as Λ is standardised. Since $V(I)$ is contained in B' , we obtain $V(I) \subseteq N_{G_0^+}(P)$. Since $V(I)$ intersects P_I (as I is nonempty by definition of $\mathcal{I}$), $N_{G_0^+}(P)$ intersects P_I , and thus $N_{G_0^+}(P_I)$ intersects P . Otherwise, by Claim 2, P is disjoint to $N_{G_0^+}(V(I) \setminus P_I)$. Since P intersects $N_{G_0^+}(V(I))$, we find that P intersects $N_{G_0^+}(P_I)$, as desired. Thus, in all cases, P intersects $N_{G_0^+}(P_I)$. This completes the proof of the claim. ■

Claim 4. For each pair $I, I' \in \mathcal{I}$, if $N_{G_0^+}(V(I))$ intersects $V(I')$, then $N_{G_0^+}(P_I)$ intersects $P_{I'}$.

Proof. If $N_{G_0^+}(V(I))$ intersects $P_{I'}$, then by Claim 3, $P_{I'}$ intersects $N_{G_0^+}(P_I)$, as desired. If $N_{G_0^+}(P_I)$ intersects $V(I')$, then P_I intersects $N_{G_0^+}(V(I'))$. By Claim 3, P_I intersects $N_{G_0^+}(P_{I'})$, and $P_{I'}$ intersects $N_{G_0^+}(P_I)$, as desired. Thus, it remains only to consider the case when $N_{G_0^+}(V(I) \setminus P_I)$ intersects $V(I') \setminus P_{I'}$.

Presume, for a contradiction, that P_I is distinct to $P_{I'}$. Since $N_{G_0^+}(V(I) \setminus P_I)$ intersects $V(I') \setminus P_{I'}$, both $V(I) \setminus P_I$ and $V(I') \setminus P_{I'}$ are nonempty. Thus, by definition of $P_I, P_{I'}$, we have that $P_I, P_{I'} \in \mathcal{P}_B$, and that $V(I)$ intersects P_I and $V(I')$ intersects $P_{I'}$. So $I \cup I'$ intersects both $G'_0[P_I]$ and $G'_0[P_{I'}]$. Since $P_I, P_{I'} \in \mathcal{P}_B$, $G'_0[P_I], G'_0[P_{I'}] \in \mathcal{B} \subseteq \mathcal{H}$. By uniqueness of $P_I, P_{I'}$ (by Claim 1) and since $P_I, P_{I'}$ are distinct, $G'_0[P_I]$ does not intersect I' and $G'_0[P_{I'}]$ does not intersect I .

Since $N_{G_0^+}(V(I) \setminus P_I)$ intersects $V(I') \setminus P_{I'}$ and since I and I' are connected, there exists (nonempty) connected components I_0, I'_0 of $I - (V(I) \cap P_I)$ and $I' - (V(I') \cap P_{I'})$

respectively such that $N_{G_0^+}(V(I_0))$ intersects $V(I'_0)$. Note that $I_0, I'_0 \in \mathcal{I}$.

Since $V(I_0), V(I'_0) \subseteq B' \subseteq V(G'_0)$, it follows that $N_{G'_0}(V(I_0))$ intersects $V(I'_0)$. By definition of I_0 and I'_0 , they are disjoint to $G'_0[P_I]$ and $G'_0[P_{I'}]$ respectively. Thus and since $G'_0[P_I]$ does not intersect $I' \supseteq I'_0$ and $G'_0[P_{I'}]$ does not intersect $I \supseteq I_0$, $I_0 \cup I'_0$ is disjoint to $G'_0[P_I] \cup G'_0[P_{I'}]$.

So:

1. $I_0 \subseteq I$,
2. $I'_0 \subseteq I'$,
3. $I_0 \cup I'_0$ is disjoint to $G'_0[P_I] \cup G'_0[P_{I'}]$,
4. $N_{G'_0}(V(I_0))$ intersects $V(I'_0)$, and
5. $I \cup I'$ intersects both $G'_0[P_I]$ and $G'_0[P_{I'}]$

Because $P_I, P_{I'} \in \mathcal{P}_0^+$ are distinct, $G'_0[P_I], G'_0[P_{I'}]$ are disjoint. Since $G'_0[P_I], G'_0[P_{I'}] \in \mathcal{B} \subseteq \mathcal{H}$ and since $I, I', I_0, I'_0 \in \mathcal{I}$, this contradicts the choice of S .

Thus, we can conclude that P_I is $P_{I'}$. Noting that $V(I)$ and $V(I')$ intersect $P_I = P_{I'}$ (as I, I' are nonempty, by definition of $\mathcal{I}$), we find that $P_I = P_{I'}$ is nonempty. Thus, $N_{G_0^+}(P_I)$ intersects $P_I = P_{I'}$, as desired. This completes the proof of the claim. $\blacksquare$

We are now finally ready to construct the raise. Recall that $\mathcal{PR}' = \mathcal{PR} - S = (Y, X \cup X_S, G', G_0^+, W, D, H', \mathcal{J}', \phi, \mathcal{P}_0^+)$, where $X_S = \bigcup_{s \in S} J'_s$. Note that $X_S \subseteq V(H')$. Thus, observe that $V(G') = (V(G_0^+) \setminus B') \sqcup X_S \sqcup (V(H') \setminus X_S)$. For each $v \in V(H') \setminus X_S$, let I_v be the connected and nonempty subgraph of U' induced by the vertices $x \in B'$ such that $v \in J'_x$. Since $v \notin X_S$, observe that $V(I_v)$ is disjoint to S , and thus $I_v \in \mathcal{I}$. Let $P'_v := P_{I_v}$. So $P'_v \in \mathcal{P}_0^+$, and either $V(I_v) \subseteq P'_v$, or $P'_v \in \mathcal{P}_B$ and P'_v intersects $V(I_v)$. Either way, since I_v is nonempty, $V(I_v)$ intersects P'_v . Thus, by definition of I_v , there exists $x_v \in P'_v \cap V(I_v) \subseteq P'_v \cap B'$ such that $v \in J'_{x_v}$. Since $\mathcal{P}_0^+$ is a partition of G_0^+ , for each $v \in V(G_0^+)$, there exists exactly one $P_v \in \mathcal{P}_0^+$ such that $v \in P_v$.

Recall that $V(G') = (V(G_0^+) \setminus B') \sqcup X_S \sqcup (V(H') \setminus X_S)$. Define a map $\tau : V(G') \mapsto \mathcal{P}_0^+$ as follows. For each $v \in V(G_0^+) \setminus B'$, set $\tau(v) := P_v$. For each $v \in X_S$, there exists $s \in S \subseteq B'$ such that $v \in J'_s$. Set $\tau(v) := P_s$. For each $v \in V(H') \setminus X_S$, set $\tau(v) := P'_v$. If $v \in V(G_0^+) \setminus B'$, recall that $\tau(v) = P_v$ and $v \in P_v$. So $v \in \tau(v)$. If $v \in X_S$, recall that $\tau(v) = P_s$, where $s \in S \subseteq B'$, $v \in J'_s$, and $s \in P_s = \tau(v)$. If $v \in V(H') \setminus X_S$, recall that $\tau(v) = P'_v$, $x_v \in B' \cap P'_v = B' \cap \tau(v)$, and $v \in J'_{x_v}$. Thus, if $v \in V(H')$, there exists $x \in B' \cap \tau(v)$ such that $v \in J'_x$. It remains only to show that for each pair of adjacent $u, v \in V(G')$ such that $\phi(u), \phi(v) \in V(G - (X \cup X_S))$, $N_{G_0^+}(\tau(u))$ intersects $\tau(v)$.

First, consider the case when $u, v \in V(G_0^+) \setminus B'$. Then $\tau(u) = P_u$ and $\tau(v) = P_v$, where

$u \in P_u$ and $v \in P_v$. Since $u, v \in V(G_0^+) \setminus B'$ and since $G' = G_0^+ \cup H'$, observe that u, v are adjacent in G_0^+ . Thus, $v \in N_{G_0^+}(u) \subseteq N_{G_0^+}(P_u) = N_{G_0^+}(\tau(u))$. Since $v \in P_v = \tau(v)$, we find that $N_{G_0^+}(\tau(u))$ intersects $\tau(v)$, as desired.

Next, consider the case when $u, v \in V(H')$ and are adjacent in H' . Since W is disjoint to H' , observe that $u = \phi(u), v = \phi(v) \in V(G - (X \cup X_S))$. In particular, neither u nor v is in X_S . So $u, v \in V(H' - X_S)$ (as $X_S \subseteq H'$), and are adjacent in $H' - X_S$. Since $u, v \in V(H' - X_S) = V(H') \setminus X_S$, recall that $\tau(u) = P'_u = P_{I_u}$ and $\tau(v) = P'_v = P_{I_v}$. Since u, v are adjacent in $H' - X_S \subseteq H'$, I_u and I_v intersect. Thus, $I_{uv} := I_u \cup I_v$ is connected (and nonempty and disjoint from S), so $I_{uv} \in \mathcal{I}$. Let $P'_{uv} := P_{I_{uv}}$. Recall that $\tau(u) = P'_u$ and $\tau(v) = P'_v$ intersect $V(I_u)$ and $V(I_v)$ respectively, and hence $\tau(u), \tau(v)$ intersect $V(I_{uv}) \subseteq N_{G_0^+}(V(I_{uv}))$. Thus, by Claim 3, $N_{G_0^+}(P'_{uv})$ intersects both $\tau(u), \tau(v)$. Hence, if either $P'_{uv} = \tau(u)$ or $P'_{uv} = \tau(v)$, we find that $N_{G_0^+}(\tau(u))$ intersects $\tau(v)$, as desired. So we may assume that P'_{uv} is distinct from both $P'_u = \tau(u)$ and $P'_v = \tau(v)$.

Since P'_u intersects $V(I_u) \subseteq V(I_{uv})$ and since P'_v intersects $V(I_v) \subseteq V(I_{uv})$, by the uniqueness of P'_{uv} (by Claim 1), neither $P'_u \in \mathcal{B}$ nor $P'_v \in \mathcal{B}$. Thus (by Claim 1), $V(I_u) \subseteq P'_u$ and $V(I_v) \subseteq P'_v$. Since I_u and I_v intersect, $P'_u, P'_v \in \mathcal{P}_0^+$ intersect. Thus, $P'_u = \tau(u)$ is $P'_v = \tau(v)$. Hence (and since $P'_u = P'_v$ is nonempty as $V(I_u), V(I_v) \subseteq P'_u = P'_v$ are nonempty), $\tau(u)$ intersects $\tau(v)$. So $N_{G_0^+}(\tau(u))$ intersects $\tau(v)$, as desired.

Since $G' = G_0^+ \cup H'$, it remains only to consider the case when $u, v \in V(G_0^+)$, are adjacent in G_0^+ , and at least one of u, v are in B' . Without loss of generality, say $u \in B'$. As before, we find that $u \in V(H') \setminus X_S$. So $\tau(u) = P'_u = P_{I_u}$. Observe that $u \in V(I_u)$ (as quasi-vortex-decompositions are planted), and thus $v \in N_{G_0^+}(V(I_u))$. Since $v \in P_v$, by Claim 3, $N_{G_0^+}(P'_u)$ intersects P_v . If $v \in G_0^+ \setminus B'$, $\tau(v) = P_v$, and thus $N_{G_0^+}(P'_u) = N_{G_0^+}(\tau(u))$ intersects $P_v = \tau(v)$, as desired. So it remains only to consider when $u, v \in B'$. In this case, we have $\tau(u) = P'_u = P_{I_u}$ and $\tau(v) = P'_v = P_{I_v}$. Noting that $v \in V(I_v)$, $N_{G_0^+}(V(I_u))$ intersects $V(I_v)$. By Claim 4, $N_{G_0^+}(P'_u) = N_{G_0^+}(\tau(u))$ intersects $P'_v = \tau(v)$, as desired. Hence, in all cases, $N_{G_0^+}(\tau(u))$ intersects $\tau(v)$. Thus, τ is a raise for $\mathcal{PR}'$. This completes the proof. $\square$

We now prove Lemma 7.18, which we restate here.

Lemma 7.18. *Let $b, k, n \in \mathbb{N}$, $w, w', q \in \mathbb{R}^{\geq 0}$, and $d \in \mathbb{R}^{>0}$ with $d \geq w$, and let Γ be an almost-embedding of an n -vertex pseudograph G . If Γ admits a (b, k, w, w', q) -partitioned-reduction $\mathcal{PR}$, then there exists a $(b, k, d, w', 28(k+1)n/d + q)$ -partitioned-reduction $\mathcal{PR}'$ of Γ that admits a raise.*

Proof. By Lemma 7.35, there exists a (b, k, d, w', q) -partitioned-reduction $\mathcal{PR}''$ of Γ that admits a break of size at most $7n/d$. By Lemma 7.37 (with $w := d$ and $q' := 7n/d$),

there exists a $(b, k, d, w', 28(k+1)n/d + q)$ -partitioned-reduction $\mathcal{PR}'$ of Γ that admits a raise. $\square$

This completes the proof of Theorem 2.9 (n -vertex K_h -minor-free graphs are $O(\sqrt{n})$ -blowups of treewidth 3 graphs). As mentioned in Section 2.1.2, n -vertex planar graphs are not $O(\sqrt{n})$ -blowups of trees (or forests, as we can assume that the base graph is connected). This leaves the open question ‘are n -vertex K_h -minor-free graphs $O(\sqrt{n})$ -blowups of treewidth 2 graphs’. For the same reason as argued in Section 6.1, if true (for every $h \in \mathbb{N}$), it would imply that these graphs are $O(\sqrt{n})$ -blowups of apex treewidth 1 graphs; that is, graphs for which we can delete a vertex to obtain a forest. A key case is whether n -vertex planar graphs are $O(\sqrt{n})$ -blowups of apex treewidth 1 graphs.

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